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# GOVERNMENT OF
# MADHYA PRADESH PUBLIC
# WORKS DEPARTMENT

# SCHEDULE OF RATES
## *FOR*
# ROAD & BRIDGE WORKS

## IN FORCE FROM APRIL 11th, 2025

*ISSUED BY -*

## ENGINEER-IN-CHIEF
## PUBLIC WORKS DEPARTMENT
## M.P. BHOPAL

**Price: Rs. 1000/-**

**(Excluding Postage and forwarding Charges etc.)**

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# FOREWORD

The Schedule of Rates (SOR) for Roads and Bridge Works, published periodically by the Public Works Department (PWD), is a comprehensive document designed to facilitate the execution of works. It also serves as a valuable reference for various government departments and public sector undertakings.

Since its inception, the PWD's Schedule of Rates has been periodically revised to align with evolving technologies and prevailing market rates. The previous edition was published on 25th March 2022. Since then, there have been significant fluctuations in the prices of labour and materials, along with the introduction of new construction materials and techniques aimed at accelerating project execution. Consequently, revising the existing SOR became essential to reflect these developments. This updated edition incorporates modern machinery, advanced construction materials, and innovative techniques, wherever applicable.

Special attention has been given to ensuring the accuracy of this SOR. Emphasis has been placed on integrating modern technologies, updating survey items for greater precision in Detailed Project Reports (DPRs), and including provisions to support maintenance activities—thereby enhancing the durability of existing infrastructure. Environmental sustainability has also been a key focus during its preparation. This SOR has been developed based on the Standard Data Book 2019.

This edition retains most items from the SOR 2022 while incorporating all addendums issued to date. The nomenclature of existing items has been reviewed and updated where necessary, and similar items have been grouped together for ease of reference.

Every effort has been made to ensure the accuracy of this document; however, inadvertent errors or omissions may still occur. If any discrepancies are noticed, users are encouraged to bring them to the attention of this office.

I extend my heartfelt gratitude to the following individuals for their valuable contributions in bringing this Schedule of Rates into effect: Er. B.P. Bourasi (Chief Engineer), Er. A.P. Rane (Chief Engineer), Er. Ajay Anand Likhar (Superintending Engineer), Er. K.K. Lachhe (Superintending Engineer), Er. R.D. Choudhary (Superintending Engineer), Er. Sunil Dahariya (Superintending Engineer, E&M), Er. Javed Shakil (Executive Engineer), Er. Manoj Gupta, General Manager,MPRDC, Er. Avinash Shivariya (Executive Engineer), Smt. Savita Malviya (Executive Engineer), Er. Kuldeep Singh (Executive Engineer), Er. Naveen Malhotra (Executive Engineer), Er.G.S. Saxena (Assistant Engineer), Er. Devendra Kumar Anuragi (Assistant Engineer), Er. Ashwani Kumar Dinkar (Sub Engineer, E&M). I would like to extend my thanks to all Chief Engineers and their team for valuable suggestions.

I am sure that this Schedule of Rates will be quite useful to all concerned in the construction industry in general and PWD in particular.

Suggestions for further improvement of this Schedule of Rates for Road and Bridge Works are highly appreciated.

This SOR is available on the Department's website www.mppwd.gov.in and shall be effective from April 11th, 2025.

(**Er. K.P.S. Rana)**
**Engineer-in-Chief**
**M.P. Public Works Department**

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# INTRODUCTION

The current Schedule of Rates (SOR) was issued on 25.03.2022. However, it did not incorporate advanced road and bridge construction techniques developed after 2020 and the rates of several items have since changed. Therefore, the department has prepared this revised SOR, incorporating various updates and reforms. This SOR includes rates for items of Electrification on Roads and Bridges.

Previously, tree-cutting was included as a standard site clearance activity, leading to the destruction of many trees. In this SOR, tree-shifting has been introduced as an alternative to tree-cutting, which will help preserve thousands of trees and contribute to environmental conservation. Additionally, new items for tree plantation have been added, specifying the planting of trees ranging from 1 to 3 meters in height on both sides of the road. These trees will be maintained under strict contractor's supervision for 3 to 5 years, thereby enhancing greenery and ecological balance.

To promote water conservation, this SOR also includes provisions for rainwater harvesting and the recharge of existing systems. These measures will play a crucial role in maintaining groundwater levels at project sites.

Additionally, the soil shall be sourced from an approved quarry or borrow area, and excavation shall be carried out as per the approved plan from the Engineer-in-Charge. Subsequently, the borrow area may be developed into a Lok Sarovar.

Furthermore, modern road construction techniques utilizing recycled materials from old roads have been incorporated. These techniques include Full Depth Reclamation (FDR), White Topping, Ultra-High-Performance Concrete (UHPC), Glass Fiber Reinforced Polymer (GFRP), Waste Plastic, Geomesh, and Stamped Concrete. Implementing these methods will reduce costs, save time, and mitigate various construction-related challenges. Items related to White Topping have also been added to the SOR.

In view of the importance of quality control, the rates for important testing equipment required for laboratory and the lab testing rates, test frequency table have also been included in this SOR.

Several other innovative road construction techniques have been included, providing engineers with the flexibility to adopt modern, efficient, and sustainable practices.

All road and bridge construction work shall be carried out as per the norms specified in IRC codes, IS codes, and ASTM specifications. However, the specifications provided by the Ministry of Road Transport and Highways (MoRT&H) shall take precedence over all other codes.

This revised SOR is a significant step towards sustainable and efficient infrastructure development, ensuring environmental protection and the adoption of modern construction technologies.

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**This SOR is based on Standard Data book 2019 published by MORT&H for analysis of Rates for Road and Bridge Works.**

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# "Guidelines for Ensuring Quality in Road Construction Works"

The Public Works Department, Madhya Pradesh (MPPWD) undertakes construction and upgradation of roads, including both flexible (bituminous) and rigid (cement concrete) pavements. MPPWD is committed to maintaining high quality in all works. These guidelines aim to ensure proper materials, methods, and skilled manpower are used as per approved standards. They form part of the Schedule of Rates (SOR) and must be followed along with IRC, MoRT&H, and departmental specifications. The contractor shall execute all works as per relevant IRC/MoRT&H guidelines and specifications for quality assurance and control.

1. The contractor shall construct trial stretches of bituminous and cement concrete pavement as per MoRT&H/IRC specifications and obtain approval from the Engineer-in-Charge before undertaking regular works.
2. The contractor shall set up a field laboratory at the project site, as per the provisions of the agreement, and deploy qualified and experienced technical staff for quality control and testing activities.
3. The contractor shall deploy technically qualified personnel for execution and supervision of works, in accordance with the terms of the agreement.
4. Only such plant and equipment as are specified and approved shall be deployed for execution of the works. No work shall be executed using substandard or unapproved machinery.
5. All materials proposed to be incorporated in the works shall be approved by the Engineer-in-Charge before use. Necessary samples shall be submitted for approval as per specifications.
6. The contractor shall maintain records of all tests carried out in the field laboratory/departmental laboratory/NABL-accredited laboratories as per the prescribed frequency. Further, records of all inspection reports, Request for Inspections (RFIs), and approvals shall be maintained properly as part of the quality control records.
7. Appropriate items have been incorporated in the Schedule of Rates (SOR) to encourage adoption of innovative technologies such as Full Depth Reclamation (FDR), "Lok Sarovar" (ponds created in borrow areas), roadside groundwater recharge pits, etc.
8. The contractor shall comply with all applicable labour laws, environmental laws, and local statutory provisions during execution of works.
9. The contractor shall adopt sustainable and environmentally responsible practices in accordance with the Environmental Management Plan (EMP) specified in the DPR/agreement.

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# GENERAL NOTES
# ROAD WORKS

1. References mentioned herein shall be applicable to all sections to the extent the context, permits and are intended to supplement the provisions in the particular section. In case of any discrepancy/ deviation, the provisions in the particular section shall take precedence.

2. In absence of any stipulation to the contrary, unit rates for various items of works are for completing the works to the requirements of the specifications including full compensation for all operations and inclusive of all labour, materials, royalties, lease rent, wastage, temporary work, plant, equipment overhead charges and profit, unless specified otherwise.

**3. Interpretations**

The Engineer-in- Chief M.P. Public works Department shall be the sole deciding authority as to the meaning, interpretation and implications for various provisions in this publication. His decision shall be final and binding.

3.1 Wherever any reference is made to any Indian Standard, or any other specification e.g. ASTM, it shall be taken as reference to the latest edition with all amendments issued thereto. In the event of any variation between the MoRT&H specifications (adopted) and the Indian Standard, the former shall take precedence over the latter.

**4. Definitions**

The following terms and expressions in the SOR shall have the meaning or implication hereby assigned to them unless otherwise specified elsewhere.

4.1 (i) "Specifications" would refer to the "Specifications for road and bridge works" (5th Revision) published by the Indian Roads Congress April 2013, on behalf of the Government of India, Ministry of Road Transport & Highway (Roads Wing) and other relevant publications.

(ii) "Clause" would refer to clauses of the specifications referred to in note (i) above.

4.2 "Table" would refer to the table of the aforesaid specifications.

4.3 Contractor: The Contractor shall mean an individual or firm or company whether incorporated or not undertaking the works and shall include the legal personal representatives of such individual or the persons composing such firm or company, or the successors of such individual or firm or company and the permitted assignees of such individual or firm of company.

4.4 Engineer-in-Charge: The "Engineer-in Charge" means the Executive Engineer of MP PWD who shall supervise and shall be in-charge of the work.

4.5 Superintending Engineer means officer in-charge of the circle.

4.6 Chief Engineer means officer in-charge of Zone.

4.7 Engineer-in-chief means head of department of M.P. PWD.

4.8 Site: "The site" shall mean the land/or other places on, in, into or through which the work is to be executed under the contract or any adjacent land, path or street through which the work is to be executed under the contract, or any adjacent land, path or street which may be allotted or used for the purpose of carrying out the contract.

4.9 Department: "Department" mean M.P. Public Works Department.

**5. Measurements**

Mode of measurements shall be as per provisions contained in the relevant clauses of the specifications, unless specified otherwise.

5.1 The order of dimensions shall be consistent and in the sequence of length, width and height or depth or thickness.

5.2 Rounding off: Rounding off where required shall be done in accordance with IS: 2-1960. The number of significant places rounded in the rounded off value should be as specified.

5.3 Schedule of rates is based on the following basic rates of materials: -

| Sr. No. | Material | Basic rate (Excluding GST) |
|---|---|---|
| 1 | Cement/OPC | Rs. 5300/- Per M. T. |
| 2 | HT Wire | Rs. 58000/- Per M. T. |
| 3 | Structural Steel | Rs. 44500/- Per M. T. |
| 4 | Bitumen VG-40 (Bulk) | Rs. 52912/- Per M. T. Ex. Refinery Koyali |
| 5 | Bitumen VG-30 (Bulk) | Rs. 50162/- Per M. T. Ex. Refinery Koyali |
| 6 | Bitumen VG-10 (Bulk) | Rs. 47862/- Per M. T. Ex. Refinery Koyali |
| 7 | Bitumen Emulsion | Rs. 42099/- Per M. T. Ex. Refinery Koyali |

5.4 The rates include all such activities are as described in the relevant clauses of the Specifications, unless specified otherwise.

5.5 Rates include provisions of necessary precautionary devices and other arrangements etc. for traffic control, e.g. provision of caution boards, red lights, watchmen flags and flagmen but do not include construction of temporary diversion.

5.6 Detailed measurements for steel and concrete for items of RCC. approach slab, RCC. Railing and RCC wearing coat shall also be recorded in M.B. However, the payment shall be regulated as per the item and rates given in the SOR

5.7 The measurement of rock excavation are to be done as per the relevant clause of "Specification of Road and Bridge works" of MoRT&H 5th Revision. All serviceable rock excavated shall be issued to the contractor at the rate of Rs.100/- per Cum.

## 6 FIELD LABORATORIES

The work covers the provision and maintenance of an adequately equipped field laboratory as required for site control on the quality of materials and the works. The Contractor shall arrange to provide fully furnished and adequately equipped field laboratory. The field laboratory shall preferably be located adjacent to the site office of the Engineer and provided with amenities like a qualified lab assistant, material engineer, water supply, electric supply etc. as approved by the Engineer–in-charge and as per clause 120 of the MoRT&H 5th Revision.

## 7. Materials

7.1 Samples of all materials to be used on the work shall be got approved by the contractor from the Engineer-in-Charge well in time. The approved samples duly authenticated and sealed shall be kept in the custody of the Engineer-in-Charge till the completion of the work. All materials to be provided by the contractor shall be brand new and as per the samples approved by the Engineer-in-Charge.

Additionally, the soil shall be sourced from an approved quarry or borrow area, and excavation shall be carried out as per the approved plan from the Engineer-in-Charge. Subsequently, the borrow area may be developed into a Lok Sarovar.

7.2 For Echological balance, Environmental Management and Bio Engineering. It is necessary to construct road side ground water recharge pits at the discretion of the Engineer-in-Charge.

7.3 For work, if the amount of contract exceeds Rs. 200.00 lacs contractor shall provide an accommodation of at least 20 sqm where all the approved material shall be kept for display under lock and key of Engineer-in-Charge. It is the contractor's duty to obtain lease/permit for extraction of minor minerals. The contractor will not be entitled to any excuse, whatsoever on account of any delay in obtaining minor minerals to be used on the work.

7.4 Materials obtained by the contractor from the sources approved by the Department shall be subjected to the mandatory tests. Where such materials do not conform to the relevant specifications, the matter shall be taken up by the Engineer-in-Charge for appropriate action against the defaulters. In all such cases, necessary documents in original and proof of payment relating to the procurement of materials shall be made available by the contractor to the Engineer-in-Charge.

7.5 Samples, whether submitted for approval to govern bulk supplies or required for testing before use and also the sample of materials bearing 'Standard mark if required for testing, shall be provided free of cost by the contractor. All other incidental expenditure to be incurred for testing of samples e.g. packaging, sealing, transportation, loading, unloading etc. shall be borne by the contractor.

7.6 Materials stored at site, depending upon the individual characteristics, shall be protected from atmospheric effects due to rain, sun, wind and moisture to avoid deterioration.

7.7 It will not be obligatory on the part of engineer-in-charge to provide any assistance in obtaining lease/permits for excavation of minor minerals.

7.8 **Sub Grade**

(i) It is mandatory to follow the provisions of section 300 of Specification for Roads and Bridge works (Fifth Revision).
(ii) No sub grade soil should be procured from the right of way of the road under construction, if found so, work should be rejected.
(iii) In sub grade work such soil should not be used as sub grade material: -
(a) which is specified as OL, OI, OH or PT in accordance with IS :1498
(b) Free Soiling Index is greater than 50 %
(c) Liquid limit and plasticity index is greater than 50 and 25 respectively
(d) Maximum Dry Density is below 17.5 kn Cubic Meter
(e) CBR value of soil below 7 CBR or as directed by the Engineer-in-Charge.

(iv) Provisions of Section 305.4.1 to 305.4.3 for strengthening, widening and reconstruction of road. It is mandatory as per the Specification for Roads and Bridges (Fifth Revision).
(v) It is mandatory to take the initial ground level and send it to the Superintending Engineer one month before starting of the sub-grade work.
(vi) Plan and profile shall be approved by the competent authority.

7.9 **Sub Base**

(i) It is mandatory to follow the provisions of section 401 to 410 of Specification for Roads and Bridge works (Fifth Revision).
(ii) Sub base material passes through a 425-micron sieve, should have a liquid limit of less than 25 and a plasticity index less then 6 and other physical requirements according to table 400-2 of Specification for Roads and Bridges (Fifth Revision).
(iii) Compaction of the compacted layer of sub base should not be less then 98% according to IS 2720 (Part-8)
(iv) Drainage Layer in Sub Base (CRM) should be minimum 100 mm thick in full width of the road.

7.10 **Base course and surface course**

(i) It is mandatory to follow the provisions of section 501 to 520 of Specification for Roads and Bridge works (Fifth Revision).
(ii) Bitumen used for execution shall be procured from any government refinery e.g. IOCL, HPCL, BPCL etc.

A duplicate register as per format here under shall be maintained at site of work. Extra certified copies of the entries for each month/each running bill shall be submitted to the Engineer-in Charge by the Contractor. The original register shall also be submitted to the Engineer-in Charge on completion of the work by the Contractor.

(iii) All invoices shall be verified through the respective refineries' websites.
(iv) The record of bitumen procured shall be maintained in following format:-

| S. No | Date of receipt of Bitumen | Quantity in Tonne | Name and address of firm, from whom purchased. | Signature of contractor or his authorized representative | Signature of authorized representative of Engineer-in-charge | Result of Bitumen test | Remarks |
|---|---|---|---|---|---|---|---|
| | | | | | | | |

(v) Following number of test shall be conducted prior to use of bitumen.

7.10.1 Test for Determination of Specific Gravity Test as per IS 1202- 1978

7.10.2 Test for Determination of Penetration as per IS 1203 - 1978

7.10.3 Test for Determination of Residue of Specified Penetration as per IS 1204-1978

7.10.4 Test for Softening Point as per IS 1205-1978

7.10.5 Test for Viscosity as per IS 1206-1978 (Part I, II & III)

7.10.6 Test for Equi viscous Temperature (EVT) as per IS 1207-1978

7.10.7 Ductility test as per IS 1208-1978

7.10.8 Flash Point and Fire Point as per IS 1209-1978

7.10.9 Float Test as per IS 1210-1978

7.10.10 Water Content Test as per IS 1211-1978

7.10.11 Loss on Heating Test as per IS 1212-1978

7.10.12 Matter Insoluble in Toluene Test as per IS 1215-1978

7.10.13 Solubility in Carbon Di-sulphide or Trichloroethylene as per IS 1216-1978

7.10.14 Volatile Matter Content Test as per IS 1220-1978

7.10.15 Test for Solubility

(vi) Design mix of Bituminous item approved by the Engineer-in-charge before 7 days from the date of commencement of work.

7.11 **Concrete Pavement**

(i) It is mandatory to follow the provisions of section 601 to 602 of Specification for Roads and Bridge works (Fifth Revision), for design of concrete mix follow IRC:44 and IRC:58 and commencement of work according to IRC :15.

(A) **Use of VSI Crushed Sand / Artificial Sand/ Fine Aggregates.**

1. VSI Crushed sand / Artificially manufactured sand/fine aggregates hereinafter referred to as "Crushed Sand" shall be as defined under C1.3.1.2 of IS 383-2016.
2. The properties of "Crushed Sand" shall conform to the provisions of IS 383:2016.

3. The "Crushed Sand" shall be free of dust and other Deleterious material.

4. The "Crushed Sand" shall be manufactured using "Automatic Vertical Shaft Impactor "type Crusher only.

5. The quantity of Microfines (Particles below 75 microns) in "Crushed Sand" shall not be more than 7%.

6. The Contractor shall intimate the Engineer -in-Charge regarding the source of supply of "crushed sand". The source of supply of "crushed sand" shall be got approved by the Executive Engineer (Engineer-In-charge) prior to the start of work.
7. Each Load of crushed sand whenever brought on site shall be tested for "Fineness modulus". Fineness modulus shall be within permissible limits, If it doesn't fall within acceptable limits, it shall be rejected.
8. The Test of Compressive strength of concrete / Mortar using "Crushed sand" shall be carried out in presence of Department's Engineer as given below...

a) 100% Cube testing in presence of Sub Engineer/ field engineer

b) 25% Cube testing in the presence of Sub Divisional Officer/Assistant Resident Engineer
c) 5% Cube testing in the presence of Executive Engineer (Engineer-in-charge) /Resident Engineer.

9. The flakiness index and elongation index tests shall be within permissible limits.

10. The Concrete mix design for each grade of concrete using "Crushed sand" shall be carried out only in Government Quality Control laboratory and the same Mix Design shall be adopted.
11. As far as possible freshly produced "Crushed sand" shall be used. Stored "Crushed sand" shall not be used.
12. For Plastering purpose, if the use of crushed sand is proposed, it shall be used with addition of super plasticizers at the rate of 100ml/bag of cement without any extra cost to Government.
13. The following tests shall be carried out for the use of "Crushed sand".
a) Sieve analysis
b) Specific gravity
c) Water absorption
d) Bulk density
e) Alkali aggregate reaction

f) Soundness
g) Deleterious Material
h) Organic impurities
i) Micro fines Content
j) Test for silt and clay.
k) Fineness Modulus test.

14. Necessary Bond regarding the use of "Crushed sand" shall be submitted by the Contractor clearly stating that. if any defects are observed during execution and in defect liability period, the same shall be rectified at his own risk and cost.

15. Grading zone I and II mentioned under Clause 6.3 table 9 of fine aggregates in IS 383:2016 shall only be used for concreting.

16. Reversible drum type / Batch mix plant (pan type) concrete mixer shall be used for concrete.

17. "Crushed sand" shall not be used for prestressed concrete works.

(B) All concrete work shall be done as provided in section 1700 of "Specifications for Road and Bridge works" (Ministry of Road Transport & Highway) 5th Revision unless otherwise provided in the contract.
(C) Finishing of concrete by plastering the surface shall not be done without obtaining written permission from the Executive Engineer. No extra for plastering shall be payable. Light touching up and rubbing the uneven surfaces by carborundum stone shall be done within the specified rates.

(D) Minimum 28 days before commencement of actual work the construction of trial length is mandatory according to MORT&H specification.

(E) At any condition bitumen should not be used as sealant for sealing of the concrete joint. It is mandatory to use approved make silicon sealants.

18. The actual quantities of materials shall be as per job mix formula for bituminous works. Job mix design shall be approved by Engineer-in-charge prior to execution of work.

19. The measurement of bituminous works shall be cross checked by the actual weight of the mix, used at the site. The contractor shall submit the weigh slips in support of the bituminous work.

8. (i) **Trial Length for Rigid pavement**

The trial length shall be constructed at least 7 days in advance of the proposed date of commencement of work. At least 30 days prior to the construction of the trial length, the Contractor shall submit for the Engineer's approval a "Method Statement" giving detailed description of the proposed materials, plant, equipment, mix

proportions, and procedure for batching, mixing, laying, compaction and other construction procedures. The Engineer shall also approve the location and length of trial construction which shall be a minimum of 100 m length laid in two days and for full width of the pavement. The trial length shall be outside the main works. The trial length shall contain the construction of at least one transverse construction joint involving hardened concrete and freshly laid Dry Lean Concrete sub-base. The construction of trial length shall be repeated till the Contractor proves his ability to satisfactorily construct the Dry Lean Concrete sub-base and subsequently same length of PQC is to be constructed. Joint, i.e cutting of joints and filling them with poly sulphide sealant to the satisfaction of Engineer-in-Charge.

**(ii) Trial Length for flexible pavement**

Additionally, contractor in case of DBM and BC shall construct at least 10 days in advance of proposal date of commencement of work. At least 15 days. prior to the construction of the trial length, the Contractor shall submit for the Engineer's approval a "Method Statement" giving detailed description of the proposed materials, plant, equipment, mix proportions and procedure for batching, mixing, laying, compaction and other construction procedures. The Engineer shall also approve the location and length of trial construction which shall be a minimum of 50 m length laid in two days and for full width of the pavement. including sub-grade and sub base. The trial length shall be outside the main works.

9. **STEEL**

(i) TMT Steel confirming to IS-1786 can be used in place of Tor Steel.

(ii) Mild steel confirming to IS: 432:1982

(iii) Tor steel confirming to IS: 1786:2008

(iv) Solid Round Glass Fiber Reinforced Polymer (GFRP) bars confirming to IS: 18256:2023

(v) Steel used as reinforcement steel shall be measured as per clause 1608 of Specifications of Roads and Bridge works, MORT&H.

**10. Testing**

The rates also include the element of testing of samples of various materials brought by the contractor for use on the work, as well as other necessary tests for items of work as stipulated in the specifications, Frequency of such tests to be carried out must not be less than the prescribed frequencies. Copies of registers, containing records of tests shall have to be presented along with running account bills. Register (original) shall have to be submitted along with the final bill. Tests shall have to be conducted by the contractor's Engineer under the supervision of the Engineer-in-Charge or his authorized representatives.

The contractor shall have to establish a field laboratory at the site of work, in other cases, testing of construction materials should be got done from any of the testing laboratories of the various Government Departments, Government/Semi Govt. under takings and Technical Institutes, Engineering College, Polytechnic, I.T.I. etc. However, it is mandatory to get done 10% of each type of test through NABL laboratory and 10% of each type of test Circle/Zonal laboratory Contractor shall bear the cost of testing charges.

10.1 The work should not be accepted in any case, if the contractor fails to observe the instructions of the department, regarding testing of materials.

10.2 Before making any payment, it will be the responsibility of the officers making payment to assure that all tests are as per prescribed frequencies, have been carried out, and shall certify that the test results are in conformity with the requirement.

10.3 If tests are not conducted to the prescribed frequency, the Engineer-in-Charge should reject that part of the work.

11. The "Site order book" shall be in custody of departmental supervisory staff. The Engineer-in Charge or his authorized representative may record his instructions in this book, which shall be noted by the contractor or his authorized representative for compliance.

12. A duplicate register as per format here under shall be maintained at site of work. Extract certified copies of the entries for each month/each running bill shall be submitted to the Engineer-in Charge by the Contractor. The original register shall also be submitted to the Engineer-in Charge on completion of the work by the Contractor.

| S. No | Date of receipt of cement | No of bags/ bulks | Name and address of firm, from whom purchased. | Signature of contractor or his authorized represent ative | Signature of authorized represent ative of Engineer-in-charge | Result of test for initial and final setting time. | Result of tests for compre ssive strengt h of cement | Re m ar ks |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | | |

13. Where contract provides for cement to be arranged by the contractor himself, only I.S. Marked cement of relevant I.S. specifications shall be allowed to be used in the work subject to the following tests. The arrangement for necessary equipment and testing shall have to be made by the contractor himself at site, to be decided by the Engineer-in-Charge All expenses shall be borne by the contractor, any lot of cement brought to site by the contractor, would be permitted to be used in the work only after the satisfactory results of the tests, under the supervision of the Engineer-in-Charge or his authorized representative. The record of the tests result shall be maintained in the register referred in subsequent para

13.1 Test for initial and final setting time as per IS: 3536 1996, 1 Test for 10 tones or part thereof upto 10 Tones.

13.2 Test for determination of compressive strength of cement as per IS:3536-1996, 1 Test for 50 Tones or part thereof upto 10 Tones.

14. The aggregates to be used for all cement concrete items shall be crushed in the mechanical crushers.

15. Nothing extra shall be payable on account of use of Narmada Sand or sand brought from any other quarry/river. Sand should conform to the requirements of IS: 1542 and IS: 2166 for using it in construction work.

16. Rubble available from excavation of hard rock/ordinary rock shall be the property of the contractor subject to recovery of Rs. 100/- per cum. of the quantity of the rock excavated.

17. The aggregate available by milling process, minimum 30% quantity to be reused in bituminous layer. This quantity shall be the property of the contractor subject to recovery of Rs. 700/- per cum.

18. The girth of trees shall be measured at 1.00 Meter (One Meter) above ground level.

19. All wood obtained from the tree shall be property of the government and shall be deposited at P.W.D. store or as per direction of Engineer-in-Charge

20. If any new item is not included in SOR and is proposed to be executed in any ongoing work, proposal with full justification thereof along with financial implication and detailed rate analysis must be submitted to Superintending Engineer for obtaining approval as extra item. If any Non S.O.R. Item is to be executed in any proposed project, rate analysis with full justification thereof must be approved by the Superintending Engineer and shall be included in BOQ under sub head of Non SOR items. Tender shall be invited on amount inclusive of SOR and Non SOR items and in such cases Non SOR items shall become schedule items for that particular tender, however it is made clear that such Non SOR items shall not be assumed as sanctioned rates for other agreements.

21. In case of any contradiction in the provisions of the MoRT&H specifications (adopted) and this document the provisions of the latter would take precedence.

22. The rates as given in this schedule for all items are final, binding and conclusive. In case of doubt and printing mistakes if any, the decision of the Engineer-in-Chief M.P.PWD will be final and binding.

23. In case of any discrepancies in rates of any SOR item included in BOQ and that provided in this SOR, provision of latter shall take precedence.

24. The following over head charges have been taken in deriving the rates:-

a) For Road works 10%
b) For Bridge works 20%

# GENERAL NOTES
# BRIDGE WORKS

1) **FOUNDATION**

i) All works below ground level or lowest water level (LWL), whichever is higher shall be termed asfoundation work.

ii) Annular concrete around the footing should be provided up to the top of the footing or to a maximum height of 1.5 meters, whichever is lower. The quantity of annular filling concrete shall be measured with a uniform offset of 0.15 meters around the footing.

iii) The contractor shall maintain a core box for each exploratory boring, as well as a sample of the actual boring at every pier location.

iv) For bridges costing ₹20.00 crore or more, the test pile shall be tested using the bi-directional method in accordance with relevant codes. For projects up to ₹20.00 crore, the static loading method shall be adopted. Contractor shall conduct pile integrity test in before casting pile cap in presence of Sub Divisional officer in charge

v) Back filling up to original bed level shall be done as per clause 304.3.7 of "Specification for Roadand Bridges". No extra shall be payable on this account.

vi) The rates of well sinking include the charges of labour, plant, cost of P.O.L. and other materials and accessories. The rates also include the cost of diversion of channel, making of island, if required to be done for laying the cutting edge, kerb and steining. The permissible tilt and shift in well in its final position shall be as per IRC-78-2014. The rates include the cost of rectification of tilt and shift in excess of permissible limit. No extra over these rates are payable for sinking of well.

vii) For the purpose of the computation of the volume in well sinking, the cubic content of well shall be measured by the portion contained in outer diameter of well steining. Projection of R.C.C. kerb, if any shall not be measured

2) **SUB-STRUCTURE**

i) The part of the bridge structure below the (a) Soffit level of the deck slab/beams and or (b) Springing level for arch spans, but above the ground level or L.W.L. which ever is higher, shall be treated as sub structure of the bridge part.

ii) Complete RCC box section will also be considered as sub-structure.

iii) The rates of bearing are inclusive of the charges required for testing of bearing, procurement,transport, if required, greasing while storing and at the time of fixing etc.

iv) The use of A.C. Pipes for weep holes is permitted only in PCC/RCC/Brick masonry sub-structure. In stone masonry inbuilt weep holes of size 80mm x150mm shall be constructed and no extra for this work is payable. No deduction for the recess or for pipes due to weep holes shall be made in the measurement of stone/brick masonry, PCC and RCC.

v) The bearings should be procured only from those manufactures who have not been disqualified byMORT&H or by MP PWD.

vi) For Hume pipe culverts, Hume pipes shall be ISI marked or DGS&D certified only and a certificate should invariably be produced to the satisfaction of Engineer-in-Charge before getting the pipesfixed at site

3) **SUPER-STRUCTURE**

i) The work above (a)Soffit level for deck slabs/beams and (b)Springing level for arch span, including kerbs, railing, expansion joints, beams, slabs etc. shall be termed as super structure of the bridge part

1) Properly designed form work shall be used for super structure. In case of steel form work the thickness of steel plate shall not be less than 3 mm. The form work shall be adequately stiffened by brackets and angles not more than 15 cm. apart in such a manner that it does not get distorted during handling and placing/vibration of concrete.

2) Centering made up of steel trusses below soffit shall not be supported in recess made in sub- structure. Providing safe centering shall be solely contractor's responsibility. The contractor shall remove all bolts, anchors protruding beyond the pier/pier-cap after removal of centering.

3) Contractor shall have the option to adopt launching of super-structure but in such cases prior permission of the department should be obtained by getting launching arrangement approved by the Chief Engineer. It is clarified that the payment to the contractor in such case shall be limited to the payment he would have received by carrying out the work through conventional method involving centering shuttering, cast in situ concreting and prestressing, etc. including all the ancillary arrangements.

4) The rate of prestressing of H.T. Steel is complete rate including hire and running expenses of plant and machineries, labour involved in stressing operations, anchorages and ducts or sheathing etc., complete including grouting.

5) The rates of wearing coat are inclusive of reinforcement provided, construction joints between panelsand the form work at the panels. Construction joints in panel shall be 10 mm. wide and shall be filled with pre-molded joint filler. Filling of expansion joints between span over piers/abutment or at articulation is not included in the rate.

6) The payment of railing includes the construction of required intermediate posts and end posts. The length of railing for payment purposes on each side separately, shall be measured from the outeredges of end posts.

7) Preferrence should be given for preparation of D.P.R. of Flyovers and grade separaters etc. use of High Performance Concrete and Ultra High-Performance.

4) **DEFINITIONS**

a) Major Bridge: Having total length between faces of abutments at cap level 60 M. and above.

b) Minor Bridge: Having total length between faces of abutments at cap level, 6 M. and above but less than 60 M.

c) Culverts: Having total length less than 6 M.

5) **SPECIFICATIONS**

i) The specifications of works shall be in accordance with 5th revision of "Specifications for Road and Bridge works" (Ministry of Road Transport & Highway) unless otherwise provided in the contract.

6) **CONCRETE**

a) All concrete work shall be done as provided in section 1700 of "Specifications for Road and Bridge works" (Ministry of Road Transport & Highway) 5th Revision unless otherwise provided in the contract.

b) Finishing of concrete by plastering the surface shall not be done without obtaining written permission from the Executive Engineer. No extra for plastering shall be payable. Light touching up and rubbing the uneven surfaces by carborundum stone shall be done within the specified rates.

7) **STEEL**

i) TMT Steel confirming to IS-1786 can be used in place of Tor Steel.

ii) Steel used as reinforcement steel shall be measured as per clause 1608 of Specifications of Road and Bridge works, MORT&H.

8) **CEMENT**

i) If the cement is brought by the contractor, the contractor shall be responsible for ensuring the quality of cement and would be bound to conduct any or all tests at his cost for ensuring its quality as per directions of the Engineer-in-Charge. The theoretical consumption of cement shall be worked out at On daily basis and shall be verified with actual cement brought to the construction site.

9) **MASONRY WORK**

i) All the stone masonry work shall be strictly as per "Specification for Road & Bridge works" of MORT&H (5th Revision).

ii) In place of stone headers, precast concrete blocks in M-15 Concrete shall be used. Cast-in-situ concrete headers are not permitted.

iii) Generally, for all stone masonry subjected to exposure of water flow (e.g. piers, abutments, returns etc.) C.R. Masonry first sort shall be used unless otherwise provided in the approved drawing.

iv) In case where width of stone masonry is more than one meter, the central portion of stone masonry (Hearting) shall be done with uncoursed random rubble masonry. Payment for the C.R. Masonry will be limited to 1/2-meter width on either faces and the balance will be paid as uncoursed Random Rubble Masonry.

10) **ISSUE RATES OF MATERIALS**

i) Materials, as may be required to be issued to the contractor by department shall be specified in the
N.I.T. with place of delivery and their issue rates. The issue rate all such material shall be decided bythe concerned Chief Engineer.

11) **LEAD AND LIFTS**

i) The rates in all items of this S.O.R. are inclusive of all lead, lift and transportation of material except otherwise specified in the item. No extra on this account is payable unless otherwise specifically mentioned in any particular item.

12) **ROYALTY AND OTHER TAXES**

i) Royalty, Octroi duty and all other taxes except GST are included in the rates. No extra on this account shall be payable. Unless otherwise provided in the contract.

13) **DISMANTLING**

i) The rates include the complete cost of dismantling and shifting away the dismantled material to place at site of work as directed by Engineer-in-Charge and the rates also include the serviceable dismantled material to be properly stacked at river bank at a location directed by the Engineer-in- Charge.



14) **MODE OF MEASUREMENTS**

i) The mode of measurements shall be as per provisions contained in the relevant clauses of thespecifications unless specified otherwise.

ii) The rates include all such activities as are described in the relevant clauses of the Specifications,unless specified otherwise.

iii) Rates include provisions of necessary precautionary devices and other arrangements etc. for traffic control, e.g. provision of caution boards, red lights, watchmen flags and flagmen but do not include construction of temporary diversion.

iv) The rates do not include the work of trial pits, which are to be measured and paid separately.

v) The contractor is bound to install at the site of work a testing laboratory as provided in the specifications as per clause 120 of MoRT&H specification (5$^{th}$ Revision).

vi) Detailed measurements for steel and concrete for items of R.C.C. approach slab, R.C.C. Railing and
R.C.C. wearing coat shall also be recorded in M.B. However, the payment shall be regulated as perthe item and rates given in the S.O.R.

vii) The measurement of rock excavation is to be done as per the relevant clause of "Specification of
Road and Bridge works" of MORT&H 5th Revision. All serviceable rock excavated shall be issuedto the contractor at the rate of Rs.100/- per Cum.

# GENERAL CONDITIONS OF THE ELECTRICAL WORKS

## UILITY SHIFTING, INTERNAL & EXTERNAL ELECTRIFICATION WORK

1. Work should be executed as per list of approved Electrical material. If required material is not available material list approved by department then prior approval from S.E. E/M Circle Bhopal must be taken before execution of work.

2. If The work not completed as per relevant specification of CPWD/MPPWD/Indian electricity rules/Indian Electricity act/ concerning DISCOM/Transco rules/ IS code for desired electrical works, the same will be rejected.

3. The contractor/agency shall extent all reasonable facilities and co-operation to the various other agencies (Like Road & Bridge civil contractor, concerning DISCOM/Transco any local body etc.) and sub contractors working at the site simultaneously, so the entire work can proceed smoothly for successful completion.

4. The contractor/agency shall have to arrange all facilities for the inspection /testing of material /work at site.

5. It is the responsibility of contractor to get prior approval for all material, before procurement at site in writing from Engineer-in-Charge or competent authority.

6. The successful tendered shall make his own arrangements for transport of all materials.

7. The installation should be tested on completion of the work by the contractor in presence of the Assistant Engineer E/M or authorized personnel and his remark should be recorded in the completion certificate. The completion certificate will be submitted by the contractor or his authorized representative, after handing over/taken over of the work to concerned agency duly signed by competent authority.

8. The work "as described", "as specified", " as shown", "as directed", " as approved" or "as required" shall be meant, as described in specification, schedule of quantities, other tender documents and as directed or approved by the Engineer-in-Charge .

9. In addition to all the conditions of the NIT, aforementioned, General conditions will be binding on the contractor and shall form part of the agreement to be executed by the contractor, in addition to the condition of the contract in the prescribed form.

10. The Contractor Shall Have "A" Class and appropriate Class Valid Electrical Contractor registration in DISCOM & MPPWD .

11. Material/equipment be dispatched to construction site directly from the manufacturer/factory or their authorized distributor/dealer.

12. The work shall be carried as per CPWD/MPPWD Electrical specification. IS specification shall be followed for items where CPWD/MPPWD specifications are not available.

"लोक निर्माण से लोक कल्याण"

13. The contractor/agency shall make his own arrangement for site store and supply of water & electricity at their own expense as required for the execution of work.

14. The contractor/agency will have to work according to approved drawing of work, If any changes appears due to civil structure drawings, The contractor/agency will have to get the drawings re-approved before completion of agreed work or contracted work from Executive Engineer (Electrical) or Superintending Engineer (Electrical) M.P.P.W.D. Bhopal .

15. The contractor/agency shall submit the test reports of all material to be used at site issued by NABL accredited Lab before execution of work at his own cost. In addition to this, if Engineer-in-Charge needs to perform any routine tests for material to be used at site, the same will be the responsibility of contractor on contractors cost. If cost of testing not borne by contractor, the same will be deducted from his bills.

16. The contractor/agency shall submit guarantee/warrantee documents, equipment user's manual to the Engineer-in-Charge, E/M before submitting bills.

17. Guarantee certificate of the Electrical/Electronic equipment/material, shall be given by manufacturing company, should be provided by the contractor from the date of completion/commissioning. All defect noticed during the DLP, shall be made good/rectified/replaced by the contractor at his own cost and risk.

18. The sample/Guaranteed technical particulars(GTP) of the approved materials to be used by the contractor shall be submitted in advance & get permission by Engineer-In-Charge(E/M) / superintending engineer(E/M) M.P.P.W.D. before use. No any electrical approved material will be used by contractor/agency Without permission by Engineer-In-Charge(E/M) / superintending engineer(E/M).

19. If contractor used non-approved Electrical material, Such Materials cost will not be payable. Contractor will not able to claim the cost of such non-approved electrical installations.

20. The Contractor shall be fully responsible for any accident, damages, losses occurs due to negligence during execution of work. No compensations shall be made by the department.

21. The Contractor shall be fully responsible for taking all sort of safety measures to avoid any Mishap/accident.

22. It is the responsibility of contractor to get approval/certificate from Electrical safety department/

Electrical licensing board Govt. of M.P. for electrical installation before handing over to department.

# CHAPTER: 1.0

# (CARRIAGE OF MATERIALS)

| S.No. | Materials | Unit | Rate (Rs.) | | | | | | | | Remarks |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | 1 km | 2 km | 3 km | 4 km | 5 km | Beyond 5 km up to 10km Addl. per km | Beyond 10 km up to 20km Addl. per km | Beyond 20km per addl. Km. | |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 |
| 1.1 | **Transportation By Mechanical Transport including loading, unloading and stacking.** | | | | | | | | | | |
| 1.1.1 | Loose moorum/sand/earth/ surkhi/ cement/stone dust/hot mix asphalt material. | cum | 70.8 | 78.4 | 82.3 | 86.2 | 91.1 | 4.3 | 3.2 | 2.7 | The rates will be applicable to net quantities after deduction of prescribed percentage for voids mentioned in the Table at the bottom of the Chapter |
| 1.1.2 | Excavated/compacted ordinary & other soil. | cum | 87.2 | 97.0 | 101.9 | 105.9 | 112.7 | 5.3 | 3.9 | 3.2 | |
| 1.1.3 | Boulders. | cum | 141.1 | 154.9 | 162.7 | 169.6 | 180.3 | 8.3 | 6.3 | 5.1 | |
| 1.1.4 | Stone aggregate (Metal) | cum | 76.5 | 84.3 | 88.2 | 92.1 | 98.0 | 4.6 | 3.4 | 2.8 | |
| 1.1.5 | Masonry stones. | cum | 63.7 | 70.6 | 73.5 | 76.5 | 81.3 | 3.8 | 2.8 | 2.3 | |
| 1.1.6 | Coal/Fuel wood/iron work/steel/G.I. Sheets/pipes/lime/machinery etc. | tonne | 62.7 | 68.6 | 72.5 | 75.5 | 80.4 | 3.8 | 2.8 | 2.3 | |
| 1.1.7 | Steel | tonne | 62.7 | 68.6 | 72.5 | 75.5 | 80.4 | 3.8 | 2.8 | 2.3 | |
| 1.1.8 | timber | cum | 80.4 | 88.2 | 93.1 | 97.0 | 102.9 | 4.8 | 3.6 | 2.9 | |
| 1.1.9 | Tar/Paint/Bitumen etc. | M.T. | 71.5 | 78.4 | 82.3 | 86.3 | 91.1 | 4.3 | 3.2 | 2.7 | |

**Table**

| S.No. | Standard size of Aggregate | Percentage reduction in volume computed by stack measurements to arrive at the volume to be paid for |
|---|---|---|
| 1. | 75 mm and 63 mm | 12.50 |
| 2. | 45 mm and 26.5 mm | 10.00 |
| 3. | 22.4 mm, 13.2 mm 11.2 mm and 6.7 mm | 5.00 |
| 4. | Fine aggregate | 5.00 |

| | | | | |
|---|---|---|---|---|
| 1.2 | Supply of mineral aggregate like broken stone/crushed stone(crushed in mechanical crusher) at road site including all lead and stacking etc. complete. | | | |
| | (i) | 75 mm standard size broken stone | cum | 796.00 |
| | (ii) | 63 mm standard size broken stone | cum | 796.00 |
| | (iii) | 53mm standard size broken stone | cum | 796.00 |
| | (iv) | 45mm standard size crushed stone | cum | 796.00 |
| | (v) | 26.5mm standard size crushed stone | cum | 798.00 |
| | (vi) | 22.4mm standard size crushed stone | cum | 798.00 |
| | (vii) | 13.2mm standard size crushed stone | cum | 738.00 |
| | (viii) | 11.2mm standard size crushed stone | cum | 738.00 |
| | (ix) | 6.7mm standard size crushed stone | cum | 471.00 |
| 1.3 | Supply of following material including all lead and stacking etc. complete. | | | |
| | (i) | Crusher stone dust | cum | 300.00 |
| | (ii) | Sand/Shingle/Kanker/Laterite | cum | 300.00 |
| | (iii) | Moorum with CBR not less than 20 and P.I. Not exceeding 6. | cum | 200.00 |

# CHAPTER: 2.0

## SITE CLEARANCE

| Item No. | | Descriptions | Unit | Rate |
|---|---|---|---|---|
| 2.1 | | **Cutting of Trees, including Cutting of Trunks, Branches and Removal**<br>Cutting of Trees, including Cutting of Trunks, Branches and Removal Cutting of trees, including cutting of trunks/branches in suitable size removal of stumps/roots, and stacking of serviceable material with all lifts and lead up to storage yard and filling earth in the depression/pit. | | |
| | (i) | Girth from 300 mm to 600 mm | each | 325.00 |
| | (ii) | Girth from 600 mm to 900 mm | each | 434.00 |
| | (iii) | Girth from 900 mm to 1800 mm | each | 883.00 |
| | (iv) | Girth above 1800 mm | each | 1640.00 |
| 2.2 | | **Clearing and Grubbing Road Land**<br>Clearing and grubbing road land including uprooting rank vegetation, grass, bushes, shrubs, saplings and trees having girth up to 300 mm, removal of stumps of trees cut earlier and disposal of unserviceable materials and stacking of serviceable material up to a lead of 1000 meters including removal and disposal of top organic soil not exceeding 150 mm in thickness. | | |
| | A | In area of light jungle | hectare | 29409.00 |
| | B | In area of thorny jungle | hectare | 23080.00 |
| 2.3 | | **Dismantling of Structures**<br>Dismantling of existing structures like culverts, bridges, retaining walls and other structure comprising of masonry, cement concrete, wood, steel, including T&P and scaffolding wherever necessary, sorting the dismantled material, disposal of unserviceable material and stacking the serviceable material with all lifts and lead up to 1000 meters. | | |
| | (i) | Lime /Cement Concrete | | |
| | A | Lime Concrete, cement concrete grade M-10 and below | cum | 529.00 |
| | B | Cement Concrete Grade M-15 & M-20 | cum | 628.00 |
| | C | Prestressed / Reinforced cement concrete grade M-25 & above | cum | 1616.00 |
| | (ii) | Dismantling Brick / Tile work | | |
| | A | In lime mortar | cum | 311.00 |
| | B | In cement mortar | cum | 411.00 |
| | C | In mud mortar | cum | 271.00 |
| | D | Dry brick pitching or brick soling | cum | 251.00 |
| | (iii) | Dismantling Stone Masonry | | |
| | A | Rubble stone masonry in lime mortar | cum | 351.00 |
| | B | Rubble stone masonry in cement mortar. | cum | 411.00 |
| | C | Rubble Stone Masonry in mud mortar. | cum | 311.00 |
| | D | Dry rubble masonry | cum | 291.00 |
| | E | Dismantling stone pitching/ dry stone spalls. | cum | 272.00 |
| | F | Dismantling boulders laid in wire crates including opening of crates and stacking dismantled materials. | cum | 311.00 |

| Item No. | | | Descriptions | Unit | Rate |
|---|---|---|---|---|---|
| | (iv) | | Wood work wrought framed and fixed in frames of trusses up to a height of 5 m above plinth level | cum | 726.00 |
| | (v) | | Steel work in all types of sections up to a height of 5 m above plinth level. | | |
| | | A | Including dismembering | tone | 1897.00 |
| | | B | Excluding dismembering. | tone | 1444.00 |
| | | C | Extra over item No (V) A and (V) B for cutting rivets. | tone | 14.00 |
| | (vi) | | Scraping of bricks dismantled from Brick work including stacking. | | |
| | | A | In lime/Cement mortar | 1000 numbers | 1745.00 |
| | | B | In mud mortar | 1000 numbers | 623.00 |
| | (vii) | | Scraping of Stone from dismantled stone masonry | | |
| | | A | In cement and lime mortar | cum | 700.00 |
| | | B | In Mud mortar | cum | 148.00 |
| | (viii) | | Scarping plaster in lime or cement mortar from brick/ stone masonry | sqm | 22.00 |
| | (ix) | | Removing all type of Hume pipes and stacking within a lead of 1000 meters including earthwork and dismantling of masonry works. | | |
| | | A | Up to 600 mm dia | meter | 334.00 |
| | | B | Above 600 mm to 900 mm dia | meter | 433.00 |
| | | C | Above 900 mm | meter | 532.00 |
| 2.4 | | | **Dismantling of Flexible Pavements**<br>Dismantling of flexible pavements thickness more than 100 mm and disposal of dismantled materials up to a lead of 1000 meters, stacking serviceable and unserviceable materials separately. | cum | 323.00 |
| 2.5 | | | **Dismantling of Cement Concrete Pavement**<br>Dismantling of cement concrete pavement by mechanical means using pneumatic tools, breaking to pieces not exceeding 0.02 cum in volume and stock piling Serviceable material at designated locations and disposal of unserviceable materials up to a lead of 1000 meters, | cum | 1296.00 |
| 2.6 | | | **Dismantling Guard Rails**<br>Dismantling guard rails by manual means and disposal of dismantled material with all lifts and up to a lead of 1000 meters, stacking serviceable materials and unserviceable materials separately. | meter | 78.00 |
| 2.7 | | | **Dismantling Kerb Stone**<br>Dismantling kerb stone by manual means and disposal of dismantled material with all lifts and up to a lead of 1000 meter | meter | 15.00 |
| 2.8 | | | **Dismantling Kerb Stone channel**<br>Dismantling kerb stone channel by manual means and disposal of dismantled material with all lifts and up to a lead of 1000 meter | meter | 21.00 |

| Item No. | Descriptions | Unit | Rate |
|---|---|---|---|
| 2.9 | **Dismantling Kilometer/ Boundary Stone**<br>Dismantling of Kilometer/ Boundary stone including cutting of earth, foundation and disposal of dismantled material with all lifts and lead up to 1000 m and back filling of pit. | | |
| | A 5th KM stone | each | 528.00 |
| | B Ordinary KM Stone | each | 309.00 |
| | C Hectometer Stone | each | 62.00 |
| | D Boundary Stone | each | 62.00 |
| 2.10 | **Dismantling of Fencing**<br>Dismantling of barbed wire fencing/ wire mesh fencing including cutting of earth, foundation concrete, posts, back filling of pit by manual means including disposal of dismantled material with all lifts and up to a lead of 1000 meters, stacking serviceable material and unserviceable material separately. | meter | 66.00 |
| 2.11 | **Dismantling of CI Water Pipe Line**<br>Dismantling of CI water pipe line up to 600 mm dia including disposal with all lifts and lead up to 1000 meters and stacking of serviceable material and unserviceable material separately under supervision of concerned department. | meter | 181.00 |
| 2.12 | **Removal of Cement Concrete Pipe of Sewer Gutter**<br>Removal of cement concrete pipe of sewer gutter up to 1500 mm dia under the supervision of concerned department including disposal with all lifts and up to a lead of 1000 meters and stacking of serviceable and unserviceable material separately but excluding earthexcavation and dismantling of masonry works. | meter | 209.00 |
| 2.13 | **Removal of Telephone / Electric Poles and Lines**<br>Removal of telephone / Electric poles including excavation and dismantling of foundation concrete and lines under the supervisionof concerned department, disposal with all lifts and up to a lead of 1000 meters and stacking the serviceable and unserviceable material separately. | each | 229.00 |

# CHAPTER: 3.0

# EARTH WORK, EROSION CONTROL AND DRAINAGE

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| 3.1 | **Excavation in Soil by Manual Means**<br>Excavation for roadway in soil including loading and unloading in/from truck for carrying of suitable cut earth to embankment site and disposing off unsuitable earth with all lifts and lead up to 1000 meters as per clauses of section-300. | cum | 210.00 |
| 3.2 | **Excavation in ordinary rock by Manual Means**<br>Excavation in ordinary rock including loading and unloading in/from in a truck and carrying of excavated material to embankment site with all lifts and leads up to 1000 meters as per relevant clauses of section-300 | cum | 314.00 |
| **Note:-** | **Item No.3.1 & 3.2 can be executed only after prior written approval by C.E.** | | |
| 3.3 | **Excavation in Soil using Hydraulic Excavator and Tippers with disposal up to 1000 meters.**<br>Excavation for roadwork in soil with hydraulic excavator including cutting and loading in tippers, trimming bottom and side slopes, in accordance with requirements of lines, grades and cross sections, and transporting suitable earth to the embankment location and disposing off unsuitable earth with all lifts and lead up to 1000m as per relevant clauses of section-300 of specification. | cum | 79.00 |
| 3.4 | **Excavation in Ordinary Rock using Hydraulic Excavator and Tippers with disposal up to 1000 meters.**<br>Excavation for roadway in ordinary rock with hydraulic excavator including cutting and loading in tippers, transporting to embankment site with in all lifts and lead up to 1000 m, trimming bottom and side slopes in accordance with requirements of lines, grades and cross sections as per relevant clauses of section-300 of specification. | cum | 177.00 |
| 3.5 | **Excavation in Hard Rock with disposal up to 1000 meters**<br>Excavation for roadway in hard rock with rock breakers i/c breaking rock or by drilling, blasting and breaking, trimming of bottom and side slopes in accordance with requirements of lines, grades and cross sections, loading and disposal of cut road with all lifts and leads up to 1000 meters and as per relevant clauses of section-300 of specification.<br>**Note: - 1. The Quality and Availability of rock shall be cheeked before affording credit.** | cum | 297.00 |
| 3.6 | **Excavation in Hard rock (Blasting prohibited)** | cum | 588.00 |
| 3.7 | **Excavation in Hard rock (Controlled Blasting)** | cum | 513.00 |
| 3.8 | **Excavation in Marshy Soil**<br>Excavation for roadway in marshy soil with hydraulic excavator including cutting and loading in tippers and disposal with all lifts and lead up to 1000 meters, trimming of bottom and side slopes in accordance with requirements of lines, grades and cross sections and as per relevant clauses of section-300 of specification. | cum | 165.00 |
| 3.9 | **Removal of Unserviceable Soil with Disposal up to 1000 meters**<br>Removal of unserviceable soil including excavation, loading and disposal up to 1000 meters lead but excluding replacement by suitable soil which shall be paid separately as per clause 305. | cum | 71.00 |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| 3.10 | Pre-splitting of Rock Excavation Slopes (Carrying out excavation in hard rock to achieve a specified slope of the rock face by controlled use of explosives and blasting accessories in properly aligned and spaced drill holes, collection of the excavated rock by a dozer, loading in tipper by a front-end loader and disposing of the material with all lifts and lead up to 1000 m, all as specified in clause No. 303) | sqm | 116.00 |
| 3.11 | **Scarifying Existing Granular Surface to a 50 mm by Manual Means** Scarifying the existing granular road surface to a depth of 50 mm and disposal of scarified material within all lifts and leads up to 1000 meters. (refer clause two of General Notes) | sqm | 39.00 |
| 3.12 | **Scarifying existing bituminous surface to a depth of 50 mm by mechanical means** Scarifying the existing bituminous road surface to a depth of 50 mm and disposal of scarified material with in all lifts and lead up to 1000 meters. | sqm | 10.00 |
| 3.13 | **Embankment/ Sub Grade Construction with Material Obtained from Borrow Pits** Construction of embankment with approved material having CBR>7 obtained from borrow pits with all lifts and leads, transporting to site, spreading, grading to required slope and compacting to meet requirement of table 300-1, 300-2 and as per relevant clauses of section-300. | cum | 298.00 |
| 3.14 | **Construction of Embankment with Material Deposited from Roadway Cutting** Construction of embankment with approved materials having CBR>7 deposited at site from roadway cutting and excavation from drain or foundation of structures graded and compacted to meet requirement of table 300-1, 300-2 and as per relevant clauses of section-300. | cum | 60.00 |
| 3.15 | Stripping and Storing Top Soil (Stripping, storing of top soil by road side at 15 m interval and re-application on embankment slopes, cut slopes and other areas in localities where the available embankment material is not conducive to plant growth) | cum | 54.00 |
| 3.16 | Turfing with Sods (Furnishing and laying of the live sods of perennial turf forming grass on embankment slope, verges or other locations shown on the drawing or as directed by the engineer including preparation of ground, fetching of sods and watering) | sqm | 30.00 |
| 3.17 | Seeding and Mulching (Preparation of seed bed on previously laid top soil, furnishing and placing of seeds, fertilizer, mulching material, applying bituminous emulsion at the rate of 0.23 liters per sqm and laying and fixing jute netting, including watering for 3 months all as per clause 308) | sqm | 162.00 |
| 3.18 | **Surface Drains in Soil** Construction of unlined surface drains of average cross-sectional area 0.40 sqm in soil to specified lines, grades, levels and dimensions to the requirement of clause 301 and 309. Suitable Excavated material to be used in embankment within a lead of 50 meters as per relevant clauses of section-300. | meter | 72.00 |
| 3.19 | **Sub Surface Drains with Perforated Pipe** Construction of subsurface drain with perforated pipe of 100 mm internal diameter of metal/ asbestos cement/ cement concrete/PVC, closely jointed, perforations ranging from 3 mm to 6 mm depending upon size of material surrounding the pipe, with 150 mm bedding | meter | 409.00 |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| | below the pipe and 300 mm cushion above the pipe, cross section of excavation 450(w) x 550 (d) mm. Suitable Excavated material to be utilized in roadway at site | | |
| 3.20 | **Aggregate Sub- Surface Drains**<br>Construction of aggregate sub surface drain 300 mm x 450 mm with aggregates conforming to table 300-4, suitable excavated material to be utilized in roadway. | meter | 225.00 |
| 3.21 | **Construction of Rock fill Embankment**<br>Construction of rock fill embankment with broken hard rock fragments of size not exceeding 300 mm laid in layers not exceeding 500 mm thick including filling of surface voids with stone spalls, blinding top layer with granular material, rolled with vibratory road roller, all complete as per clause 313 | cum | 102.00 |
| 3.22 | **Embankment Construction with Fly ash/Pond ash available from coal or lignite burning Thermal Plants as waste material**.<br>Construction of embankment with fly ash conforming to table 1 of IRC: SP: 58 - 2001 obtained from coal or lignite burning thermal power stations as waste material, spread and compacted in layer of 200 mm thickness each at OMC, all as specified in IRC: SP: 58-2001 and as per approved plans. | cum | 243.00 |

# CHAPTER: 4.0

SUB-BASES, BASES
( NON-BITUMINOUS)
AND SHOULDERS

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| 4.1 | Granular Sub-base with Close Graded Material (Table: - 400-1) | cum | 1114.00 |
| A | Plant Mix Method (Construction of granular sub-base by providing close graded Material, mixing in a mechanical mix plant at OMC, carriage of mixed Material to work site, spreading in uniform layers with motor grader on prepared surface and compacting with vibratory power roller to achieve the desired density, complete as per clause 401) | | |
| 4.2 | **Crusher Run Macadam Sub Base/Base Course**<br>Providing crushed stone aggregate, depositing on a prepared surface by hauling vehicles, spreading and mixing with a motor grader, watering and compacting with a vibratory roller as per clause 410 to form a layer of sub-base/Base in accordance to Table No. 400-14, 400-15 and as per clause of section 407 of specifications. | | |
| (i) | For 53 mm maximum size | cum | 1186.00 |
| (ii) | For 37.5 mm maximum size | cum | 1198.00 |
| 4.3 | **Wet Mix Macadam**<br>Providing, laying, spreading and compacting graded stone aggregate to wet mix macadam specification including premixing the Material with water at OMC in mechanical mix<br>plant carriage of mixed Material by tipper to site, laying in uniform layers with paver in sub- base / base course on well prepared surface and compacting with vibratory roller to achieve the desired density in accordance to Table No. 400-12, 400-13 and as per clause of section 406 of specifications. | | |
| A | By Sensor Paver | Cum | 1451.00 |
| 4.4 | Deduction for item No.4.3 if sensor paver is not used. | Cum | 100.00 |
| 4.5 | **Cement Stabilization for Improving Embankment**<br>Laying and spreading available soil in the embankment on a prepared surface, pulverizing, mixing the spread soil in place with Soil Stabilizer & Binder Spreader with 2 percent slaked lime using Binder spreader Machine, having minimum content of 70 per cent of CaO, grading with motor grader and compacting with the road roller at OMC to the desired density to form a layer of improved sub grade (with prior permission of CE.) | | |
| A | By Mechanical Means | cum | 441.00 |
| B | By Manual Means | cum | 423.00 |
| 4.6 | **Lime Stabilization for Improving Subgrade**<br>Laying and spreading available soil in the sub-grade on a prepared surface, pulverizing, mixing the spread soil in place with Soil Stabilizer & Binder Spreader with 2 percent slaked lime using Binder spreader Machine, having minimum content of 70 per cent of CaO, grading with motor grader and compacting with the road roller at OMC to the desired density to form a layer of improved sub grade | | |
| A | By Mechanical Means | cum | 483.00 |
| B | By Manual Means | cum | 463.00 |
| 4.7 | **Cement Stabilization for Improving Subgrade** | | |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| | Laying and spreading available soil in the sub-grade on a prepared surface, pulverizing, mixing the spread soil in place with Soil Stabilizer & Binder Spreader with 2 percent cement using Binder spreader Machine, grading with motor grader and compacting with the road roller at OMC to the desired density to form a layer of improved sub grade | | |
| A | By Mechanical Means | cum | 475.00 |
| B | By Manual Means | cum | 455.00 |
| 4.8 | **Lime Treated Soil for Sub-Base**<br>Providing, laying and spreading soil on a prepared sub grade, pulverizing, as per requirement of Table 400-3 mixing the spread soil in place with rotavator with 3 % slaked lime with minimum content of 70% of CaO, to achieve CBR>30 grading with motor grader and compacting with the road roller at OMC to achieve at least 98% of the max dry density to form a layer of sub base as per clause 402 of specification. | cum | 621.00 |
| 4.9 | **Cement Treated Soil Sub Base/ Base**<br>Providing, laying and spreading soil on a prepared sub grade, pulverizing, adding the designed quantity of cement to the spread soil, mixing in place with rotavator, grading with the motor grader and compacting with vibrator roller at OMC to achieve the desired unconfined compressive strength and to form a layer of sub-base/base as per clause 403 of specification. | cum | 719.00 |
| 4.10 | Cement Treated Crushed Rock or combination as per clause 403.2 and table 400.4 in Sub base/ Base<br>Providing, laying and spreading Material on a prepared sub grade, adding the designed quantity of cement to the spread Material, mixing in place with rotavator, grading with the motor grader and compacting with vibrator roller at OMC to achieve the desired unconfined compressive strength and to form a layer of sub-base/base as per clause 403 of specification. | | |
| (i) | For Sub-Base course | | |
| A | Plant Mix Method (Using by Mechanical Paver) | cum | 1395.00 |
| B | By Mix in Place Method ((Using by Soil Stabilizer) | cum | 1345.00 |
| (ii) | For Base course | | |
| A | Plant Mix Method (Using by Mechanical Paver) | cum | 1395.00 |
| B | By Mix in Place Method ((Using by Soil Stabilizer) | cum | 1345.00 |
| 4.11 | **Crushed Cement Concrete Sub-base / Base**<br>Breaking and crushing of material obtained by breaking damaged cement concrete slabs to size range not exceeding 75 mm as specified in table 400-9 transporting the aggregates obtained from breaking of cement concrete slabs at a lead of 1.0 km., laying and compacting the same as sub base/ base course, constructed as per clause 405 of specification. | cum | 257.00 |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| 4.12 | **Lime, Fly ash stabilized soil sub-base**<br>Construction of Sub-base using lime - fly ash admixture with granular soil, free from organic matter/ deleterious material or clayey silts and low plasticity clays having PI between 5 and 20 and liquid limit less than 25 and commercial dry lime, slaked at site or pre-slaked with CaO content not less than 50%, fly ash to conform to gradation as per clause 4.3 of IRC: 88-1984, lime + fly ash content ranging between 10 to 30%, the minimum un-confined compressive strength and CBR value after 28 days curing and 4 days soaking to be 7.5kg/sq, cm and 25% respectively, all as specified in IRC: 88-1984. | cum | 586.00 |
| 4.13 | **Inverted Choke** (Construction of inverted choke by providing, laying, spreading and compacting screening B type/ coarse sand of specified grade in uniform layer on a prepared surface with motor grader and compacting with power roller etc.) | cum | 5186.00 |
| 4.14 | **Making 50 mm wide Furrows**<br>Making 50 mm wide furrows, 450 to the center line of the road and at one-meter interval in the existing thin bituminous wearing coarse including sweeping and disposal of excavated material within 1000 meters lead | | |
| a) | 25 mm deep furrow cutting | sqm | 10.00 |
| b) | 50 mm deep furrow cutting | sqm | 13.00 |
| 4.15 | **Construction of Median and Island with Soil Taken from Roadway Cutting**<br>Construction of Median and Island above road level with approved material deposited at site from roadway cutting and excavation for drain and foundation of structures, spread, graded and compacted as per clause 408 of specification | cum | 150.00 |
| 4.16 | Construction of Shoulders with approved material/selected soil i/c excavation all lifts & leads i/c grading to required slope & camber of 4% and compacting on OMC using vibratory roller of 80 to 100 kN static weight. Field compaction shall be done in corresponding layers of carriage way thickness and meet the requirements of MDD to achieve compaction (98% -100%) as per relevant clause of 408. | | |
| A | Hard Shoulders (CBR value >12) | cum | 398.00 |
| 4.17 | Construction of Median and Island with Soil Taken from Borrow Areas construction of median and Island above road level with approved material brought from borrow pits, spread, sloped and compacted as per clause 408 of specification. | cum | 262.00 |
| 4.18 | **Footpaths and Separators**<br>Construction of footpath/ separator by providing a 150 mm compacted granular sub base as per clause 410 and 25-mm thick cement concrete grade M15, over laid with precast concrete tiles in cement mortar 1:3 including provision of all drainage arrangements but excluding kerb channel.<br>Note- The above shall be included in the DPR and executed with prior written permission from the Chief Engineer. | sqm | 486.00 |
| | **Full Depth Reclamation** | | |
| 4.19 | Stabilization of in-situ (existing pavement crust) or soil or otherwise sub base/ base course up to the required depth by cold in-situ recycling using chemical additives/ otherwise: Providing | cum | 3495.00 |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| | pulverizing, spreading, milling and mixing of chemical additives at the appropriate rate as per job mix design in accordance with IRC: SP: 72-2015 and IRC-37-2018. Cementitious additive@ rate of minimum 4- 7% should be spread on the existing pavement using a mobile truck mounted containerized cement/ additive spreader with micro processor-controlled weighing and spreading system. The additive spreader shall have variable working width sufficient to cover whole pavement lane. The in-situ stabilization process shall be carried out by a mobile and self propelled stabilizer/ reclaimer of working width of 2.4m with minimum engine horse power of 440 kw with a variable working depth up to 50 cm. the resultant stabilized mix then would be profiled to the required grade, level and thickness using motor grader and the mix would be compacted using 20 tone pad foot roller in combination with smooth wheel roller to achieve desired proctor density as per IRC 37-2018 and complete in all respect and curing with water as required including all materials, labour and machinery etc. the entire in-situ process would be carried out in single pass with milling and pulverizing of damaged asphalt pavement/ soil/aggregates/ soil-aggregate mixture to the desired depth and with simultaneous addition of additives and water with machine integrated spray bars fitted on the wheeled self-propelled and vibratory pad foot roller to achieve therequired density. | | |
| 4.20 | Providing and laying Stress Absorbing Membrane Interlayer (SAMI)/bitumen impregnated geotextile layer after clearing of the road surface geotextile conforming to physical requirements of specification clause 703, laid over a tack coat of paving grade Bitumen 80-100 penetration, laid at the rate of 1.05 kg per sqm over thoroughly cleaned and repaired surface to provide a water resistant membrane and crack retarding layer paving fabric to be free of wrinkling and folding and to be laid before cooling of tack coat, brooming and rolling of surface with pneumatic roller to maximize paving fabric contact with pavement surface. | sqm | 75.00 |

# CHAPTER: 5.0

# BASES AND SURFACE COURSES (BITUMINOUS)

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| 5.1 | **Prime Coat Over WMM/WBM** | | |
| | Providing and applying primer coat with catonic bitumen emulsion on prepared surface of granular Base including cleaning of road surface and spraying primer at the rate of 0.70 to 1.00 kg/sqm using mechanical means as per clause 502 of specifications. | sqm | 44.00 |
| | (i) Providing and applying primer coat with cutback MC 30 bitumen emulsion on prepared surface of granular Base including cleaning of road surface and spraying primer at the rate of 0.60 to 0.90 kg/sqm using mechanical means as per clause 502 of specifications. | sqm | 39.00 |
| 5.2 | (ii) **Prime Coat over Stabilized soil bases / Crusher Run Macadam** | | |
| | Providing and applying primer coat with SS1 grade bitumen emulsion or prepared surface of granular Base including clearing of road surface and spraying primer at the rate of 0.90 to 1.20 kg/sqm using mechanical means. | sqm | 55.00 |
| | **(i)** Providing and applying primer coat with cutback MC 70 bitumen emulsion or prepared surface of granular Base including clearing of road surface and spraying primer at the rate of 0.90 to 1.20 kg/sqm using mechanical means. | sqm | 55.00 |
| 5.3 | **(ii)** Providing and applying tack coat with cationic bitumen emulsion using emulsion pressure distributor on the prepared bituminous/granular surface cleaned with mechanical broom and as per relevant clauses of section-503 of specifications. | | |
| | (i) @ 0.25 kg per sqm (normal bituminous surfaces) | sqm | 16.00 |
| | (ii) @0.30 kg per sqm (dry & hungry bituminous surfaces /Granular surfaces treated with primer) | sqm | 17.00 |
| | (iii) @ 0.35 kg per sqm (non-bituminous surfaces) cement concrete pavement. | sqm | 18.00 |
| | (iv) @ 0.40 kg per sqm (non-bituminous surfaces)granular base not primed. | sqm | 21.00 |
| 5.4 | Providing and applying tack coat with low viscosity bitumen of VG 10 grade using bitumen pressure sprayer on the prepared bituminous/granular surface cleaned with mechanical broom and as per relevant clauses of section-503 of specifications. | | |
| | (i) @ 0.25 kg per sqm (normal bituminous surfaces) | sqm | 16.00 |
| | (ii) @0.30 kg per sqm (dry & hungry bituminoussurfaces /granular surfaces treated with primer) | sqm | 20.00 |
| | (iii) @ 0.35 kg per sqm (non-bituminous surfaces/cement concrete pavement.) | sqm | 22.00 |
| | (iv) @ 0.40 kg per sqm (non-bituminous surfaces) granular base not primed. | sqm | 23.00 |
| | **Note: - this item can be executed only after prior written approval by CE.** | | |
| 5.5 | **Bituminous Macadam** | | |
| | Providing and laying bituminous macadam using crushed aggregates of specified grading premixed with bituminous binder, transported to site, laid over a previously prepared surface with Electronic Sensor paver finisher to the required grade, level and alignment and rolled as per clauses 501.6 and 501.7 to achieve the desired compaction and as | | |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| | per clause 504 of specifications. | | |
| | a) Using Batch Mix Plant of appropriate capacity and Hydraustatic Paver finisher with Sensor. | | |
| | (i) for Grading I (40 mm nominal size) bitumen content3.4% with bitumen 60/70 (VG 30) | cum | 5781.00 |
| | (ii) for Grading II (19 mm nominal size) bitumen content 3.5% with bitumen 60/70 (VG 30) | cum | 5829.00 |
| | (ii) for Grading II (19 mm nominal size) bitumen content 3.5% with bitumen 60/70 (VG 40) | cum | 5929.00 |
| | b) Using Drum Mix Plant of appropriate capacity and paver finisher. | | |
| | (i) for Grading I (40 mm nominal size) bitumen content3.4% with bitumen 60/70 (VG 30) | cum | 5281.00 |
| | (ii) for Grading (19 mm nominal size) bitumen content 3.5% with bitumen 60/70 (VG 30) | cum | 5329.00 |
| | (ii) for Grading II (19 mm nominal size) bitumen content 3.5% with bitumen 60/70 (VG 40) | cum | 5429.00 |
| 5.6 | **Dense Graded Bituminous Macadam** Providing and laying dense bituminous macadam using crushed aggregates of specified grading, premixed with bituminous binder @ 4.0 to 4.5% by weight of total mix of mix and filler, transporting the hot mix to work site, laying with an Electronic Sensor paver finisher with sensor control to the required grade, level and alignment, rolling with smooth wheeled, vibratory and tandem rollers to achieve the desired compaction as per clause 505 of specification complete in all respects. | | |
| | a) Using Batch Mix Plant of appropriate capacity | | |
| | (i) for Grading I (37.5 mm nominal size) Bitumen Content minimum 4% with bitumen 60/70 (VG 30) | cum | 7585.00 |
| | (ii) for Grading I (37.5 mm nominal size) Bitumen Content minimum 4% with bitumen 30/40 (VG 40) | cum | 7892.00 |
| | (iii) for Grading I (37.5 mm nominal size) Bitumen Content minimum 4.3% with CRMB 60 | cum | 8442.00 |
| | (iv) for Grading II (26.5 mm nominal size) Bitumen Content minimum 4.5% with bitumen 60/70 (VG 30) | cum | 8259.00 |
| | (v) for Grading II (26.5 mm nominal size) Bitumen Content minimum 4.5% with bitumen 30/40 (VG 40) | cum | 8605.00 |
| | (vi) for Grading II (26.5 mm nominal size) Bitumen Content minimum 4.7% with CRMB 60 | cum | 9018.00 |
| | b) Using Drum Mix Plant of appropriate capacity. | | |
| | (i) for Grading I (37.5 mm nominal size) Bitumen Content minimum 4%with bitumen 60/70 (VG 30) | cum | 7085.00 |
| | (ii) for Grading I (37.5 mm nominal size) Bitumen Content minimum 4% with bitumen 30/40 (VG 40) | cum | 7392.00 |
| | (iii) for Grading I (37.5 mm nominal size) Bitumen Content minimum 4.3% with CRMB 60 | cum | 7942.00 |
| | (iv) for Grading II (26.5 mm nominal size) Bitumen Content minimum 4.5% with bitumen 60/70 (VG 30) | cum | 7759.00 |
| | (v) for Grading II (26.5 mm nominal size) Bitumen Content minimum 4.5% with bitumen 30/40 (VG 40) | cum | 8105.00 |
| | (vi) for Grading II (26.5 mm nominal size) Bitumen Content minimum 4.7% with CRMB 60 | cum | 8518.00 |

| Item No. | | Description | Unit | Rate |
|---|---|---|---|---|
| 5.7 | | **Bituminous Concrete**<br>Providing and laying bituminous concrete using crushed aggregates of specified grading, premixed with bituminous binder @ 5.4 to 5.6 % by weight of total mix of mix and filler, transporting the hot mix to work site, laying with an Electronic Sensor paver finisher with sensor control to the required grade, level and alignment, rolling with smooth wheeled, vibratory tandem and pneumatic tyre rollers to achieve the desired compaction as per clause 507 of specification. | | |
| | a) | Using Batch Mix Plant of appropriate capacity | | |
| | (i) | for Grading-I (50-65 mm thickness) Bitumen Content minimum 5.2% with 60/70 bitumen (VG-30) | cum | 9487.00 |
| | (ii) | for Grading-I (50-65 mm thickness) Bitumen Content minimum 5.2% with 30/40 bitumen (VG-40) | cum | 9895.00 |
| | (iii) | for Grading I (50 mm thickness) Bitumen Content minimum 5.4% with CRMB-60 | cum | 10334.00 |
| | (v) | for Grading-II (30-40 mm thickness) Bitumen Content minimum 5.4% with 60/70 bitumen (VG-30) | cum | 9760.00 |
| | (vi) | for Grading-II (30-40 mm thickness) Bitumen Content minimum 5.4% with 30/40 bitumen (VG-40) | cum | 10185.00 |
| | (vii) | for Grading II (30-40 mm thickness) Bitumen Content minimum 5.6% with CRMB -60 | cum | 10629.00 |
| | b) | Using Drum Mix Plant of appropriate capacity. | | |
| | (i) | for Grading-I (50-65 mm thickness) Bitumen Content minimum 5.2% with 60/70 bitumen (VG-30) | cum | 8987.00 |
| | (ii) | for Grading-I (50-65 mm thickness) Bitumen Content minimum 5.2% with 30/40 bitumen (VG-40) | cum | 9395.00 |
| | (iii) | for Grading I (50-65 mm thickness) Bitumen Content minimum 5.4% with CRMB -60 | cum | 9834.00 |
| | (v) | for Grading-II (30-40 mm thickness) Bitumen Content minimum 5.4% with 60/70 bitumen (VG-30) | cum | 9260.00 |
| | (vi) | for Grading-II (30-40 mm thickness) Bitumen Content minimum 5.4% with 30/40 bitumen (VG-40) | cum | 9685.00 |
| | (vii) | for Grading II (30-40 mm thickness) Bitumen Content minimum 5.6% with CRBM-60 | cum | 10129.00 |
| | | Note:- The aggregate Avilable by milling process (30% of the total exvated quantity by milling process) will be used in item no 5.6 & 5.7. this quantity shall be the property of the contractor subject to recovery of Rs. 700/- per cum. | | |
| 5.8 | | **Surface Dressing**<br>Providing and laying surface dressing as wearing course in single coat using crushed stone aggregates of specified size on a layer of bituminous binder laid on prepared surface and rolling with 8-10 tone smooth wheeled steel roller all complete as per clause 509 of specification. | | |
| | I | with 19 mm nominal chipping size & bitumen @1.2kg per sqm. | sqm | 94.00 |

| Item No. | | Description | Unit | Rate |
|---|---|---|---|---|
| | II | with 13 mm nominal chipping size & bitumen @ 1.0 kgper sqm. | sqm | 74.00 |

**NOTE: - Normally 13 mm chipping size should be used. second coat, if specified, should be provided as per clause 509.3.6 of specifications.**

5.9 **Open - Graded Premix Surfacing**

Providing, laying and rolling of open - graded premix surfacing of 20 mm thickness composed of 13.2 mm to 5.6 mm aggregates either using penetration grade bitumen or cut-back or emulsion to required line, grade and level to serve as wearing course on a previously prepared base, including mixing in a suitable plant, laying and rolling with a smooth wheeled roller 8-10 tone capacity, finished to required level and grades all complete as per clause 510 of specification.

| | | | |
|---|---|---|---|
| (i) | Case - I: Mechanical method using Penetration grade Bitumen VG 30 and HMP of appropriate capacity not less than75 tones/hour. | sqm | 121.00 |
| (ii) | Case - II: Open-Graded Premix Surfacing usingcationic Bitumen Emulsion | sqm | 135.00 |

5.10 **Close Graded Premix Surfacing/Mixed Seal Surfacing** (Mechanical means using HMP of appropriate capacity not lessthan 75 tones /hour.)

Providing, laying and rolling of close-graded premix surfacing material of 20 mm thickness composed of 11.2 mm to 0.09 mm (Type-A) or 13.2 mm to 0.09 mm (Type-B) aggregates using penetration grade bitumen to the required line, grade and level to serve as wearing course on a previously prepared base, including mixing in a suitable plant, laying and rolling with a Smooth wheeled roller 8-10 tonne capacity & finishing to required level and grade all complete as per clause 508 of specifications.

**with 60/70 butumen (VG-30)**

| | | | |
|---|---|---|---|
| i) | Type A | sqm | 157.00 |
| ii) | Type B | sqm | 139.00 |

5.11 **Seal Coat**

Providing and laying seal coat, sealing the voids in a bituminous surface laid to the specified levels, grade and cross fall using Type A and B seal coats, all complete as per clause 511 of Specification.

| | | | |
|---|---|---|---|
| (i) | Case - I: Type A Providing and laying of liquid seal coat comprising of an application of layer of bitumen binder followed by a cover of stony chips, All complete as per clause 511 of specification. | sqm | 71.00 |
| (ii) | Case - II: Type B (Providing and laying of premix sand seal coat using HMP of appropriate capacity not less than 75 tonnes/ hours using crushed stone chipping (aggregate shall pass 2.36mm sieve and retained to 180-micron sieve) and penetration bitumen VG 30 of suitable grade all complete as per clause 511 of specification. | sqm | 50.00 |

5.12 **Pavement Milling**

Milling of existing Bituminous surface to a specified depth using Milling machine including disposal of removed material within all lifts and leads upto 1000 metres

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| | Up to 50 mm | sqm | 48.00 |
| | (ii) 50 mm to 100 mm | sqm | 53.00 |
| | (iii) 100 mm to 150 mm | sqm | 58.00 |
| | Note: - This item shall be included in the DPR and executed with prior written permission from CE MPPWD. | | |
| 5.13 | **Mastic Asphalt**<br>Providing and laying 25 mm thick mastic asphalt wearing course with paving grade bitumen meeting the requirements given in table 500-39, prepared by using mastic cooker and laid to required level and slope after cleaning the surface, including providing antiskid surface with bitumen precoated fine-grained hard stone chipping of 13.2 mm nominal size at the rate of 0.005cum per 10 sqm and at an approximate spacing of 10 cm center to center in both directions, pressed into surface when the temperature of surfaces not less than $100^0$C, protruding 1 mm to 4 mm over mastic surface, all complete as per clause 516 of specification. | sqm | 937.00 |
| 5.14 | **Slurry Seal** | | |
| | (i) Providing and laying slurry seal consisting of a mixture of fine aggregates, Portland cement filler, bituminous emulsion and water on road surface including cleaning of surface, mixing of slurry seal in a suitable mobile plant, laying and compacting to provide even riding surface all complete as per clause 512 of Specifications. | | |
| | (i) 5.0 mm thickness Type-III (6-8) | sqm | 70.00 |
| | (ii) 3.0 mm thickness Type-II (4-6) | sqm | 50.00 |
| | (iii) 1.5 mm thickness Type-I (2-3) | sqm | 30.00 |
| 5.15 | **Recycling of Bituminous Pavement with Central Recycling Plant**<br>Recycling pavement by cold milling of exiting bituminous layers, planning the surface after cold milling, reclaiming excavated material to the extent of 30 % of the required quantity, hauling and stock piling the reclaimed material near the central recycling plant after carrying out necessary checks and evaluation, adding fresh material including rejuvenators as required, mixing in a hot mix plant, transporting and laying at site and compacting to the required grade, level and thickness, all as specified in clause 519 of specification. | cum | 4829.00 |
| 5.16 | **Sand Asphalt Base Course**<br>Providing, laying and rolling sand-asphalt base course composed of sand, mineral filler and bituminous binder on a prepared sub- grade or sub-base to the lines, levels, grades and cross sections as per the drawings including mixing in a plant of suitable type and capacity, | cum | 9368.00 |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| | transporting, laying, compacting and finishing complete as per clause 506 of specification. | | |
| 5.17 | **Crack Prevention Courses** | | |
| | (i) Stress Absorbing Membrane (SAM) crack width less than 6 mm Providing and laying of a stress absorbing membrane over a cracked road surface, with crack width below 6 mm after cleaning with a mechanical broom, using modified binder complying with clause 517, sprayed at the rate of 10-12 kg per 10 sqm and spreading 5.6 mm crushed stone aggregates @ 0.11 cum per 10 sqm with hydraulic chip spreader, sweeping the surface for uniform spread of aggregates and surface finished to conform to clause 902 of specification all complete as per clause 517 of specification. | sqm | 69.00 |
| | (ii) Stress Absorbing Membrane (SAM) with crack width 6 mm to 9 mm Providing and laying a single coat of a stress absorbing membrane over a cracked road surface, with crack width 6 to 9 mm after cleaning with a mechanical broom, using modified binder complying with clause 517, sprayed at the rate of 11 kg per 10 sqm and spreading 11.2 mm crushed stone aggregates @ 0.12 cum per 10 sqm, sweeping the surface for uniform spread of aggregates and surface finished to conform to clause 902. | sqm | 83.00 |
| | (iii) Stress Absorbing Membrane (SAM) crack width above 9mm and cracked area above 50 %<br>Providing and laying a single coat of a stress absorbing membrane over a cracked road surface, with crack width above 9 mm and cracked area above 50 % after cleaning with a mechanical broom, using modified binder complying with clause 517, sprayed at the rate of 15 kg per 10 sqm and spreading 11.2 mm crushed stone aggregates @ 0.12 cum per 10 sqm, sweeping the surface for uniform spread of aggregates and surface finished to conform to clause 902. | sqm | 111.00 |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| | (iv) Case - IV: Bitumen Impregnated Geotextile<br>Providing and laying a bitumen impregnated geotextile layer after cleaning the road surface, geotextile conforming to requirements of clause 704.3, laid over a tack coat with 1.05 kg per sqm of paving grade bitumen 80 - 100 penetration (VG-10) and constructed to the requirement of clause 704.4.5 | sqm | 263.00 |
| 5.18 | **Recipe Cold Mix**<br>(Providing and laying of premix of crushed stone aggregates and emulsion binder, mixed in a batch type cold mixing plant, laid over prepared surface, by paver finisher, rolled with a pneumatic tyred roller initially and finished with a smooth steel wheel roller, all as per clause 518.3) | | |
| | (i) 25 mm thickness | cum | 7759.00 |
| | (ii) 40 mm thickness | cum | 10950.00 |
| | (iii) 75 mm thickness | cum | 12865.00 |
| 5.19 | Providing, laying and rolling of close-graded premix surfacing / mixed seal surfacing material of 20 mm thickness composed of 13.2 mm to 0.09 mm (Type-B) aggregates using penetration grade bitumen, including PCC as a integral part of it, to the required line, grade and level to serve as wearing course on a previously prepared base, including mixing in a suitable plant, laying and rolling with a Smooth wheeled roller 8-10 tonne capacity, and finishing to required level and grade and as per relevant clauses of section-500. | | |
| (i) | Type-B aggregate | | |
| A | with 60/70 bitumen | Cum. | 6050.00 |
| 5.20 | Providing and laying Micro Surfacing Course as per IRC SP 81:2008 on existing structural sound & distressed pavement surface. (It is to be applied over an existing pavement surface which is structurally sound, but the surface is showing signs of premature ageing, aggregate loss, high degree of polishing, oxidation / hungry surface) comprising of aggregates confirming to section 514 of MoRT&H Specification. The Modified Bitumen Emulsion @ 10%, Filler cement @ 2%, Additive @ 0.5% and water as per requirement, aggregates as per Table 500-27. All these ingredients are mixed with required quantity of water to prepare semi fluid mass. The mix shall be spread mechanically using | | |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| | Microsurfacing Paver. The Cost includes cleaning of surface with Air compresser, laying and compacting to provide even riding surface and shall be rolled by Pneumatic Tyre Roller and ensure excessive movement does not occur, including cost of all materials, labour, usage charges of machinery, lead, lifts, loading, unloading, stacking, transporting, etc complete as per technical specification. | | |
| i) | Type II grading (4 mm to 6 mm thickness) | sqm | 163.00 |
| ii) | Type III grading (6 mm to 8mm thickness) | sqm | 184.00 |
| | NOTE This item shall be executed only after prior approval of the C.E. | | |
| 5.21 | labour only for Laying and rolling of Bituminous courses i/c primer and tack coat (excluding cost of bitumen & metal) with a smooth wheeled roller 8-10 tonne capacity i/c hire charges & running expenses, finished to required level and grades and as per relevant clauses of section-500. | | |
| i) | 20mm Open Graded Premix Carpet | sqm | 23.00 |
| ii) | 75mm Built-up Spray Grout | sqm | 26.00 |
| 5.22 | **Bituminous Concrete with waste plastic** | | |
| ***(i)*** | **Bituminous Concrete Grading -I with waste plastic** (Providing and laying bituminous concrete with higher capacity batch type hot mix plant using crushed aggregates of specified grading, premixed with bituminous binder @ 5.2 per cent of mix using 8 per cent waste plastic of Bituminous Binder and filler, transporting the hot mix to work site, laying with a hydrostatic paver finisher with sensor control to the required grade, level and alignment, rolling with smooth wheeled, vibratory and tandem rollers to achieve the desired compaction as per MoRT&H specification clause No. 507 complete in all respects) | cum | 8451.00 |
| ***(ii)*** | **Bituminous Concrete Grading -II with waste plastic** (Providing and laying bituminous concrete with higher capacity batch type hot mix plant using crushed aggregates of specified grading, premixed with bituminous binder @ 5.4 per cent of mix using 8 per cent waste plastic of Bituminous Binder and filler, transporting the hot mix to work site, laying with a hydrostatic paver finisher with sensor control to the required grade, level and alignment, rolling with smooth wheeled, vibratory and tandem rollers to achieve the desired compaction as per MoRT&H specification clause No. 507 complete in all respects) | **cum** | 8702.00 |
| 5.23 | **20 mm thick open-grade premix carpet using Bituminous Binder mixing with processed waste plastic by replacing 8% Bitumen binder by weight. (as per IRC; SP:98-2013).** Providing laying and | | |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| | rolling of 20 mm thick open-graded premix carpet (thickness not less than 20 mm at any place) composed of 13.2 mm graded aggregate size @ 0. 18 cum per 10 sqm &11.2 mm graded size 0.09 cum per 10 sqm using with bituminous binder @ 13.43 kg per 10 sqm (i.e 92% of standard bitumen) and 8 % waste processed plastic by weight shall be used for coating aggregate after its heating as per IRC:SP:98-2013, laying to required line, grade and level to serve as wearing course on a previously prepared base, including mixing in suitable plant, laying and rolling with a three wheel 8 to 10 ton static roller capacity, finished to required level and grades as per technical specification clause 508 to be seal coat of either type A, type B or type C as per technical specification<br>Note- The above shall be included in the DPR and executed with prior written permission from the E-IN-C MPPWD.<br>(i) Bitumen (VG-30) | sqm | 114.00 |
| 5.24 | Deduction for not using Electronic Sensor Paver for the relevant Item. | Cum | 250.00 |

# CHAPTER: 6.0

# CEMENT CONCRETE PAVEMENTS

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| 6.1 | **Dry Lean Cement Concrete Sub- base**<br>Construction of dry lean cement concrete Sub- base over a prepared sub-grade or drainage layer with coarse and fine aggregate conforming to IS: 383, the size of coarse aggregate not exceeding 25 mm, aggregate cement ratio not to exceed 15:1, aggregate gradation after blending to be as per table 600-1, cement content not to be less than 150 kg/ cum, optimum moisture content to be determined during trial length construction, concrete strength not to be less than 10 Mpa at 7 days, mixed in a batching plant, transported to site, laid with a paver finisher with electronic sensor, compacting with 8- 10 tonnes vibratory roller, finishing and curing as per clause 601 of specification complete in all respect | cum | 2712.00 |
| 6.2 | Deduct from Item No.6.1 above if paver finisher with electronic sensor is not used and laying, (The acceptance criteria regarding level, thickness, surface regularity, texture finish, strength of concrete and all other quality control measures shall be the same as in case of machine laid work.)<br><br>**Note: - This item is to be executed with prior written permission from the C.E.** | cum | 300.00 |
| 6.3 | **Cement Concrete Pavement**<br>Construction of un-reinforced, dowel jointed, plain cement concrete pavement with M-40 grade concrete over a prepared sub base with cement , coarse and fine aggregate conforming to IS 383, maximum size of coarse aggregate not exceeding 25 mm, mixed in a batching and mixing plant as per approved mix design, transported to site, laid with electronic sensor slip form paver, spread, compacted and finished in a continuous operation including provision of contraction, expansion, construction and longitudinal joints, joint filler, separation membrane, sealant primer, joint sealant, debonding strip, admixtures as approved, curing compound, finishing to lines and grades as per drawing as per IRC-15-2011 and as per relevant clauses of section 602 of specifications complete but excluding cost of steel in dowel bar and tie rods etc. including groove cutting for width of 4 mm- 6mm and 1/3rd depth of PQC within 8-10 hrs of placing/ finishing of PQC including cleaning of groove and providing approved sealant filling etc. as per clause 602.2.10 and 600.10 and IRC-57-2006 as per drawing and as per specification. | cum | 5769.00 |
| 6.4 | **Cement Concrete Pavement**<br>Construction of un-reinforced, dowel jointed, plain cement concrete pavement with M-30 grade concrete over a prepared sub base with cement , coarse and fine aggregate conforming to IS 383, maximum size of coarse aggregate not exceeding 25 mm, mixed in a batching and mixing plant as per approved mix design, transported to site, laid with electronic sensor slip form paver, spread, compacted and finished in a continuous operation including provision of contraction, expansion, construction and longitudinal joints, joint filler, separation membrane, sealant primer, joint sealant, debonding strip, admixtures as approved, curing compound, finishing to lines and grades as per drawing as per IRC-15-2011 and as per relevant clauses of section 602 of specifications complete but excluding cost of steel in dowel bars and tie rods etc. including groove cutting for | cum | 5532.00 |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| | width of 4 mm- 6mm and 1/3rd depth of PQC within 8-10 hrs of placing/ finishing of PQC including cost of cutting including cleaning of groove and providing approved sealant filling etc. as per clause 602.2.10 and 600.10 and IRC-57-2006 as per drawing and as per specification. | | |
| 6.5 | Deduct from Item No.6.3 & 6.4 above if slip form paver with electronic sensor is not used and laying, compaction is done by fix form paver having 60 mm vibratory needle @ 2 Nos. per metre. (The acceptance criteria regarding level, thickness, surface regularity, texture finish, strength of concrete and all other quality control measures shall be the same as in case of machine laid work.)<br>**Note: - This item is to be executed with prior written permission from the C.E.** | cum | 800.00 |
| 6.6 | **Rolled Cement Concrete Base**<br>Construction of rolled cement concrete base course with coarse and fine aggregate conforming to IS:383, the size of coarse aggregate not exceeding 25 mm with minimum, aggregate cement ratio not to exceed 15:1 and minimum cement content of 200 kg/cum, aggregate gradation to be as per table 600-3 after blending, mixing in batching plant at optimum moisture content, transporting to site, laying with a paver with electronic sensor, compacting with 8-10 tonnes smooth wheeled vibratory roller to achieve, the designed flexural strength, finishing and curing complete. | cum | 3087.00 |
| 6.7 | Deduct from Item No.6.6 above if paver with electronic sensor is not used and laying, compaction is done by any other method. (The acceptance criteria regarding level, thickness, surface regularity, texture finish, strength of concrete and all other quality control measures shall be the same as in case of machine laid work.) .<br>**Note: - This item is to be executed with prior written permission from the C.E.** | cum | 150.00 |
| 6.8 | Providing and fixing Steel in dowel bar and tie rod including craddle, laps etc. complete required as per design. | | |
| | (a) Mild steel confirming to IS: 432:1982 | MT | 57734.00 |
| | (b) Tor steel confirming to IS: 1786:2008 | MT | 58734.00 |
| 6.9 | **Cement Concrete Pavement M-40 grade for Conventional White topping. (Thickness 200 mm or more).**<br>Providing and laying Conventional white Topping as per IRC: SP:76-2015 including polymeric synthetic fibre as per ASTM (1116) such as polyester/polyethylene/ polypropylene / Glass fibre at 9.25 kg/cum with groove cutting grid 1m x 1m to a depth of 1/3 of the Thickness and of width3-5 mm with sealing joint as per design including cost of all material Over bituminous surface (Existing bituminous surface prepared if necessary, either by milling or leveling course to be done with BM/DBM/DLC to correct surface profile and to be paid separately) as Cement Concrete Pavement, Construction. of un-reinforced, 'dowel Jointed, plain cement concrete pavement M-40 grade concrete over a prepared sub base, coarse and fine aggregate conforming to IS 383. maximum size of coarse aggregate not | cum | 7600.00 |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| | exceeding 31.5 mm, mixed' in a batching and mixing plant as per approved mix design. transported to site, laid with a slip form paver, spread, compacted and finished, in a continuous operation including provision of expansion Joint, construction joint separation membrane admixtures as approved, curing compound, finishing to lines and grades as per drawing as per IRC 15 -2017 and as per relevant clauses of MoRT&H Specifications of section 600. Complete but excluding cost of steel in dowel bar and tie rods including joint sealing complete. | | |
| 6.10 | Cement Concrete Pavement M-40 grade for Thin white topping (Thickness greater than 100 mm and less than 200 mm)<br>Providing and laying Conventional white Topping as per IRC: SP:76-2015 including polymeric synthetic fibre as per ASTM (1116) such as polyester / polyethylene / polypropylene / Glass fibre at 9.25 kg/cum with groove cutting grid 1m x 1m to a depth of 1/3 of the Thickness and of width 3-5 mm as per design including cost of all material Over bituminous surface (Existing bituminous surface prepared if necessary, either by milling or leveling course to be done: with. BM/DBM/DLC to correct surface profile and to be paid separately) as Cement Concrete Pavement, Construction, of un-reinforced, 'dowel Jointed, plain cement concrete pavement M-40 grade concrete over a prepared sub base. coarse and fine aggregate conforming to IS 383, maximum size of coarse aggregate not exceeding 31.5 mm. mixed' in a batching and mixing plant as per approved mix design. transported to site, laid with a slip form paver.spreading, compacted and finished. In a continuous operation including. Provision of expansion Joint. construction joint, separation membrane. admixtures as approved, curing compound. finishing to lines and grades as per drawing as per IRC 15 2017 and as per relevant clauses of MoRT&H Specifications of section 600. Complete but excluding cost of steel in dowel bar and tie rods including joint sealing complete. | cum | 7695.00 |
| 6.11 | **Cement concrete pavement M-40 grade for Ultra Thin white Topping (Thickness equal to or less than 100 mm)**<br>Providing and laying Conventional white Topping as per IRC: SP:76-2015 including polymeric synthetic fibre as per ASTM (1116) such as polyester / polyethylene / polypropylene / Glass fibre at 9.25 kg/cum with groove cutting grid 1m x 1m to a depth of 1/3 of the Thickness and of width3-5 mm with sealing joint as per design including cost of all material Over bituminous surface (Existing bituminous surface prepared if necessary, either by milling or leveling course to be done with BM/DBM/DLC to correct surface profile and to be paid separately) as Cement Concrete Pavement, Construction of un-reinforced, 'dowel Jointed. plain cement concrete pavement M-40 grade concrete over a prepared sub base, coarse and fine aggregate conforming to IS 383 maximum size of coarse aggregate not exceeding 31.5 mm, mixed' in a batching and mixing plant as per approved mix design. transported to site, laid with a slip form paver, spread, compacted and finished, in a continuous operation including. Provision of expansion Joint, construction joint, separation membrane admixtures as approved, curing compound, finishing to lines and grades as per drawing as per IRC 15 2017 and as per relevant clauses of MoRT&H Specifications of section 600. Complete but excluding cost of steel in dowel bar and tie rods including joint sealing complete. | cum | 7790.00 |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| 6.12 | Providing groove cutting for width of 4 mm- 6mm and 1/3rd depth of PQC White topping within 8-10 hrs of placing/ finishing of PQC Including cost of cutting. machine, generators, Handling Operation and Maintenance of machinery, blades, coolant etc. including cleaning of groove and providing approved sealant filling etc. as per clause 602.2.10 and 600.10 and IRC-57-2006, including cost of conveyance of all materials labour, lead and lift charges and all other incidental charges as per drawing and as per specification. | MT | 101.00 |
| 6.13 | Providing and mixing fiber mesh stealth e3 Fiber mesh 150 multi-dimensional graded multifilament 100% virgin macro polypropylene/ polymeric fibers contain no reprocessed olefin materials and are specifically manufactured for use as concrete secondary reinforcement. Application per cubic meter shall be @ 0.9kg per cum of concrete. Fiber mesh stealth e3 shall confirm to ASTM C 1116 Type 111.4.106 | cum | 396.00 |
| 6.14 | Providing and laying factory made coloured chamfered edge Cement Concrete paver blocks of required strength, thickness and size/shape, made by table vibratory method using PU mould, laid in required colour and pattern over 50mm thick compacted bed of stone dust, compacting and proper embedding/laying of inter locking paver blocks into the bedding layer through vibratory compaction by using plate vibrator, filling the joints with sand and cutting of paver blocks as per required size and pattern, finishing and sweeping extra sand including locking edges with M 15 cement concrete in footpath, light traffic parking etc. complete as per direction of Engineer-in-Charge. | | |
| | (i) 100mm thick C.C. paver block of M-35 grade with approved colour, design and pattern. | sqm | 837.00 |
| | (ii) 80mm thick C.C. paver block of M-35 grade with approved colour, design and pattern. | sqm | 748.00 |
| 6.15 | Providing and laying reinforced cement concrete pipe NP-4 for service ducts below concrete pavement. | | |
| | a) 300 mm Dia | metre | 850.00 |
| | b) 600 mm Dia | metre | 4250.00 |
| 6.16 | Providing, fixing and lowering, laying and jointing Double Wall Corrugated (DWC) Pipes of HDPE in 6m. length bar with IS-14930-II in prepared trenches in grade and level including cost of couplers and testing of pipes at factory as per the stipulation in relevant IS codes etc. complete. | | |
| | i) 145mm outer diameter / 123mm inner diameter | metre | 399.00 |
| | ii) 200mm outer diameter / 174mm inner diameter | metre | 611.00 |
| **6.17** | **Construction of Base/Sub-base of pavement with lean concrete - fly ash.** (Construction of Base/sub-base using cement, sand, fly ash and coarse aggregates proportioned as per table 4 of IRC: 74/1979 and with water content ratio, slump and compressive strength as defined in the said table, mix prepared in a batching and mixing plant and compacted with a vibratory roller 8-10 tonnes capacity within the time limit laid down vide clause 7.6.3 of IRC: 74-1979, construction joints properly formed at the end of day's work, cured for 14 days, all as specified in IRC: 74-1979 and as per approved plans.) | cum | 2576.00 |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| **6.18** | **Cement - Fly ash concrete pavement.** (Construction reinforced, dowel jointed, plain cement concrete pavement over a prepared sub base with 43 grade cement, coarse and fine aggregate conforming to IS 383, maximum size of coarse aggregate not exceeding 25 mm, replacing cement by fly ash to the extent of 15% and sand by 10%, mixed in a batching and mixing plant as per approved mix design, transported to site, laid with a fixed form or slip form paver, spread, compacted and finished in a continuous operation including provision of contraction, expansion, construction and longitudinal joints, joint filler, separation membrane, sealant primer, joint sealant, debonding strip, dowel bar, tie rod, admixtures as approved, curing compound, finishing to lines and grades as per drawing) | cum | 5406.00 |
| 6.19 | Supplying and providing continuously deformed Steel Fiber Reinforcement in RCC for pavement, heavy duty roads and in parking areas, undulating steel fibers with aspect ratio between 50 to 80 (Diameter between 0.45 to 1 mm and length between 12.5 to 60 mm) manufactured from cold drawn wire and having a minimum tensile strength of 1100N/Sq. mm, as per CE Certification label mark, non-glued/glued, non-balling, conforming to ASTM A 820/A 820 M-06, compatible with all construction chemicals used for the batching, laying and curing of concrete. The fibers shall be added to the concrete at specified dosage and shall be mixed with concrete for sufficient time to ensure uniform distribution and pumping of concrete. Addition of steel fibers in cement concrete supported slab/underground structure/precast items (like drains, manhole covers, walls, paver blocks, pipes, tunnel segments) etc. dose as per approved designs and directions of Engineer-in-charge. Dose of Steel shall be as per IRC SP: 46-2013 updated Amendment | kg | 98.00 |
| 6.20 | Supplying and providing Glass fibers chopped strands of 12mm length at 900 grams per/cum for mixing in RCC work to gain benefit of anti-cracking. Glass Fibers conform to properties as per ASTM c1666@-M-08 Density:2.68 +0.3g/m3 Tensile strength: 1-1.7 MPA Filament Diameter:31x10-118x10 inch AR Value: As per EN 14649 Dose of Fiber shall be as per IRC SP: 46-2013 updated Amendment. | kg | 250.00 |

# CHAPTER: 7.0

# GEOSYNTHETICS AND REINFORCED EARTH

| | | | |
|---|---|---|---|
| 7.1 | **Sub- Surface Drain with Geotextiles**<br>Construction of sub surface drain 200 mm dia using geotextiles treated with carbon black with physical properties as given in clause 702.2.3 formed in to a stable network and a planar geocomposite structure, joints wrapped with geotextile to prevent ingress of soil, all as per clause 702 and approved drawings including excavation and backfilling. | metre | 997.00 |
| 7.2 | **Narrow Filter Sub- Surface Drain**<br>Construction of a narrow filter sub- surface drain consisting of porous or perforated pipe laid in narrow trench surrounded by a geotextile filter fabric, with a minimum of 450 mm overlap of fabric and installed as per clause 702.3 and 309.3.5 including excavation and backfilling | metre | 677.00 |
| 7.3 | **Laying Boulder Apron in Crates of Synthetic Geogrids** Providing, preparing and laying of geogrid crated apron 1 m x 5 m, 600 mm thick including excavation and backfilling with baffles at 1 metre interval, made with geogrids having characteristics as per clause 704.2, joining sides with connectors/ring staples, top corners to be tie tensioned, placing of suitable cross interval ties in layers of 300 mm connecting opposite side with lateral braces and tied with polymer braids to avoid bulging, constructed as per clause 704.3. filled with stone with minimum size of 200 mm and specific gravity not less than 2.65, packed with stone spalls, keyed to the foundation recess in case of sloping ground and laid over a layer of geotextile to prevent migration of fines, all as per clause 704 and laid as per clause 2503.3 and approved design. | cum | 3783.00 |
| 7.4 | **Reinforced Earth Retaining Wall**<br>Reinforced earth retaining walls have four main components as under: (i) Excavation for foundation, foundation concrete and cement concrete grooved seating in the foundation for facing elements (facia material (ii) Assembling, joining with facing elements and laying of the reinforcing elements.). (iii) Facia material and its placement. (iv) Earthfill with specified material as per specification to be retained by the wall. (This will be paid extra) | | |
| | (i) Assembling, joining and laying of reinforcingelements. | | |
| | With reinforcing element of polymeric strips. | | |
| | Glass reinforced polymer/fibre reinforced polymer/polymeric strips<br>(Rate as per item 7.7) | | |
| | With reinforcing elements of synthetic geogrids<br>(Rate as per item 7.6 or 7.7 which ever is applicable) | | |
| (ii) | Facing elements of RCC M35 | | |
| | (a) Pre-cast reinforced concrete panel | sqm | 1890.00 |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| | (b) Pre-cast concrete Hollow Blocks | sqm | 1690.00 |
| (iii) | Earth fill with specified material as per relevantspecifications. (The earth fill material shall be clean, free drawing granular with high friction and low cohesion, noncorrosive, coarse grained with not 10 percent of particle passing 75-micron sieve free of any deleterious matter, chlorides, salt, acids alkaline mineral oil, fungus and microbes and shall be of specified PH value.) (a) | cum | 312.00 |
| 7.5 | Glass reinforced polymer/fibre reinforced polymer/ polymeric strips | | |
| | (i) Synthetic Geogrid Ultimate tensile strength- 30 kN/m | Rm | 37.00 |
| | (ii) Synthetic Geogrid Ultimate tensile strength- 40 kN/m | Rm | 41.00 |
| | (iii) Synthetic Geogrid Ultimate tensile strength- 50 kN/m | Rm | 46.00 |
| | (iv) Synthetic Geogrid Ultimate tensile strength- 60 kN/m | Rm | 56.00 |
| | (v) Synthetic Geogrid Ultimate tensile strength- 75 kN/m | Rm | 62.00 |
| | (vi) Synthetic Geogrid Ultimate tensile strength- 85 kN/m | Rm | 70.00 |
| | (vii) Synthetic Geogrid Ultimate tensile strength- 100 kN/m | Rm | 75.00 |
| 7.6 | **Bi-axial extruded high modulus polypropylene geogrid** | | |
| | (i) Supplying & laying of bi-axial extruded high modulus polypropylene geogrid conforming to MORT&H specification for base/sub-base reinforcement having minimum tensile strength 15kN/m in the longitudinal and transverse direction, with 5kN/m and 7kN/m tensile strength at 2% and 5% strain respectively in the longitudinal and transverse direction, junction efficiency not less than 95% and with 38mm X 38mm mesh opening. | sqm | 175.00 |
| | (ii) Supplying & laying of bi-axial extruded high modulus polypropylene geogrid conforming to MORT&H specification for base/sub-base reinforcement having minimum tensile strength 20kN/m in the longitudinal and transverse direction, with 7kN/m and 14kN/m tensile strength at 2% and 5% strain respectively in the longitudinal and transverse direction, junction efficiency not less than 95% and with 38mm X 38mm mesh opening. | sqm | 221.00 |
| | (ii) Supplying & laying of bi-axial extruded high modulus polypropylene geogrid conforming to MORT&H specification | sqm | 261.00 |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| | for base/sub-base reinforcement having minimum tensile strength 30kN/m in the longitudinal and transverse direction, with 10.5kN/m and 21kN/m tensile strength at 2% and 5% strain respectively in the longitudinal and transverse direction, junction efficiency not less than 95% and with 38mm X 38mm mesh opening. | | |
| | (iv) Supplying & laying of bi-axial extruded high modulus polypropylene geogrid coforming to MORT&H specification for base/sub-base reinforcement having minimum tensile strength 40kN/m in the longitudinal and transverse direction, with 14kN/m and 28kN/m tensile strength at 2% and 5% strain respectively in the longitudinal and transverse direction, junction efficiency not less than 95% and with 38mm X 38mm mesh opening. | sqm | 268.00 |
| 7.7 | Supplying and laying high strength flexible Polymer Strip (HSFG) as soil reinforcement / basal reinforcement as per MoRT&H 3100 and IRC 113, made of high tenacity polyester core with polyethylene coating with minimum Long-Term Design Strength (LTDS) of more than 50% of ultimate tensile strength at 30-degree Celcius corresponding to 12 % strain etc. complete and as directed by Engineer - In - Charge. | | |
| | (i) Polymer Strip Ultimate tensile strength- 100 KN/cum | sqm | 108.00 |
| | (ii) Polymer Strip Ultimate tensile strength- 150 KN/cum | sqm | 121.00 |
| | (iii) Polymer Strip Ultimate tensile strength- 200 KN/cum | sqm | 135.00 |
| | (iv) Polymer Strip Ultimate tensile strength- 250 KN/cum | sqm | 148.00 |
| | (v) Polymer Strip Ultimate tensile strength- 300 KN/cum | sqm | 161.00 |
| | (vi) Polymer Strip Ultimate tensile strength- 350 KN/cum | sqm | 175.00 |
| | (vii) Polymer Strip Ultimate tensile strength- 400 KN/cum | sqm | 188.00 |
| | (viii) Polymer Strip Ultimate tensile strength- 500 KN/cum | sqm | 200.00 |
| | (ix) Polymer Strip Ultimate tensile strength- 600 KN/cum | sqm | 215.00 |
| | (x) Polymer Strip Ultimate tensile strength- 800 KN/cum | sqm | 228.00 |
| | (xi) Polymer Strip Ultimate tensile strength- 900 KN/cum | sqm | 241.00 |
| | (xii) Polymer Strip Ultimate tensile strength- 1000 KN/cum | sqm | 255.00 |
| | (xiii) Polymer Strip Ultimate tensile strength- 1100 KN/cum | sqm | 268.00 |
| | (xiv) Polymer Strip Ultimate tensile strength- 1200 KN/cum | sqm | 275.00 |
| | (xv) | | |
| 7.8 | Supplying & laying of drainage composite for use behind walls, between two different fills, alongside drains of road, below concrete | sqm | 188.00 |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| | lining of canals etc. Geocomposite for planar drainage, realized by thermobonding a draining core in extruded monofilaments with two filtering nonwoven geotextiles that may also be working as separation or protecting layers. The draining three-dimensional core will have a "W" configuration as longitudinal parallel channels. Minimum thickness to be 7.2mm, with two filtering UV stabilized polypropylene nonwoven geotextile of minimum thickness of 0.90 mm having pores of 150 micron and tensile strength of 8.0 kN/m that will be working as separation or protecting layer, geocomposite having in plane flow capacity of 2.1 L / (m.s) at hydraulic gradient of 1.0 & 20 kpa pressure and tensile strength of 18 kN/m , with mass per unit area of 740 gsm, supplied in the form of roll for easy transportation to site of work as per detailed specification all complete as per directions of Engineer-in-Charge. | | |
| 7.9 | Supplying & laying of drainage composite for use behind walls, between two different fills, alongside drains of road, below concrete lining of canals etc. Geocomposite for planar drainage, realized by thermobonding a draining core in extruded monofilaments with two filtering nonwoven geotextiles that may also be working as separation or protecting layers. The draining three-dimensional core will have a "W" configuration as longitudinal parallel channels. Minimum thickness to be 7.2mm, with two filtering UV stabilized polypropylene nonwoven geotextile of minimum thickness of 0.75 mm having pores of 150 micron and tensile strength of 8.0 kN/m that will be working as separation or protecting layer, geocomposite having in plane flow capacity of 2.1 L / (m.s) at hydraulic gradient of 1.0 & 20 kpa pressure and tensile strength of 18 kN/m , with mass per unit area of 740 gsm, supplied in the form of roll for easy transportation to site of work as per detailed specification all complete as per directions of Engineer-in-Charge. | sqm | 155.00 |
| 7.10 | Providing and placing in position precast, reinforced cement concrete Earth Retaining Wall as per design and shape, using coarse aggregate and fine aggregate derived from natural sources, Portland Pozzolana/ Ordinary Portland/ Portland slag cement, admixtures in recommended proportions as per IS: 456-2000 and standard like | | |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| | indian standard, international standards viz. Japanese industrial Standards (JIS)/ US Standard/EU standard/having of strength not less than M-50. Using Thermo-Mechanically Treated bars. (TMT/TMX) FE 500D. Precast reinforced cement concrete. Rate includes all cost of material, labor charges, Cost of reinforcement, formwork and lifting accessories, and transportation up to works site. Rate also includes unloading the material at site, all lead and lift including making necessary holes of required sizes for carrying through service lines etc., providing steel hooks for lifting, etc. complete. | | |
| (i) | Earth Retaining Wall- 800 mm Height | RM | 4543.00 |
| (ii) | Earth Retaining Wall- 1000 mm Height | RM | 5139.00 |
| (iii) | Earth Retaining Wall- 1200 mm Height | RM | 5913.00 |
| (iv) | Earth Retaining Wall- 1800 mm Height | RM | 8552.00 |
| (v) | Earth Retaining Wall- 2000 mm Height | RM | 9254.00 |
| (vi) | Earth Retaining Wall- 2250 mm Height | RM | 12133.00 |
| (vii) | Earth Retaining Wall- 2500 mm Height | RM | 13074.00 |

# CHAPTER: 8.0

# TRAFFIC SIGNS, MARKINGS AND OTHER ROAD APPURTENANCES

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| 8.1 | **Cast in Situ Cement Concrete M20 kerb**<br>Construction of cement concrete kerb with top and bottom width 115 and 165 mm respectively, 250 mm high in M 20 grade PCC onM-15grade 150 mm thick, or with kerb laying machine, foundation having 50 mm projection beyond kerb stone, kerb stone laid with kerb laying machine, foundation concrete laid manually, all complete as per clause 409 | | |
| | A Using Concrete Mixer | metre | 360.00 |
| | B Using Concrete Batching and Mixing Plant | metre | 390.00 |
| 8.2 | **Cast in Situ Cement Concrete M 20 Kerb with Channel**<br>Construction of cement concrete kerb with channel with top and bottom width 115 and 165 mm respectively, 250 mm high in M 20 grade PCC on M15 grade foundation 150 mm thick, kerb channel 300 mm wide, 50 mm thick in PCC M20 grade, sloped towards the kerb, kerb stone with channel laid with kerb laying machine, foundation concrete laid manually, all complete as per clause 409 | metre | 390.00 |
| | A Using Concrete Mixer | metre | 495.00 |
| | B Using Concrete Batching and Mixing Plant | | 525.00 |
| 8.3 | **Retro- reflectorised Traffic signs**<br>Providing and fixing of retro-reflectorised cautionary, mandatory and informatory sign board as per IRC 67-2022 made of high intensity Micro-Prismatic Grade Sheeting (Type IV) vide IRC: 67:2022 clause 6.7.3.2 and clause 801.3.3 of MoRT &H fixed over Aluminum composite material sheet with thermoplastic core of Low density polyethylene (LDPE) between two thick skins/sheets of aluminum with overall thickness of 4mm and aluminum skin of thickness 0.3 on both side, the ACM shall conform to Table 6.1 of IRC:67:2022 and Table 800-1 of MoRT &H specification and High Intensity Micro Prismatic Grade Sheeting shall conform to Table 6.6 of IRC:67:2022 and Table 800-3 of MoRT &H specification of in the clause 6.8 of the IRC 67-2022 and clause 801.3.7 of MoRT&H message legend numerical letter etc. and borders shall be screen printed on high intensity micro prismatic sheeting shall conform to clause 801.3.7 with digital printing ink or of cutout from transparent durable overlay or cutout from the same type of reflecting sheeting for cautionary/ mandatory/ signs, supported on MS sign post of size 75mm x75mm x 6mm of 3.60 meter over all length and fix to ground by means of properly designed foundation of dimension 450x450x600mm with M20 grade concrete. The ACM sheet shall be fixed to the post with four minimum four number breakaway bolts. & supported with a back-support frame of 25mmx25mmx3mm angle. 7 years warranty for Retro Reflection Sheeting to be provided from the original sheeting manufacture brand like 3M/Avery Dennison/ Orafol Make as per clause 6.9 of IRC: 67:2022 & a certified copy for three years for outdoor exposure report from an independent test lab as per clause 6.7 of IRC: 67:2022 for the product offered shall be submitted by the contractor.<br>Note: - Manufacture name and date of manufacture is to be printed (Embossed) at the backside of the board. | | |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| | i) 120 cm equilateral triangle | each | 6346.00 |
| | ii) 90 cm equilateral triangle | each | 4531.00 |
| | iii) 75 cm equilateral triangle | each | 3818.00 |
| | iv) 60 cm equilateral triangle | each | 3717.00 |
| | v) 120 cm circular | each | 9723.00 |
| | vi) 90 cm circular | each | 6431.00 |
| | vii) 75 cm circular | each | 5137.00 |
| | viii) 60 cm circular | each | 4078.00 |
| | ix) 80 cm x 75 cm rectangular | each | 6689.00 |
| | x) 80 cm x 60 cm rectangular | each | 5391.00 |
| | xi) 60 cm x 50 cm rectangular | each | 4193.00 |
| | xii) 60 cm x 45 cm rectangular | each | 3994.00 |
| | xiii) 60 cm x 60 cm square | each | 4593.00 |
| | xiv) 120 cm high octagon | each | 10136.00 |
| | xv) 90 cm high octagon | each | 6663.00 |
| | xvi) 75 cm high octagon | each | 5298.00 |
| 8.4 | **Retro- reflectorised Traffic signs**<br>Providing and fixing of retro-reflectorised cautionary, mandatory and informatory sign board as per IRC 67-2022 made of high intensity Micro-Prismatic Grade Sheeting (Type XI) vide IRC: 67:2022 clause 6.7.4.3 and clause 801.3.3 of MoRT&H fixed over Aluminum composite material sheet with thermoplastic core of Low density polyethylene (LDPE) between two thick skins/sheets of aluminum with overall thickness of 4mm and aluminum skin of thickness 0.3 on both side, the ACM shall conform to Table 6.1 of IRC:67:2022 and Table 800-1 of MoRT&H specification and High Intensity Micro Prismatic Grade Sheeting shall conform to Table 6.6 of IRC:67:2022 and Table 800-3 of MoRT&H specification of in the clause 6.8 of the IRC 67-2022 and clause 801.3.7 of MoRT&H message legend numerical letter etc. and borders shall be screen printed on high intensity micro prismatic sheeting shall conform to clause 801.3.7 with digital printing ink or of cutout from transparent durable overlay or cutout from the same type of reflecting sheeting for cautionary/ mandatory/ signs, supported on MS sign post of size 75mm x75mm x 6mm of 3.60 meter over all length and fix to ground by means of properly designed foundation of dimension 450x450x600mm with M20 grade concrete. The ACM sheet shall be fixed to the post with four minimum four number breakaway bolts. & supported with a back-support frame of 25mmx25mmx3mm angle. 10 years warranty for Retro Reflection Sheeting to be provided from the original sheeting manufacture brand like 3M/Avery Dennison/ Orafol Make as per clause 6.9 of IRC: 67:2022 & a certified copy for three years for outdoor exposure report from an independent test lab as per clause 6.7 of IRC: 67:2022 for the product offered shall be submitted by the | | |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| | contractor.<br>Note: - Manufacture name and date of manufacture is to be printed (Embossed) at the backside of the board. | | |
| | i) 120 cm equilateral triangle | each | 6446.00 |
| | ii) 90 cm equilateral triangle | each | 4700.00 |
| | iii) 75 cm equilateral triangle | each | 3938.00 |
| | iv) 60 cm equilateral triangle | each | 3827.00 |
| | v) 120 cm circular | each | 9923.00 |
| | vi) 90 cm circular | each | 6531.00 |
| | vii) 75 cm circular | each | 5250.00 |
| | viii) 60 cm circular | each | 4178.00 |
| | ix) 80 cm x 75 cm rectangular | each | 6800.00 |
| | x) 80 cm x 60 cm rectangular | each | 5500.00 |
| | xi) 60 cm x 50 cm rectangular | each | 4280.00 |
| | xii) 60 cm x 45 cm rectangular | each | 4064.00 |
| | xiii) 60 cm x 60 cm square | each | 4685.00 |
| | xiv) 120 cm high octagon | each | 10280.00 |
| | xv) 90 cm high octagon | each | 6743.00 |
| | xvi) 75 cm high octagon | each | 5388.00 |
| 8.5 | **Direction and Place Identification signs upto 0.9 sqm size board.**<br>Providing and fixing of retro-reflectorised cautionary, mandatory and informatory sign board as per IRC 67-2022 made of high intensity Micro-Prismatic Grade Sheeting (Type IV) vide IRC: 67:2022 clause 6.7.3.2 and clause 801.3.3 of MoRT&H fixed over Aluminum composite material sheet with thermoplastic core of Low density polyethylene (LDPE) between two thick skins/sheets of aluminum with overall thickness of 4mm and aluminum skin of thickness 0.3 on both side, the ACM shall conform to Table 6.1 of IRC:67:2022 and Table 800-1 of MoRT&H specification and High Intensity Micro Prismatic Grade Sheeting shall conform to Table 6.6 of IRC:67:2022 and Table 800-3 of MoRT&H specification of in the clause 6.8 of the IRC 67-2022 and clause 801.3.7 of MoRT&H message legend numerical letter etc. and borders shall be screen printed on high intensity micro prismatic sheeting shall conform to clause 801.3.7 with digital printing ink or of cutout from transparent durable overlay or cutout from the same type of reflecting sheeting for cautionary/ mandatory/ signs, supported on 2 nos. of MS sign post of size 75mm x75mm x 6mm of 3.60 meter over all length and fix to ground by means of properly designed foundation of dimension 450x450x600mm with M20 grade concrete. The ACM sheet shall be fixed to the post with four minimum four number breakaway bolts. & supported with a back-support frame of 25mmx25mmx3mm angle. 7 years warranty for Retro Reflection Sheeting to be provided from the original sheeting manufacture brand like 3M/Avery Dennison/ Orafol Make as per clause 6.9 of IRC: 67:2022 & a certified copy for three years for | sqm | 9027.00 |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| | outdoor exposure report from an independent test lab as per clause 6.7 of IRC: 67:2022 for the product offered shall be submitted by the contractor.<br><br>Note: - Manufacture name and date of manufacture is to be printed (Embossed) at the backside of the board. | | |
| 8.6 | **Direction and Place Identification signs upto 0.9 sqm size board.**<br>Providing and fixing of retro-reflectorised cautionary, mandatory and informatory sign board as per IRC 67-2022 made of high intensity Micro-Prismatic Grade Sheeting (Type XI) vide IRC: 67:2022 clause 6.7.4.3 and clause 801.3.3 of MoRT&H fixed over Aluminum composite material sheet with thermoplastic core of Low density polyethylene (LDPE) between two thick skins/sheets of aluminum with overall thickness of 4mm and aluminum skin of thickness 0.3 on both side, the ACM shall conform to Table 6.1 of IRC:67:2022 and Table 800-1 of MoRT&H specification and High Intensity Micro Prismatic Grade Sheeting shall conform to Table 6.6 of IRC:67:2022 and Table 800-3 of MoRT&H specification of in the clause 6.8 of the IRC 67-2022 and clause 801.3.7 of MoRT&H message legend numerical letter etc. and borders shall be screen printed on high intensity micro prismatic sheeting shall conform to clause 801.3.7 with digital printing ink or of cutout from transparent durable overlay or cutout from the same type of reflecting sheeting for cautionary/ mandatory/ signs, supported on 2 nos. of MS sign post of size 75mm x75mm x 6mm of 3.60 meter over all length and fix to ground by means of properly designed foundation of dimension 450x450x600mm with M20 grade concrete. The ACM sheet shall be fixed to the post with four minimum four number breakaway bolts. & supported with a back-support frame of 25mmx25mmx3mm angle. 10 years warranty for Retro Reflection Sheeting to be provided from the original sheeting manufacture brand like 3M/Avery Dennison/ Orafol Make as per clause 6.9 of IRC: 67:2022 & a certified copy for three years for outdoor exposure report from an independent test lab as per clause 6.7 of IRC: 67:2022 for the product offered shall be submitted by the contractor.<br>Note: - Manufacture name and date of manufacture is to be printed (Embossed) at the backside of the board. | sqm | 9200.00 |
| 8.7 | **Direction and Place Identification signs more than 0.9 sqm size board.**<br>Providing and fixing of retro-reflectorised cautionary, mandatory and informatory sign board as per IRC 67-2022 made of high intensity Micro-Prismatic Grade Sheeting (Type IV) vide IRC: 67:2022 clause 6.7.3.2 and clause 801.3.3 of MoRT&H fixed over Aluminum composite material sheet with thermoplastic core of Low density polyethylene (LDPE) between two thick skins/sheets of aluminum with overall thickness of 4mm and aluminum skin of thickness 0.3 on both side, the ACM shall conform to Table 6.1 of IRC:67:2022 and Table 800-1 of MoRT&H specification and High Intensity Micro Prismatic Grade Sheeting shall conform to Table 6.6 of IRC:67:2022 and | sqm | 14497.00 |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| | Table 800-3 of MoRT&H specification of in the clause 6.8 of the IRC 67-2022 and clause 801.3.7 of MoRT&H message legend numerical letter etc. and borders shall be screen printed on high intensity micro prismatic sheeting shall conform to clause 801.3.7 with digital printing ink or of cutout from transparent durable overlay or cutout from the same type of reflecting sheeting for cautionary/ mandatory/ signs, supported on 2 nos. of MS sign post of size 75mm x75mm x 6mm of 3.60 meter over all length and fix to ground by means of properly designed foundation of dimension 450x450x600mm with M20 grade concrete. The ACM sheet shall be fixed to the post with four minimum four number breakaway bolts. & supported with a back-support frame of 25mmx25mmx3mm angle. 7 years warranty for Retro Reflection Sheeting to be provided from the original sheeting manufacture brand like 3M/Avery Dennison/ Orafol Make as per clause 6.9 of IRC: 67:2022 & a certified copy for three years for outdoor exposure report from an independent test lab as per clause 6.7 of IRC: 67:2022 for the product offered shall be submitted by the contractor.<br>Note: - Manufacture name and date of manufacture is to be printed (Embossed) at the backside of the board. | | |
| 8.8 | **Direction and Place Identification signs more than 0.9 sqm size board.**<br>Providing and fixing of retro-reflectorised cautionary, mandatory and informatory sign board as per IRC 67-2022 made of high intensity Micro-Prismatic Grade Sheeting (Type XI) vide IRC: 67:2022 clause 6.7.4.3 and clause 801.3.3 of MoRT&H fixed over Aluminum composite material sheet with thermoplastic core of Low density polyethylene (LDPE) between two thick skins/sheets of aluminum with overall thickness of 4mm and aluminum skin of thickness 0.3 on both side, the ACM shall conform to Table 6.1 of IRC:67:2022 and Table 800-1 of MoRT&H specification and High Intensity Micro Prismatic Grade Sheeting shall conform to Table 6.6 of IRC:67:2022 and Table 800-3 of MoRT&H specification of in the clause 6.8 of the IRC 67-2022 and clause 801.3.7 of MoRT&H message legend numerical letter etc. and borders shall be screen printed on high intensity micro prismatic sheeting shall conform to clause 801.3.7 with digital printing ink or of cutout from transparent durable overlay or cutout from the same type of reflecting sheeting for cautionary/ mandatory/ signs, supported on 2 nos. of MS sign post of size 75mm x75mm x 6mm of 3.60 meter over all length and fix to ground by means of properly designed foundation of dimension 450x450x600mm with M20 grade concrete. The ACM sheet shall be fixed to the post with four minimum four number breakaway bolts. & supported with a back-support frame of 25mmx25mmx3mm angle. 10 years warranty for Retro Reflection Sheeting to be provided from the original sheeting manufacture brand like 3M/Avery Dennison/ Orafol Make as per clause 6.9 of IRC: 67:2022 & a certified copy for three years for outdoor exposure report from an independent test lab as per clause 6.7 of IRC: 67:2022 for the product offered shall be | sqm. | 15506.00 |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| | submitted by the contractor.<br>**Note: - Manufacture name and date of manufacture is to be printed (Embossed) at the backside of the board.** | | |
| 8.9 | Direction and Place Identification signs more than 0.9 sqm size board. Providing and fixing of retro-reflectorised cautionary, mandatory and informatory sign board as per IRC 67-2012 made of high intensity Micro-Prismatic Grade Sheeting (Type XI) vide IRC:67:2012 clause 6.7.3.2 and clause 801.3.3 of MoRT&H fixed over Aluminum composite material sheet with thermoplastic core of Low density polyethylene (LDPE) between two thickskins/sheets of aluminum with overall thickness of 4mm and aluminum skin of thickness 0.3 on both side, the ACM shall conform to Table 6.1 of IRC:67:2012 and Table 800-1 of MoRT&H specification and High Intensity Micro Prismatic Grade Sheeting shall conform to Table 6.6 of IRC:67:2012 and Table 800-3 of MoRT&H specification of in the clause IRC 6.8of the IRC 67-2012 and clause 801.3.7 of MoRT&H message legend numerical letter etc. and borders shall be screen printed on high intensity micro prismatic sheeting shall conform to clause801.3.7 with digital printing ink or of cutout from transparent durable overlay or cutout from the same type of reflecting sheeting for cautionary/ mandatory/ signs, supported on 2 nos. of MS sign post of size 100 x100 x 3mm Square Pipe of 3.60 meter over all length and fix to ground by means of properly designed foundation of dimension 450x450x600mm with M20 grade concrete. The ACM sheet shall be fixed to the post with four minimum four number breakaway bolts. & supported with a back support frame of 25x50x1.25mm tubular frame with the help of suitable sizes pop rivettes of 6mm or by the help of G I / Nickle Chome plated nut & bolts to be hammered after tightening. 10 years warranty for Retro Reflection Sheeting to be provided from the original sheeting manufacture brand like 3M/Avery Dennison/ Orafol Make as per clause 6.9 of IRC: 67:2012 & a certified copy for three years for outdoor exposure report from an independent test lab as per clause 6.7 of IRC: 67:2012 for the product offered shall be submitted by the contractor.<br>Note:- Manufacture name and date of manufacture is to be printed (Embossed) at the backside of the board. | sqm | 19950.00 |
| 8.10 | Retro- reflectorised Traffic signs Providing and fixing of retro-reflectorised cautionary, mandatory and informatory sign board as per IRC 67-2012 made of high intensity Micro-Prismatic Grade Sheeting (Type XI) vide IRC: 67:2012 clause 6.7.3.2 and clause 801.3.3 of MoRT&H fixed over Aluminum composite material sheet with thermoplastic core of Low density polyethylene (LDPE) between two thick skins/sheets of aluminum with overall thickness of 4mm and aluminum skin of thickness 0.3 on both side, the ACM shall conform to Table 6.1 of IRC:67:2012 and Table 800-1 of MoRT&H specification and High Intensity Micro Prismatic Grade Sheeting shall conform to Table 6.6 of IRC:67:2012 and Table 800-3 of MoRT&H specification of in the clause IRC 6.8 of the IRC 67-2012 and clause 801.3.7 of MoRT&H message legend numerical letter etc. and borders shall be screen printed on high intensity micro prismatic sheeting shall conform to clause 801.3.7 with digital printing ink or of cutout from transparent durable overlay or cutout from the same type of reflecting sheeting for cautionary/ mandatory/ signs, supported on MS sign post of size 80 x80 x 3mm Square pipe of 3 meter over all | | 19950.00 |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| | length and fix to ground by means of properly designed foundation of dimension 450x450x600mm with M20 grade concrete. The ACM sheet shall be fixed to the post with four minimum four number breakaway bolts. & supported with a back support frame of 25x50x1.25mm tubular frame with the help of suitable sizes pop rivettes of 6mm or by the help of G I / Nickle Chome plated nut & bolts to be hammered after tightening. 10 years warranty for Retro Reflection Sheeting to be provided from the original sheeting manufacture brand like 3M/Avery Dennison/ Orafol make as per clause 6.9 of IRC: 67:2012 & a certified copy for three years for outdoor exposure report from an independent test lab as per clause 6.7 of IRC: 67:2012 for the product offered shall be submitted by the contractor.<br>Note:- Manufacture name and date of manufacture is to be printed (Embossed) at the backside of the board. | | |
| | 90 cm equilateral triangle | each | 7950.00 |
| | 60 cm equilateral triangle | each | 6900.00 |
| | 60 cm circular | each | 7300.00 |
| | 80 cm x 60 cm rectangular | each | 8000.00 |
| | 60 cm x 45 cm rectangular | each | 7500.00 |
| | 60 cm x 60 cm square | each | 6800.00 |
| | 90 cm high octagon | each | 9000.00 |
| 8.11 | Direction and Place Identification signs more than 0.9 sqm size board. Providing and fixing of retro-reflectorised cautionary, mandatory and informatory sign board as per IRC 67-2012 made of high intensity Micro-Prismatic Grade Sheeting (Type IV) vide IRC:67:2012 clause 6.7.3.2 and clause 801.3.3 of MoRT&H fixed over Aluminum composite material sheet with thermoplastic core of Low density polyethylene (LDPE) between two thickskins/sheets of aluminum with overall thickness of 4mm and aluminum skin of thickness 0.3 on both side, the ACM shall conform to Table 6.1 of IRC:67:2012 and Table 800-1 of MoRT&H specification and High Intensity Micro Prismatic Grade Sheeting shall conform to Table 6.6 of IRC:67:2012 and Table 800-3 of MoRT&H specification of in the clause IRC 6.8of the IRC 67-2012 and clause 801.3.7 of MoRT&H message legend numerical letter etc. and borders shall be screen printed on high intensity micro prismatic sheeting shall conform to clause801.3.7 with digital printing ink or of cutout from transparent durable overlay or cutout from the same type of reflecting sheeting for cautionary/ mandatory/ signs, supported on MS sign post of size 100 x100 x 3mm Square Pipe of 3.60 meter over all length and fix to ground by means of properly designed foundation of dimension 450x450x600mm with M20 grade concrete. The ACM sheet shall be fixed to the post with four minimum four number breakaway bolts. & supported with a back support frame of 25x50x1.25mm tubular frame with the help of suitable sizes pop rivettes of 6mm or by the help of G I / Nickle Chome plated nut & bolts to be hammered after tightening. 7 years warranty for Retro Reflection Sheeting to be provided from the original sheeting manufacture brand like 3M/Avery Dennison/ Orafol Make as per clause 6.9 of IRC: 67:2012 & a certified copy for three years for outdoor exposure report from an independent test lab | sqm. | 19500.00 |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| | as per clause 6.7 of IRC: 67:2012 for the product offered shall be submitted by the contractor.<br>**Note:- Manufacture name and date of manufacture is to be printed (Embossed) at the backside of the board.** | | |
| 8.12 | Retro- reflectorised Traffic signs Providing and fixing of retro-reflectorised cautionary, mandatory and informatory sign board as per IRC 67-2012 made of high intensity Micro-Prismatic Grade Sheeting (Type IV) vide IRC: 67:2012 clause 6.7.3.2 and clause 801.3.3 of MoRT&H fixed over Aluminum composite material sheet with thermoplastic core of Low density polyethylene (LDPE) between two thick skins/sheets of aluminum with overall thickness of 4mm and aluminum skin of thickness 0.3 on both side, the ACM shall conform to Table 6.1 of IRC:67:2012 and Table 800-1 of MoRT&H specification and High Intensity Micro Prismatic Grade Sheeting shall conform to Table 6.6 of IRC:67:2012 and Table 800-3 of MoRT&H specification of in the clause IRC 6.8 of the IRC 67-2012 and clause 801.3.7 of MoRT&H message legend numerical letter etc. and borders shall be screen printed on high intensity micro prismatic sheeting shall conform to clause 801.3.7 with digital printing ink or of cutout from transparent durable overlay or cutout from the same type of reflecting sheeting for cautionary/ mandatory/ signs, supported on MS sign post of size 80 x80 x 3mm Square pipe of 3 meter over all length and fix to ground by means of properly designed foundation of dimension 450x450x600mm with M20 grade concrete. The ACM sheet shall be fixed to the post with four minimum four number breakaway bolts. & supported with a back support frame of 25x50x1.25mm tubular frame with the help of suitable sizes pop rivettes of 6mm or by the help of G I / Nickle Chome plated nut & bolts to be hammered after tightening. 7 years warranty for Retro Reflection Sheeting to be provided from the original sheeting manufacture brand like 3M/Avery Dennison/ Orafol make as per clause 6.9 of IRC: 67:2012 & a certified copy for three years for outdoor exposure report from an independent test lab as per clause 6.7 of IRC: 67:2012 for the product offered shall be submitted by the contractor.<br>Note:- Manufacture name and date of manufacture is to be printed (Embossed) at the backside of the board. | | 19500.00 |
| | 90 cm equilateral triangle | each | 7680.00 |
| | 60 cm equilateral triangle | each | 6650.00 |
| | 60 cm circular | each | 7050.00 |
| | 80cm x 60cm rectangular | each | 7700.00 |
| | 60cm x 45 cm rentangular | each | 7400.00 |
| | 60cm x 60 cm square | each | 6400.00 |
| | 90cm high octagon | each | 8600.00 |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| 8.13 | Providing and Fixing kerbstone with LED Lighting System as per following specifications:<br><br>Endimensions with vertical mounting legs with 10mm horizontal square shape mild steel reinforcements. Illuminated kerb stone having visible dimensions:-500mm x 150mm x 150mm.(as per irc - 86-1983) 80mm legs having ms reinforcements shal be embedded in BT/concrete road surface.the kerb stone should have kerb angle of 45 degree, with rubber plugs at both ends of the kerbstones (2 nos)having load carrying capacity - vertical: 1465kg, horizontal: 2349 (certified by Govt. Recoganized bodies)having thermal stability -: -40 to +120 degree celcius (certified by govt. Recoganized bodies. Illuminated kerb stones including: ingrace protection and connector rating IP67, electrical wire (as per is: 694), led strip (as per Im: 80) standard. Including for electrical fixing -FRP junction box consisting of 10 amp. Switch module power supply (SMPS) having input voltage 230v ac, output voltage 36v dc, 2 pole mch. Illuminated kerb stones having warranty for 3 years, illuminated kerbstone shall be fixed into cast-in-situ cement concrete of m20 grade at base, back filling and grooves, including dismantling, fixing, removal of dismantled material and cleaning etc. Complete. The colour of kerb stone shall be provided as directed by Engineer-in-Charge. | RM | 7411.00 |
| 8.14 | Providing and Repairing /painting i/c post and pasting Retro reflective Type XI on old existing informatory boards vide clause IRC: 67:2022, message (legends, numerals letter.) borders for the informatory & other sign boards shall be cut out item durable transparent overlay film or cut out from the same reflective sheeting only. Cut-outs shall be from durable transparent overlay materials as specified by the sheeting manufacturer and shall be bonded with the sheeting in the manner specified by the manufacturer and as directed by Engineer-in-Charge. Sheet shall be fixed on the existing board by de-greasing either by acid or hot alkaline etching and all scale dirt removed to obtain smooth plain surface before application of high intensity grade cube corner retro reflective sheeting (Type XI) all complete. 10 years Warranty for Retro Reflective Sheeting from the original sheeting manufacturer brand like 3M/Avery Dennison/ Orafol Make as per clause 6.9 of IRC:67:2022 & a certified copy. of three years outdoor exposure report as per clause 6.7 of IRC:67:2022 for the product offered shall be submitted by the contractor. | sqm | 5297.00 |
| 8.15 | Providing and Repairing /painting i/c post and pasting Retro reflective Type XI on old existing overhead & cantilever boards vide clause IRC: 67:2022, message (legends, numerals letter.) borders for the informatory & other sign boards shall be cut out item durable transparent overlay film or cut out from the same reflective sheeting only. Cut-outs shall be from durable transparent overlay materials as specified by the sheeting manufacturer and shall be bonded with the sheeting in the manner specified by the manufacturer and as directed by Engineer-in-Charge. Sheet shall be fixed on the existing board by de-greasing either by acid or hot alkaline etching and all scale dirt removed to obtain smooth plain surface before application of high intensity grade cube corner retro reflective sheeting (Type XI) all complete. 10 years Warranty for | sqm. | 13900.00 |

| Item No. | Description | Unit | Rate |
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| | Retro Reflective Sheeting from the original sheeting manufacturer brand like 3M/Avery Dennison/ Orafol Make as per clause 6.9 of IRC:67:2022 & a certified copy. of three years outdoor exposure report as per clause 6.7 of IRC:67:2022 for the product offered shall be submitted by the contractor. | | |
| 8.16 | Overhead sign Board Providing and erecting work overhead signs made of high intensity Micro-Prismatic Grade Sheeting (Type IV/ Type XI) vide IRC:67:2012 clause 6.7.3.2 and clause 801.3.3 of MoRT&H fixed over Aluminum composite material sheet with thermoplastic core of Low density polyethylene (LDPE) between two thickskins/sheets of aluminum with overall thickness of 4mm and aluminum skin of thickness 0.3 on both side, the ACM shall conform to Table 6.1 of IRC:67:2012 and Table 800-1 of MoRT&H specification and High Intensity Micro Prismatic Grade Sheeting shall conform to Table 6.6 of IRC:67:2012 and Table 800-3 of MoRT&H specification of in the clause IRC 6.8of the IRC 67-2012 and clause 801.3.7 of MoRT&H message legend numerical letter etc. and borders shall be screen printed on high intensity micro prismatic sheeting shall conform to clause801.3.7 with digital printing ink or of cutout from transparent durable overlay or cutout from the same type of reflecting sheeting for overhead board supported on MS Vertical post of pipe of 300 mm NB (325 OD) x 5 mm thick pipe x 6 meter length two Post and fix to ground by means of properly designed foundation of dimension 1000x1000x2500 mm with M25 grade concrete including Iron bar 16mm & 12 mm. overhead structure made of 48 OD & 42 OD NB Pipes, Medium Class. The board shall be riveted on M S angle frame 32 x 32 x3 mm and clamped by flats as required to hold them rigidly in position and type as per structural design requirement and approved plans. The post frame and other steel parts shall be painted by one coat of steel primer and two coats of epoxy paint as per IRC 67-2012. 7 years warranty for Type IV sheet/ 10 years warranty for Type XI Retro Reflection Sheeting to be provided from the original sheeting manufacture brand like 3M/Avery Dennison/ Orafol Make as per clause 6.9 of IRC: 67:2012 & a certified copy for three years for outdoor exposure report from an independent test lab as per clause 6.7 of IRC: 67:2012 for the product offered shall be submitted by the contractor.<br>Note:- Manufacture name and date of manufacture is to be printed (Embossed) at the backside of the board. | each | 630000.00 |
| 8.16 (A) | Foundation work i.e. excavation, M 25 grade concrete, reinforcement shall be paid vide relevant item. | | |
| 8.16 (B) | Structure steel work including panting and erection installation etc. complete. | MT | 110000.00 |
| 8.16 (C) | Information Board including provision of base frame Type IV retro reflection sheeting. | sqm | 11000.00 |
| 8.16 (D) | Information Board including provision of base frame Type XI retro reflection sheeting. | sqm | 12000.00 |
| 8.17 | **Metal Beam Crash Barrier** | | 11000.00 |
| | A Type - A, "W" : Metal Beam Crash Barrier (Providing and erecting a "W" metal beam crash barrier comprising of 3 mm thick corrugated sheet metal beam rail, 70 cm above road/ground level, fixed on ISMC series channel vertical post, 150 x 75 x 5 mm spaced 2 m | metre | 3863.00 |

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| | centre to centre, however 5 spaces of 600mm each between vertical post and 5 spaces of 900mm each between vertical post will be kept adjacent to RCC parapet/ concrete crash barrier in accordance to figure 9.14 of the IRC-SP: 73-2018, 1.8 m high, 1.1 m below ground/road level, all steel parts and fitments to be galvanised by hot dip process, all fittings to conform to IS:1367 and IS:1364, metal beam rail to be fixed on the vertical post with a spacer of channel section 150 x 75 x 5 mm, 330 mmlong complete as per clause 811) and treatment for metal beam cresh barrier on approach shall be modified eccentric loader terminal arrangement as shown in figure 9.12 of IRC- SP : 73-2018 End portion of metal beam crash barrier shall be flared outside by 1.25 mtr. Distance in last section of 8 mtr. Length. Departure side it shall be trailing terminal (TT) arrangement as shown in figure 9.13 of the IRC SP-73-2018. End portion of metal beam crash barrier shall be flared outside by 0.35 mtr. Distance in last section of 8 mtr. Length | | |
| | B - Type - B, "THRIE" : Metal Beam Crash Barrier (Providing and erecting a "Thrie" metal beam crash barrier comprising of 3 mm thick corrugated sheet metal beam rail, 85 cm above road/ground level, fixed on ISMC series channel vertical post, 150 x 75 x 5 mm spaced 2 m centre to centre, however 5 spaces of 600mm each between vertical post and 5 spaces of 900mm each between vertical post will be kept adjacent to RCC parapet/ concrete crash barrier in accordance to figure 9.14 of the IRC-SP: 73-2018 2 m high with 1.15 m below ground level, all steel parts and fitments to be galvanised by hot dip process, all fittings to conform to IS:1367 and IS:1364, metal beam rail to be fixed on the vertical post with a space of channel section 150 x 75 x 5 mm, 546 mm long complete as per clause 811) and treatment for metal beam cresh barrier on approach shall be modified eccentric loader terminal arrangement as shown in figure 9.12 of IRC-SP : 73-2018 End portion of metal beam crash barrier shall be flared outside by 1.25 mtr. Distance in last section of 8 mtr. Length. Departure side it shall be trailing terminal (TT) arrangement as shown in figure 9.13 of the IRC SP-73-2018. End portion of metal beam crash barrier shall be flared outside by 0.35 mtr. Distance in last section of 8 mtr. Length.<br>Note: - The manufacture or the contractor may submit other than specified design and get it approved subjected to conformity to the requirements of EN1317 part-2. The manufacture or the contractor shall demonstrate the compliance of these requirements by submission of videos of actual crash testing along with authenticated crash test reports. The crash test must have been conducted in accredited crash test laboratories/Institution, as per MORT&H circular RW/NH-29023/2/2019-S&R (P&B) dated 01 Jan 2020. | metre | 5393.00 |
| 8.18 | **Flexible Crash Barrier, Wire Rope Safety Barrier**<br>Providing and erecting a wire rope safety barrier with vertical posts of medium weight RS Joist (ISMB series) 100 mm x 75 mm (11.50 kg/m), 1.50 m long 0.85 m above ground and 0.65 m below ground level, split at the bottom for better grip, embedded in M 15 grade cement concrete 450 x 450 x 450 mm, 1.50 m center to center and with 4 horizontal steel wire rope 40 mm dia and anchored at terminal posts 15 m apart. Terminal post to be embedded in M 15 grade cement concrete foundation 2400 x 450 x 900 mm (depth), strengthened by a strut of RS joist 100 x 75 mm, 2 m long at $45^0$ inclination and a tie 100 x 8 mm, 1.50 m long at the bottom, all | metre | 2338.00 |

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| | embedded in foundation concrete as per approved design and drawing, rate excluding excavation and cement concrete. | | |
| 8.19 | **Reinforced Cement Concrete Crash Barrier**<br>Providing of an Reinforced cement concrete crash barrier at the edges of the road, approaches to bridge structures, constructed with M-40 grade concrete with HYSD reinforcement conforming to IRC:21 and dowel bars 25 mm dia, 450 mm long at expansion joints filled with pre-moulded asphalt filler board, keyed to the structure on which it is built and installed as per design given in the enclosure to MoRT&H circular No. RW/NH - 33022/1/94-DO III dated 24 June 1994 as per dimensions in the approved drawing and at locations directed by the Engineer, all as specified.<br>(i) M 40 grade concrete | metre | 3294.00 |
| 8.20 | **Reinforced Cement Concrete New Jersey**<br>Providing of an Reinforced cement concrete New Jersey median to bridge structures and medians, constructed with M-25 grade concrete with HYSD reinforcement conforming to IRC:21 and dowel bars 25 mm dia, 450 mm long at expansion joints filled with pre-moulded asphalt filler board, keyed to the structure on which it is built and installed as per design givenin the enclosure to MoRT&H circular No. RW/NH - 33022/1/94-DO III dated 24 June 1994 as per dimensions in the approved drawing and at locations directed by the Engineer, all as specified.<br>(i) M 25 grade concrete | metre | 3394.00 |
| 8.21 | **Road Delineators**<br>Supplying and installation road delineators (road way indicators). as per MORT&H Clause 806. The structure shall be made in roll forming process having height of 800-900 mm above the ground level, width not less than 100 mm and shall extend not more than 200-300mm below the ground while being installed, buried or pressed in to the ground in confirmation with IRC:79: 2019, are as directed by the Engineer-in-Charge. The guide post with pure polyester powder coating with minimum 40 microns thickness for protection against corrosion, on top of which Type XI retro reflective sheeting confirming to IRC-67 and ASTMD-4956 shall be pasted on both sides complying to IRC:79:2019. The delineator should consist of minimum retro reflective unit exposed area of 330 cm2 white color, full cube corner micro prismatic non-metallic retro reflective sheeting on each side conforming with IRC 67:2022 and meeting the coefficient of retro reflection values as per ASTM D 4956 Type XI label specification. The delineator shall have grooves across the-length to make the reflective sheets vandal-proof. | each | 890.00 |
| 8.22 | **Road Delineators**<br>Supplying and installation of delineators (road way indicators, hazard markers, object markers), 80-100 cm high above ground level, painted black and white in 15 cm wide stripes, fitted with 80 x 100 mm rectangular or 75 mm dia circular reflectorised panels at the top, buried or pressed into the ground and confirming toIRC-79 and the drawings. | each | 283.00 |
| 8.23 | **Road Markers/Road Stud with Lense Reflector**<br>Providing and fixing single mound twin shanked molded Road Markers/Road Stud made of poly carbonate or ABS body and shall | each | 309.00 |

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| | support a load of 13635 Kg tested in accordance to ASTM D 4280 and complying lo specifications of Section-5 of IRC:35-2015 and clause 804.2.2 of MORT&H and Section-5 of IRC:35-2015. he reflecting panels shall consist of number of lenses containing single or dual prismatic cubes capable of providing total internal reflection of the light entering the lens face. The slope or retro-reflecting surface should be 35 + 5 degree to base. The reflective marker shall be fixed to the road surface using the epoxy/ adhesives and the procedure recommended by the manufacturer. No nails shall be used to affix the markers so Cubes capable of that they do not pose safety hazard on the road. The contractor shall submit a two-year replacement warranty. for satisfactory field performance including stipulated retro-reflectance of the reflecting panel. (Make- 3M /AVERY DENNISON/ ORAFOL / ALCOLITEA / AUTOMARK) | | |
| 8.24 | **Solar powered Road Marker (Solar Stud)**<br>Supplying and fixing solar raised Pavement markers made of polycarbonate molded body with circular shape, solar powered, LED self Illumination in active mode, 360-degree Illumination and reflective panels with micro prismatic lens capable of providing total internal reflection of the light entering the lens face in passive mode. The marker shall support a load of 20000 kg tested in accordance to ASTM D 4280, The marker should be resistant to dust and water ingress according to IP 65 standards and should withstand temperatures in the range of 0 c to 70 c color of lighting could be provided in red or yellow (amber) as per requirement and typical frequency of blinking is 1Hz. There should be current losses of less than 20 microamperes at 2.4 v in sleep charging mode to enhance the life of the marker and a full charge should provide for a minimum autonomy of 50 hours. The height width and length of the marker shall not less than 10 mm x 100 mm x 100 mm. Also, the surface diameter of the marker shall not be less than 100 mm respectively. The weight of the marker shall not exceed 0.5 kilograms fixing will be by drilling holes on the road for the shanks to go inside, without nails and using epoxy resin based adhesive and complete as directed by the engineer. (Make- 3M /AVERY DENNISON/ ORAFOL / ALCOLITEA ) | each | 1771.00 |
| 8.25 | Providing white and colour washing to road side trees in 2 coats in 2 bands of white and one red band at centre of 30 cm height each including preparing the surface, cost of material, conveying etc. complete. | each | 57.00 |
| 8.26 | Providing and Fixing of Median Marker that are made of tough, high impact resistant, injection-molded, thermoplastic body with an isosceles trapezoidal structure of length, width and height not less than 15cm, 10cm and 10cm respectively and thickness not less than 1.8mm, the body structure shall be rounded at its acute angle, all the corners and edges. The plastic used for molding the Median Marker shall have a minimum Notched Izod Impact strength value of 600 J/m at room temperature, when tested in accordance with ASTM D256 and shall retain at least 70% of this value when subjected to accelerated weathering for 1000hrs as per ASTM G155 or UL746C. The logo of the manufacturer shall be embossed on either side of the body in the injection molding process. The Median Marker shall | each | 300.00 |

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| | have rectangular-shaped, fluorescent yellow color retro-reflective sheeting of size not less than 8.5cm*8.5cm and with fully reflective micro prismatic cube corners as its retro-reflective elements as per IRC 67 2022 and ASTM D4956 type XI specifications reflectivity values. The retro-reflective sheeting shall be one or both sides of the Median Marker and shall be edge protected with no exposed edges which will prevent edge lifting, vandalism, sheeting damage, etc. The Median Marker shall be fixed by a combination of epoxy adhesive and grouting. (Make- 3M /AVERY DENNISON/ ORAFOL/) | | |
| 8.27 | Providing and fixing reflective fluorescent tape on old road sign board by following the procedure recommended by manufacturer. Tape shall be consisting of white or colored sheeting having smooth outer surface which has the property of retro-reflection over its entire surface. It shall be weather resistant and show color fastness. It shall be new and unused and shall show no evidence of cracking, scaling, pitting, blisting, edge lifting and shall have negligible shrinkage or expansion including cleaning / scraping and painting of existing board etc. complete. (Guidance on the recommended application of each class of tape may be taken from IRC-67) | sqm | 5297.00 |
| 8.28 | Providing and applying Cold plastic paint two component rolled on surfacing material solvent free, high build two pack seamless, tough, skid resistant, material with has property of attaining 2.00mm thickness in single coat application white (or as required color) based on glass and color retaining acralic cross linking resin system for colored road surfacing including surface cleaning and cost of all material etc. complete. (All inclusive on bitumen or concrete surface) work shall carried out as per IRC 35-2015 clause 2.2) | sqm | 1516.00 |
| 8.29 | Providing and fixing precast concrete post delinators, post of size 130 x 90 x 1300 milimetre consisting of 4 B.T. wires with 14,000 Kg/cm2 C.C. M-20 grade 16mm dia. hold fast, including excavation and C.C. 1:4:8 foundation block, oil painting post with 3 coats and Red fluorescent tape 75 x 75mm and white floursent border 20mm wide around the red tape on all the four sides with necessary adhesive and lamination etc. complete. | each | 545.00 |
| 8.30 | Supplying rolled steel channel post of 100 x 50 mm (ISMC) drilled on top and painted with one coat of epoxy primer and two coats of epoxy finish paint having black and white bands of 25 cm last part of the post 0.75 m below ground level is painted with three coats of red lead paint with 2 nos. of high tensile G.I. bolt and nut of size 10 mm Dia. of 25 mm long including transportation etc. complete. Spec. No.: As per IRC-67, and MoRT&H. Circular No. RW/NH-33023/31/88 D.O.III Dated 2-5-1994 | RM | 1443.00 |
| 8.31 | Supplying M.S. Angle iron post of size 65 mm x 65 mm x 6 mm drilled on top and painted with one coat of epoxy primer and two coats of epoxy finish paint having black and white bands of 25 cm, last part of the post 0.75 m below ground level is painted with three coats of red lead paint with 2 nos. of high tensile G.I. bolt and nut of size 10 mm dia. of 25 mm long including transportation etc. complete. | RM | 1347.00 |

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| | Spec. No.: As per IRC-67, and MoRT&H. Circular No. RW/NH-33023/31/88 D.O.III Dated 2-5-1994 | | |
| 8.32 | Providing and Laying of 2 or more sets (6 Nos. each set) Dual Ribs Pattern Rumble Markings on roads with vehicular speeds of more than 50 Km/hr laid as per the spacing recommended for each set under Table 11.1 and Fig. 11.2 of IRC 35:2015 using Plastirib 2K, or equivalent 2 component cold plastic profiled material Each Dual Rib pattern rumble marking shall have width of 500 mm. Each dual rib pattern rumble marking shall have 2 ribs each of Width of 100 mm and height 20 mm. Gap between two ribs shall be 300 mm. Each rumble marking shall be made complete with drop on Soild Plus reflective glass beads @ 300 gm / One Square Metre spread throughout the surface of marking. The finished surface to be levelled, uniform and free from streaks and holes, including surface cleaning and cost of all materials etc. complete. | sqm | 4333.00 |
| 8.33 | **Anti - Glare Devices in Median** | | |
| | (A) **Plantation**<br>Plantation of shrubs and plants of approved species in the median. apart from cutting off glare from vehicle coming from opposite direction, these plants provide a pleasant environment and are eco-friendly. The rate for this item is available in the chapter 11 on horticulture. | metre | 2767.00 |
| | (B) Anti - Glare Screen with 25 mm steel pipe framework fixed with circular and rectangular vans (Providing and erecting an anti - glare screen with 25 mm dia vertical pipes fabricated and framed in the form of panels of one metre length and 1.75 mtr height fixed with circular vane 250 mm dia at top and rectangular vane 600 x 300 mm at the middle, made out of steel sheet of 3 mm thickness, end vertical pipes of the panel made larger for embedding in foundation concrete, applying 2 coats of paint on all exposed surfaces, all as per approved design and drawings.) | metre | 850.00 |
| | **(c)** Anti - Glare Screen with Rectangular Vane of MS sheet (Providing and erecting anti - glare screen with rectangular vanes of size 750 x 500 mm made from MS sheet, 3 mm thick and fixed on MS angle 50 x 50 x 6 mm at an angle of 450 to the direction of flow of traffic, 1.5 m center to center, top edge of the screen 1.75 m above ground level, vertical post firmly embedded in cement concrete foundation 0.60 m below ground level, applying 2 coats of paint on exposed faces, all complete as per approved design and drawings) | metre | 847.00 |
| 8.34 | **Unbreakable Glare Screen**<br>Supply and installation of unbreakable glare screens on metal or concrete crash barriers with a center-to-center distance ranging from 700mm to 1700mm. The glare screen should be made of EVA material, with a height of 650mm and a width of 290mm. It must be securely bolted to the existing barrier; if no barrier is available, separate MS posts shall be installed as per the instructions of the Engineer-in-Charge. The cost for the MS posts will be paid separately | each | 4103.00 |

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| 8.35 | **Cable Duct Across the Road**<br>Providing and laying of a reinforced cement concrete pipe duct, 300 mm dia, across the road (new construction), extending from drain to drain in cuts and toe of slope to toe of slope in fills, constructing head walls at both ends, providing a minimum fill of granular material over top and sides of RCC pipe as per IRC:98-2011, bedded on a 0.3 m thick layer of granular material free of rock pieces, outer to outer distance of pipe at least half dia of pipe subject to minimum 450 mm in case of double and triple row ducts, joints to be made leak proof, invert level of duct to be above higher than ground level to prevent entry of water and dirt, all as per IRC: 98 - 2011 and approved drawings. | | 847.00 |
| | Single Row for one utility service | metre | 1900.00 |
| | Double Row for two utility services | metre | 3422.00 |
| | Triple Row for three utility services | metre | 5097.00 |
| 8.36 | **Kilo Metre Stone**<br>Reinforced cement concrete M15grade kilometre stone of standard design as per IRC:8-1980, fixing in position including painting and printing etc | | 3422.00 |
| | (i) 5th kilometre stone (precast) | each | 3686.00 |
| | (ii) Ordinary Kilometer stone (Precast) | each | 2131.00 |
| | (iii) Hectometer stone (Precast) | each | 621.00 |
| 8.37 | **Boundary pillar**<br>Reinforced cement concrete M15 grade boundary pillars of standard design as per IRC:25-1967, fixed in position including finishing and lettering but excluding painting. | each | 546.00 |
| 8.38 | **Road Marking with Hot Applied Thermoplastic Compound with Reflectorizing Glass Beeds on Bituminous surface.**<br>Providing and applying (Cleaning of Bituminous surface) thermoplastic compound having composition of binder (Resin) minimum 18 %, titanium dioxide minimum 10 % and intermixed glass beed, minimum 30% of brand like Asian, Burger or equivalent make of 2.5 mm thick including drop down reflectorizing, glass beeds @ 250 gms per sqm area thickness of 2.5 mm is exclusive of surface applied glass beeds as per IRC:35.The finished surface to be level, uniform and free from streaks and holes, all complete as per clause 803 of specifications.<br>Note: - The guarantee of Thermoplastic should be minimum 3 Years. | sqm | 440.00 |
| 8.39 | **Road Marking with Hot Applied Thermoplastic Compound with Reflectorizing Glass Beeds on Concrete surface.**<br>Providing and applying (Cleaning of concrete surface and after applying Concrete primer) thermoplastic compound having composition of binder (Regin) minimum 18 %, titanium dioxide | sqm | 465.00 |

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| | minimum 10 % and intermixed glass beed, minimum 30% of brand like Asian, Burger or equivalent make of 2.5 mm thick including drop down reflectorizing, glass beeds @ 250 gms per sqm area thickness of 2.5 mm is exclusive of surface applied glassbeeds as per IRC:35 along with concrete primer. The finished surface to be level, uniform and free from streaks and holes, all complete as per clause 803 of specifications.<br>Note: - The guarantee of Thermoplastic should be minimum 3 Years. | sqm | 300.00 |
| 8.40 | **Printing/Painting new letter and figures of any shade**<br>Printing/Painting new letter and figures of any shade with synthetic enamel<br>paint black or any other approved colour to give an even shade as per as direction of Engineer-in-charge. | | |
| 8.41 | **Painting Two Coats on New Concrete Surfaces**<br>Painting two coats after filling the surface with synthetic enamel paint in all shades on new plastered concrete surfaces | sqm | 95.00 |
| 8.42 | **Painting on Steel Surfaces**<br>Providing and applying two coats of ready-mix paint of approved brand on steel surface after through cleaning of surface to give an even shade | sqm | 90.00 |
| 8.43 | Painting on Wood Surfaces (Providing and applying two coats of ready-mix paint of approved brand on wood surface after through cleaning of surface to give an even shade) | sqm | 79.00 |
| 8.44 | **Painting Lines, Dashes, Arrows etc on Roads in Two Coats on New Work**<br>Painting lines, dashes, arrows etc on roads in two coats on new work with ready mixed road Grade-I marking paint (cold applied reflective paint) conforming to IS:164 on concrete/bituminous surface, including cleaning the surface of all dirt, dust and other foreign matter, demarcation at site and traffic control. | | 90.00 |
| | (i) Over 10 cm in width | sqm | 141.00 |
| | (ii) Up to 10 cm in width | sqm | 118.00 |
| 8.45 | **Painting Lines, Dashes, Arrows etc on Roads in Two Coats on Old Work**<br>Painting lines, dashes, arrows etc on roads in two coats on old work with ready mixed Grade-I road marking paint (cold applied reflective paint) confirming to IS: 164 on bituminous / concrete surface, concrete including cleaning the surface of all dirt, dust and other foreign matter, demarcation at site and traffic control. | | sqm |
| | **(i)** Over 10 cm in width | sqm | 98.00 |
| | (ii) Up to 10 cm in width | sqm | 107.00 |
| 8.46 | **G.I Barbed Wire Fencing 1.2 metre high**<br>Providing and fixing 1.2 metres high GI barbed wire (weighing 9.38 kg per 100 mts.) fencing with 1.8 m angle iron posts 40 mm x 40 mm x 6 mm placed every 3 metres center to center founded in M15 grade cement concrete, 0.6 metre below ground level, every | metre | 291.00 |

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| | 15th post, last but one end post and corner post shall be strutted on both sides and end post on one side only and provided with 9 horizontal lines and 2 diagonals interwoven with horizontal wires, fixed with GI staples, turn buckles etc complete as per clause 808. | | |
| 8.47 | **G.I Barbed Wire Fencing 1.8 metre high**<br>Providing and fixing 1.8 metres high GI barbed wire (weighing 9.38 kg per 100 mts.) fencing with 2.4 m angle iron posts 50 mm x 50 mm x 6 mm placed every 3 metres center to center founded in M15 grade cement concrete, 0.6 metre below ground level, every 15th post, last but one end post and corner post shall be strutted on both sides and end post on one side only and provided with 12 horizontal lines and 2 diagonals interwoven with horizontal wires, fixed with GI staples, turn buckles etc complete as per clause 808. | metre | 467.00 |
| 8.48 | **Fencing with weaved/welded steel wire Fabric (weighing 4 kg per sqm.)75 mm x 50 mm (Suggestive)**<br>Providing 1.20-metre-high fencing with angle iron posts 50mmx 50mmx6mm at 3 metre center to center with 0.40 metre embedded in M15 grade cement concrete, corner, end and every 10th post tobe strutted, provided with **weaved/**welded steel wire 75 mm x 25 mm mesh (Weighing 4 kg per sqm) and fixed to iron posts by flatiron 50 x 5 mm and bolts etc. complete in all respects. | metre | 644.00 |
| 8.49 | **Tubular Steel Railing on Medium Weight steel channel (ISMC series) 100 mm x 50 mm**<br>Providing, fixing and erecting 50 mm dia steel pipe **(weighing 3.56 kg per mts.)** railing in 3 rows on medium weight steel channels (ISMC series) 100 mm x 50 mm, 1.2 metres high above ground, 2 m center to center, painting etc. complete as per approved drawings. | metre | 1682.00 |
| 8.50 | **Tubular Steel Railing on Precast RCC posts, 1.2 m high above ground level**<br>Providing, fixing and erecting 50 mm dia **(weighing 3.56 kg per mts.)** steel pipe railing in 3 rows on precast M20 grade RCC vertical posts1.8 meters high (1.2 m above GL) with 3 holes 50 mm dia for pipe, fixed 2 meters center painting etc., complete as per approved drawing. | metre | 1472.00 |
| 8.51 | Providing fixing and erecting /Pre-cast RCC information board including reinforcement with RCC M25 Posts (2 Nos) of size 150 x150x3300 mm RCC M-25 plate of size 1300x900 x 50mm the post shall be with groove at top of size 900x60x40mm. So that the plate is fixed inside it. Both posts are fixed jointly with M-15 concrete of size 600x1500x750 excavation in all types of soil including painting of Post and plate with synthetic enamel paint in all shades and also including providing and fixing High Intensity micro Prismatic retro reflective sheeting (Type IV) 50 mm wide around both side of the board and two places in each post (Preferably at 30cm and 150 cm respectively above road level) All work shall be as per MORT&H specifications and as per drawing including all material, labour, lead and lift, T&P etc. complete. | each | 13500.00 |

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| 8.52 | **Providing and Fixing at Black spot/ hair pin turns, junction, ghat section locations for Safety of road traffic from fatal accidents and saving human life can be made easier by using patented RADAR based innovative technology equipment. This system works using RADAR based high frequency waves which detects moving objects coming towards it from a defined distance and activates blinking of LED Lamp along with High Volume Honking Horn.**<br>This device installation will involve following activities & materials: Excavation of one number 1050 mm deep, 500 mm width X 500 mm length earth pit with PCC M15 upto the depth of 75 mm at bottom as Base Pad for fixing one number G.I. Steel pipe (ISI marked) 3 inch (80 mm) diameter, 3 mm wall thickness, 4.500 Meter long, with clamps at the top for fixing Solar Cell, square shaped steel base (anchor) plate 5 mm thick, 300 x 300 mm centrally welded at bottom of G. I. steel pipe, with 4 nos 16 mm dia holes for bolting in cement concrete (concrete ratio 1:1.5:3, Grade M20). One number steel box type housing (weather proof type) approx weight 15 Kgs painted and clamped on pipe at minimum 2.500 Meter height above ground surface for fitting of electronic software equipment of Radar along with warning unit. Solar Cell 1 number, suitable for power supply to 23 Watt LED lamp and 110 dB honking horn with wiring.<br>23 Watts white light LED Lamp 1 number (ISI marked), 110 dB High Decible Warning Horn with Loud Sound 1 number (ISI marked), Control Switch suitable to operate the system. Electrical Switch Board (ISI marked) for 220 Volt AC Supply to device along with suitable wiring arrangement. 200 x 300 mm size name plate OR identification plate of steel, painted and clamped on pole. Concreting for fixing pole (in the ratio 1:1.5:3 Grade M20) over bouldering in earth pit and muffing (skirt) 300 mm above ground and 500 x 500 mm surrounding the pole to avoid water logging and rusting. Barbed wire wrapping around pole upto 600 mm length below main housing unit to avoid theft etc. Stay sets to avoid leaning of pole can be provided as per requirement of the Engineer-in-Charge.<br>Note :- This item shall be executed before prior permission of the Chief Engineer. | each | 325000.00 |
| 8.53 | Providing and installation of Road Restraint System i.e Single Roller Crash Barrier System. Single Roller System comprising of H1 & H2 post of hot dipped galvanized steel of minimum 139.8 mm diameter (H1-2200 mm and H2-720 mm), wall thickness 4.5 mm, main post spaced at an interval of 1000 mm, installed at the edge of earthen/hard/paved shoulder of the road, in 300 mm dia/square pit of depth minimum 1240 mm, fixed with PCC M15 grade concrete. Safety frame rail from metal sheet of 3.2 mm thick are fixed to main posts. shock absorbing rollers 345 mm diameter and 480 mm height made of polypropylene- EVA, post caps, locker, pipes and connection material. Distance between main post to short post shall be 500 mm c/c, the height of upper edge rail is 960 m m above road surface. All steel parts/frame shall be S235JR or equivalent SS400, the post shall be S235JR hot dipped galvanized. The system shall be installed as per drawing & specification enclosed, complete with 5 years free replacement warranty on all the material/ parts against any manufacturing defects and deformation but excluding damages due to accidents, if any.<br>Note 1: This item shall be included in the DPR and executed only with | RM | 73400.00 |

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|---|---|---|---|
| | prior written permission from the E-in-C MPPWD: | | |
| 8.54 | Bamboo Crash Barrier (Providing and erecting a 4 bamboo beam crash barrier comprising of bamboo poles having dia 80mm (±10mm), wall thickness of 10 mm, 0.85 m above road/ground level, fixed on clubbing of 3 bamboo vertical post, spaced 1.6 m post to post, 1.8 m high, 0.95 m below ground/road level, all steel parts and fitments to be galvanised by hot dip process, all fittings to conform to 4.8 grade steel, clubbing of 4 bamboo beam rail to be fixed on the vertical post with a spacer of dia 110 mm (+ 10mm), 300 mm long. All bamboo poles are well Seasoned, Treated with Creosote Oil through Vacuum Pressure Impregnation (VPI) method and High Density Polyethylene (HDPE) coated with coating thickness of Imm. End portion of approach side MELT Arrangement (Modified eccentric Loader Terminal) and End portion of departure side TT Arrangement (Trailing Terminal) as shown in figure 9.12 & 9.13 of the IRC SP-73-2018 respectively. Including Yellow or White reflective strip to be affixed front face of heam and post. ( 25 ^ * 50 mm on Beam 500mm c/c, 50*50mm on post). | RM | 3990.00 |
| 8.55 | Providing and Laying of 15mm high Single Rib Pattern Transverse Bar Markings with 2 component cold plastic with specialised cold plastic Primer coating rolled on surfacing material solvent free, high build two pack seamless, tough, skid resistant, for material to reduce speed. Cold Plastic road marking material shall be applied in single monolithic application to create rumble strip ratsed section of 15 mm high, 300 mm wide confirming to clause 3.7 of IRC 99, across the entire carriageway. Each rumble marking shall be made complete with specialised high refractive index glass beads of minimum reflectivity index of 1.5. The glass bead shall be 250gm/sqm. Type 1 beads (Rofer MoRT&H Specification Clause803.4.2) which are a constituent of the basic cold plastic compound vide Table 800-9 ( MoRT&H Specification) and Type 2 beads are those which are to be sprayed on the surface vide Clause 803.6.4 ( MoRT&H Specification), Pavement Marking Performance such as Day/Night visibility, Dry/Wet visibility and skid resistance shall be confirming to clause 15.5 of IRC 35. The finished surface to be levelled, uniform and free from streaks and holes, to be applied on the edge lines including surface cleaning and cost of all materials stc. Complete. | sqm | 6120.00 |
| 8.56 | Providing and fixing of Rumble Strip made up of TPU polyurethane (PU) Material size 500L x 100W x 15H mm and fixing of Rumble Strip in Bituminous grout/ Epoxy and Screwing the same with M.S Screw 15 x100mm etc. i/c removing asphalt at the site, Material, labour for fixing grouting and for controlling traffic during fixing etc. complete. | RM | 1650.00 |
| 8.57 | Providing and fixing Aluminum-backed flexible prismatic sheeting consisting of yellow colored flexible prismatic sheeting with non-metallic prismatic lens as retro reflective elements and conforming to ASTM D 4956-09 Type VI specification values for reboundable sheeting. AFP sheeting shall be of 1ft width with a 50 micron thick Aluminum (Al) foil with pressure sensitive adhesive and liner. AFP sheeting shall have screen printed arrow/slant line pattern in black color on top & shall not crack when tested for flexibility as per section S2.2.2 of ASTM D 4956-09. The AFP sheeting shall be applied with a | sqm | 9621.00 |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| | neoprene contact adhesive with Polychloroprene as base & the edges of the product shall be sealed all around with a two-part epoxy based structural adhesive and shall be extremely resistant to peel-off & shall confirm to IRC 79-2019. The AFP shall be provided with govt test report from ARAI/ CRRI/ ICAT and report confirming to coefficient of retro reflection, Flexibility and impact resistance test as per ASTM which shall be submitted to engineer-in-charge before installation/ award of work. | | |
| 8.58 | Providing and fixing advance warning speed arrester stud made out of poly carbonate/ABS molded body and highly reflective panels with micro prismatic lens capable of providing total internal reflection of the light entering the lens face with retro reflectance and chromaticity values shall be as given in the detailed specifications and the product shall conform to minimum ASTM D4280 and also meet all the testing conditions of ASTM D4280 listed for RPM (Raised Pavement Marker). The testing and test report shall be as per IRC 35 clause 5.1.2. The speed arrester should be fixed to the road surface using the adhesive recommended by the manufacturer. The length, height and width of the body shall not be less than 150 mm, 9 mm and 90 mm with a tolerance of +/-1 mm. The marker shall support a load of 30000 Kg tested in accordance with ASTM D4280. The area of each side of micro prismatic reflective lensshall not be less than 15.0 sq.cm. The slope or retro – reflective surface shall be 25 + 2 degree to base. It should have one Shank of not less than 25 mm diameter and not less than 40 mm length molded with the body for anchorage. The certificate confirming to initial CIL value for reflective lens not less than 125 for red lens, 250 for yellow lens & 400 for white lens at 0° entrance and 0.2° observation angle as per ASTM D4280 shall be provided for better visibility & improving road safety for commuters. | each | 641.00 |
| 8.59 | Providing and fixing speed attenuators designed with a thick shell made of HDPE/ polyethylene. Filled with water absorbs light front impacts, and filled with sand absorbs bigger front impacts. Mainly suggested for installation in junctions, detours, streets, avenues, and highways as a crash cushion. The outer body of the attenuator is both UV resistant and weather resistant and pasted with retro- reflective sheeting of Grade Type IV and above. It includes a safety and filling valve with hermetic sealing that prevents water loss by evaporation. It's capable of carrying a ballast of water or sand. It doesn't have sharp corners and metallic parts in case of impacts or collisions. It should be | each | 19000.00 |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| | either Orange or yellow in Colour for visibility during the daytime and it must have two arrow-shaped strip of retro-reflective sheeting Type IV or above to increase its visibility in night | | |
| 8.60 | Resilient Road Delineators<br>Supply and installation of resilient road delineators (roadway indicators) made from highly flexible and durable EVA material. The delineators have a total height of 820mm and a diameter of 100mm. Each delineator should be equipped with three reflective sheets spaced evenly, meeting the retroreflectivity requirements according to ASTM D 4956 Tyupe XI label specifications. The delineator will feature grooves along its length to protect the reflective sheets from vandalism. After digging a 100mm hole, the delineator should be fixed with an anchor and screw, then grouted as per the Engineer's instructions. | each | 3579.00 |
| 8.61 | Providing fixing and erecting/Pre-cast RCC information board including reinforcement with RCC M25 Posts (2 Nos) of size 150 x150x3300 mm RCC M-25 plate of size 1300x900 x 50mm the post shall be with groove at top of size 900x60x40mm. So that the plate is fixed inside it. Both posts are fixed jointly with M-15 concrete of size 600x1500x750 excavation in all types of soil including painting of Post and plate with synthetic enamel paint in all shades and also including providing and fixing High Intensity micro Prismatic retro reflective sheeting (Type IV) 50 mm wide around both side of the board and two places in each post (Preferably at 30cm and 150 cm respectively above road level) All work shall be as per MORT&H specifications and as per drawing including all material, labour, lead and lift. T&P etc. complete. | each | 17500.00 |

# CHAPTER: 9.0

# CULVERTS, BOX CELL & DRAINS

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| **9.1** | **PCC M-15 in Foundation**<br>Plain cement concrete M-15 mix with crushed stone aggregate 40 mm nominal size mechanically mixed, placed in foundation and compacted by vibration including curing for 14 days. as per MORT&H Specification 409. | cum | 4225.00 |
| **9.2** | Providing and Laying Reinforced Cement Concrete Pipe NP4/prestrssed concrete pipe on first class bedding in single row. Providing and Laying Reinforced cement concrete pipe NP4/prestrssed concrete pipe for culverts on first class bedding of granular material (cost of bedding included) in single row including fixing collar with cement mortar 1:2 but excluding excavation, protection works, backfilling, concrete and masonry works in head walls and parapets. | | |
| | A 600 mm dia | metre | 4250.00 |
| | B 1000 mm dia | metre | 6732.00 |
| | C 1200 mm dia | metre | 8554.00 |
| | D 1500 mm dia | metre | 11200.00 |
| **9.3** | Providing and Laying Reinforced Cement Concrete Pipe NP 4 /prestrssed concrete pipe on first class bedding in double row. Providing and Laying Reinforced cement concrete pipe NP4 /prestrssed concrete pipe for culverts on first class bedding of granular material (cost of bedding included) in double row including fixing collar with cement mortar 1:2 but excluding excavation, protection works, backfilling, concrete and masonry works in head walls and parapets. | | |
| | A 600 mm dia | metre | 8500.00 |
| | B 1000 mm dia | metre | 13621.00 |
| | C 1200 mm dia | metre | 17277.00 |
| | D 1500 mm dia | metre | 22500.00 |
| **9.4** | Providing and Laying Reinforced cement concrete pipe NP4 / prestrssed concrete pipe for culverts on first class bedding of granular material (cost of bedding included) in triple row including fixing collar with cement mortar 1:2 but excluding excavation, protection works, backfilling, concrete and masonry works in head walls and parapets and as per relevant clauses of section-2900. | | |
| | A 600 mm dia | metre | 12750.00 |
| | B 1000 mm dia | metre | 23196.00 |
| | C 1200 mm dia | metre | 28662.00 |
| | D 1500 mm dia | metre | 33500.00 |
| **9.5** | Add for every additional row in item 9.4 above | | |
| | A 600 mm dia | metre | 4250.00 |
| | B 1000 mm dia | metre | 6732.00 |
| | C 1200 mm dia | metre | 8554.00 |
| | D 1500 mm dia | metre | 11200.00 |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| **9.6** | Providing concrete craddle bedding in M-15 grade concrete as per clause 2900 and as per section 1700 and 2900. | cum | 4225.00 |
| **9.7** | Deduction for not providing the first-class bedding (in case sand is available in the bed or the concrete craddle is provided as per item 9.6 above) in item No.9.2., 9.3, 9.4 & 9.5 above. | cum | 514.00 |
| 9.8 | Providing P.C.C. M-15 Nominal mix for hume pipe culverts in foundation and sub structure etc. as per drawing and as per relevant clauses of section 1500, 1700, 2100 and 2900. | cum | 4425.00 |
| **9.9** | Providing and placing in position precast reinforced cement concrete U-shape drain as per design and shape, using coarse aggregate and fine aggregate derived from natural sources, Ordinary Portland cement, admixtures as per IS: 456-2000 and standard like India standard, international standards viz. Japanese Industrial Standards (JIS)/ US standard/EU standard/ having of strength not less than M-50. Using Thermo- Mechanically Treated bars. (TMT/TMX) FE 500D or more conforming to IS 1786 reinforcement bar (minimum 95 kg per cum) and capable of carrying a wheel load of min 25 Tones for heavy vehicle movement. Rate includes all cost of material, labor charges at the plant, cost of reinforcement, formwork and lifting accessories, and transportation upto work at the site. making necessary holes of required sizes for carrying through service lines etc., providing steel hooks for lifting, etc. the work to be executed as per the direction of Engineer in-charge. | | |
| (i) | Pre-Cast U Shape Drain 300x300mm wall thick.70 mm | RM | 2182.00 |
| (ii) | Pre-Cast U Shape Drain 450x450 mm wall thick.80 mm | RM | 2873.00 |
| (iii) | Pre-Cast U Shape Drain 600x600 wall thick. 90 mm | RM | 3914.00 |
| (iv) | Pre-Cast U Shape Drain 750x750 mm wall thick.100 mm | RM | 4728.00 |
| (v) | Pre-Cast U Shape Drain 900x900 mm wall thick.120 mm | RM | 6636.00 |
| (vi) | Pre-Cast U Shape Drain – 1200x1200 mm wall thick.120 mm | RM | 8566.00 |
| 9.10 | Providing and placing in position precast reinforced cement concrete U-shape drain as per design and shape, using coarse aggregate and fine aggregate derived from natural sources, Ordinary Portland cement, admixtures as per IS: 456-2000 and standard like India standard, international standards viz. Japanese Industrial Standards (JIS)/ US standard/EU standard/ having of strength not less than M-50. Using Thermo- Mechanically Treated bars. (TMT/TMX) FE 500D or more conforming to IS 1786 reinforcement bar (minimum 95 kg per cum) and capable of carrying a wheel load of min 5 Tones for heavy vehicle movement. Rate includes all cost of material, labour charges at the plant, cost of reinforcement, formwork and lifting accessories, and transportation upto work at the site. making necessary holes of required sizes for carrying through service lines etc., providing steel hooks for lifting, etc. the work to be executed as per the direction of Engineer in- charge. | | |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| (i) | Pre-Cast U Shape Drain 300x300mm Wall Th 70 mm | RM | 1885.00 |
| (ii) | Pre-Cast U Shape Drain 450x450 mm Wall Th 80 mm | RM | 2308.00 |
| (iii) | Pre Cast U Shape Drain 600x600 Wall Th 90 mm | RM | 3221.00 |
| (iv) | Pre-Cast U Shape Drain 750x750 mm Wall Th 100 mm | RM | 3951.00 |
| (v) | Pre-Cast U Shape Drain 900x900 mm Wall Th 120 mm | RM | 5132.00 |
| 9.11 | Providing and fixing factory-made precast RCC-U-shape drain drain covers, having of strength not less than M-50, using Thermo-Mechanically Treated bars. (TMT/TMX) FE 500D or more conforming to IS 1786 reinforcement bar (minimum 95 kg per cum) and capable of carrying a wheel load of 25.0 Tones for heavy vehicle movement, of size Lid for U-300x300 mm. As per standard size reinforced with 8 mm dia main longitudinal & 6 mm nos cross-sectional T'M T. hoop bars, including providing 50 mm dia perforations @ 2 nos. including providing edge binding with M.S. flats of size 50 mm x 1.80 mm complete, all as per direction of Engineer-in-charge | | |
| (i) | Pre-Cast RCC U Shape drain Lid for U-300x300 mm | each | 1550.00 |
| (ii) | Pre-Cast RCC U Shape drain Lid for U-450x450 mm | each | 1990.00 |
| (iii) | Pre-Cast RCC U Shape drain Lid for U-600x600 mm | each | 2426.00 |
| (iv) | Pre-Cast RCC U Shape drain Lid for U-750x750 mm | each | 2898.00 |
| (vi) | Pre-Cast RCC U Shape drain Lid for U-900x900 mm | each | 3215.00 |
| (vii) | Pre-Cast RCC U Shape drain Lid for U-1200x1200 mm | each | 6593.00 |
| 9.12 | Providing and fixing factory-made precast RCC U-shape drain (FUJI) drain covers, having of strength not less than M-50 using Thermo-Mechanically Treated bars (TMT/TMX) FE 500D or more conforming to IS 1786 reinforcement bar (minimum 95 kg per cum)and capable of carrying a wheel load of 5.0 Tones for heavy vehicle movement, of size Lid for U-300x300 mm. As per standard size reinforced with 8 mm dia main longitudinal & 6 mm nos cross-sectional T'M T. hoop bars, including providing 50 mm dia perforations @ 2 nos. including providing edge binding with M.S. flats of size 50 mm x 1.60 mm complete, all as per direction of Engineer-in-charge | | |
| (i) | Pre-Cast U Shape Drain Cover 300x300 mm | RM | 1390.00 |
| (ii) | Pre-Cast U Shape Drain Cover 450x450 mm | RM | 1674.00 |
| (iii) | Pre-Cast U Shape Drain Cover 600x600 mm | RM | 1977.00 |
| (iv) | Pre-Cast U Shape Drain Cover 750x750 mm | RM | 2255.00 |
| (v) | Pre-Cast U Shape Drain Cover 900x900 mm | RM | 2438.00 |
| 9.13 | Providing and placing in position precast reinforced cement concrete Flume as per design and shape, using coarse aggregate and fine aggregate derived from natural sources, Portland Pozzolana/ Ordinary Portland/ Portland Slag cement, admixtures in recommended proportions as per IS: 456-2000 and standard like India standard, International standards viz. Japanese Industrial Standards (JIS)/ US | | |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| | standard/EU standard/ having of strength not less than M-50. Using Thermo Mechanically Treated bars. (TMT/TMX) FE 500D or more conforming to IS 1786 reinforcement bar and capable of carrying a wheel load of heavy vehicle movement, Precast reinforced cement concrete drains using, Stormwater/roadside drain, etc. Rate includes all cost of material, labor charges at the plant, cost of reinforcement, formwork and lifting accessories, and transportation up to work at the site. Making necessary holes of required sizes for carrying through service lines etc. providing steel hooks for lifting, etc. the work executed as per the direction of Engineer-in-Charge. | | |
| A | **PRE-CAST FLUME HEIGHT 900MM (Minimum reinforcement 95 kg per Cum)** | | |
| (i) | Storm Water Drain (900x600) MM | RM | 5355.00 |
| (ii) | Storm Water Drain (900x700) MM | RM | 5514.00 |
| (iii) | Storm Water Drain (900x800) MM | RM | 5708.00 |
| (iv) | Storm Water Drain (900x900) MM | RM | 5866.00 |
| (v) | Storm Water Drain (900x1000) MM | RM | 6023.00 |
| (vi) | Storm Water Drain (900x1100) MM | RM | 6206.00 |
| (vii) | Storm Water Drain (900x1200) MM | RM | 6334.00 |
| (viii) | Storm Water Drain (900x1300) MM | RM | 6494.00 |
| (ix) | Storm Water Drain (900x1400) MM | RM | 6693.00 |
| (x) | Storm Water Drain (900x1500) MM | RM | 7530.00 |
| (xi) | Storm Water Drain (900x1600) MM | RM | 7703.00 |
| (xii) | Storm Water Drain (900x1700) MM | RM | 7877.00 |
| (xiii) | Storm Water Drain (900x1800) MM | RM | 8050.00 |
| B | **PRE-CAST FLUME HEIGHT 1200MM (Minimum reinforcement 95 kg per Cum)** | | |
| (i) | Storm Water Drain (1200x700) MM | RM | 6846.00 |
| (ii) | Storm Water Drain (1200x800) MM | RM | 7074.00 |
| (iii) | Storm Water Drain (1200x900) MM | RM | 7232.00 |
| (iv) | Storm Water Drain (1200x1000) MM | RM | 7390.00 |
| (v) | Storm Water Drain (1200x1100) MM | RM | 7548.00 |
| (vi) | Storm Water Drain (1200x1200) MM | RM | 7706.00 |
| (vii) | Storm Water Drain (1200x1300) MM | RM | 7928.00 |
| (viii) | Storm Water Drain (1200x1400) MM | RM | 8126.00 |
| (ix) | Storm Water Drain (1200x1500) MM | RM | 8288.00 |
| (x) | Storm Water Drain (1200x1600) MM | RM | 8449.00 |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| (xi) | Storm Water Drain (1200x1700) MM | RM | 8610.00 |
| (xii) | Storm Water Drain (1200x1800) MM | RM | 8805.00 |
| (xiii) | Storm Water Drain (1200x1900) MM | RM | 9783.00 |
| (xiv) | Storm Water Drain (1200x2000) MM | RM | 10010.00 |
| (xv) | Storm Water Drain (1200x2100) MM | RM | 10185.00 |
| (xvi) | Storm Water Drain (1200x2200) MM | RM | 10358.00 |
| (xvii) | Storm Water Drain (1200x2300) MM | RM | 10566.00 |
| (xviii) | Storm Water Drain (1200x2400) MM | RM | 10740.00 |
| C | **PRE-CAST FLUME HEIGHT 1300MM (Minimum reinforcement 95 kg per Cum)** | | |
| (i) | Storm Water Drain (1300x800) MM | RM | 7775.00 |
| (ii) | Storm Water Drain (1200x900) MM | RM | 7934.00 |
| (iii) | Storm Water Drain (1300x1000) MM | RM | 8125.00 |
| (vi) | Storm Water Drain (1300x1100) MM | RM | 8283.00 |
| (v) | Storm Water Drain (1300x1200) MM | RM | 8475.00 |
| (vi) | Storm Water Drain (1300x1300) MM | RM | 8633.00 |
| (vii) | Storm Water Drain (1300x1400) MM | RM | 8862.00 |
| (viii) | Storm Water Drain (1300x1500) MM | RM | 9057.00 |
| (ix) | Storm Water Drain (1300x1600) MM | RM | 9218.00 |
| (x) | Storm Water Drain (1300x1700) MM | RM | 9414.00 |
| (xi) | Storm Water Drain (1300x1800) MM | RM | 9574.00 |
| (xii) | Storm Water Drain (1300x1900) MM | RM | 9736.00 |
| (xiii) | Storm Water Drain (1300x2000) MM | RM | 10851.00 |
| (xiv) | Storm Water Drain (1300x2100) MM | RM | 11026.00 |
| (xv) | Storm Water Drain (1300x2200) MM | RM | 11233.00 |
| (xvi) | Storm Water Drain (1300x2300) MM | RM | 11407.00 |
| (xvii) | Storm Water Drain (1300x2400) MM | RM | 11581.00 |
| (xvii) | Storm Water Drain (1300x2500) MM | RM | 11789.00 |
| (xix) | Storm Water Drain (1300x2600) MM | RM | 12017.00 |
| D | **PRE-CAST FLUME HEIGHT 1400MM (Minimum reinforcement 95 kg per Cum)** | | |
| (i) | Storm Water Drain (1400x800) MM | RM | 8578.00 |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| (ii) | Storm Water Drain (1400x900) MM | RM | 8735.00 |
| (iii) | Storm Water Drain (1400x1000) MM | RM | 8927.00 |
| (iv) | Storm Water Drain (1400x1100) MM | RM | 9086.00 |
| (v) | Storm Water Drain (1400x1200) MM | RM | 9278.00 |
| (vi) | Storm Water Drain (1400x1300) MM | RM | 9540.00 |
| (vii) | Storm Water Drain (1400x1400) MM | RM | 9702.00 |
| (viii) | Storm Water Drain (1400x1500) MM | RM | 9895.00 |
| (ix) | Storm Water Drain (1400x1600) MM | RM | 10058.00 |
| (x) | Storm Water Drain (1400x1700) MM | RM | 10278.00 |
| (xi) | Storm Water Drain (1400x1800) MM | RM | 10253.00 |
| (xii) | Storm Water Drain (1400x1900) MM | RM | 10449.00 |
| (xiii) | Storm Water Drain (1400x2000) MM | RM | 11562.00 |
| (xiv) | Storm Water Drain (1400x2100) MM | RM | 11770.00 |
| (xv) | Storm Water Drain (1400x2200) MM | RM | 11944.00 |
| (xvi) | Storm Water Drain (1400x2300) MM | RM | 12152.00 |
| (xvii) | Storm Water Drain (1400x2400) MM | RM | 12360.00 |
| (xviii) | Storm Water Drain (1400x2500) MM | RM | 12534.00 |
| (xix) | Storm Water Drain (1400x2600) MM | RM | 12796.00 |
| (xx) | Storm Water Drain (1400x2700) MM | RM | 12970.00 |
| (xxi) | Storm Water Drain (1400x2800) MM | RM | 13178.00 |
| E | **PRE-CAST FLUME HEIGHT 1500MM (Minimum reinforcement 95 kg per Cum)** | | |
| (i) | Storm Water Drain (1500x800) MM | RM | 11266.00 |
| (ii) | Storm Water Drain (1500x900) MM | RM | 9013.00 |
| (iii) | Storm Water Drain (1500x1000) MM | RM | 9120.00 |
| (iv) | Storm Water Drain (1500x1100) MM | RM | 9278.00 |
| (v) | Storm Water Drain (1500x1200) MM | RM | 9470.00 |
| (vi) | Storm Water Drain (1500x1300) MM | RM | 9679.00 |
| (vii) | Storm Water Drain (1500x1400) MM | RM | 9855.00 |
| (viii) | Storm Water Drain (1500x1500) MM | RM | 10117.00 |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| (ix) | Storm Water Drain (1500x1600) MM | RM | 8335.00 |
| (x) | Storm Water Drain (1500x1700) MM | RM | 10474.00 |
| (xi) | Storm Water Drain (1500x1800) MM | RM | 10670.00 |
| (xii) | Storm Water Drain (1500x1900) MM | RM | 10865.00 |
| (xiii) | Storm Water Drain (1500x2000) MM | RM | 11061.00 |
| (xiv) | Storm Water Drain (1500x2100) MM | RM | 12205.00 |
| (xv) | Storm Water Drain (1500x2200) MM | RM | 12413.00 |
| (xvi) | Storm Water Drain (1500x2300) MM | RM | 12622.00 |
| (xvii) | Storm Water Drain (1500x2400) MM | RM | 12830.00 |
| (xviii) | Storm Water Drain (1500x2500) MM | RM | 13038.00 |
| (xix) | Storm Water Drain (1500x2600) MM | RM | 13212.00 |
| (xx) | Storm Water Drain (1500x2700) MM | RM | 13474.00 |
| (xxi) | Storm Water Drain (1500x2800) MM | RM | 13890.00 |
| (xxii) | Storm Water Drain (1500x2900) MM | RM | 14098.00 |
| (xxiii) | Storm Water Drain (1500x3000) MM | RM | 14273.00 |
| F | **PRE-CAST FLUME HEIGHT 1600MM (Minimum reinforcement 95 kg per Cum)** | | |
| (i) | Storm Water Drain (1600x800) MM | RM | 9390.00 |
| (ii) | Storm Water Drain (1600x900) MM | RM | 9582.00 |
| (iii) | Storm Water Drain (1600x1000) MM | RM | 9740.00 |
| (iv) | Storm Water Drain (1600x1100) MM | RM | 9932.00 |
| (v) | Storm Water Drain (1600x1200) MM | RM | 10124.00 |
| (vi) | Storm Water Drain (1600x1300) MM | RM | 10316.00 |
| (vii) | Storm Water Drain (1600x1400) MM | RM | 10578.00 |
| (viii) | Storm Water Drain (1600x1500) MM | RM | 10739.00 |
| (ix) | Storm Water Drain (1600x1600) MM | RM | 10934.00 |
| (x) | Storm Water Drain (1600x1700) MM | RM | 11130.00 |
| (xi) | Storm Water Drain (1600x1800) MM | RM | 11326.00 |
| (xii) | Storm Water Drain (1600x1900) MM | RM | 11522.00 |
| (xiii) | Storm Water Drain (1600x2000) MM | RM | 12694.00 |
| (xiv) | Storm Water Drain (1600x2100) MM | RM | 12902.00 |
| (xv) | Storm Water Drain (1600x2200) MM | RM | 13110.00 |
| (xvi) | Storm Water Drain (1600x2300) MM | RM | 13319.00 |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| (xvii) | Storm Water Drain (1600x2400) MM | RM | 13527.00 |
| (xviii) | Storm Water Drain (1600x2500) MM | RM | 13702.00 |
| (xix) | Storm Water Drain (1600x2600) MM | RM | 13963.00 |
| (xx) | Storm Water Drain (1600x2700) MM | RM | 14172.00 |
| (xxi) | Storm Water Drain (1600x2800) MM | RM | 14380.00 |
| (xxii) | Storm Water Drain (1600x2900) MM | RM | 14588.00 |
| (xxiii) | Storm Water Drain (1600x3000) MM | RM | 14762.00 |
| 9.14 | **Pre-Cast Box Culvert**<br>Providing and placing in position precast cement concrete Box-Culvert as per design and shape, using coarse aggregate and fine aggregate derived from natural sources, ordinary Portland/ admixtures as per design mix IS: 456-2000 and standard like Indian standard international standard viz. japaness industrial standard (JIS) /US standard/ EU standard having of strength not less than M-50, using Thermo-Mechanically Treated bars,(TMT/TMX) FE 500D or more conforming to IS 1786 reinforcement bar (Minimum reinforcement 95 kg per Cum) and capable of carrying a wheel load of 32.5 Tones with earth cushion 0.20 m to 0.00 m for heavy vehicle movement. Rate includes all cost of material, labor charges at the plant, cost of reinforcement, formwork and lifting accessories, and transportation up to work at the site. Making necessary holes of required sizes for carrying through service lines etc. providing steel hooks for lifting etc. the work to be executed as per the direction of Engineer in- charge. | | |
| (i) | Box Culvert Size 500x500 mm Th 120mm | RM | 8654.00 |
| (ii) | Box Culvert Size 600x600 mm Th 120mm | RM | 9677.00 |
| (iii) | Box Culvert Size 800x800 mm Th 120 mm | RM | 11807.00 |
| (iv) | Box Culvert Size 600x1000 mm Th 130 mm | RM | 13588.00 |
| (v) | Box Culvert Size 900x1000 mm Th 130 mm | RM | 15480.00 |
| (vi) | Box Culvert Size 1000x1000 mm Th 130 mm | RM | 16167.00 |
| (vii) | Box Culvert Size 1200x1000 mm Th 130 mm | RM | 17457.00 |
| (viii) | Box Culvert Size 800x1400 mm Th 130 mm | RM | 15314.00 |
| (ix) | Box Culvert Size 900x1400 mm Th 130 mm | RM | 15825.00 |
| (x) | Box Culvert Size 1000x1400 mm Th 130 mm | RM | 16445.00 |
| (xi) | Box Culvert Size 1100x1400 mm Th 130 mm | RM | 17040.00 |
| (xii) | Box Culvert Size 1200x1400 mm Th 130 mm | RM | 17550.00 |
| (xiii) | Box Culvert Size 1300x1400 mm Th 150 mm | RM | 18789.00 |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| (xiv) | Box Culvert Size 1400x1400 mm Th 150 mm | RM | 19383.00 |
| (xv) | Box Culvert Size 1500x1400 mm Th 150 mm | RM | 19978.00 |
| (xvi) | Box Culvert Size 2000x1200 mm Th 180 mm | RM | 32481.00 |
| (xvii) | Box Culvert Size 2000x1500 mm Th 180 mm | RM | 34306.00 |
| (xviii) | Box Culvert Size 2000x1800 mm Th 180 mm | RM | 36061.00 |
| (xix) | Box Culvert Size 2000x2000 mm Th 180 mm | RM | 37321.00 |
| (xx) | Box Culvert Size 2500x1500 mm Th 220 mm | RM | 41782.00 |
| (xxi) | Box Culvert Size 2500x1800 mm Th 220 mm | RM | 43782.00 |
| (xxii) | Box Culvert Size 2500x2000 mm Th 220 mm | RM | 45130.00 |
| (xxiii) | Box Culvert Size 2500x2500 mm Th 220 mm | RM | 48434.00 |
| 9.15 | Construction of RCC drain 1.0m x 1.0m inside dimension lined with 150 mm thick RCC, M-20 side walls, 150 mm thick RCC M-20 bottom slab, over 100 mm thick (Av.) levelling course in PCC M-15 including excavation, dressing of sides and bottom, provision of HYSD reinforcement consisting of 8 mm dia horizontal bars @ 200 mm c/c and U-shaped 10 mm dia vertical bars @ 120 mm c/c including cutting. bending and binding wire, placing in position, with shuttering, Weep holes at drainage layer level and concreting by using concrete mixer, compaction by vibration etc complete as per direction of the Engineer-in-Charge. | RM | 4011.00 |
| 9.16 | Construction of RCC drain 0.75 m x 0.75 m inside dimension lined with 135 mm thick RCC, M-20 side walls, 135 mm thick RCC M-20 bottom slab, over 100 mm thick (Av.) levelling course in PCC M-15 including excavation, dressing of sides and bottom, provision of HYSD reinforcement consisting of 8 mm dia horizontal bars @ 200 mm c/c and U-shaped 10 mm dia vertical bars @ 150 mm c/c including cutting. bending and binding wire, placing in position, with shuttering. Weep holes at drainage layer level and concreting by using concrete mixer, compaction by vibration etc complete as per direction of the Engineer-in-Charge. | RM | 2773.00 |
| 9.17 | Construction of RCC drain 0.60 m x 0.60 m inside dimension lined with 120 mm thick RCC, M-20 side walls, 120 mm thick RCC M-20 bottom slab, over 100 mm thick (Av.) levelling course in PCC M-15 including excavation, dressing of sides and bottom, provision of HYSD reinforcement consisting of 8 mm dia horizontal bars @ 200 mm c/c and U-shaped 8 mm dia vertical bars @ 150 mm c/c including cutting. bending and binding wire, placing in position, with shuttering, Weep holes at drainage layer level and concreting by using concrete mixer, compaction by vibration etc complete as per direction of the Engineer-in-Charge. | RM | 2132.00 |

# CHAPTER: 10.0

# MAINTENANCE OF ROADS

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| 10.1 | Repair to pot holes and removal of loose material, trimming of sides, cleaning of surface by providing tack coat with bitumen emulsion, 20 mm thick pre-mix carpet using catonic bitumen emulsion @ 21.5 kg per 10 sqm and seal coat type B with bitumen emulsion @ 10kg per 10 sqm as per technical specification and compacting with three wheeled 80-100 kN Static Roller. | sqm. | 307.00 |
| 10.2 | Repair to pot holes and removal of loose material, trimming of sides, cleaning of surface by providing Primer coat, tack coat, 20 mm thick pre-mix carpet material having binder content @ 14.6 kg per sqm and seal coat type B having bitumen content @ 6.8 kg per 10 sqm as per technical specification and compacting with three wheeled 80-100 kN Static Roller. | sqm. | 298.00 |
| 10.3 | Filling Pot- holes and Patch Repairs with - Bituminous concrete, 40mm. (Removal of all failed material, trimming of completed excavation to provide firm vertical faces, cleaning of surface, painting of tack coat on the sides and base of excavation as per clause 503, back filling the pot holes with hot bituminous material as per clause 504, compacting, trimming and finishing the surface to form a smooth continuous surface, all as per clause 3004.2) | | |
| (i) | for grading I Material | sqm | 336.00 |
| (ii) | for grading II Material | sqm | 346.00 |
| 10.4 | Repair to pot holes by removal of failed material, trimming the sides to vertical and levelling the bottom, cleaning the same with compressed air or any appropriate method filled with 75mm B.M, applying bitumen tack coat at the bottom and vertical sides as per as per MoRT&H specification clause 3004.1 and 3004.2 | cum | 8939.00 |
| 10.5 | Built-Up-Spray Grout<br>Providing, laying and rolling of built-up-spray grout layer over prepared base consisting of a two-layer composite construction of compacted crushed coarse aggregates using motor grader for aggregates key stone chips spreader may be used with application of bituminous binder after each layer, and with key aggregates placed on top of the second layer to serve as a Base conforming to the line, grades and cross-section specified, the compacted layer thickness being 75 mm. | sqm | 284.00 |
| 10.6 | Repair of Pot Holes, Patch Repair<br>Providing, laying and rolling of built-up-spray grout layer over prepared base consisting of composite construction of compacted crushed coarse aggregates and key stone chips with application of bituminous binder after each layer, and with key aggregates placed on top of the final layer to serve as a Base conforming to the line, grades and cross-section specified, (number of layers of coarse aggregate shall be as per site requirement). | cum | 3781.00 |
| 10.7 | Providing and Laying Pressure Sensitive Technology ( PST ) based ready mix compound for repair potholes, manufactured from aggregates of 4-12 mm, treated with special grade bitumen, Pressure Sensitive Polymer( PSP) and Cross Linking Chemical (CLC) agents etc, technically evaluated for repairs on bitumenous/concrete surfaces, confirming to CRRI and IRC norms for pressure sensitive polymer products for repairs of Potholes on | sqm | 938.00 |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| | road/ Pavements, havinfg upto 25mm thick layer on base of compacted aggregates ( Crusher Brocken ) of 50mm thickness including labours, material, tools and tampering etc complete. | | |
| 10.8 | Providing and laying Pressure Sensitive Technology ( PST ) based ready mix compound for repairing potholes, manufactured from aggregates of 4-12mm, treated compound having mix of bituminous antistriping compound, stabilizer, Pressure Sensitive Polymer and water repellent compound like oil soluble polyacid etc. having good consistency for laying, technically evaluated for repairs on bituminous/concrete surface confirming to CRRI and IRC norms for pressure sensitive polymer products for repairs fo potholes on road/ pavement having upto 25mm thick ready thickness including labours, material, tools and tampering etc. complete. (Undertaking shall be taken from the supplier before starting the work) | sqm | 881.00 |
| 10.9 | Providing 50 mm thick Full grout bituminous treatment including supplying all materials, prepairing the exisisting road surface, applying tack coat @ 0.735 kg / sqm and laying the required thickness of metal layer, heating and spraying bitumen of specified Grade at the rate of 5.00 kg / sqm, spreading chips and laying seal coat of specified Grade of bitumen of 1.70 Kg/sqm and compacting, making the surface good etc. complete (For only Patch Repair works / For Road work)<br>(VG-30 bulk bitumen rates are considered to arrive at rates) | sqm | 715.00 |
| 10.10 | Providing 50 mm thick semi grout bituminous treatment including supplying all materials, prepairing the exisisting road surface, applying tack coat @ 0.735 kg / sqm and laying the required thickness of metal layer, heating and spraying bitumen of specified Grade at the rate of 2.75 kg / sqm, spreading chips and laying seal coat of specified Grade of bitumen of 1.25 Kg/sqm and compacting, making the surface good etc. complete (For only Patch Repair works / For Road work)<br>(VG-30 bulk bitumen rates are considered to arrive at rates) | sqm | 552.00 |
| 10.11 | Pot Hole filling with Jet Patcher-Repairs to potholes 50 mm upto 100 mm by using semidense bituminous concrete in cold mix by using emulsion including making the potholes /patches in regular shape and size. Removing loose pockets, dewatering in potholes, tack coats of 25 Kilogram/10 Sq.Mt. with RS1 emulsion etc.by use of advanced jetpatching technolgy machine of approved make and specification having capacity 5.50 TO 8.00 cubic meters with auxillary engine with direct drive to heavy duty blower, Rotary Positive Displacement Blower with direct drive and separate airline silencer. (Capacity – 15 PSI, Flow rate 1370/cum/hr). cleaning as directed by Engineer – in-Charge etc. complete (Emulsion @ 5% by weight of mix) | | |
| (a) | Upto 50mm | sqm | 1317.00 |
| (b) | 50mm-100mm | sqm | 1867.00 |
| 10.12 | Repair to cement concrete road by providing laying & Fixing mortar-based patch master & Liquid rapid harding chemical including cleaning of concrete surface and applying of special bounding liquid. | sqm | 3550.00 |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| 10.13 | Repair of joint Grooves with Epoxy Mortar<br>Repair of spalled joint grooves of contraction joints, construction joint, longitudinal joints and expansion joints in concrete pavements using epoxy mortar or epoxy concrete. | metre | 1543.00 |
| 10.14 | Repair of old Joints Sealant<br>Removal of existing sealant and re-sealing of contraction, construction longitudinal or expansion joints in concrete pavement with fresh sealant material | metre | 49.00 |
| 10.15 | Removing all loose concrete which is detached from the substrate, by using a mechanical wire brush, complete with removing dust by compressed air and dampening the surface with water without puddles. | sqm | 30.00 |
| 10.16 | Applying one coat of bonding material complying to ASTM C881 Type-2, Grade-2, class-C like Anchorex EPB or equivalent on the cleaned substrate by spray/roller @ 10 sq m per ltr | sqm | 20.00 |
| 10.17 | Mixing and applying of 30mm thickness Repair material confirming to IRC SP 83-2018 clause 12.1.2.a and 12.1.4 an ultra fast strength concrete repair mortar which does not require external water curing and attains compressive strength of 40N/mm2 @ 5 hours from placing as per IS 516 such as Strextra EFS 4, or equivelent after diluting it with 10mm aggregates at a proportion of 1:0:5 (concrete 10mm aggregates); mixing and placing the mortar as per manufactures data sheet over the prepared surface of the pavement, complete along strengthening, no water curing and opening the road to traffic within 5-6 hours of placing(the last bag of mortar) | sqm | 655.00 |
| 10.18 | Providing and applying a dry-shake surface non- metallic floor surface hardner of mohs hardness 6 or more like Tapcrete 333 or equivelent over the repair mortar Strextra R-20 or equivalt while the mortar is still green that means the mortar has not achieved final set, consumption @ 7kg/sq.m. applied as per manufactures data sheet, inclusive with finishing material labour. | sqm | 117.00 |
| 10.19 | Crack stitching with inclined tie bar (cross Stitching) used for cracks in reasonably good condition in order to arrest movement of slab and slabs pieces by making primer verticals holes,, alternative inclined hole, cleaning the hole through oil free compressor and filled with epoxy resin in enough quality that will coat HYSD bar confirming IS 1786 of 12 mm dia in every hole also groone shall be made along with line cracks display spalling and filled with a low viscocity resin or fire epoxy mortar following the method specified IRC SP 83:2008. | RM | 650.00 |
| 10.20 | Stapling<br>Crack stitching with V-bars (Stapling) to be used for cracks in reasonably good condition in order to arrest movement of slabs and slab pieces by making the position of vertical holes of dia 30 mm at a distance of 228 mm form the crack at spacing of 600 mm center | RM | 455.00 |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| | to center, drilling holes, Slit removing, Devris cleaning from the holes, roughen the sites of hole and slit by sand blasting, sand paper. Insert the steel bar (FE 500), fill the holes and silt with epoxy mortar (1:3 epoxy: sand) up to 10mm above the top surface of steel bar. Filling with non-shrinkable concrete or any other equivalent material may be filled up to the top level of PQ2 and fill the sides of the groove with low visco sity epoxy following the IRC SP 83:2018. | | |
| 10.21 | Providing and laying a woven Polypropylene Geotextile of Type-I, II & III manufactured from black UV stabilised Polypropylene Tape yarn on the prepared subgrade as the separator cum reinforcement with necessary overlap as per drawing. (For use this item SE/CE permission is obligatory.) confirming to IRC SP 59 -2019 | | |
| a) | Providing and laying a fibre glass geogrid of Tensile strength 50 kN/m manufactured from high quality glass fibre strands and coated with a polymer modified bitumen as reinforcement to Asphalt overlay flexible pavements and Asphalt over distressed rigid P.C.C. and Bituminous surface confirming to IRC SP 59 - 2019 | sqm | 220.00 |
| b) | Providing and laying a fibre glass stsgrid of Tensile strength 100 kN/m manufactured from high quality glass fibre strands and coated with a polymer modified bitumen as reinforcement to Asphalt overlay flexible pavements and Asphalt over distressed rigid P.C.C. and Bituminous surface IRC SP 59 -2019 | sqm | 339.00 |
| c) | Providing and laying a fibre glass geogrid of Tensile strength 200 kN/m manufactured from high quality glass fibre strands and coated with a polymer modified bitumen as reinforcement to Asphalt overlay flexible pavements and Asphalt over distressed rigid P.C.C. and Bituminous surface IRC SP 59 -2019 | sqm | 426.00 |
| 10.22 | Supply and laying mortar based instant patch master for cemented or bituminous based roads with rapid setting mortar adhesive and rapid hardner imported chemicals @ 1sqm x 5mm thickness. | sqm | 3550 |
| 10.23 | Removing & Resetting existing Heavy-Duty Cobble/Inter locking stones/Pavers of thickness 60/75/80/100mm using Cement and 40mm thick sand bed (average thickness) and compacting with plate vibrator to achieve final desired compaction & setting of pavers to its final level, including cost of materials, labour and usage charges of machinery complete as per specifications. (The cost of new paver/stones required shall be added separately) | sqm | 484 |
| 10.24 | Removal of existing sealant and re sealing of contraction. Longitudinal or expansion joints in concrete pavement with fresh sealant material as directed by Engineer incharge as per specification including all material, labour, etc. complete | RM | 34.00 |
| 10.25 | Restoration of rain cuts with soil, having CBR>12, clearing the loose soil, benching for 300mm width laying fresh material in layers not exceeding 250 mm and compaction with power rammer to achieve 97% compaction to restore the original alignment, level and slopes as per drawings and MoRT&H specification clause 3002 | cum | 125.00 |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| 10.26 | Maintenance of Hard Shoulder (stripping excess soil) (Stripping excess soil from the shoulder surface to achieve the approved level and compacting with plate compactor) | sqm | 17.00 |
| 10.27 | Stripping excess soil from the shoulder surface to achieve the approved level and compacting with smooth wheel roller, vibratory roller, hand roller, plate vibrator, plate compactor at OMC to achive atlist 97% of the Maximum dry density in accordance to MoRT&H specification clause 3003.3 | sqm | 21.00 |
| 10.28 | Making up loss of material/irregularities on shoulders upto 100 mm thickness to the design level by adding fresh approved soil having CBR >12 and compacting it with appropriate equipment at OMC as per MoRT&H specification clause 3003 | sqm | 52.00 |
| 10.29 | Providing and placing empty cement/fertilizer bags (having capacity not less than 50 kg each) filled with mixure of good earth/Moorum, Gravels etc. for restoration of raincuts & road, bridge section / Ghat section for erosion control & traffic safety etc as per direction of Engineer-in-Charge including all materials, labour etc complete. | bag | 15.00 |
| 10.30 | Land Slide Clearance in soil<br>Clearance of land slides in soil and ordinary rock by a bull-dozer D 80 A-12, 180 HP and disposal of the same on the valley side | cum | 61.00 |
| 10.31 | Land slide Clearance in Hard Rock Requiring Blasting<br>Clearing of land slide in hard rock requiring blasting for 50% of the boulders and disposal of the same on the valley side. | cum | 118.00 |
| 10.32 | Providing painting, figuring and numbering as per IRC specifications complete with Synthetic Enamal Paint for 5th Km. Stone. | | |
| | (i) Two coats on new work | each | 385.00 |
| | (i) One coat on old work. | each | 293.00 |
| 10.33 | Providing painting, figuring and numbering as per IRC specifications complete with Synthetic Enamal Paint for ordinary Km. Stone. | | |
| | (i) Two coats on new work | each | 154.00 |
| | (ii) One coat on old work. | each | 99.00 |
| 10.34 | Providing painting, figuring and numbering as per IRC specifications complete with Synthetic Enamal Paint for 0.2 Km. Stones or Boundary Stones. | | |
| | (i) Two coats on new work | each | 39.00 |
| | (ii) One coat on old work. | each | 25.00 |
| 10.35 | Providing painting, figuring & numbering to major bridges, linear water-way exceeding 30m.Complete with Synthetic Enamel Paint. | | |
| | (i) Two coats on new work. | each | 268.00 |
| | (ii) One coat on old work. | each | 214.00 |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| 10.36 | Providing painting to angle Iron posts of road sign boards complete with Synthetic Enamel Paint. | | |
| | (i) Two coats on new work. | each | 114.00 |
| | (ii) One coat on old work. | each | 105.00 |
| 10.37 | Providing and applying two or more cost acrylic emulsion paint, having VOC (Volatile Organic Compound) content less than 50 grams/ litre, of approved brand and manufacture, including applying additional coats wherever required, to achieve even shade and colour with cow dung paint confirming to Is specification equivalent to 15 15489.2013 to unplastered concrete surface after cleaning of surface of dirt, dust, oil, grease, efflorescence etc. | sqm | 84.00 |
| 10.38 | Providing and applying one or more coat acrylic emulsion paint, having VOC (Volatile Organic Compound) content less than 50 grams/litre, of approved brand and manufacture, including applying additional coats wherever required, to achieve even shade and colour with cow dung paint confirming to 15 specification equivalent to IS 15489.2013 to unplastered concrete surface after cleaning of surface of dirt, dust, oil, grease, efflorescence etc. | sqm | 56.00 |
| 10.39 | **Bitumenous Helipad**<br>Construction of Bituminous Material Helipad including clearing and Grubbing, cutting, trimming, loading of earth material existing ground with hydraulic excavator for avg. depth up to 0.70 to 0.90 m and filling, levelling of ground with approved material from borrow pit having CBR > 7 includes all lifts and leads for avg depth of 0.3 m to 0.50 m. Construction of helipad also comprises of different layers of Crusher Run Macadam of thickness 150 mm, wet mix Macadam of thickness 250 mm, Dense Bituminous Macadam of thickness as specified (includes Prime coat, tack coat) and Bituminous Concrete of thickness 30 mm. The operation includes of Non-bituminous layers comprises of Mixing, spreading, laying in uniform layers through motor grader and watering at OMC, compaction with vibratory roller to 1 achieve the desired density as per clause 406 & 407 of MORT&H respectively. The operation of bituminous layers includes mixing of appropriate material, and binder content through Drum mix plant, transporting the hot mix material to the work site, laying with a Electronic sensor paver finisher with sensor control to required grade, level and Rolling with smooth wheeled, Vibratory and tandom rollers to achieve the desired compaction as per clause 505 and 507 of MORTandH respectively. Also provide marking of H and Outer circle with hot applied thermoplastic compound with reflectorising glass beeds on bitumenous surface as per standard drawing & Instructed by Engineer-in-Charge. | | |
| (i) | BT Helipad 30 m dia (DGBM 75 mm) | each | 896009.00 |
| (ii) | BT Helipad 25 m dia (DGBM 50 mm) | each | 610855.00 |
| (ii) | BT Helipad 15 m dia (DGBM 50 mm) | each | 233409.00 |
| 10.40 | **Concrete Helipad**<br>Construction of Bituminous Material Helipad including clearing and Grubbing, cutting, trimming, loading of earth material existing | | |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| | ground with hydraulic excavator for avg. depth up to 0.3 m and filling, levelling of ground with approved material from borrow pit having CBR > 7 includes all lifts and leads for avg depth of 0.3 m. Construction of helipad also comprises of different layers of Crusher Run Macadam of thickness 200 mm, Base concrete with Plain Cement Concrete (PCC) gradre-M-15 of thickness 100 mm, Plain Cement Concrete of Grade M-25 of thickness 250 mm. The operation includes of CRM layers comprises of Mixing, spreading, laying in uniform layers through motor grader and watering at OMC, compaction with vibratory roller to achieve the desired density as per clause 407 of MORT&H respectively. The operation of Concrete layers includes mixing of appropriate size of material, and mix material aat the work site, laying to required grade, level and to use vibrator to achieve the desired compaction as per specification. Also provide marking of H and Outer circle with hot applied thermoplastic compound with reflectorising glass beeds on bitumenous surface as per standard drawing & Instructed by Engineer-in-Charge. | | |
| (i) | CC Helipad 30 m dia | each | 1665682.00 |
| (ii) | CC Helipad 25 m dia | each | 1144807.00 |
| (ii) | CC Helipad 15 m dia | each | 426015.00 |
| 10.41 | **Construction of Murrom Road with Dust topping** Construction of Murrom Road including clearing and Grubbing, cutting, trimming, loading of earth material existing ground with hydraulic excavator for avg. depth up to 0.3 m and filling, levelling of ground with approved material from borrow pit having CBR > 7 includes all lifts and leads for avg depth of 0.3 m. Construction of road also comprises of layers of Crusher Run Macadam/Dust topping of thickness 100 mm, The operation includes of Mixing, spreading, Laying in uniform layers through motor grader and watering at OMC, compaction with vibratory roller to achieve the desired density as per clause 407 of MORT&H respectively.(For minimum length of 100 m) | RM | 1168.00 |
| 10.42 | Construction of Bitumenous Road Construction of Bituminous Material Road including clearing and Grubbing, cutting,trimming, loading of earth material existing ground with hydraulic excavator for avg. depth up to 0.3 m and filling, levelling of ground with approved material from borrow pit having CBR > 7 includes all lifts and leads for avg depth of 0.3 m with Avg. width of 7m. Construction of Road also comprises of different layers with avg. width of 5.5 m includes Crusher Run Macadam of thickness 15 cm, wet mix Macadam of thickness 225 mm, Dense Bituminous Macadam of thickness 50 mm (includes Prime coat, tack coat) and Bituminous Concrete of thickness 30 mm. The operation includes of Non-bituminous layers comprises of Mixing, spreading, laying in uniform layers through motor grader and watering at OMC, compaction with vibratory roller to achieve the desired density as per clause 406 & 407 of MORT&H respectively. The operation of bituminous layers includes mixing of appropriate material, and binder content through Drum mix plant, transporting the hot mix | metre | 7280.00 |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| | material to the work site, laying with an Electronic sensor paver finisher with sensor control to required grade, level and Rolling with smooth wheeled, Vibratory and tandom rollers to achieve the desired compaction as per clause 505 and 507 of MORTandH respectively. Also provide road marking on either edge with hot applied thermoplastic compound with reflectorising glass beeds on bitumenous surface as per standard drawing & Instructed by Engineer-in-Charge. (For minimum length of 100 m) | | |
| 10.43 | Hire Charges of Mechanical Machineries i/c Fuel & driver on Hourly basis (Minimum 4 hours in day) | | |
| (i) | Hydraulic Excavator (JCB Or Similar) | Per Hour | 1100.00 |
| (ii) | Tractor with Dozar/grader attachment | Per Hour | 700.00 |
| (iii) | Soil Compactor /tandem wheel vibratory roller | Per Hour | 1700.00 |
| (iv) | Static Roller 8 tonne | Per Hour | 600.00 |
| 10.44 | Application of 2-3 coats of cow dung mix with water manually up to uniform consistency and applying it by brooming to prepare uniform surface for helipad \|landing I/c labour, cow dung watering & all necessary complete as directed by Engineer in-charge (inner radius of 12.5 to 15 m) | each | 12000.00 |
| 10.45 | Lime painting in 2-3 coats over prepared cow dung surface for helipad \|lime painting includes H-marking 75 cm wide and outer circle marking 50cm wide as per drawing and as directed by Engineer in-charge | each | 3000.00 |
| 10.46 | Supplying & Sprinkling of Water over ground by mechanical means includes water tanker with mounted water sprinkler system to \|facilitate non-dust surface before and at the time of landing of helicopter as directed by Engineer in-charge | Per 1000 Ltr | 500.00 |
| 10.47 | Transport of Material (Balli, Barricades/post pipe & Other Material) by mechanical Means including Loading & Unloading both Of Ballies/ steel Barricades/post/pipe and other material from PWD store/Forest depo/ PHE Store or any other place to vehicle and supplying it to the programme ground. | | |
| (i) | H.C.V. (GVW upto 20 T) | | |
| | A) Up to 5 KM | Per Trip | 4000.00 |
| | B) Beyond 5 Km Per KM | Per KM | 100.00 |
| (ii) | L.C.V. (GVW upto 7.5 T) | | |
| | A) Up to 5 KM | Per Trip | 3000.00 |
| | B) Beyond 5 Km Per KM | Per KM | 80.00 |
| (iii) | By Tractor Trolly | | |
| | \|A) Up to5 KM | Per | 2000.00 |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| | | Trip | |
| | B) Beyond 5 Km Per KM | Per KM | 70.00 |
| 10.48 | **Bamboo Barricading Fixing**<br>Fixing of vertical post of hollow pipe/balies etc of required Hight i/c excavation of pits of required size in any soil avg. depth 30 cm to \|rigidity fix pipe /post/ballies with Fixing and tying of bamboo in horizontal layers of uniform spacing to vertical post of hollow pipe/ballies in barricading. Dismantling & stacking the same \|post/balies/pipe/bamboo after use as per directed by Engineer-in-Charge. (Item is i/c material for fixing & excluding cost of bamboo). | | |
| (i) | For Three Layer | M | 30.00 |
| (ii) | For Four Layer | M | 40.00 |
| 10.49 | Fixing of welded mesh/chain link jali on ballies/barricading as directed \|by Engineer in-charge and removing & stacking of same. (i/c all necessary material for fixing but excluding cost of mesh/chain link and transportation) | M | 20.00 |
| 10.50 | Fixing of Steel Drop gate /Barrier i/c excavation of pits for stand and stopper post, fixing of posts on pit & filling balancing bin with sufficient weight as directed by Engineer in-charge. (i/c all material T&P and removal staking of same refilling of pits etc.) | each | 500.00 |
| 10.51 | (a) Fixing of steel barricading i/c cost of binding material all complete. | M | 25.00 |
| | (b) Removing & stacking the same after use as directed by Engineer-in-Charge | M | 30.00 |
| 10.52 | Uniform preparation of surface by manual spreading of crushed stone/crusher stone dust/moorum/copara or similar material, 20mm to required thickness i/c manual levelling & dressing at site etc complete | cum | 280.00 |
| 10.53 | **Rainwater Harvesting System**<br>Involving construction of roadside rainwater harvesting system involves the development of percolation tanks across the basin using earthen structures.<br>Recharge tank inner side of diameter 1.5 metre including excavation upto 2.50 m depth and placing perforated NP4 1500 mm dia. pipe 3.00 m length, above 0.5 m ground level 150 mm diameter bore will be drilled up to 15.00 meter the ground water table or the sand layer, whichever is higher. Additionally, a 100 mm diameter perforated PVC pipe will be installed up to RCC cover and to the bore depth to facilitate water infiltration.<br>Inside the percolation tank, a recharge zone will be prepared with multiple filtration layers to enhance groundwater recharge which shall be consist of 1. Stone Boulder Layer: (5 cm to 20 cm in size) 1 m height forming the base. 2. Gravel Layer: (5 mm to 10 mm in size) 0.5 m height placed over the boulder layer. 3. Brick Bats Layer: Positioned above the gravel layer to aid filtration 0.5 m height. 4. Coarse Sand Layer: (1.5 mm to 2 mm in size) 0.50 m | each | 66816.00 |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| | height forming the topmost layer for improved filtration efficiency. Precast R.C.C. perforated drain covers Providing and fixing factory made precast RCC perforated drain covers, having concrete of strength not less than M-25, of size 2.00 m dia., including reinforcement and with 8 mm dia four no longitudinal and 9 nos cross sectional T.M.T. Hoop bars, including providing50 mm dia perforations @ 100 to 125 mm c/c, including providing edge binding with M.S. flats of size 50 mmx1.6 mm complete, recharge tank cover all around Gabion wall upto 0.6 m width and 0.5 m height all as per direction of Engineer-in- Charge.<br>To effectively collect and channel rainwater, the ditch and furrow method will be implemented across the basin, directing water toward the percolation pit for efficient groundwater recharge. | | |
| 10.54 | Rainwater Harvesting with brick masonry 0.2 m thick perforation with PVC pipe 100mm dia @ 0.50 m centre to centre all around the recharge pit including excavation upto 2.00 m depth with bore hole 150 mm dia and 15 m depth filled with gravels 3-6 mm size, recharge pit filled boulder 5-20 cm and 0.5 m depth. over the boulders layer gravel 5-10 mm size depth 0.5 m, over the gravels layer core sand 0.5 m depth. | | |
| (i) | 1.00 m x 1.00 m | each | 29300.00 |
| (ii) | 1.50 m x 2.00 m | each | 44300.00 |
| (iii)) | 1.50 m x 3.00 m | each | 54800.00 |
| (iv) | 2.00 m x 3.00 m | each | 62500.00 |
| 10.55 | Rainwater Harvesting 2.00 m x 2.00 m size with gabion wall 0.5 m thick with wire crates including excavation upto 2.00 m depth with bore hole 150 mm dia and 15 m depth filled with gravels 3-6 mm size, recharge pit filled boulder 5-20 cm and 0.5 m depth. over the boulders layer gravel 5-10 mm size depth 0.5 m, over the gravels layer core sand 0.5 m depth. | sqm | 28000.00 |
| 10.56 | Rainwater Harvesting 2.00 m x 2.00 m size including excavation upto 2.00 m depth with bore hole 150 mm dia and 15 m depth filled with gravels 3-6 mm size, recharge pit filled boulder 5-20 cm and 0.5 m depth. over the boulders layer gravel 5-10 mm size depth 0.5 m, over the gravels layer core sand 0.5 m depth. | sqm | 21000.00 |
| 10.57 | Rainwater Harvesting 0.30 m dia upto 3.5m depth drill by tractor mounded are manually shall be field with gravel and core sand including exevated material deposed off with all lead and lift. | each | 7500.00 |
| Note:- (i) Annual Maintaince upto Defect Laibilty Period from Item No 10.52 to 10.56, 10 % extra amount shall be paid per year before rainyseason.<br>(ii) Item no 10.52 to 10.56 shall be monitered and checked by GPS photograph this photograph shall be submitted to department and kept upto defect laibilty period. | | | |
| 10.58 | Providing Korbandi with approved material (CBR >12%) obtained from borrow pits with all lifts & leads, transporting to site, spreading, grading to required slope and as per specification with width of 0.6m and Average depth of 10 cm. including all material, labour, etc. complete. | RM | 11.00 |
| 10.59 | Repair to pot holes by removal of failed material, trimming the sides to vertical and levelling the bottom, cleaning the same with | sqm. | 647.00 |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| | compressed air or any appropriate method, filled with Bituminous macadam material having binder content 3.4% applying bitumen emulsion tack coat at the bottom and on sides covered by 20 mm thick type A or B close Graded Surfacing or Mixed Seal Surfacing, compecting with Three wheeled 80-100 KN static Roller as per technical specifications including all material, labour, etc. complete. (Note: - Payment shall be release after Sealing.) | | |
| 10.60 | Providing and repairing by WMM layer including filling up of pot holes, ruts and rectifying corrugated surface, damaged edges and ravelling as per technical specification with depth 50mm to 100 mm including all material, labour, etc. complete. | sqm. | 135.00 |
| 10.61 | The maintenance of all type of drains including erosion, repair, clearing, cleaning, reshaping, regrading, deepening of side drains as well as catch water drains as per technical specification including all material, labour, etc. complete. | RM | 14.00 |
| 10.62 | White washing with lime & Color washing of tree up to the height of 1.5 mts. from the road crown. The top & bottom band will be 0.30 mts white washed, middle band 0.30 mts. With color wash including cleaning for the preparation of surface as directed by Engineer-in-charge. Including Fevicol adhesive @ 2.5 % and 0.3% indigo of lime quantity including all material, labour, etc. complete. | | |
| | (i) Girth from 300 mm to 600 mm | each | 11.00 |
| | (ii) Girth from 600 mm to 900 mm | each | 18.00 |
| | (iii) Girth from 900 mm to 1800 mm | each | 33.00 |
| | (iv) Girth above 1800 mm | each | 61.00 |
| 10.63 | Construction of boulder/stone/rubble wall as per drawing and specifications in mudmorter or dry, for protection work, temporary diversion etc. including all materials, labour etc complete. | cum | 1221.00 |
| 10.64 | Cleaning and repairing vegetation/ garbage / soil / sand etc near the kerbs of bridges, parapets of culverts etc including cleaning of all water sprouts (on each side separately.) | RM | 4.00 |
| 10.65 | White washing with white cement, two coats on parapet walls, wheel guards and kerbs etc including preparation of surface by cleaning scraping etc. as directed by Engineer-in-charge. Including Fevicol adhesive @ 2.5 % and 0.3% indigo of lime quantity including cost of all material etc. including all material, labour, etc. complete. | sqm. | 25.00 |
| 10.66 | Suppiving and applying STAMPED CONCRETE SYSTEM by approved make blended colour hardener in pigments, cement and graded quartz silica, special conditioning admixtures, UV stabilized, of group 1/group 2 @ 2.93 kg per sqm. In first pass 65 % of required qty shall be applied and then 35% shall be applied where there is lack of colour. It shall be leveled and finished by bull float followed by steel floater to get smooth finish. Supplying and applying Bomanite India Release Agent, UV stabilized, pre dispersed and anticaking @ 1.22 kg per 10 sqm by brush broadcasting on the aren to be stamped. Stamping with Bomanite India stamping tools as per decided Bomanite India pattern which | sqm. | 754.00 |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| | includes placing of stamp in correct position, pressing with rubber mallet removing the stamping tool after confirmation that imprint has appeared. Providing and applying Bomanite India Acrylic sealer coat to the stamped area including removing release agent with water jet if required, after curing applying first coat of sealer using either spray pump, brush or roller @ 7.43 sqm per lit. After complete drying of first cont applying second coat @ 7.43 sqm per lit. allowing to dry for at least 4 hours etc. complete. Work should be executed by authorised Bomonite applicator.<br>Note-1: M-30 Grade Ready Mix Concrete shall be laid as per specification and shall be paild separately. | | |
| 10.67 | Supplying and mixing of Coloured Concrete System IPS/Broom Finish by approved make pre dispersed, UV stabilized, Integral colour bag as per shade Group-1 (Buskin, Camel, Chamois, Light Copper, Light Melon, Quail Gray, Rhuberb, Salmon, Seal Gray, Sierra Rose, Sunset) one bag/m3 concrete, finishing either IPS plain finish or broom finish including laying and finishing. Providing and applying Bomanite India Acrylic sealer coat to the area, after curing applying first coat of sealer using either spray pump, brush or roller @ 7.43 sqm per lit. After complete drying of first coat applying second coat @ 7.43 sqm per lit, allowing to dry for at least 4 hours etc. job complete.<br>Note-1: M-30 Grade Ready Mix Concrete shall be laid as per specification and shall be paild separately. | | |
| | a) 50 mm | sqm. | 508.00 |
| | b)75 mm | sqm. | 578.00 |
| 10.68 | Supplying and mixing of Coloured Concrete System IPS/Broom Finish by approved make pre dispersed, UV stabilized, Integral colour bag as per shade Group-2 (Apricot, Coal Gray, Cool Mauve, Comstack, Golden Rod, Granola, Gunmetal Gray, Maize, Rust Green, Paprika, Red Clay, Siena, Thristle Brown) one bag/m3 concrete, finishing either IPS plain finish or broom finish including laying and finishing. Providing and applying Bomanite India Acrylic sealer coat to the area, after curing applying first coat of sealer using either spray pump, brush or roller @ 7.43 sqm per lit. After complete drying of first coat applying second coat @ 7.43 sqm per lit, allowing to dry for at least 4 hours etc. job complete.<br>Note-1: M-30 Grade Ready Mix Concrete shall be laid as per specification and shall be paild separately. | | |
| | a) 50 mm | sqm. | 721.00 |
| | c) 75 mm | sqm. | 899.00 |
| 10.69 | Supplying and mixing of COLOURED CONCRETE SYSTEM WITH STAMPING by approved make pre dispersed, uv stabilized, Integral colour bag as per shade Group-1 (Buskin, Camel, Chamois, Light Copper, Light Melon, Quail Gray, Rhuberb, Salmon, Seal Gray, Sierra Rose, Sunset) one bag/m3 of concrete, firushing IPS plain finish. Supplying and applying Bomanite India Release agent. UV stabilized, pre dispersed and anticaking @1.22 kg per 10 sqm by brush broadcasting on the | | |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| | area to be stamped. Stamping with Bomanite India stamping tools as per decided pattern which includes placing of stamp in correct position, pressing with rubber mallet removing the stamping tool after confirmation that imprint has appeared. Providing and applying Bomanite India Acrylic sealer coat to the stamped area including removing release agent with water jet if required, after curing applying first coat of sealer using either spray pump, brush or roller @ 7.43 sqm per lit. After complete drying of first coat applying second coat @ 7.43 sqm per lit., allowing to dry for at least 4 hours etc. complete.<br>Note-1: M-30 Grade Ready Mix Concrete shall be laid as per specification and shall be paild separately. | | |
| | a) 50 mm | sqm. | 621.00 |
| | b) 75 mm | sqm. | 691.00 |
| 10.70 | Supplying and mixing of COLOURED CONCRETE SYSTEM WITH STAMPING by approved make pre dispersed, uv stabilized, Integral colour bag as per shade Group-2 (Apricot, Coal Gray, Cool Mauve, Comstack, Golden Rod, Granola, Gunmetal Gray, Maize, Rust Green, Paprika, Red Clay, Siena, Thristle Brown) one bag/m3 of concrete, finishing IPS plain finish. Supplying and applying Bomanite India Release agent, UV stabilized, pre dispersed and anticaking @ 1.22 kg per 10 sqm by brush broadcasting on the area to be stamped. Stamping with Bomanite India stamping tools as per decided pattern which includes placing of stamp in correct position, pressing with rubber mallet removing the stamping tool after confirmation that imprint has appeared. Providing and applying Bomanite India Acrylic sealer coat to the stamped area including removing release agent with water jet if required, after curing applying first coat of sealer using either spray pump, brush or roller @ 7.43 sqm per lit. After complete drying of first coat applying second coat @ 7.43 sqm per lit, allowing to dry for at least 4 hours etc. complete. | | |
| | a) 50 mm | sqm. | 834.00 |
| | b) 75 mm | sqm. | 1011.00 |
| | A 50 mm | sqm. | 2469.00 |
| | b) 75 mm | sqm. | 2652.00 |
| 10.71 | Providing and laying Plain/Reinforced cement concrete as per drawing and technical specifications and as directed by Engineer-in-charge | | |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| | (A)Concrete with plant (Drum Type) and placing | | |
| | (i) | | |
| | (ii) PCC Grade M15 | cum | 3925.00 |
| | (iii) PCC Grade M20 | cum | 4083.00 |
| | (iv) RCC Grade M20 | cum | 4110.00 |
| | (v) PCC Grade M25 | cum | 4489.00 |
| | (vi) RCC Grade M25 | cum | 4648.00 |
| | (vii) PCC Grade M30 | cum | 4528.00 |
| | (viii) RCC Grade M30 | cum | 4731.00 |
| 10.72 | Centering and shuttering including strutting, propping etc. and removal of form work. | sqm | 250.00 |
| 10.73 | Brick work with well burnt chimney bricks in bulls pattern trench kiln, crushing strength not less than 25kg /cm2 and water absorption not more than 20% i/c curing scaffolding etc complete. | | |
| | (i) Cement mortar 1:4 (1 cement : 4 sand) | cum | 5252.00 |
| | (ii) Cement Mortar 1:6 (1 cement:6 sand). | cum | 5061.00 |
| 10.74 | Structural steel work in single section fixed with or without connecting plate including cutting, hoisting, fixing in position and applying a priming coat of approved steel primer all complete. | kg | 91.00 |
| 10.75 | Structural steel work using M.S. flats, angles, channels I-section, H-section etc. riveted, bolted or welded in built up sections, trusses and framed work, including cutting, hoisting, fixing in position and applying a priming coat of approved steel primer all complete. | kg | 103.00 |
| 10.76 | Steel work in built up M.S. tubularsection (round, square or rectangular hollow tubes etc.) trusses/frame work etc. including cutting, hoisting, fixing in position and applying a priming coat of approved steel primer, including welding and bolted with special shaped washersetc. complete. | kg | 117.00 |
| 10.77 | Welding by gas or electric plant including transportation of plant | cm | 5.00 |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| | at site etc.complete | | |
| 10.78 | Providing and fixing stainless steel (Grade 304) railing made of Hollowtubes, channels, plates etc. including welding, grinding, buffing, polishing and making curvature (wherever required) and fitting the same with necessary stainless steel nuts and bolts complete i/c fixing the railing with necessary accessories and stainless steel dash fasteners, stainless steel bolts etc., of required size, on the top of the floor or the side of waist slab with suitable arrangement as per approval of Engineerin-charge. (For payment purpose only weight of stainless steel members shall be considered excluding fixing accessories such as nuts, bolts, fasteners etc.) | kg | 582.00 |
| 10.79 | Granite work gang saw cut (polished and machine cut) of thickness 18mm for wall lining (veneer work), backing filled with a grout of average 12 mm thick in cement mortar 1:3 (1 cement: 3 sand) including pointing with white cement mortar 1:2 (1 white cement: 2 marble dust) with an admixture of pigment to match the marble shade: (To be secured to the backing by means of cramps, which shall be paid for separately). | sqm | 2675.00 |
| 10.80 | 15 mm cement plaster of mix | | |
| | (i) 1:4 (1 cement : 4 sand) | sqm | 184.00 |
| | (ii) 1:6 (1 cement : 6 sand) | sqm | 168.00 |
| 10.81 | Providing and applying white cement based putty of average thickness 1 mm, of approved brand over the plastered wall surface to prepare the surface even and smooth i/c all cost of material, labour and scaffold etc in all position complete. Note : Putty shall be applied only on the internal walls, for application on the exterior walls prior permission of Chief Engineer will be required. | sqm | 74.00 |

# CHAPTER: 11.0

# HORTICULTURE

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| 11.1 | **Spreading of Sludge Farm Yard Manure or/and good Earth** (Spreading of sludge farm yard manure or/ and good earth in required thickness (cost of sludge, farm- yard manure or/and good earth to be paid for separately)) | cum | 33.00 |
| 11.2 | Grassing with 'Doobs' Grass (Grassing with 'Doobs' grass including watering and maintenance of the lawn for 30 days or more till the grass forms a thick lawn free from weeds and fit for moving including supplying good earth if needed) | | |
| (i) | In rows 15 cm apart in either direction | sqm | 22.00 |
| (ii) | In rows 7.5 cm apart in either direction | sqm | 37.00 |
| 11.3 | Making Lawns including Ploughing and Dragging with 'Swagha' Breaking of Clod (Making lawns including ploughing and breaking of clod, removal of rubbish, dressing and supplying doobs grass or any other variety of grass roots and planting at 15 cm apart, including supplying and spreading of farm yard manure at rate of 0.18 cum per 100 sqm) including as directed by Engineer-in-charge | | |
| | (i) For one year maintenance | sqm | 220.00 |
| | (ii) For three year maintenance | sqm | 280.00 |
| | (iii) For five year maintenance | sqm | 400.00 |
| 11.4 | Maintenance of Lawns or Turfing of Slopes (Maintenance of lawns or Turfing of slopes (rough grassing) for a period of one year including watering etc) | sqm | 130.00 |
| | A) Maintenance of Lawns with Fine Grassing for the First Year Payment shall be made First year 50% Second year 20% third year 30 % for DLP Three Years for the plants servive at that time shall be deducted for plants found dead after Three Years. | sqm | 150.00 |
| 11.5 | B) Maintenance of Lawns with Fine Grassing for the First Year Payment shall be made First year 30% Second year 10% third year 10 % fourth year 10% and fifth year 40% for DLP Five Years for the plants servive at that time. Shall be deducted for plants found dead after Five Years. | sqm | 350.00 |
| 11.6 | Turfing Lawns with American/Carpet Grass, fine Grassing including Ploughing, Dressing (Turfing lawns with fine grassing including ploughing, dressing including breaking of clods, removal of rubbish, dressing and supplying doobs grass roots at 10 cm apart, including supplying and spreading of farm yard manure at rate of0.6 cum per 100 sqm) including maintenance one year. | sqm | 280.00 |
| 11.7 | a) Planting Flowering Plants and Shrubs in Central Verge | km | 37332.00 |
| | b) Maintenance of Flowering Plants and Shrubs in Central Verge for FiveYear Payment shall be made First year 50% Second year 20% third year 30 % for DLP Three Years for the plants servive at that time. Shall be deducted for plants found dead after Three Years. | km | 400623.00 |
| | C) Maintenance of Flowering Plants and Shrubs in Central Verge for FiveYear Payment shall be made First year 30% Second year 10% third year 10 % fourth year 10% and fifth year 40% for DLP Five Years for the plants servive at that time shall be deducted for plants found dead after Five Years. | km | 300623.00 |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| | Note Item no 11.6 to 11.7 shall be monitered and checked by GPS photograph this photograph shall be submitted to department and kept upto defect laibilty period. | | |
| 11.8 | **Planting Permanent Hedges including Digging of Trenches**<br>Planting permanent hedges including digging of trenches, 60 cm wide and 45 cm deep, refilling the excavated earth mixed with farmyard manure, supplied at the rate of 4.65 cum per 100 metres and supplying and planting hedge plants at 30 cm apart. | metre | 219.00 |
| 11.9 | **Planting of Trees and their Maintenance for three Year** Planting of various varities such as Pipal (Ficus religiose), Bargad (Ficus benghalensis), Neem (Azadirachta indica), Ashok (Saraca Asoca), Kadam (Neolamarckia cadamba), Aam (Mango), Gulmohar (Delonix regia), Amaltass (Hibiscus rosa-sinensis), Bougainvillea) (Bougainvillea glabra), Lantana (Lantana camara) etc. of trees as approved by the Engineer-in-charge by the road side (Avenue trees) in 0.60 m dia holes, 1 m deep dug in the ground, mixing the soil with decayed farm yard/sludge mannure, planting the saplings, backfilling the trench, watering, and maintaining the plants excluding tree guard. | | |
| (i) | Minimum Plant Height 1.00 M | Each | 2250.00 |
| (ii) | Minimum Plant Height 1.50 M | Each | 2750.00 |
| (iii) | Minimum Plant Height 2.00 M | Each | 3250.00 |
| (iv) | Minimum Plant Height 2.50 M | Each | 3750.00 |
| (v) | Minimum Plant Height 3.00 M | Each | 4750.00 |
| | **Note:-(i) Payment shall be made First year 50% Second year 30% third year 20% for DLP Three Years for the plants servive at that time. Shall be deducted for plants found dead after Three Years.** | | |
| 11.10 | **Planting of Trees and their Maintenance for Five Year** Planting of various varities such as Pipal (Ficus religiose), Bargad (Ficus benghalensis), Neem (Azadirachta indica), Ashok (Saraca Asoca), Kadam (Neolamarckia cadamba), Aam (Mango), Gulmohar (Delonix regia), Amaltass (Hibiscus rosa-sinensis), Bougainvillea) (Bougainvillea glabra), Lantana (Lantana camara) etc. of trees as approved by the Engineer-in-charge by the road side (Avenue trees) in 0.60 m dia holes, 1 m deep dug in the ground, mixing the soil with decayed farm yard/sludge mannure, planting the saplings, backfilling the trench, watering, and maintaining the plants excluding tree guard. | | |
| (i) | Minimum Plant Height 1.00 M | Each | 3250.00 |
| (ii) | Minimum Plant Height 1.50 M | Each | 3750.00 |
| (iii) | Minimum Plant Height 2.00 M | Each | 4250.00 |
| (iv) | Minimum Plant Height 2.50 M | Each | 4750.00 |
| (v) | Minimum Plant Height 3.00 M | Each | 5750.00 |
| | **Note:-(i) Payment shall be made First year 30% Second year 10% third year 10 % fourth year 10% and fifth year 40% for DLP Five Years for the plants servive at that time. Shall be deducted for plants found dead after Five Years.**<br>(ii) Item no 11.09 to 11.10 shall be monitered and checked by GPS photograph this photograph shall be submitted to department and kept upto defect laibilty period. | | |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| 11.11 | Renovation Lawns including, Weeding, Forking the Ground, Top Dressing with Forked Soil (Renovation lawns including, weeding, forking the ground, top dressing with forked soil, watering and maintenance the lawns, for 30 days or more, till the grass forms a thick lawn, free from weeds, and fit for moving and disposal of rubbish as directed, including supplying good earth, if needed but excluding the cost of well decayed farm yard manure) | sqm | 17.00 |
| 11.12 | Supply at Site Well Decayed Farm Yard Manure (Supply at site of work well decayed farm yard manure, from any available source, approved by the Engineer-in-Charge including screening and stacking) | cum | 15.00 |
| 11.13 | **Half Brick Circular Tree Guard, in 2nd class Brick, internal diametre 1.25 metres, and height 1.2 metres, above ground and 0.20 metre below ground**<br>Half brick circular tree guard, in 2nd class brick, internal diametre 1.25 metres, and height 1.2 metres, above ground and 0.20 metre below ground, bottom two courses laid dry, and top three courses in cement mortar 1:6 (1 cement 6 sand) and the intermediate courses being in dry honey comb masonry, as per design complete. | each | 2488.00 |
| 11.14 | **Edging with 2nd class Bricks, laid dry lengthwise**<br>Edging with 2nd class bricks, laid dry lengthwise, including excavation, refilling, consolidation, with a hand packing and spreading nearly surplus earth within a lead of 50 metres. | metre | 51.00 |
| 11.15 | **Making Tree Guard 53 cm dia and 1.3 m high as per design from empty bitumen drum** (Making tree guard 53 cm dia and 1.3 m high as per design from empty bitumen drum, slit suitably to permit sun and air, (supplied by the department at stock issue rate) including providing and fixing 2 nos MS sheet rings 50 x 0.5 mm with rivets, complete in all respect) | each | 872.00 |
| 11.16 | **Making Tree Guard 53 cm dia and 2 metres high as per design from empty bitumen drums** (Making tree guard 53 cm dia and 2 metres high as per design from empty bitumen drums, slit suitably to permit sun and air, ( supplied by the department at stock issue rate) including providing and fixing four legs 40 cm long of 30 x 3 mm MS riveted to tree guard and providing and fixing 2 nos MS sheet rings 50 x 0.5 mm with rivets complete in all respects) | each | 1810.00 |
| 11.17 | **Wrought Iron and Mild Steel Welded Work (Wrought iron and mild steel welded work)** (using angles, square bars, tees and channel grills, grating frames, gates and tree guards of any size and design etc. including cost of screens and welding rods or bolts and nuts complete fixed in position but without the cost of excavation and concrete for fixing which will be paid separately) | quintal | 8338.00 |
| 11.18 | **Tree Guard with MS Iron** (Providing and fixing MS iron tree guard 60 cm dia and 2 metre high above ground level formed of 4 Nos (25 x 6 mm) and 8 Nos (25 x 3 mm) vertical MS riveted to 3 Nos (25 x 6 | each | 1975.00 |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| | mm) iron rings in two halves, bolted together with 8 mm dia and 30 mm long bolts including painting two coats with paint of approved brand over a coat of priming, complete in all respects.) | | |
| 11.19 | **Tree Guard with MS Angle Iron and Steel Wire** (Providing and fixing tree guard 0.60 metre square, 2.00-metre-high fabricated with MS angle iron 30 x 30 x 3 mm, MS iron 25 x 3 mm and steel wire3 mm dia welded and fabricated as per design in two halves bolted together) | each | 2619.00 |
| 11.20 | RCC Pre Cast Tree Guard<br>Providing and fixing RCC precast tree guard of 2.00 meters height in a square shape with a minimum thickness of 50 mm. The tree guard shall be installed by excavating up to 0.50 meters depth and secured in place with M-15 grade concrete all around up to 0.15 meters thickness. Adequate openings shall be provided to ensure proper air circulation and sunlight exposure for the tree's growth. | | |
| (i) | 600 mm x600 mm | each | 1500.00 |
| (ii) | 750 mm x750 mm | each | 1800.00 |
| (iii) | 1000 mmX 1000 mm | each | 2100.00 |
| 11.21 | **Compensatory Afforestation** (Planting trees as compensatory afforestation at the rate of 290 trees per hectare at a spacing of 6 m by grubbing and leveling the ground upto a depth of 150 mm, digging holes 0.9 m dia, 1 m deep, mixing farm yard/sludge manure with soil, planting of sapling 2 m high with 25 cm dia stem, backfilling the hole and watering) | hectare | 69757.00 |
| 11.22 | **Tree Guard with MS Angle Iron and Steel Wire**<br>Providing and fixing tree guard 0.60 metre square, 2.00-metre-high fabricated with MS angle iron 30 x 30 x 3 mm, MS iron 25 x 3 mm and steel wire3 mm dia welded and fabricated as per design in two halves bolted together. | each | 1975.00 |
| 11.23 | **Tree Shifting and their Maintenance for Three Year**<br>Transplantation of Trees including trimming the branches of full-grown tree as per the requirement, giving a wax treatment of the trimmed area of branches, uprooting the tree i/c trimming and transporting the uprooted tree to the specified site of transplantation as directed by the engineer-in-Charge up to 12 Kms. lead including all lifts, transplanting, the uprooted tree by digging a required size of pit, filling it with fresh garden soil and manure as per the requirement & maintaining it for a period of 4 months by watering, weeding, manuring etc complete. | | |
| | A Trees having trunk girth upto 1.00 M. | Per Tree | 25236.00 |
| | B Trees having trunk girth more than 1.00 M. upto 1.5 M. | Per Tree | 27854.00 |
| | C Trees having trunk girth more than 1.5 M. to 2.00 M. | Per Tree | 30472.00 |
| | D Trees having trunk girth more tha 2.00 M. | Per Tree | 32566.00 |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| **Note: -** | Payment shall be made First year 50% Second year 30% third year 20% for DLP Three Years for the plants servive at that time shall be deducted for plants found dead after Three Years. | | |
| 11.24 | **Tree Shifting and their Maintenance for Five Year** Transplantation of Trees including trimming the branches of full-grown tree as per the requirement, giving a wax treatment of the trimmed area of branches, uprooting the tree i/c trimming and transporting the uprooted tree to the specified site of transplantation as directed by the engineer-in-Charge up to 12 Kms. lead including all lifts, transplanting, the uprooted tree by digging a required size of pit, filling it with fresh garden soil and manure as per the requirement & maintaining it for a period of 4 months by watering, weeding, manuring etc complete. | | |
| A | Trees having trunk girth upto 1.00 M. | Per Tree | 30236.00 |
| B | Trees having trunk girth more than 1.00 M. upto 1.5 M. | Per Tree | 32854.00 |
| C | Trees having trunk girth more than 1.5 M. to 2.00 M. | Per Tree | 35472.00 |
| D | Trees having trunk girth more tha 2.00 M. | Per Tree | 37566.00 |
| **Note: -** | Payment shall be made First year 30% Second year 10% third year 10 % fourth year 10% and fifth year 40% for DLP Five Years for the plants servive at that time shall be deducted for plants found dead after Five Years.<br>Note:- (ii) Item no 11.23 & 11.24 shall be monitered and checked by GPS photograph this photograph shall be submitted to department and kept upto defect laibilty period. | | |
| 11.25 | Providing and Fixing RCC Fencing pole, (pole size 0.15 x 0.15 and spacing between two poles minimum @ 2.5m and height of pole 2.5m and minimum excavation from the ground level 0.3 x 0.3x 0.6 and filled with M-15 concrete) 1.8 m fencing height with GI chain link fencing Mesh 100x100x2.5 mm Dia wire with GI Barbed wire at top and bottom including labour charges for fixing and all expences for complete work | RM | 322.00 |
| 11.26 | Providing and Fixing Barbed Wire Type A- The Barbs shall have four points and shall be formed by Twisting Two point wires each two turns, tightly around both the line wires making all together four complete turns line wire 2.5 mm point wire 2.5 mm distance between two barbs 75mm | KG | 85.00 |
| 11.27 | Providing and Fixing Barbed Wire Type A- The Barbs shall have four points and shall be formed by Twisting Two point wires each two turns, tightly around both the line wires making all together | KG | 83.00 |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| | four complete turns line wire 2.5 mm point wire 2.5 mm distance between two barbs 150mm | | |
| 11.28 | Providing and Fixing Barbed Wire Type A- The Barbs shall have four points and shall be formed by Twisting Two point wires each two turns, tightly around both the line wires making all together four complete turns line wire 2.5 mm point wire 2.0 mm distance between two barbs 75mm | KG | 85.00 |
| 11.29 | Providing and Fixing Barbed Wire Type A- The Barbs shall have four points and shall be formed by Twisting Two point wires each two turns, tightly around both the line wires making all together four complete turns line wire 2.5 mm point wire 2.0 mm distance between two barbs 150mm | KG | 85.00 |
| 11.30 | RCC Fencing Poles 1.6 meter Height | Each | 168.00 |
| 11.31 | RCC Fencing Poles 1.8 meter Height | Each | 184.00 |
| 11.32 | RCC Fencing Poles 2.0 meter Height | Each | 200.00 |
| 11.33 | RCC Fencing Poles 2.3 meter Height | Each | 248.00 |
| 11.34 | RCC Fencing Poles 2.45 meter Height | Each | 266.00 |
| 11.35 | RCC Fencing Ples Bend Type-I 2.45+0.45 mtr | Each | 340.00 |
| 11.36 | RCC Fencing Ples Bend Type-II 3.00+0.75 mtr | Each | 350.00 |

# CHAPTER: 12.0

# FOUNDATIONS

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| 12.1 | **Excavation for Structures**<br>Earth work in excavation of foundation of structures as per drawing and technical specification, including setting out, construction of shoring and bracing, removal of stumps and other deleterious matter, dressing of sides and bottom as per relevant clauses of section 300 & 2100 in | | |
| I | Ordinary soil (including back filling with approved material) | | |
| (i) | Depth upto 3 m | cum | 79.00 |
| (ii) | Depth 3 m to 6 m | cum | 87.00 |
| (iii) | Depth above 6m | cum | 98.00 |
| II | Ordinary rock (not requiring blasting) | | |
| (i) | Depth upto 3 m | cum | 239.00 |
| (ii) | 3 m to 6 m depth | cum | 251.00 |
| (iii) | Above 6 m depth | cum | 269.00 |
| III | Hard rock (requiring blasting) | cum | 297.00 |
| IV | Hard rock (blasting Prohibited) | cum | 1030.00 |
| V | Marshy soil | cum | 165.00 |
| VI | Back Filling in Marshy Foundation Pits | cum | 470.00 |
| 12.2 | Add extra as follows in the rates of the above items ifdewatering is resorted to | | |
| a) | upto 3 m depth | | 50% extra |
| b) | beyond 3 m to 6 m depth | | 75% extra |
| c) | Above 6 m depth | | 100% extra |
| 12.3 | Providing Plain cement concrete M-15 nominal mix in foundation as per relevant clauses of sections 1500, 1700 and 2100. | Cum | 4225.00 |
| 12.4 | Sand Filling in Foundation Trenches as per Drawing &Technical Specification. | cum | 1445.00 |
| 12.5 | Providing & Filling Annular Space Around Footing in Rock upto top of rock with Lean cement concrete M-15 nominal mix in foundation 1.5 m in soft rock and 0.60 m in hard rock with crushed stone aggregate 40 mm nominal size and as per relevant clauses of sections 1500, 1700 and 2100.<br>Note: - Only footing size with 15cm off set all around to be considered for pavement. | cum | 4225.00 |
| 12.6 | Providing and laying Plain/Reinforced cement concrete in open foundation including form work shuttering etc. complete as per drawing and technical specifications and as per relevant clauses of sections 1500, 1700 & 2100 with. | | |
| (i) | Concrete with plant (Drum Type) and placing | | |
| A | PCC Grade M15 | cum | 3925.00 |
| B | PCC Grade M20 | cum | 4083.00 |
| C | RCC Grade M20 | cum | 4110.00 |

| Item No. | | Description | Unit | Rate |
|---|---|---|---|---|
| | D | PCC Grade M25 | cum | 4489.00 |
| | E | RCC Grade M25 | cum | 4648.00 |
| | F | PCC Grade M30 | cum | 4528.00 |
| | G | RCC Grade M30 | cum | 4731.00 |
| | H | RCC Grade M35 | cum | 4907.00 |
| | I | RCC Grade M 40 | cum | 5206.00 |
| | J | RCC Grade M 45 | cum | 5324.00 |
| | (i) | With Batching Plant, (Pan Type or Twin Shaft type) Transit Mixer and Concrete Pump | | |
| | A | PCC Grade M15 | cum | 4425.00 |
| | B | PCC Grade M20 | cum | 4583.00 |
| | C | RCC Grade M20 | cum | 4610.00 |
| | D | PCC Grade M25 | cum | 4989.00 |
| | E | RCC Grade M25 | cum | 5148.00 |
| | F | PCC Grade M30 | cum | 5028.00 |
| | G | RCC Grade M30 | cum | 5231.00 |
| | H | RCC Grade M35 | cum | 5407.00 |
| | I | RCC Grad M 40 | cum | 5706.00 |
| | J | RCC Grad M 45 | cum | 5824.00 |
| | (ii) | With Batching Plant, (Pan Type or Twin Shaft type)Transit Mixer and placing with Boom placer | | |
| | A | PCC Grade M15 | cum | 4625.00 |
| | B | PCC Grade M20 | cum | 4783.00 |
| | C | RCC Grade M20 | cum | 4810.00 |
| | D | PCC Grade M25 | cum | 5189.00 |
| | E | RCC Grade M25 | cum | 5348.00 |
| | F | PCC Grade M30 | cum | 5228.00 |
| | G | RCC Grade M30 | cum | 5431.00 |
| | H | RCC Grade M35 | cum | 5607.00 |
| | I | RCC Grade M 40 | cum | 5906.00 |
| | J | RCC Grade M 45 | cum | 6024.00 |
| 12.7 | | Providing and laying cutting edge of mild steel weighing 40 kg per metre for well foundation complete as per drawing and technical specification. | tonne | 95178.00 |
| 12.8 | | Plain/Reinforced cement concrete, in well foundation complete as per drawing and technical specification | | |
| | (i) | Concrete with plant (Drum Type) and placing | | |
| | A | Well curb/ Well staining / Well cap | | |
| | i | RCC M20 Grade | Cum | 4755.00 |
| | ii | RCC M25 Grade | Cum | 5355.00 |
| | iii | RCC M30 Grade | Cum | 5450.00 |
| | iv | RCC M35 Grade | Cum | 5653.00 |
| | v | RCC M40 Grade | Cum | 5998.00 |
| | vi | RCC M45 Grade | Cum | 6108.00 |
| | vii | Bottom Plug | | |

| Item No. | | Description | Unit | Rate |
|---|---|---|---|---|
| | viii | PCC Grade M20 | Cum | 4288.00 |
| | ix | PCC Grade M25 | Cum | 4545.00 |
| | x | PCC Grade M30 | Cum | 4585.00 |
| | xi | PCC Grade M35 | Cum | 4877.00 |
| | | Intermediate plug / Top plug | | |
| | i | PCC Grade M20 | Cum | 4210.00 |
| | ii | PCC Grade M25 | Cum | 4306.00 |
| | iii | PCC Grade M30 | Cum | 4343.00 |
| | (ii) | With Batching Plant, (Pan Type or Twin Shaft type)Transit Mixer and Concrete Pump | | |
| | A | Well curb/ Well staining / Well cap | | |
| | i | RCC M20 Grade | Cum | 5255.00 |
| | ii | RCC M25 Grade | Cum | 5855.00 |
| | iii | RCC M30 Grade | Cum | 5950.00 |
| | iv | RCC M35 Grade | Cum | 6153.00 |
| | v | RCC M40 Grade | Cum | 6498.00 |
| | vi | RCC M45 Grade | Cum | 6608.00 |
| | vii | Bottom Plug | | |
| | viii | PCC Grade M20 | Cum | 4788.00 |
| | ix | PCC Grade M25 | Cum | 5045.00 |
| | x | PCC Grade M30 | Cum | 5085.00 |
| | xi | PCC Grade M35 | Cum | 5377.00 |
| | C | Intermediate plug / Top plug | | |
| | (i) | PCC Grade M20 | cum | 4415.00 |
| | (ii) | PCC Grade M25 | cum | 4806.00 |
| | (iii) | PCC Grade M30 | cum | 4843.00 |
| | **(III)** | With Batching Plant, (Pan Type or Twin Shaft type)Transit Mixer and placing with Boom placer | | |
| | A | Well curb/ Well staining / Well cap | | |
| | (i) | RCC M20 Grade | cum | 5455.00 |
| | (ii) | RCC M25 Grade | cum | 6055.00 |
| | (iii) | RCC M30 Grade | cum | 6150.00 |
| | (iv) | RCC M35 Grade | cum | 6353.00 |
| | (v) | RCC M40 Grade | cum | 6698.00 |
| | (vi) | RCC M45 Grade | cum | 6808.00 |
| | B | Bottom Plug | | |
| | (i) | PCC Grade M20 | cum | 4988.00 |
| | (ii) | PCC Grade M25 | cum | 5245.00 |
| | (iii) | PCC Grade M30 | cum | 5285.00 |
| | (iv) | PCC Grade M35 | cum | 5577.00 |
| | C | Intermediate plug / Top plug | | |
| | (i) | PCC Grade M20 | cum | 4910.00 |
| | (ii) | PCC Grade M25 | cum | 5006.00 |
| | (iii) | PCC Grade M30 | cum | 5043.00 |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| 12.9 | Sinking well circular/ D type / D type twin / square/ hexagonal/ octagonal for foundation to levels as per drawing including dredging, dewatering and drop chiselling as may be necessary, including sinking by loading with necessary kentledges and other usual means for the type of work i/c all the needed plant and machinery etc. complete as directed and removal of boulders or tree trunks etc. complete as per drawing and technical specifications. (Note: Depth of sinking shall be reckoned from bed level or excavated level, if so or water level whichever is higher) | | |
| | A Sandy soil | | |
| | (i) Depth below bed level up to 3 m | cum | 267.00 |
| | (ii) Beyond 3m to 10m depth | cum | 384.00 |
| | (iii) Beyond 10m to 20m | cum | 851.00 |
| | (iv) Beyond 20m to 30 m | cum | 1842.00 |
| | (v) Beyond 30m to 40 m | cum | 4101.00 |
| | B Clayey soil | | |
| | (i) Depth below bed level up to 3 m | cum | 388.00 |
| | (ii) Beyond 3m to 10m depth | cum | 809.00 |
| | (iii) Beyond 10 m to 20 m | cum | 1878.00 |
| | (iv) Beyond 20m to 30 m | cum | 3883.00 |
| | (v) Beyond 30m to 40 m | cum | 8648.00 |
| 12.10 | Sand filling in wells complete as per drawing and technical specifications as per clause 1210. | cum | 2062.00 |
| 12.11 | Providing steel liner 10 mm thick for curbs and 6mm thick for steining of wells including fabricating and setting out as per detailed drawing as per section 1200 & 1900. | tonne | 84008.00 |
| 12.12 | Bored cast-in-situ M35 grade R.C.C. pile excluding reinforcement complete as per drawing and technical specifications and removal of excavated earth with all lifts and lead up to 1000 m. (Pile diameter-750 mm) as per specification 1100, 1600 and 1700. | metre | 6260.00 |
| 12.13 | Bored cast-in-situ M35 grade R.C.C. pile excluding reinforcement complete as per drawing and technical specifications and removal of excavated earth with all lifts and lead up to 1000 m. (Pile diameter-1000 mm) as per specification 1100, 1600 and 1700. | metre | 9984.00 |
| 12.14 | Bored cast-in-situ M35 grade R.C.C. pile excluding reinforcement complete as per drawing and technical specifications and removal of excavated earth with all lifts and lead up to 1000 m. (Pile diameter-1200 mm) as per specification 1100, 1600 and 1700. | metre | 12928.00 |
| 12.15 | Bored cast-in-situ M35 grade R.C.C. pile excluding reinforcement complete as per drawing and technical specifications and removal of excavated earth with all lifts and lead up to 1000 m. (Pile diameter-1500 mm) as per specification 1100, 1600 and 1700. | metre | 15550.00 |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| 12.16 | Driven cast-in-place vertical M35 grade R.C.C. pile excluding reinforcement complete as per drawing and & Technical Specification (Pile diameter - 750 mm) as per specification 1100, 1600 and 1700. | metre | 7843.00 |
| 12.17 | Driven cast-in-place vertical M35 grade R.C.C. piles excluding reinforcement complete as per drawing and & Technical Specification (Pile diameter - 1000 mm) as per specification 1100, 1600 and 1700. | metre | 9984.00 |
| 12.18 | Driven cast-in-place vertical M35 grade R.C.C. piles excluding reinforcement complete as per drawing and & Technical Specification (Pile diameter - 1200 mm) as per specification 1100, 1600 and 1700. | metre | 12928.00 |
| 12.19 | Driven precast vertical M35 grade R.C.C. piles excluding reinforcement complete as per drawing and & Technical Specification (Pile Diameter=500 mm) as per specification 1100, 1600 and 1700. | metre | 2723.00 |
| 12.20 | Driven precast vertical M35 grade R.C.C. piles excluding reinforcement complete as per drawing and & Technical Specification (Pile Diameter=750 mm) as per specification 1100, 1600 and 1700. | metre | 4395.00 |
| 12.21 | Driven precast vertical M35 grade R.C.C. piles excluding reinforcement complete as per drawing and & Technical Specification (Pile Diameter=1000 mm) as per specification 1100, 1600 and 1700. | metre | 5030.00 |
| 12.22 | Driven precast vertical M35 grade R.C.C. piles excluding reinforcement complete as per drawing and & Technical Specification (Size of pile - 300 mm x 300 mm) as per specification 1100, 1600 and 1700. | metre | 1876.00 |
| 12.23 | Driven precast vertical M35 grade R.C.C. piles excluding reinforcement complete as per drawing and & Technical Specification (Size of pile - 500 mm x 500 mm) as per specification 1100, 1600 and 1700. | metre | 3181.00 |
| 12.24 | Driven precast vertical M35 grade R.C.C. piles excluding reinforcement complete as per drawing and & Technical Specification (Size of pile - 750 mm x 750 mm) as per specification 1100, 1600 and 1700. | metre | 5348.00 |
| 12.25 | Driven vertical steel piles complete as per drawing and Technical Specification (Section of the pile - H Section steel column 400 x 250 mm (ISHB Series) as per specification 1100, 1600 and 1700. | metre | 6140.00 |
| 12.26 | Driven vertical steel piles complete as per drawing and Technical Specification (Section of the pile - H Section steel column 450 x 250 mm (ISHB Series) as per specification 1100, 1600 and 1700. | metre | 6660.00 |
| 12.27 | Conducting Static Pile Test as per IS 2911 part 4 and IRC 75:2024 on Intial (2.5 tune of Design load) & Routine (1.5 tunnes | | |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| | of Design Load) pile, day using following methods. | | |
| | a) By using kentiledge method as per IS 2911 part 4 clause No 7.1.3 (a) | MT | 1100.00 |
| | b) By using Reaction pile method as per IS 2911 part 4 clause No 7.1.3 (b) | MT | 1150.00 (Including mobilized equipment) |
| | (c ) By using Rock Anchor method as per IS 2911 Part 4 clause No 7.1.3(C ) | MT | 1250.00 (Including mobilized equipment) |
| 12.28 | | MT | 700.00 |
| | Conducting cyclic static load test as per IS 2911 part 4 and IRC 75-2024ON Intial & Routine pile by using following methods. | | |
| | a) By using Kenti ledge method | MT | 1200.00 |
| | b) By using Reaction pile method. | MT | 1300.00 |
| 12.29 | Dynamic Pile load test<br>Conducting Dyamic pile load test on routine pile IRC 78-2024 Appendix-6 (Part-I) | MT | 350.00 |
| 12.30 | Directional Test<br>Conducting pile testby using Directional method as IRC 78-2024 Appendix -07 clause no 709.2.4<br>(Mostly conducting more then 1500 MT capacity of pile | MT | 1500.00 |
| 12.31 | Cement concrete for reinforced concrete in pile cap complete as per drawing and Technical Specification and as per relevant clauses of sections 1100, 1500 & 1700. | | |
| | A RCC Grade M20 | cum | 4534.00 |
| | B RCC Grade M25 | cum | 5054.00 |
| | C RCC Grade M30 | cum | 5136.00 |
| | D RCC Grade M35 | cum | 5312.00 |
| | E RCC Grade M40 | cum | 5611.00 |
| | F RCC Grade M45 | cum | 5729.00 |
| 12.32 | Levelling course for Pile cap PCC in M-15 below Pile cap asper drawing and as per section 1100 and 1700. | cum | 4225.00 |
| 12.33 | Supplying, fitting and placing un-coated HYSD bar reinforcement in foundation complete as per drawing and technical specifications and as per relevant clauses of section 1600. | | |
| | FE 500 | MT | 67252.00 |
| | FE 500D/FE 550/FE 550D | MT | 68252.00 |
| | FE 600 | MT | 78395.00 |
| | Corrosion Resistant Steel | MT | 78895.00 |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| 12.34 | Solid Round Glass Fiber Reinforced Polymer (GFRP) bars confirming to IS: 18256:2023<br>Note: - This item shall be executed as a temperature/ surface reinforcement | kg | 207.00 |
| 12.35 | Supplying, fitting and placing un-coated Mild steel reinforcement complete in foundation as per drawing andtechnical specification and as per relevant clauses of section 1600. | MT | 64480.00 |
| 12.36 | Providing fusion Bonded Epoxy Coating (FBEC) to TMT Bars as per IS:13620-1993 specifications for a thickness of 175micron & permissible variation of 50micron including testing of coating. FREQUENCY OF TEST OF FUSION BONDED EPOXY COAT a) Coating thickness shall be tested at a) frequency of not less than one full length bar every twenty bars for each size<br>b) Continuity of coating shall be determined by testing one full length bar in every twenty bars for each size<br>c) Coating thickness over the whole of the coated bar section shall be determined by sectioning and examining one bar in every 20t for each size.<br>d) For testing adhesion of coating, samples shall be selected from each size according to the frequency given as: For all types of bar: Minimum 1 for every diameter or as directed by Engineer-in-Charge. (The item is to be operated with prior approval of Superintending Engineer) | | |
| | (i) 8 mm | tonne | 19000.00 |
| | (ii) 10 mm | tonne | 15000.00 |
| | (iii) 12 mm | tonne | 13000.00 |
| | (iv) 16mm | tonne | 10000.00 |
| | (v) 20 mm | tonne | 8500.00 |
| | (vi) 25 mm | tonne | 7250.00 |
| | (vii) 32 mm | tonne | 5750.00 |
| 12.37 | Taking, exploratory boring 100mm dia at the location of piersand abutments or for high embankments in approaches in all types of strata as per IRC-78-1983 and section 2400 of specifications. | | |

| Item No. | | Description | Unit | Rate |
|---|---|---|---|---|
| | a) | Upto 1.0 m below bed level | metre | 2586.00 |
| | b) | Beyond 1.0 m depth upto 5.0 m | metre | 2705.00 |
| | c) | Beyond 5.0 m depth | metre | 2822.00 |
| 12.38 | | Providing and laying 1.5 m deep in rock and 1.5 m above rock 25mm dia Tor steel dowel bar in foundation including drilling 65mm dia bore hole in rock necessary bending, hooking tying reinforcement in position and grouting etc. complete as per drawing and specifications. | each | 1108.00 |
| 12.39 | | Providing and fixing pressure relief pipes of 110 mm diametre of PVC as per drawing for R.C.C Raft, galaries returns, abutments, wing wall etc complete. | RM | 479.00 |
| 12.40 | | Vertical Rock Anchor for Foundation Stability including all accessories for stressing, High tensile steel wires/strands stressing operations and grouting complete as per drawing and Technical Specifications | RM | 20000.00 |

# CHAPTER: 13.0

# SUB-STRUCTURE

| Item No. | | Description | Unit | Rate |
|---|---|---|---|---|
| 13.1 | | Plain/Reinforced cement concrete in sub-structure complete as per drawing and technical specifications | | |
| | A | PCC Grade M15 | | |
| | | Concrete with plant (Drum Type) and placing | cum | 3980.00 |
| | | With Batching Plant, (Pan Type or Twin Shaft type)Transit Mixer and Concrete Pump | cum | 4480.00 |
| | | With Batching Plant, (Pan Type or Twin Shaft type)Transit Mixer and placing with Boom placer | cum | 4780.00 |
| | B | PCC Grade M20 | | |
| | | Concrete with plant (Drum Type) and placing | cum | 4395.00 |
| | | With Batching Plant, (Pan Type or Twin Shaft type)Transit Mixer and Concrete Pump | cum | 4895.00 |
| | | With Batching Plant, (Pan Type or Twin Shaft type)Transit Mixer and placing with Boom placer | cum | 5195.00 |
| | C | PCC Grade M25 | | |
| | | Concrete with plant (Drum Type) and placing | | |
| | | 1 Height up to 5 m | cum | 4777.00 |
| | | 2 Height 5 to 10 m | cum | 4969.00 |
| | | With Batching Plant, (Pan Type or Twin Shaft type)Transit Mixer and Concrete Pump | | |
| | | 1 Height up to 5 m | cum | 5277.00 |
| | | 2 Height 5 to 10 m | cum | 5469.00 |
| | | With Batching Plant, (Pan Type or Twin Shaft type)Transit Mixer and placing with Boom placer | | |
| | | 1 Height up to 5 m | cum | 5577.00 |
| | | 2 Height 5 to 10 m | cum | 5769.00 |
| | D | PCC Grade M30 | | |
| | | Concrete with plant (Drum Type) and placing | | |
| | | 1 Height up to 5 m | cum | 4819.00 |
| | | 2 Height 5 to 10 m | cum | 5012.00 |
| | | With Batching Plant, (Pan Type or Twin Shaft type)Transit Mixer and Concrete Pump | | |
| | | 1 Height up to 5 m | cum | 5319.00 |
| | | 2 Height 5 to 10 m | cum | 5512.00 |
| | | With Batching Plant, (Pan Type or Twin Shaft type)Transit Mixer and placing with Boom placer | | |
| | | 1 Height up to 5 m | cum | 5519.00 |
| | | 2 Height 5 to 10 m | cum | 5812.00 |
| | E | RCC Grade M20 | | |
| | | Concrete with plant (Drum Type) and placing | | |
| | | 1 Height up to 5 m | cum | 4384.00 |
| | | 2 Height 5 to 10 m | cum | 4510.00 |
| | | 3 Height 10 to 15 M | cum | 4775.00 |

| Item No. | | Description | Unit | Rate |
|---|---|---|---|---|
| | | 4 Height Above 15 M | cum | 4975.00 |
| | | With Batching Plant, (Pan Type or Twin Shaft type)Transit Mixer and Concrete Pump | | |
| | | 1 Height up to 5 m | cum | 4884.00 |
| | | 2 Height 5 to 10 m | cum | 5010.00 |
| | | 3 Height 10 to 15 M | cum | 5275.00 |
| | | 4 Height Above 15 M | cum | 5475.00 |
| | | With Batching Plant, (Pan Type or Twin Shaft type)Transit Mixer and placing with Boom placer | | |
| | | 1 Height up to 5 m | cum | 5184.00 |
| | | 2 Height 5 to 10 m | cum | 5310.00 |
| | | 3 Height 10 to 15 M | cum | 5575.00 |
| | F | RCC Grade M25 | | |
| | | Concrete with plant (Drum Type) and placing | | |
| | | 1 Height up to 5 m | cum | 4946.00 |
| | | 2 Height 5 to 10 m | cum | 5124.00 |
| | | 3 Height 10 to 15 M | cum | 5276.00 |
| | | 4 Height Above 15 M | cum | 5576.00 |
| | | With Batching Plant, (Pan Type or Twin Shaft type)Transit Mixer and Concrete Pump | | |
| | | 1 Height up to 5 m | cum | 5446.00 |
| | | 2 Height 5 to 10 m | cum | 5624.00 |
| | | 3 Height 10 to 15 M | cum | 5776.00 |
| | | 4 Height Above 15 M | cum | 6076.00 |
| | | With Batching Plant, (Pan Type or Twin Shaft type)Transit Mixer and placing with Boom placer | | |
| | | 1 Height up to 5 m | cum | 5746.00 |
| | | 2 Height 5 to 10 m | cum | 5924.00 |
| | | 3 Height 10 to 15 M | cum | 6076.00 |
| | | 4 Height Above 15 M | cum | 6376.00 |
| | G | RCC Grade M30 | | |
| | | Concrete with plant (Drum Type) and placing | | |
| | | 1 Height up to 5 m | cum | 5033.00 |
| | | 2 Height 5 to 10 m | cum | 5189.00 |
| | | 3 Height 10 to 15 M | cum | 5410.00 |
| | | 4 Height Above 15 M | cum | 5610.00 |
| | | With Batching Plant, (Pan Type or Twin Shaft type)Transit Mixer and Concrete Pump | | |
| | | 1 Height up to 5 m | cum | 5533.00 |
| | | 2 Height 5 to 10 m | cum | 5689.00 |

| Item No. | | Description | Unit | Rate |
|---|---|---|---|---|
| | | 3 Height 10 to 15 M | cum | 5910.00 |
| | | 4 Height Above 15 M | cum | 6110.00 |
| | | With Batching Plant, (Pan Type or Twin Shaft type) Transit Mixer and placing with Boom placer | | |
| | | 1 Height up to 5 m | cum | 5833.00 |
| | | 2 Height 5 to 10 m | cum | 5989.00 |
| | | 3 Height 10 to 15 M | cum | 6210.00 |
| | | 4 Height Above 15 M | cum | 6410.00 |
| | H | RCC Grade M35 | | |
| | | Concrete with plant (Drum Type) and placing | | |
| | | 1 Height up to 5 m | cum | 5219.00 |
| | | 2 Height 5 to 10 m | cum | 5344.00 |
| | | 3 Height 10 to 15 M | cum | 5531.00 |
| | | 4 Height Above 15 M | cum | 5731.00 |
| | | With Batching Plant, (Pan Type or Twin Shaft type)Transit Mixer and Concrete Pump | | |
| | | 1 Height up to 5 m | cum | 5719.00 |
| | | 2 Height 5 to 10 m | cum | 5844.00 |
| | | 3 Height 10 to 15 M | cum | 6031.00 |
| | | 4 Height Above 15 M | cum | 6231.00 |
| | | With Batching Plant, (Pan Type or Twin Shaft type)Transit Mixer and placing with Boom placer | | |
| | | 1 Height up to 5 m | cum | 6019.00 |
| | | 2 Height 5 to 10 m | cum | 6144.00 |
| | | 3 Height 10 to 15 M | cum | 6331.00 |
| | | 4 Height Above 15 M | cum | 6531.00 |
| | I | RCC/PSC Grade M40 | | |
| | | Concrete with plant (Drum Type) and placing | | |
| | | 1 Height up to 5 m | cum | 5535.00 |
| | | 2 Height 5 to 10 m | cum | 5667.00 |
| | | 3 Height 10 to 15 M | cum | 5865.00 |
| | | 4 Height Above 15 M | cum | 6065.00 |
| | | With Batching Plant, (Pan Type or Twin Shaft type)Transit Mixer and Concrete Pump | | |
| | | 1 Height up to 5 m | cum | 6035.00 |
| | | 2 Height 5 to 10 m | cum | 6167.00 |
| | | 3 Height 10 to 15 M | cum | 6365.00 |
| | | 4 Height Above 15 M | cum | 6565.00 |
| | | With Batching Plant, (Pan Type or Twin Shaft type)Transit | | |

| Item No. | | Description | Unit | Rate |
|---|---|---|---|---|
| | | Mixer and placing with Boom placer | | |
| | | 1 Height up to 5 m | cum | 6335.00 |
| | | 2 Height 5 to 10 m | cum | 6467.00 |
| | | 3 Height 10 to 15 M | cum | 6665.00 |
| | | 4 Height Above 15 M | cum | 6865.00 |
| | J | RCC/PSC Grade M45 | | |
| | | Concrete with plant (Drum Type) and placing | | |
| | | 1 Height up to 5 m | cum | 5660.00 |
| | | 2 Height 5 to 10 m | cum | 5795.00 |
| | | 3 Height 10 to 15 M | cum | 5996.00 |
| | | 4 Height Above 15 M | cum | 6196.00 |
| | | With Batching Plant, (Pan Type or Twin Shaft type)Transit Mixer and Concrete Pump | | |
| | | 1 Height up to 5 m | cum | 6160.00 |
| | | 2 Height 5 to 10 m | cum | 6295.00 |
| | | 3 Height 10 to 15 M | cum | 6496.00 |
| | | 4 Height Above 15 M | cum | 6696.00 |
| | | With Batching Plant, (Pan Type or Twin Shaft type)Transit Mixer and placing with Boom placer | | |
| | | 1 Height up to 5 m | cum | 6460.00 |
| | | 2 Height 5 to 10 m | cum | 6595.00 |
| | | 3 Height 10 to 15 M | cum | 6796.00 |
| | | 4 Height Above 15 M | cum | 6996.00 |
| | K | RCC/PSC Grade M 50 | | |
| | | Concrete with plant (Drum Type) and placing | | |
| | | 1 Height up to 5 m | cum | 5785.00 |
| | | 2 Height 5 to 10 m | cum | 5922.00 |
| | | 3 Height 10 to 15 M | cum | 6128.00 |
| | | 4 Height Above 15 M | cum | 6328.00 |
| | | With Batching Plant, (Pan Type or Twin Shaft type)Transit Mixer and Concrete Pump | | |
| | | 1 Height up to 5 m | cum | 6285.00 |
| | | 2 Height 5 to 10 m | cum | 6422.00 |
| | | 3 Height 10 to 15 M | cum | 6628.00 |
| | | 4 Height Above 15 M | cum | 6828.00 |
| | | With Batching Plant, (Pan Type or Twin Shaft type) Transit Mixer and placing with Boom placer | | |
| | | 1 Height up to 5 m | cum | 6585.00 |
| | | 2 Height 5 to 10 m | cum | 6722.00 |
| | | 3 Height 10 to 15 M | cum | 6928.00 |

| Item No. | | Description | Unit | Rate |
|---|---|---|---|---|
| | | 4 Height Above 15 M | cum | 7128.00 |
| | L | RCC/PSC Grade M 55 | | |
| | | Concrete with plant (Drum Type) and placing | | |
| | | 1 Height up to 5 m | cum | 5838.00 |
| | | 2 Height 5 to 10 m | cum | 5970.00 |
| | | 3 Height 10 to 15 M | cum | 6167.00 |
| | | 4 Height Above 15 M | cum | 6355.00 |
| | | With Batching Plant, (Pan Type or Twin Shaft type) Transit Mixer and Concrete Pump | | |
| | | 1 Height up to 5 m | cum | 6338.00 |
| | | 2 Height 5 to 10 m | cum | 6470.00 |
| | | 3 Height 10 to 15 M | cum | 6667.00 |
| | | 4 Height Above 15 M | cum | 6855.00 |
| | | With Batching Plant, (Pan Type or Twin Shaft type)Transit Mixer and placing with Boom placer | | |
| | | 1 Height up to 5 m | cum | 6638.00 |
| | | 2 Height 5 to 10 m | cum | 6770.00 |
| | | 3 Height 10 to 15 M | cum | 6967.00 |
| | | 4 Height Above 15 M | cum | 7155.00 |
| | M | RCC/PSC Grade M 60 | | |
| | | Concrete with plant (Drum Type) and placing | | |
| | | 1 Height up to 5 m | cum | 5828.00 |
| | | 2 Height 5 to 10 m | cum | 5958.00 |
| | | 3 Height 10 to 15 M | cum | 6155.00 |
| | | 4 Height Above 15 M | cum | 6342.00 |
| | | With Batching Plant, (Pan Type or Twin Shaft type)Transit Mixer and Concrete Pump | | |
| | | 1 Height up to 5 m | cum | 6328.00 |
| | | 2 Height 5 to 10 m | cum | 6458.00 |
| | | 3 Height 10 to 15 M | cum | 6655.00 |
| | | 4 Height Above 15 M | cum | 6842.00 |
| | | With Batching Plant, (Pan Type or Twin Shaft type)Transit Mixer and placing with Boom placer | | |
| | | 1 Height up to 5 m | cum | 6628.00 |
| | | 2 Height 5 to 10 m | cum | 6758.00 |
| | | 3 Height 10 to 15 M | cum | 6955.00 |
| | | 4 Height Above 15 M | cum | 7142.00 |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| 13.2 | Extra for using M-50/ M-60 Grade Self-compacting Cement Concrete | cum | 665.00 |
| 13.3 | **Plain/Reinforced cement concrete in sub-structure complete as per drawing and technical specifications.**<br>RCC Grade M 55 in Paylon by Slipform Technique up to any Height | cum | 11500.00 |
| 13.4 | Supplying, fitting and placing HYSD bar reinforcement in sub-structure complete as per drawing and technical specifications and as per relevant clause of section 1600. | | |
| | FE 500 | MT | 67395.00 |
| | FE 500D/ FE 550/ FE 550D | MT | 68395.00 |
| | FE 600 | MT | 78395.00 |
| | Corrosion Resistant Steel | MT | 78895.00 |
| 13.5 | Solid Round Glass Fiber Reinforced Polymer (GFRP) bars confirming to IS: 18256:2023<br>Note: - This item shall be executed as a temperature/ surface reinforcement | Kg | 207.00 |
| 13.6 | Supplying, fitting and placing Mild steel reinforcement complete in sub-structure as per drawing and technical specification and as per relevant clause of section 1600. | tonne | 63963.00 |
| 13.7 | Providing weep holes in Brick masonry/Plain/Reinforced concrete abutment, wing wall/return wall with 100 mm dia AC pipe, extending through the full width of the structure with slope of 1V :20H towards drawing foce. Complete as per drawing and Technical specifications. | metre | 174.00 |
| 13.8 | Back filling behind abutment, wing wall and return wall complete as per drawing and Technical specification and as per relevant clauses 305 of specifications and as per appendix 6 of IRC-78. | | |
| | A Granular material | cum | 771.00 |
| | B Sandy material | cum | 1523.00 |
| 13.9 | Providing and laying of Filter media with granular materials/stone crushed aggregates satisfying the requirements laid down in clause 2504.2.2 of MoRT&H specifications to a thickness of not less than 600 mm with smaller size towards the soil and bigger size towards the wall and provided behind entire surface after abutment, return wall and wing wall to the full height compacted to a firm condition complete as per drawing and technical specification. | cum | 871.00 |
| 13.10 | Supplying, fitting and fixing in position true to line and level sliding plate bearing with PTFE surface sliding on stainless steel complete including all accessories as per drawing and Technical Specifications and BS: 5400, section 9.1 & 9.2 (for PTFE) and clause 2004 of MoRT&H Specifications. | tonne capacity | 268.00 |
| 13.11 | Supplying, fitting and fixing in position true to line and level forged steel roller bearing conforming to IRC: 83(Pt.-1) section IX and clause 2003 of MoRT&H specifications complete including all accessories as per drawing and Technical Specifications. | tonne capacity | 2062.00 |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| 13.12 | Supplying, fitting and fixing in position true to line and level elastomeric bearing conforming to IRC: 83 (Part-II) section IX and clause 2005 of MoRT&H specifications complete including all accessories as per drawing and Technical Specifications.<br><br>1)All bearings must fulfill the latest requirement of IRC-83-II-2018.<br>2) Each bearing must of its permanent Unique Identification number & manufacture logo for future traceability.<br>3) Each batch used in manufacturing of bearings must have minimum 60% polymer & maximum 5% of Ash content.<br>4) All bearings are having under the tolerance limit of IRC-83-11-2018.<br>5) No reclaimed or vulcanized wastes shall be used for the manufacturing of bearing.<br>6) EPDM or Similar candidate elastomer use shall not be permitted.<br>7) In case of any issue or special railway requirement, a sample made from same batch of minimum size (200 x 200 x 20) mm shall be tested for 2 million fatigue cycle with frequency of 2hz to 3hz. | cubic centimetre | 0.88 |
| 13.13 | Supplying, fitting and fixing in position true to line and level sliding plate bearing with stainless steel plate sliding on stainless steel plate with mild steel matrix complete including all accessories as per drawing and Technical Specifications. | tonne capacity | 257.00 |
| 13.14 | Supplying, fitting and fixing in position true to line and level POT-PTFE bearing consisting of a metal piston supported by a disc or unreinforced elastomer confined within a metal cylinder, sealing rings, dust seals, PTFE surface sliding against stainless steel mating surface, completre assembly to be of cast steel/fabricated structural steel, metal and elastomer elements to be as per IRC: 83 part-I & II respectively and other parts conforming to BS: 5400, section 9.1 & 9.2 and clause 2006 of MoRT&H Specifications complete asper drawing and approved technical specifications. | tonne capacity | 303.00 |
| 13.15 | Supplying, fitting and fixing in position true to line and level of Spherical Bearing consisting of set of concave and convex mating steel backing plate with a low friction sliding interface, flat sliding elements, guides and restraining rings; with all components conforming to approved drawing and technical specifications & Bearing Code IRC:83 Part IV - 2014 & its amendments. | MT | 275.00 |
| 13.16 | Providing and fixing in position bituminous paper bearing for slabs as per approved drawing and confirming to IS:1398. | sqm | 48.00 |
| 13.17 | Providing structural steel for sub-structure complete as per drawing and technical specifications | tonne | 86547.00 |
| 13.18 | Anchor bar for RCC slabs of submersible bridges Providng and fixing 20mm dia MS anchor bar duly welded at bottom with MS plate of size 500mm x 75mm x 10mm including all material labour charges etc. as per plate no. 7.22 SP- 20 or otherwise as | | |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| | directed by Engineer incharge. | | |
| | A) Fixed end anchor bar | Each | 603.00 |
| | B) Free end anchor bar | Each | 1192.00 |
| 13.19 | Supplying, Providing & fixing in position Stainless steel rebars of grade G 410-L Ferritic confirming to IS 16651: 2017 for RCC works of any dia. Up to 32 mm whenever directed including handling, fixing. straightening, wastage, cutting, bending, placing in position, binding complete in all respect. | | |
| | (i) Dia 6 mm to 10 mm | Mt | 158847.00 |
| | (ii) Dia 12 mm to 20 mm | MT | 154847.00 |
| | (iii) Dia 25 mm to 32 mm | MT | 152847.00 |
| 13.20 | Supplying and fixing of Hot Rolled Annealed & pickled stainless steel chequered plate having flat bottom of Grade 409M confirming to IS:6911-2017 Amendment 2 Pattern as per IS: 3502 1A, bead height minimum 0.80mm for pathway, staircase, floor for the bridge including fixing the same over channels by drilling holes in channels etc. including supply and fixing of nuts & bolts, tools & plants, labour, lead including ascend/descend complete in all respect. | | |
| | (i) 3 mm thickness | Sqm | 4122.00 |
| | (ii) 4 mm thickness | Sqm | 5394.00 |
| | (iii) 6 mm thickness | Sqm | 7905.00 |
| | (iv) 8 mm thickness | Sqm | 10432.00 |
| | (v) 10 mm thickness | Sqm | 12638.00 |
| | (vi) 12 mm thickness | Sqm | 15488.00 |
| 13.21 | Supplying Stainless Steel Hot Rolled Annealed Pickled Plates 350N/mm2 designated IRS 350 CR similar to RDSO Specifications BS-S 7.5.3.1-9 for the Structural application, ASTMA 1010/ A1010M-01E1 and Indian Standard (IS) 6911-2017 amendment 2 having chemical properties of C 0.03% Max, Mn1.5%, Si 1% max., 5 0.01% max., P 0.04% max., Ni 1.5% max, Mo 0.10-0.75% min., Cr 10.5%- 12.5%, N 0.03% max & mechanical properties 0.2% proof Stress 350MPA min, Ultimate Tensile Strength 485 MPA min% Elongation 18min, inclusive of all necessary testing & certification | MT | 175852.00 |
| 13.22 | Supplying Stainless Steel Hot Rolled Annealed Pickled Plates 450N/mm2 designated IRS 450 CR similar to RDSO Specifications BS-S 7.5.3.1-9 for the Structural application, ASTMA1010/A1010M-01E1 and Indian Standard (IS) 6911-2017 amendment 2 having chemical properties of C 0.03%Max, Mn1.5%, Si 1% max., S 0.01% max., P 0.04% max., Ni 1.5% max, Mo 0.10-0.75% min., Cr 10.5%- 12.5%, N 0.03% max & mechanical properties 0.2% proof Stress 450MPA min, Ultimate Tensile Strength 585 MPA min% Elongation 18min, inclusive of all | MT | 196523.00 |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| | necessary testing & certification. | | |
| 13.23 | Providing, fabricating, detailing and fixing at desired location using Stainless Steel Hot Rolled Annealed Pickled Plates 350N/mm2 designated IRS 350 CR similar to RDSO Specifications BS-S 7.5.3.1-9 for the Structural application, ASTMA1010/A1010M-01E1 and Indian Standard (IS) 6911-2017 amendment 2 having chemical properties of C 0.03% Max, Mn 1.5%, Si 1% max., S 0.01% max., P 0.04% max., Ni 1.5% max, Mo 0.10-0.75% min., Cr 10.5%- 12.5%, N 0.03% max & mechanical properties 0.2% proof Stress 350MPA min, Ultimate Tensile Strength 485 MPA min% Elongation 18min, inclusive of all necessary testing & certification. complete, transporting to site and erecting structural stainless steel members for all heights & at all levels including provision of necessary erection bolts, fixing bolts, nuts, washers, cleats, stiffeners, gussets, base plate, and all necessary fixtures and operations like preheating as per specifications, straightening. bending, cutting, drilling, grinding, machining if specified, welding. grinding and removing the welding burr etc, complete as directed ny Engineer-in-Charge. | MT | 389758.00 |
| 13.24 | Providing, fabricating (Fabrication Charge with all consumables, lifting and handling machineries, electricity charges and destressing of plates including temporary assembly etc.), detailing (Preparation of shop/fabrication drawing details for each section from approved good for construction drawing) and fixing at desired location using Stainless Steel Hot Rolled Annealed Pickled Plates 450 N/mm2 IRS 450 CR similar to RDSO specifications BS-S 7.5.3.1-9 ASTM A1010/A1010M 01E1 and Indian Standard (IS) 6911-2017 amendment 2 having Chemical properties of C 0.03%Max, Mn 1.5%, Si 1% max., $ 0.01% max., P 0.04% max., Ni 1.5% max, Mo 0.10-0.75% min., Cr 10.5%- 12.5%, N 0.03% max & Mechanical properties 10.2% proof Stress 450 MPA min, Ultimate Tensile Strength 585 MPA min % Elongation 18 Min, inclusive of all necessary testing & certification. complete, transporting to site (including loading/unloading from fabrication plant to site) and erecting structural stainless steel members for all heights & at all | MT | 402662.00 |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| | levels (charges for crane for erection, fuel & lubricants) including provision of necessary erection bolts, fixing bolts, nuts, washers, cleats, stiffeners, gussets, base plate, Stud, and all necessary fixtures and operations like preheating as per specifications, straightening, bending, cutting, drilling, grinding, machining if specified, welding, grinding and removing the welding burr etc.. complete as directed by Engineer-in-Charge. | | |
| 13.25 | Providing and fixing Acoustic Noise Barrier System absorptive type panels having tongue and groove for panel interlocking with total thickness of panels is 95±10 mm Panel shall be made of galvanized sheet with thickness not less than Imm. Front side of the panel shall be perforated while the back side shall remains plain. Composite panels are made of 90+10 mm thick rockwool with density of 90±10 kg/m3 and surface is over laid with glass veil/fiber glass tissue on perforated side. Powder Coating shall be done as per approved shade. Panel shall be erected between vertical post ISMB 125x125 with base plate 200x1200x12mm at high level bridge and at Embankment location the vertical post ISMB 150X150 with base plate 300x300x12mm, anchor bolts of 16mm dia including red oxide and oil paint. The Rockwool shall be provided by the hardcore manufacturer of acoustic noise barrier system. This item shall be executed as per IRC SP 130-2022 | | |
| | At embankment level with minimu 3 m height | Sqm | 9800.00 |
| | A High level Bridge/Elevated corridor with minimum2m height | Sqm | 10800.00 |
| | At Urban Section | | |
| | At embankment level with minimu 3 m height | Sqm | 9800.00 |
| | A High level Bridge/Elevated corridor with minimum2m height | Sqm | 10800.00 |

# CHAPTER: 14.0

# SUPER-STRUCTURE

14.1 Providing and Placing Reinforced/Prestressed cement concrete insuper-structure e/x reinforcement as per drawing and Technical Specification and as per relevant clauses of sections 1500, 1700 and 2300 in

| | | | |
|---|---|---|---|
| **A** | **RCC Grade M 20** | | |
| | **Concrete with plant (Drum Type) and placing** | | |
| | (i) Solid Slab | | |
| | 1 Height up to 5 m | cum | 4779.00 |
| | 2 Height 5 to 10 m | cum | 4999.00 |
| | 3 Height 10 to 15 m | cum | 5219.00 |
| | 4 Height Above 15m | cum | 5419.00 |
| | **With Batching Plant (Pan Type or Twin Type Shaft type), Transit Mixer and Concrete Pump** | | |
| | (i) Solid Slab | cum | 5279.00 |
| | 1 Height up to 5 m | cum | 5499.00 |
| | 2 Height 5 to 10 m | cum | 5719.00 |
| | 3 Height 10 to 15 m | cum | 5919.00 |
| | 4 Height Above 15m | | |
| | **With Batching Plant (Pan Type or Twin Type Shaft type), Transit Mixer and placing with Boom placer** | | |
| | (i) Solid Slab | cum | 5579.00 |
| | 1 Height up to 5 m | cum | 5799.00 |
| | 2 Height 5 to 10 m | cum | 6019.00 |
| | 3 Height 10 to 15 m | cum | 6219.00 |
| | 4 Height Above 15m | | |
| **B** | **RCC Grade M25** | | |
| | **Concrete with plant (Drum Type) and placing** | | |
| | (i) Solid Slab | cum | 5379.00 |
| | 1 Height up to 5 m | cum | 5623.00 |
| | 2 Height 5 to 10 m | cum | 5868.00 |
| | 3 Height 10 to 15 m | cum | 6068.00 |
| | 4 Height Above 15m | cum | 5379.00 |
| | **With Batching Plant (Pan Type or Twin Type Shaft type), Transit Mixer and Concrete Pump** | | |
| | (i) Solid Slab | | |
| | 1 Height up to 5 m | cum | 5879.00 |
| | 2 Height 5 to 10 m | cum | 6123.00 |
| | 3 Height 10 to 15 m | cum | 6368.00 |
| | 4 Height Above 15m | cum | 6568.00 |
| | **With Batching Plant (Pan Type or Twin Type Shaft type), Transit Mixer and placing with Boom placer** | | |
| | (i) Solid Slab | | |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| | 1 Height up to 5 m | cum | 6179.00 |
| | 2 Height 5 to 10 m | cum | 6423.00 |
| | 3 Height 10 to 15 m | cum | 6668.00 |
| | 4 Height Above 15m | cum | 6868.00 |
| **C** | **RCC Grade M 30** | | |
| | **Concrete with plant (Drum Type) and placing** | | |
| | (i) Solid Slab | | |
| | 1 Height up to 5 m | cum | 5475.00 |
| | 2 Height 5 to 10 m | cum | 5724.00 |
| | 3 Height 10 to 15 m | cum | 5973.00 |
| | 4 Height Above 15m | cum | 6173.00 |
| | (ii) For Cast in Situ RCC T beam and slab | | |
| | 1 Height up to 5 m | cum | 5724.00 |
| | 2 Height 5 to 10 m | cum | 5973.00 |
| | 3 Height 10 to 15 m | cum | 6222.00 |
| | 4 Height Above 15m | cum | 6422.00 |
| | **With Batching Plant (Pan Type or Twin Type Shaft type), Transit Mixer and Concrete Pump** | | |
| | (i) For Solid slab & voided slab | | |
| | 1 Height up to 5 m | cum | 5975.00 |
| | 2 Height 5 to 10 m | cum | 6224.00 |
| | 3 Height 10 to 15 m | cum | 6473.00 |
| | 4 Height Above 15m | cum | 6673.00 |
| | (ii) For Cast in Situ RCC T beam and slab | | |
| | 1 Height up to 5 m | cum | 6224.00 |
| | 2 Height 5 to 10 m | cum | 6473.00 |
| | 3 Height 10 to 15 m | cum | 6722.00 |
| | 4 Height Above 15m | cum | 6922.00 |
| | **With Batching Plant (Pan Type or Twin Type Shaft type),Transit Mixer and placing with Boom placer** | | |
| | (i) For Solid slab & voided slab | | |
| | 1 Height up to 5 m | cum | 6275.00 |
| | 2 Height 5 to 10 m | cum | 6524.00 |
| | 3 Height 10 to 15 m | cum | 6773.00 |
| | 4 Height Above 15m | cum | 6973.00 |
| | (ii) For Cast in Situ RCC T beam and slab | | |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| | 1 Height up to 5 m | cum | 6524.00 |
| | 2 Height 5 to 10 m | cum | 6773.00 |
| | 3 Height 10 to 15 m | cum | 7022.00 |
| | 4 Height Above 15m | cum | 7222.00 |
| D | **RCC/PSC Grade M35** | | |
| | **Concrete with plant (Drum Type) and placing** | | |
| | (i) For solid slab & Voided slab | | |
| | 1 Height up to 5 m | cum | 5574.00 |
| | 2 Height 5 to 10 m | cum | 5831.00 |
| | 3 Height 10 to 15 m | cum | 6089.00 |
| | 4 Height Above 15m | cum | 6289.00 |
| | (ii) For cast in situ RCC T Beam and slab | | |
| | 1 Height up to 5 m | cum | 6089.00 |
| | 2 Height 5 to 10 m | cum | 6346.00 |
| | 3 Height 10 to 15 m | cum | 6603.00 |
| | 4 Height Above 15m | cum | 6803.00 |
| | (iii) For cast in situ Box Girder and balanced cantilever | | |
| | 1 Height up to 5 m | cum | 6603.00 |
| | 2 Height 5 to 10 m | cum | 7118.00 |
| | 3 Height 10 to 15 m | cum | 7633.00 |
| | 4 Height Above 15m | cum | 7833.00 |
| | (iv) For RCC/PSC I beam, i/c launching | cum | 8333.00 |
| | **With Batching Plant (Pan Type or Twin Type Shaft type), Transit Mixer and Concrete Pump** | | |
| | (i) For solid slab & Voided slab | | |
| | 1 Height up to 5 m | cum | 6074.00 |
| | 2 Height 5 to 10 m | cum | 6331.00 |
| | 3 Height 10 to 15 m | cum | 6589.00 |
| | 4 Height Above 15m | cum | 6789.00 |
| | (ii) For cast in situ RCC T Beam and slab | | |
| | 1 Height up to 5 m | cum | 6589.00 |
| | 2 Height 5 to 10 m | cum | 6846.00 |
| | 3 Height 10 to 15 m | cum | 7103.00 |
| | 4 Height Above 15m | cum | 7303.00 |
| | (iii) For cast in situ Box Girder and balanced cantilever | | |
| | 1 Height up to 5 m | cum | 7103.00 |
| | 2 Height 5 to 10 m | cum | 7618.00 |
| | 3 Height 10 to 15 m | cum | 8133.00 |
| | 4 Height Above 15m | cum | 8333.00 |
| | (iv) For RCC/PSC I beam, i/c launching | cum | 8833.00 |
| | **With Batching Plant (Pan Type or Twin Type Shaft type),** | | |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| | **Transit Mixer and placing with Boom placer** | | |
| | (i) For solid slab & Voided slab | | |
| | 1 Height up to 5 m | cum | 6374.00 |
| | 2 Height 5 to 10 m | cum | 6631.00 |
| | 3 Height 10 to 15 m | cum | 6889.00 |
| | 4 Height Above 15m | cum | 7089.00 |
| | (ii) For cast in situ RCC T Beam and slab | | |
| | 1 Height up to 5 m | cum | 6889.00 |
| | 2 Height 5 to 10 m | cum | 7146.00 |
| | 3 Height 10 to 15 m | cum | 7403.00 |
| | 4 Height Above 15m | cum | 7603.00 |
| | (iii) For cast in situ Box Girder and balanced cantilever | | |
| | 1 Height up to 5 m | cum | 7403.00 |
| | 2 Height 5 to 10 m | cum | 7918.00 |
| | 3 Height 10 to 15 m | cum | 8433.00 |
| | 4 Height Above 15m | cum | 8633.00 |
| | (iv) For RCC/PSC I beam, i/c launching | cum | 9133.00 |
| E | **RCC/PSC Grade M 40** | | |
| | **Concrete with plant (Drum Type) and placing** | | |
| | (i) For solid slab & Voided slab | | |
| | 1 Height up to 5 m | cum | 5913.00 |
| | 2 Height 5 to 10 m | cum | 6185.00 |
| | 3 Height 10 to 15 m | cum | 6457.00 |
| | 4 Height Above 15m | cum | 6657.00 |
| | (ii) For cast in situ RCC& PSC T Beam and slab | | |
| | 1 Height up to 5 m | cum | 6185.00 |
| | 2 Height 5 to 10 m | cum | 6457.00 |
| | 3 Height 10 to 15 m | cum | 6728.00 |
| | 4 Height Above 15m | cum | 6928.00 |
| | (iii) For cast in situ Box Girder and balanced cantilever | | |
| | 1 Height up to 5 m | cum | 7000.00 |
| | 2 Height 5 to 10 m | cum | 7544.00 |
| | 3 Height 10 to 15 m | cum | 8087.00 |
| | 4 Height Above 15m | cum | 8287.00 |
| | (iv) For RCC/PSC I beam, i/c launching | cum | 8787.00 |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| | Note:- (for deck slab rate as per RCC/PSC T Beam and slab) | | |
| | **With Batching Plant (Pan Type or Twin Type Shaft type), Transit Mixer and Concrete Pump** | | |
| | (i) For solid slab & Voided slab | | |
| | 1 Height up to 5 m | cum | 6413.00 |
| | 2 Height 5 to 10 m | cum | 6685.00 |
| | 3 Height 10 to 15 m | cum | 6957.00 |
| | 4 Height Above 15m | cum | 7157.00 |
| | (ii) For cast in situ RCC& PSC T Beam and slab | | |
| | 1 Height up to 5 m | cum | 6685.00 |
| | 2 Height 5 to 10 m | cum | 6957.00 |
| | 3 Height 10 to 15 m | cum | 7228.00 |
| | 4 Height Above 15m | cum | 7428.00 |
| | (iii) For cast in situ Box Girder and balanced cantilever | | |
| | 1 Height up to 5 m | cum | 7500.00 |
| | 2 Height 5 to 10 m | cum | 8044.00 |
| | 3 Height 10 to 15 m | cum | 8587.00 |
| | 4 Height Above 15m | cum | 8787.00 |
| | (iv) For RCC/PSC I beam, i/c launching | cum | 9287.00 |
| | Note: - (for deck slab rate as per RCC/PSC T Beam and slab) | | |
| | **With Batching Plant (Pan Type or Twin Type Shaft type), Transit Mixer and placing with Boom placer** | | |
| | (i) For solid slab & Voided slab | | |
| | 1 Height up to 5 m | cum | 6713.00 |
| | 2 Height 5 to 10 m | cum | 6985.00 |
| | 3 Height 10 to 15 m | cum | 7257.00 |
| | 4 Height Above 15m | cum | 7457.00 |
| | (ii) For cast in situ RCC& PSC T Beam and slab | | |
| | 1 Height up to 5 m | cum | 6985.00 |
| | 2 Height 5 to 10 m | cum | 7257.00 |
| | 3 Height 10 to 15 m | cum | 7528.00 |
| | 4 Height Above 15m | cum | 7728.00 |
| | (iii) For cast in situ Box Girder and balanced cantilever | | |
| | 1 Height up to 5 m | cum | 7800.00 |
| | 2 Height 5 to 10 m | cum | 8344.00 |
| | 3 Height 10 to 15 m | cum | 8887.00 |
| | 4 Height Above 15m | cum | 9087.00 |
| | (iv) For RCC/PSC I beam, i/c launching | cum | 9587.00 |
| | Note: - (for deck slab rate as per RCC/PSC T Beam and slab) | | |
| **F** | **RCC/PSC Grade M 45** | | |
| | **Concrete with plant (Drum Type) and placing** | | |
| | (i) For solid slab & Voided slab | | |
| | 1 Height up to 5 m | cum | 5936.00 |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| | 2 Height 5 to 10 m | cum | 6214.00 |
| | 3 Height 10 to 15 m | cum | 6491.00 |
| | 4 Height Above 15m | cum | 6691.00 |
| | (ii) For cast in situ RCC& PSC T Beam and slab | | |
| | 1 Height up to 5 m | cum | 6214.00 |
| | 2 Height 5 to 10 m | cum | 6491.00 |
| | 3 Height 10 to 15 m | cum | 6768.00 |
| | 4 Height Above 15m | cum | 6968.00 |
| | (iii) For cast in situ Box Girder and balanced cantilever | | |
| | 1 Height up to 5 m | cum | 7046.00 |
| | 2 Height 5 to 10 m | cum | 7601.00 |
| | 3 Height 10 to 15 m | cum | 8155.00 |
| | 4 Height Above 15m | cum | 8355.00 |
| | (iv) For RCC/PSC I beam, i/c launching | cum | 8855.00 |
| | Note:- (for deck slab rate as per RCC/PSC T Beam and slab) | | |
| | (v) Segment Construction i/c launching | cum | 9255.00 |
| | **With Batching Plant (Pan Type or Twin Type Shaft type), Transit Mixer and Concrete Pump** | | |
| | (i) For solid slab & Voided slab | | |
| | 1 Height up to 5 m | cum | 6436.00 |
| | 2 Height 5 to 10 m | cum | 6714.00 |
| | 3 Height 10 to 15 m | cum | 6991.00 |
| | 4 Height Above 15m | cum | 7191.00 |
| | (ii) For cast in situ RCC& PSC T Beam and slab | | |
| | 1 Height up to 5 m | cum | 6714.00 |
| | 2 Height 5 to 10 m | cum | 6991.00 |
| | 3 Height 10 to 15 m | cum | 7268.00 |
| | 4 Height Above 15m | cum | 7468.00 |
| | (iii) For cast in situ Box Girder and balanced cantilever | | |
| | 1 Height up to 5 m | cum | 7546.00 |
| | 2 Height 5 to 10 m | cum | 8101.00 |
| | 3 Height 10 to 15 m | cum | 8655.00 |
| | 4 Height Above 15m | cum | 8855.00 |
| | (iv) For RCC/PSC I beam, i/c launching | cum | 9355.00 |
| | Note: - (for deck slab rate as per RCC/PSC T Beam and slab) | | |
| | (v) Segment Construction i/c launching | cum | 9755.00 |
| | **With Batching Plant (Pan Type or Twin Type Shaft type), Transit Mixer and placing with Boom placer** | | |
| | (i) For solid slab & Voided slab | | |
| | 1 Height up to 5 m | cum | 6736.00 |
| | 2 Height 5 to 10 m | cum | 7014.00 |
| | 3 Height 10 to 15 m | cum | 7291.00 |
| | 4 Height Above 15m | cum | 7491.00 |
| | (ii) For cast in situ RCC& PSC T Beam and slab | | |
| | 1 Height up to 5 m | cum | 7014.00 |
| | 2 Height 5 to 10 m | cum | 7291.00 |
| | 3 Height 10 to 15 m | cum | 7568.00 |
| | 4 Height Above 15m | cum | 7768.00 |
| | (iii) For cast in situ Box Girder and balanced cantilever | | |
| | 1 Height up to 5 m | cum | 7846.00 |
| | 2 Height 5 to 10 m | cum | 8401.00 |

| Item No. | | Description | Unit | Rate |
|---|---|---|---|---|
| | | 3 Height 10 to 15 m | cum | 8955.00 |
| | | 4 Height Above 15m | cum | 9155.00 |
| | | (iv) For RCC/PSC I beam, i/c launching | cum | 9655.00 |
| | | Note: - (for deck slab rate as per RCC/PSC T Beam and slab) | | |
| | | (v) Segment Construction i/c launching | cum | 10055.00 |
| | **G** | **PSC Grade M 50** | | |
| | | **Concrete with plant (Drum Type) and placing** | | |
| | | (i) For cast in situ Box Girder and balanced cantilever | | |
| | | 1 Height up to 5 m | cum | 7144.00 |
| | | 2 Height 5 to 10 m | cum | 7710.00 |
| | | 3 Height 10 to 15 m | cum | 8276.00 |
| | | 4 Height Above 15m | cum | 8476.00 |
| | | (ii) For RCC/PSC I beam, i/c launching | cum | 8976.00 |
| | | Note:- (for deck slab rate as per RCC/PSC T Beam and slab) | | |
| | | (iii) Segment Construction i/c launching | cum | 9376.00 |
| | | **With Batching Plant (Pan Type or Twin Type Shaft type), Transit Mixer and Concrete Pump** | | |
| | | (i) For cast in situ Box Girder and balanced cantilever | | |
| | | 1 Height up to 5 m | cum | 7644.00 |
| | | 2 Height 5 to 10 m | cum | 8210.00 |
| | | 3 Height 10 to 15 m | cum | 8776.00 |
| | | 4 Height Above 15m | cum | 8976.00 |
| | | (ii) For RCC/PSC I beam, i/c launching | cum | 9476.00 |
| | | Note: - (for deck slab rate as per RCC/PSC T Beam and slab) | | |
| | | (iii) Segment Construction i/c launching | cum | 9876.00 |
| | | **With Batching Plant (Pan Type or Twin Type Shaft type), Transit Mixer and placing with Boom placer** | | |
| | | (i) For cast in situ Box Girder and balanced cantilever | | |
| | | 1 Height up to 5 m | cum | 7944.00 |
| | | 2 Height 5 to 10 m | cum | 8510.00 |
| | | 3 Height 10 to 15 m | cum | 9076.00 |
| | | 4 Height Above 15m | cum | 9276.00 |
| | | (ii) For RCC/PSC I beam, i/c launching | cum | 9776.00 |
| | | Note: - (for deck slab rate as per RCC/PSC T Beam and slab) | | |
| | | (iii) Segment Construction i/c launching | cum | 10176.00 |
| | **H** | **RCC/PSC Grade M 55** | | |
| | | **Concrete with plant (Drum Type) and placing** | | |
| | | ((i) For cast in situ Box Girder and balanced cantilever | cum | 7260.00 |
| | | 1 Height up to 5 m | cum | 7835.00 |
| | | 2 Height 5 to 10 m | cum | 8409.00 |
| | | 3 Height 10 to 15 m | cum | 8609.00 |
| | | 4 Height Above 15m | cum | 9109.00 |
| | | (ii) For RCC/PSC I beam, i/c launching | | |
| | | Note:- (for deck slab rate as per RCC/PSC T Beam and slab) | | |
| | | (iii) Segment Construction i/c launching | cum | 9509.00 |
| | | **With Batching Plant (Pan Type or Twin Type Shaft type), Transit Mixer and Concrete Pump** | | |
| | | (i) For cast in situ Box Girder and balanced cantilever | | |
| | | 1 Height up to 5 m | cum | 7760.00 |
| | | 2 Height 5 to 10 m | cum | 8335.00 |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| | 3 Height 10 to 15 m | cum | 8909.00 |
| | 4 Height Above 15m | cum | 9109.00 |
| | (ii) For RCC/PSC I beam, i/c launching | cum | 9609.00 |
| | Note: - (for deck slab rate as per RCC/PSC T Beam and slab) | | |
| | (iii) Segment Construction i/c launching | cum | 10009.00 |
| | **With Batching Plant (Pan Type or Twin Type Shaft type), Transit Mixer and placing with Boom placer** | | |
| | (i) For cast in situ Box Girder and balanced cantilever | | |
| | 1 Height up to 5 m | cum | 8060.00 |
| | 2 Height 5 to 10 m | cum | 8635.00 |
| | 3 Height 10 to 15 m | cum | 9209.00 |
| | 4 Height Above 15m | cum | 9409.00 |
| | (ii) For RCC/PSC I beam, i/c launching | cum | 9909.00 |
| | Note: - (for deck slab rate as per RCC/PSC T Beam and slab) | | |
| | (iii) Segment Construction i/c launching | cum | 10309.00 |
| **I** | **RCC/PSC Grade M 60** | | |
| | **Concrete with plant (Drum Type) and placing** | | |
| | For cast in situ Box Girder and balanced cantilever | | |
| | 1 Height up to 5 m | cum | 7460.00 |
| | 2 Height 5 to 10 m | cum | 8035.00 |
| | 3 Height 10 to 15 m | cum | 8609.00 |
| | 4 Height Above 15m | cum | 8809.00 |
| | (ii) For RCC/PSC I beam, i/c launching | cum | 9309.00 |
| | Note:- (for deck slab rate as per RCC/PSC T Beam and slab) | | |
| | (iii) Segment Construction i/c launching | cum | 9709.00 |
| | **With Batching Plant (Pan Type or Twin Type Shaft type), Transit Mixer and Concrete Pump** | | |
| | (i) For cast in situ Box Girder and balanced cantilever | | |
| | 1 Height up to 5 m | cum | 7960.00 |
| | 2 Height 5 to 10 m | cum | 8535.00 |
| | 3 Height 10 to 15 m | cum | 9109.00 |
| | 4 Height Above 15m | cum | 9309.00 |
| | (ii) For RCC/PSC I beam, i/c launching | cum | 9809.00 |
| | Note: - (for deck slab rate as per RCC/PSC T Beam and slab) | | |
| | (iii) Segment Construction i/c launching | cum | 10209.00 |
| | **With Batching Plant (Pan Type or Twin Type Shaft type), Transit Mixer and placing with Boom placer** | | |
| | (i) For cast in situ Box Girder and balanced cantilever | | |
| | 1 Height up to 5 m | cum | 8260.00 |
| | 2 Height 5 to 10 m | cum | 8835.00 |
| | 3 Height 10 to 15 m | cum | 9409.00 |
| | 4 Height Above 15m | cum | 9609.00 |
| | (ii) For RCC/PSC I beam, i/c launching | cum | 10109.00 |
| | Note: - (for deck slab rate as per RCC/PSC T Beam and slab) | | |
| | (iii) Segment Construction i/c launching | cum | 10509.00 |
| **J** | **RCC/PSC Grade M 65** | | |
| | **Concrete with plant (Drum Type) and placing** | | |
| | (i) Solid Slab | | |
| | 1 Height up to 5 m | cum | 7760.00 |
| | 2 Height 5 to 10 m | cum | 8335.00 |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| | 3 Height 10 to 15 m | cum | 8909.00 |
| | 4 Height Above 15m | cum | 9109.00 |
| | (ii) For RCC/PSC I beam, i/c launching | cum | 9609.00 |
| | Note:- (for deck slab rate as per RCC/PSC T Beam and slab) | | |
| | (iii) Segment Construction i/c launching | cum | 10009.00 |
| | **With Batching Plant (Pan Type or Twin Type Shaft type), Transit Mixer and Concrete Pump** | | |
| | (i) For cast in situ Box Girder and balanced cantilever | | |
| | 1 Height up to 5 m | cum | 8260.00 |
| | 2 Height 5 to 10 m | cum | 8835.00 |
| | 3 Height 10 to 15 m | cum | 9409.00 |
| | 4 Height Above 15m | cum | 9609.00 |
| | (ii) For RCC/PSC I beam, i/c launching | cum | 10109.00 |
| | Note: - (for deck slab rate as per RCC/PSC T Beam and slab) | | |
| | (iii) Segment Construction i/c launching | cum | 10509.00 |
| | **With Batching Plant (Pan Type or Twin Type Shaft type), Transit Mixer and placing with Boom placer** | | |
| | (i) For cast in situ Box Girder and balanced cantilever | | |
| | 1 Height up to 5 m | cum | 8560.00 |
| | 2 Height 5 to 10 m | cum | 9135.00 |
| | 3 Height 10 to 15 m | cum | 9709.00 |
| | 4 Height Above 15m | cum | 9909.00 |
| | (ii) For RCC/PSC I beam, i/c launching | cum | 10409.00 |
| | Note: - (for deck slab rate as per RCC/PSC T Beam and slab) | | |
| | (ii) Segment Construction i/c launching | cum | 10809.00 |
| 14.2 | **High Performance Concrete**<br>Providing and placing High Performance concrete in super-structure ex/c reinforcement as per drawing and technical specifications as per relevent clauses of section 1500,1700 and 2300 as well as conferming to provision in IRC: SP 70-2016 | | |
| (i) | M 70 | cum | 7293.00 |
| (ii) | M 80 | cum | 7958.00 |
| (iii) | M 90 | cum | 8623.00 |
| 14.3 | **Ultra High-Performance Concrete**<br>Casting speical grade UHPFRC- Ultra High-Performance fiber reinforced concrete M150 in controlled enviorment by maintaining room temperature to the ambient temperature by executing casting work in factory shed with air ventilation. AC and by providing chilling plant with concrete mixture capacity of 12 cum to maintain the homogeneity along the concrete required for 1 segment of Trapezoidal shape segment of specified length for Navigational span at precast casting yard along with the provision of steam curing. casting machineries such as casting mould survey software. overhead pantries, ancillary civil work such as gentry wall of 400 M. Stacking bed to stack 240 segments. curing water tank, survey tower. Reinforcement jig, Overhead gantry of 80 MT of 25 mtrs span. Overhead Gantry of 15 MT of 25 mtrs span, Diesel genset of 250 KVA. welding machine, steel cutting, bending machines. concrete pump/ Boom placer concrete bucket with type mounted cranes etc and stacking yard to complete the entire casting of U girders withing 6-month duration. including | Cum | 125000.0 |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| | transporation in position complete. | | |
| 14.4 | Supplying, fitting and placing HYSD bars reinforcement in super-structure complete as per confirming to IS 1786:2000 drawing and technical specifications as per relevant clauses of section 1600 | | |
| | (i) FE 500 | Tonne | 67538.00 |
| | (ii) FE 500D/ FE 550/ FE 550D | Tonne | 68538.00 |
| | (iii) FE 600 | MT | 78395.00 |
| | (iv) Corrosion Resistant Steel | MT | 78895.00 |
| 14.5 | Solid Round Glass Fibre Reinforced Polymer (GFRP) Bars for Concrete Reinforcement.<br><br>Supplying, fitting & placing GFRP (Glass fibre Reinforced Polymer) bars conforming to IRC 137:2022 for structures such as Crash Barriers, Concrete drains, Concrete pavements, Approach slabs as per drawing & technical specification. (Only GFRP Bars manufactured using Vinyl Ester Resin systems and Glass fibres classified as E-CR or R that meet the requirements of IS: 18256:2023 and ASTM D578 shall be used.)<br><br>Note: - This item shall be executed as a temperature/ surface reinforcement | Kg. | 207.00 |
| 14.6 | **Cable and suspension Superstructure**<br>The work includes the supply, fabrication, installation, monitoring, and multi-stage stressing/prestressing of stay cables using a parallel strand system with 15.2 mm diameter strands and a guaranteed ultimate tensile strength (GUTS) of 1860 MPa. The stay cables shall be cut and placed in the specified profiles to provide permanent deck support.<br>The scope also includes:<br>- Installation of high-density polyethylene (HDPE) sheathing ducts.<br>- Fixing live anchorages at the deck and dead anchorages at the pylon, with steel anchor boxes embedded in the pylon as per approved drawings.<br>- Completion of all works in accordance with detailed drawings and technical specifications.<br>- Recess filling with approved materials and application of protective coating and wax after tensioning.<br>- Recording and reporting of tensioning data.<br>- Proper treatment of the projected strand ends after anchoring and covering them with specified materials.<br>- Health monitoring using a magnetic flux system with online operation and maintenance for 10 years.<br>- Testing of materials, including fatigue testing, to ensure compliance with specification | MT | 1600000/ |
| 14.7 | Providing and laying RCC in Paylon by Slipform Technique up to any Height. | | |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| (i) | RCC Grade M 55 | Cum | 11500.00 |
| 14.8 | **Structural Steel Work for Bridge Superstructure**<br>Providing and installing structural steel work for the superstructure of bridges, including Bow String, Composite Girder, Semi-Through Girder, and Plate Girder types. The work involves cutting, riveting, bolting, or welding, as well as launching with the necessary lifting arrangements, in accordance with the approved design. It also includes hoisting into position and completing all associated tasks.<br>Metallizing of steel girders shall be carried out by spraying aluminum after surface preparation through sand/grit blasting. This process will be followed by:<br>- One coat of etch primer (IS: 5666)<br>- One coat of zinc chrome primer (IS: 104)<br>- Two coats of aluminum paint (IS: 2339)<br>All work shall include labour, tools, plants (T&P), and materials, ensuring compliance with relevant specifications and procedures as per Clause 39 of IRS-B1-2001.<br>Coating Specifications:<br>- Nominal thickness of sprayed aluminum coating: 150 microns<br>- Dry film thickness (DFT) of zinc chrome primer: 25-30 microns<br>- DFT of each coat of aluminum paint: 12-14 microns<br>The structural steel shall conform to Grade Designation E250, with Quality B0 as per IS: 2062, in accordance with Section 1900 of the specifications.<br>Note: The rate for the deck slab shall be as per RCC/PSC T-Beam and slab specifications. | | |
| | Bow string | MT | 140000.00 |
| | Composite Girder | MT | 125000.00 |
| | Extra for providing steel grade E 350 in place of E 250 | MT | 4910.00 |
| | Note: - (for deck slab rate as per RCC/PSC T Beam and slab) | | |
| 14.9 | Add Extra for providing anti-corrosive treated HYSD reinforcement with Fusion Bonded Epoxy Coating (FBEC)<br>Note: - This item shall be executed only after prior approval of C.E. The measurement of this item shall be taken and recorded by the E.E. | tonne | 6672.00 |
| 14.10 | High tensile steel wires/strands including all accessories for stressing, stressing operations and grouting complete as per drawing and Technical Specifications and as per relevant clauses of section 1800. | tonne | 128142.00 |
| 14.11 | Providing and laying Cement concrete wearing coat 75 mm thick with M-30 grade concrete with cement , coarse and fine aggregate conforming to IS 383, maximum size of coarse aggregate not exceeding 19 mm, mixed in a batching and mixing plant as per approved mix design, transported to site, laid with static paver, spread, compacted and finished with power trowel in a continuous operation including provision of admixtures as approved, curing compound, finishing to lines and grades as per drawing but including cost of steel etc. including reinforcement, texuring with texuring brush complete as per drawing and Technical Specifications and as per relevant clauses of | cum | 10517.00 |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| | sections 1500, 1700 and Clause 2702 of specifications. | | |
| 14.12 | Mastic Asphalt (Providing and laying 12 mm thick mastic asphalt wearing course on top of deck slab excluding prime coat with paving grade bitumen meeting the requirements given in table 500-29, prepared by using mastic cooker and laid to required level and slope after cleaning the surface, including providing antiskid surface with bitumen precoated fine grained hard stone chipping of 9.5 mm nominal size at the rate of 0.005cum per 10 sqm and at an approximate spacing of 10 cm center to center in both directions, pressed into surface when the temperature of surfaces not less than 100 deg. C, protruding 1 mm to 4 mm over mastic surface, all complete as per clause 515 and 2702 of specifications.) | sqm | 776.00 |
| 14.13 | Heavy duty Polymer modified ASPHALTOSEAL 5mm thick MASTIC PAD (Roll size 1MX8M) conforming to ASTM D 6222-Type 1. Grade S specially designed for Bridge deck, elevated corridors, ROB. Parking decks, Bridge ramps. Padestrian subways, configured with High strength dimensionally stable heavy weight hybrid reinforcing media at core and finished suitably on top with calibrated minerals for protecting and enhancing durability equals to 1.1 sqm per sqm considering for 10% overlap) Applying with single compound primer modified water based bituminous primer like TIKIPRIME W mixed with water 1:1 ration applied @ 250 grams/sqm including transportation and application charges | sqm | 595.00 |
| 14.14 | Construction of RCC railing with precast rails and cast in situ vertical post of M30 Grade, aggregate size not exceeding 12 mm, true to line and grade, tolurence of vertical RCC post not to exceed 1 in 500, centre to centre spacing between vertical post not to exceed 2000 mm, leaving adequate space between vertical post at expansion joints of superstructure for expansion, complete as per approved drawings and technical and as per relevant clauses of sections 1500, 1600, 1700 and clause 2703 of specifications (as per MoRT&H specification drawing SD/202 or SD/305) | metre | 1746.00 |
| 14.15 | Construction of RCC railing with precast rails and cast in situ vertical post of M30 Grade in-situ with 20 mm nominal size aggregate, true to line and grade, tolurence of vertical RCC post not to exceed 1 in 500, centre to centre spacing between vertical post not to exceed 2000 mm, leaving adequate space between vertical post at expansion joints of superstructure for expansion, complete as per approved drawings and technical specifications and as per relevant clauses of sections 1500, 1600, 1700 and clause 2703 of specifications (as per MoRT&H specification drawing SD/201 or SD/304) | metre | 1999.00 |
| 14.16 | Providing, fitting and fixing mild steel railing complete as per drawing and Technical Specification and as per relevant clauses of section 1900 and 2700. | metre | 3045.00 |
| 14.17 | Drainage Spouts complete as per MoRT&H standard drawing and Technical specificationand as per clause 2705 of specifications. | each | 1227.00 |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| 14.18 | PCC M15 Grade leveling course below approach slab complete as perdrawing and Technical specification and as per relevant clauses of section 1700. | cum | 5651.00 |
| 14.19 | Reinforced cement concrete non-structural approach slab (Grade of concrete M 30) as per MoRT&H standard drawing including reinforcement and formwork complete as per drawing and Technical specification and as per relevant clauses of section 1500, 1600, 1700 and clause 2704 of specifications. | cum | 6672.00 |
| 14.20 | Providing and fixing Helical pipes, Galvanised steel sheet metal in voided concrete slabs including 20mm dia tie rod and sealing joints etc. as per section 1700 and 1800. | | |
| | A 550 mm dia | Meter | 1975.00 |
| | B 600 mm Dia. | Metre | 2050.00 |
| | C 800 mm dia | Meter | 2370.00 |
| 14.21 | **Painting on concrete surface**<br>Providing and applying 2 coats of water-based cement paint to unplastered concrete surface after cleaning the surface of dirt, dust, oil, grease, efflorescence and applying paint @ of 1 litre for 2 Sq.m. | Sqm | 84.00 |
| 14.22 | **Painting on Steel Surfaces**<br>Providing and applying two coats of ready-mix paint of approved brand on steel surface after through cleaning of surface to give an even shade | sqm | 73.00 |
| 14.23 | **Burried Joint**<br>Providing and laying a burried expansion joint, expansion gap being 20 mm, covered with 12 mm thick, 200 mm wide galvanised wieldable structural steel plate as per IS: 2062, placed symmetrical to centre line of the joint, resting freely over the top surface of the deck concrete, welding of 8 mm dia. 100 mm long galvanised nails spaced 300 mm c/c along the centre line of the plate, all as specified in clause 2604. | metre | 2000.00 |
| 14.24 | Filler joint | | |
| | (i) Providing & fixing 2 mm thick corrugated copper plate in expansion joint complete as per drawing & Technical Specification. | metre | 3675.00 |
| | (ii) Providing & fixing 20 mm thick compressible fibre boardin expansion joint complete as per drawing & Technical Specification. | sqm | 405.00 |
| | (iii) Providing and fixing in position 20 mm thick premoulded joint filler in expansion joint for fixed ends of simply supported spans not exceeding 10 m to cater for a horizontal movement upto 20 mm, covered with sealant complete as per drawing and technical specifications. | metre | 152.00 |
| | (iv) Providing and filling joint sealing compound as per drawings and technical specifications with coarse sand and 6% bitumen by weight | metre | 34.00 |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| 14.25 | **Asphaltic Plug joint**<br>Providing and laying of asphaltic plug joint to provide for horizontal movement of 25 mm and vertical movement of 2 mm, depth of joint varying from 75 mm to 100 mm, width varying from 500 mm to 750 mm (in traffic direction), covered with a closure plate of 200mm x 6mm of wieldable structural steel conforming to IS: 2062, asphaltic plug to consist of polymer modified bitumen binder, carefully selected single size aggregate of 12.5 mm nominal size and a heat resistant foam caulking/backer rod, all as per approved drawings and specifications. | metre | 1428.00 |
| 14.27 | **Compression Seal Joint**<br>Providing and laying of compression seal joint consisting of steel armoured nosing at two edges of the joint gap suitably anchored to the deck concrete and a preformed chloroprene elastomer or closed cell foam joint sealer compressed and fixed into the joint gap with special adhesive binder to cater for a horizontal movement upto 40 mm and vertical movement of 3 mm. | metre | 5542.00 |
| 14.28 | **Strip Seal Expansion Joint**<br>Providing and laying of a strip seal expansion joint catering to maximum horizontal movement upto 80 mm, complete as per approved drawings and standard specifications to be installed by the manufacturer/supplier or their authorised representative ensuring compliance to the manufacturer's instructions for installation.<br>1. The extrusion section shall be truly monolithic, and no welding shall be allowed in the section.<br>2. The minimum height of the extrusion shall be 80 mm, and the total cross-sectional area of the extrusions, including the seal-holding lip, shall be at least 1500 $mm^2$.<br>3. The minimum thickness of the seal-holding lip shall be 6 mm. However, the thickness of the extrusion throughout the section shall not be less than 10 mm.<br>4. The total cross-sectional area of the anchor loop or headed shear studs on each side of the joint shall not be less than 1600 $mm^2$ per meter length of the joint. The center-to-center spacing shall not exceed 250 mm for loop anchors and 150 mm for headed shear studs. The thickness of the anchor plate shall be at least 0.7 times the diameter of the anchor loop or 12 mm, whichever is greater.<br>5. The steel for edge beams shall conform to any of the following | metre | 5326.00 |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| | steel grades: | | |
| 14.29 | **Modular Strip / Box Seal Joint**<br>Providing and laying of a modular strip Box steel expansion joint including anchorage catering to a horizontal movement beyond 80 mm and upto 160mm, complete as per approved drawings and standard specifications to be installed by the manufacturer/supplier or their authorised representative ensuring compliance to the manufacturer's instructions for installation. | metre | 39663.00 |
| 14.30 | **Modular Strip / Box Seal Joint**<br>Providing and laying of a modular strip box seal expansion joint catering to a horizontal movement beyond 160mm and upto 240mm, complete as per approved drawings and standard specifications to be installed by the manufacturer/supplier or their authorised representative ensuring compliance to the manufacturer's instructions for installation. | metre | 48919.00 |
| 14.31 | Testing of span of bridge for deflection due to live load with platforms for loading arrangements apparatus for measurement including unloading etc. complete as per approved drawing. | tonne | 905.00 |
| 14.32 | Providing and fixing in position structural steel work for the superstructure of bridges including cutting, riveting, bolting or welding as per the approved design and including hoisting in position etc. complete as per section 1900 of specifications. | MT | 74375.00 |
| 14.33 | Providing and fixing in position Marble Plates of size 0.9 m. x 1.2 m.with inscribed details as per detailed drawings. | each | 1521.00 |
| 14.34 | Providing and fixing and erecting 40mm dia steel pipe (medium weight) railing in 3 rows on steel channel ISMC-100mmx50mm and 0.95 in height and fixed at 1.8m. center to center and duly painted etc. complete as per standard drawing No.STD-33R, dtd.11.05.1987 of MP Rajya Setu Nirman Nigam Bhopal. | RM | 2000.00 |
| 14.35 | Providing and Fixing Polycarbonate Noise Barrier having solid | Sqm | 13775.00 |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| | Polycarbonate sheet thickness not less than 8 mm shall be UV stable, weather resistant, sound insulation shall not be less than 32 dB and meets the requirement of ZTV-LSW06, EN 1793 and EN 1794. Sheet shall be fixed with steel channel of steel I-section 92x104mm, height-2.0 m Z-frame with EPDM Rubber Gasket. The whole assembly shall be fixed with MS Base plate size 220x220x16mm with 4 numbers anchor fastener- M16x250 Each Z- frame shall have spacing of 2.0 m which should be grouted with Epoxy. Steel work shall be well painted with primer etc. as per the direction of Engineer-in-charge. | | |
| 14.36 | Providing and Fixing Polycarbonate Noise Barrier having solid Polycarbonate sheet thickness not less than 8 mm shall be UV stable, weather resistant, sound insulation shall not be less than 32 dB and meets the requirement of ZTV-LSW06, EN 1793 and EN 1794. Sheet shall be fixed with steel square hollow section having vertical member 60x60x3.2 mm and horizontal member 40x40x2.6 mm, height 2.0 mtr polycarbonate sheet fixed with EPDM gasket. The whole assembly shall be fixed with MS Base plate size 250x220x12mm with 4 numbers anchor fastener-M16x150. Steel work shall be well powder coated etc. as per the direction of Engineer-in-charge. | Sqm | 13447.00 |
| 14.37 | **Painting two coats after filling** the surface with synthetic enamel paint bridge No. and span arrangements as per as directed by Engineer. | Nos. | 168.00 |
| 14.38 | **Providing and Fixing to drainage spout for disposal of rain water with** | | |
| | (i) P.V.C. pipe 110 mm dia.4kg/cm2 pressure (class II) | RM | 348.00 |
| | (i) P.V.C. pipe 160 mm dia.4kg/cm2 pressure (class II) | RM | 738.00 |

# CHAPTER 15

# Arch Bridge

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| 15.1 | Providing factory made hydraulically pressed mechanically vibrated and compacted Precast Ready-Mix controlled cement concrete of trap /granite/quartzite/gneiss metal Blocks of 520 mm top & 460 mm bottomX1000mm length and 400mm thick having various cross section areas for assembly of Arch Strip, including lifting hooks, including machined surfaced laser cut, trapezoidal steel moulds. As specified and as per IS 456:2000 including cost of all materials, preparation of casting bed, manufacturing, curing, removing from steel moulds by using crane or required machinery and stacking of blocks for preparation of arch strip of various lengths, including lead of all materials up to RMC plant, with fully automatic micro processor-based PLC with SCADA-enabaled concrete Batch Mix Plant (PAN Mixer) with Crushed sand VSI Grade. etc. complete. as directed by the Engineer-in-Charge. | | |
| | (a) M 30 | Each | 3722.00 |
| | (b) M 40 | Each | 3823.00 |
| | (c) M 50 | Each | 3944.00 |
| 15.2 | Providing factory made hydraulically pressed mechanically vibrated and compacted Precast Ready-Mix controlled cement concrete of trap /granite/quartzite/gneiss metal Socket Blocks of 520 mm top & 460 mm bottom x1000mm length and 300mm thick having various cross section areas for assembly of Arch Strip, including lifting hooks, including machined surfaced laser cut, trapezoidal steel moulds. As specified and as per IS 456:2000 including cost of all materials, preparation of casting bed, manufacturing, curing, removing from steel moulds by using crane or required machinery and stacking of blocks for preparation of arch strip of various lengths, including lead of all materials up to RMC plant, with fully automatic micro processor-based PLC with SCADA-enabaled concrete Batch Mix Plant (PAN Mixer) with Crushed sand VSI Grade. etc. complete. as directed by the Engineer-in-Charge. | | |
| | (a) M 30 | Each | 4940.00 |
| | (b) M 40 | Each | 5047.00 |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| | (c) M 50 | Each | 5128.00 |
| 15.3 | Providing and laying M-50 grade Ready Mix cement concrete of 60mm thickness, mechanically vibrated and compacted above linearly and parallelly arranged precast blocks placed on ISMC 100 base to form a layer of screed concrete over unit arch strip. Including HYFLEX steel wire rope – 6 X 36, steel core fixed with aluminum ferrules of required size and shape in individual block, for connecting blocks in unit arch strip. Including crimping of each ferrule with wire rope using embedded steel bar of 8mm diameter and 150mm in length. Including 10mm dia steel rods for fixing wire ropes. Including Z-plate acting as form work and steel separation plate of 10mm thickness over individual joint. As specified and as per IS 456:2000 including cost of all materials, preparation of level bed, curing, including lead of all materials up to RMC plant / casting yard, etc. complete. as directed by the Engineer-in-Charge. | Sqm | 5131.00 |
| 15.4 | Providing Lifting, transportation and placing activities of precast arch units and sockets from casting yard / RMC plant to construction site including all necessary lifting arrangements in hydraulic telescopic cranes of appropriate lifting capacity, transportation on flat bed trailer of required length, including lifting slings with D-shackle, pulling ratchets, including ISMC 300 steel supporting cross beam, two One Numbers of P type 10 ton capacity chain pulley , including temporary supports for lifting and transporting on trailers, including construction of leveling platform crane and trailer, including scaffolding and platform at pier / abutment top level, including necessary all manpower., etc. complete., as directed by Engineer-in-Charge. | MT | 3678.00 |

# CHAPTER- 16

## River Training and Protection works

| Item No. | | Description | Unit | Rate |
|---|---|---|---|---|
| 16.1 | | Providing and laying boulders apron on river bed for protection against scour with stone boulders weighing not less than 40 kg each complete as per drawing and Technical specification. | | |
| | A | Boulder laid dry without wire crates. | cum | 2351.00 |
| 16.2 | | Boulder apron laid in wire crates<br>(Providing and laying of boulder apron laid in wire crates made with 4mm dia GI wire conforming to IS: 280 & IS:4826 in 100mm x 100mm mesh (weaved diagonally) including 10% extra for laps and joints laid with stone boulders weighing not less than 40 kg each.) | cum | 2512.00 |
| 16.3 | | Cement concrete blocks<br>(size 0.5 x 0.5 x 0.5 m) (Providing and laying of apron with cement concrete blocks of size 0.5x0.5x0.5 m cast in-situ and made with nominal mix of M-15 grade cement concrete with a minimum cement content of 250 kg/cum as per IRC: 21-2000.) | cum | 4199.00 |
| 16.4 | | Providing and laying Pitching on slopes laid over prepared filter media including boulder apron laid dry in front of toe of embankment complete as per drawing and Technical specifications | | |
| | A | Stone/Boulder | cum | 2351.00 |
| | B | Cement Concrete blocks of size 0.3x0.3 x0.3 m cast in cement concrete of Grade M15 | cum | 4199.00 |
| 16.5 | | Providing and laying suitable designed Filter material underneath pitching in slopescomplete as per drawing and Technical specification | cum | 1591.00 |
| 16.6 | | Geotextile Filter<br>(Laying of a geotextile filter between pitching and embankment slopes on which pitching is laid to prevent escape of the embankment material through the voids of the stone pitching/cement concrete blocks as well as to allow free movement of water without creating any uplift head on the pitching.) | sqm | 240.00 |
| 16.7 | | Providing and laying Flooring complete as per drawing and Technicalspecifications laid over cement concrete bedding. | | |
| | A | Rubble stone laid in cement mortar 1:3 | cum | 5018.00 |
| | B | Cement Concrete blocks Grade M15 | cum | 5530.00 |
| | C | Cement Concrete Grade M15 | cum | 5651.00 |
| 16.8 | | Dry rubble Flooring | cum | 2886.00 |
| 16.9 | | Curtain wall complete as per drawing and Technical specification | | |
| | A | Stone masonry in cement mortar (1:3) | cum | 5616.00 |
| | B | Cement concrete Grade M15 | cum | 5651.00 |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| 16.10 | Flexible Apron: Construction of flexible apron 1 m thick comprising of loose stone boulders weighing not less than 40 kg beyond curtain wall. | cum | 2449.00 |
| 16.11 | Gabian Structure for Retaining Earth (Providing and construction of a gabain structure for retaining earth with segments of wire crates of size 7 m x 3 m x 0.6 m each divided into 1.5 m compartments by crossnetting, made from 4 mm galvanised steel wire @ 32 kg per 10 sqm having minimum tensile strength of 300 Mpa conforming to IS:280and galvanizing coating conforming to IS:4826, woven into mesh with double twist, mesh size not exceeding 100 x 100 mm, filled with boulders with least dimension of 200 mm, all loose ends to be tied with 4 mm galvanised steel wire) | cum | 2466.00 |
| 16.12 | Gabian Structure for Erosion Control, River Training Works and Protection works<br>(Providing and constructing gabain structures for erosion control, river training works and protection works with wire crates of size 2 m x 1 m x 0.3 m each divided into 1m compartments by cross netting, made from 4 mm galvanised steel wire @ 32 kg per 10 sqm having minimum tensile strength of 300 Mpa conforming to IS:280 and galvanizing coating conforming to IS:4826, woven into mesh with double twist, mesh size not exceeding 100 mm x 100 mm, filled with<br>boulders with least dimension of 200 mm, all loose ends to be securely tied with 4 mm galvanised steel wire.) | cum | 3034.00 |
| 16.13 | Providing & making Gabion structure with Mechanically Woven Double Twisted Hexagonal Shaped Wire mesh Gabion Boxes as per IS 16014:2012, MORT&H Clause 2500, of required size, Mesh Type 10x12 (D=100 mm with tolerance of ± 2%) Zinc coated, Mesh wire diameter 3.0 mm, mechanically edged/selvedge with partitions at every 1m interval and shall have minimum 10 numbers of openings per meter of mesh perpendicular to twist, tying with lacing wire of diameter 2.2 mm, supplied @3% by weight of Gabion boxes, filled with boulders with least dimension of 200 mm, as per drawing, all complete as per direction of Engineer-in-charge. | cum | 2454.00 |
| 16.14 | Laying of a fine aggregate concrete grade M30 filled fabric form for erosion protection of embankments (Embankment Erosion Protection using Fine Aggregate Concrete Filled Fabric Form Mattress system) | cum | 4422.20 |

# CHAPTER: 17.0
# REPAIR AND REHABILITATION

| | | | |
|---|---|---|---|
| 17.1 | Removal of existing cement concrete wearing coat including its disposal complete as per Technical specification without causing any detrimental effect to any part of the bridge structure and removal of dismantled material with all lifts and lead up to 1000m (Thickness 75 mm) | sqm | 283.00 |
| 17.2 | Removal of existing asphaltic wearing coat comprising of 50 mm thick asphaltic concrete laid over 12 mm thick mastic asphalt including disposal with all lift and lead up to 1000m. | sqm | 213.00 |
| 17.3 | Geniting concrete surface with cement mortar applied with compressor after cleaning surface and spraying with epoxy complete as per Technical specification | sqm | 1577.00 |
| 17.4 | Providing and inserting nipples with approved fixing compound after drilling holes for grouting as per Technical specifications including subsequent cutting/removal and sealing of the hole as necessary of nipples after completion of grouting with Cement/Epoxy | each | 133.00 |
| 17.5 | Sealing of cracks/porous concrete by injection process through nipples/Grouting complete as per Technical specification. | | |
| | A Cement Grout | kg | 65.00 |
| | B Cement mortar (1:1) Grouting | kg | 188.00 |
| 17.6 | Patching of damaged concrete surface with polymer concrete and curing compounds, initiator and promoter, available in present formulations, to be applied as per instructions of manufacturer and asapproved by the Engineer. | sqm | 1794.00 |
| 17.7 | Sealing of crack / porous concrete with Epoxy Grout by injectionthrough nipples complete as per clause 2803.1. | kg | 1628.00 |
| 17.8 | Applying epoxy mortar over leached, honey combed and spalled concrete surface and exposed steel reinforcement complete as per Technical specification | sqm | 965.00 |
| 17.9 | Removal of defective concrete, cleaning the surface thoroughly, applying the shotcrete mixture mechanically with compressed air under pressure, comprising of cement, sand, coarse aggregates, water and quick setting compound in the proportion as per clause 2807.1., sand and coarse aggregates conforming to IS: 383 and table 1of IS: 9012 respectively, water cement ratio ranging from 0.35 to 0.50, density of gunite not less than 2000 kg/cum, strength not less than 25 Mpa and workmanship conforming to clause 2807.6. | sqm | 487.00 |
| 17.10 | Providing and laying in situ/ready mix M35 grade RCC jacketing work around the existing piers of bridges etc. complete. (With fully automatic micro processor-based PLC with SCADA enabled concrete batch mix plant/ pan mixer with fine aggregates of required specification (VSI sand finely washed etc.) height of 8 to 10 meter) | cum | 7734.00 |
| 17.11 | Applying pre-packed cement- b a s e d polymer mortar of strength 45Mpa at 28 days for replacement of spalled concrete | sqm | 148.00 |
| 17.12 | Eproxy bonding of new concrete to old concrete | sqm | 1142.00 |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| 17.13 | Supply, Drilling/ Cleaning Hole and injecting hilti make ETA approved RE 500 V3 chemical excluding reinforcement. | Nos. | 421.00 |
| 17.14 | Providing and fixing of carbon fiber sheet of 430 GSM with adhesive including of hanging scaffolding of all manpower and equipment's complete in all respect. | Sqm | 8986.00 |
| 17.15 | Strengthening structural elements by providing & fixing CARBON FIBER (Make sika or BASF or FOSROC or equivalant) surface preparation: Griding concrete substrate, cleaning it with wire brush removing oil, laitance if present, rounding sharp edges to min 25mm radius etc. complete. Profiling: providing & Applying compatible Mbrace primer on prepared substrate, Filling the holes and uneven surface with high strength epoxy putty etc. complete IF required.<br>Wrapping: Providing & Applying first coat of saturant, cutting the carbon fiber sheet to size, wrapping the fiber sheet to structural element at desired orientation using tamping roller to avoid and air voids etc. complete Sand Pasting: Providing & Applying second coat of saturant after min. 12 hrs. rectify air voids if any paste the rive sand on it to make surface rough to take any further finishes (Mode of measurement: Per sq.mt of Mbrace CF 400 sheet applied and not surface area of concrete application & Quantity will be measured including wastage) The rate includes all type of material, application, equipment & scaffolding etc. complete, Rate is applicable for all height & level. | Sqm | 6260.00 |
| 17.16 | Providing external prestressing with high tensile steel wires/strands including drilling for passage of prestessing steel, all accessories for stressing and stressing operation and grouting complete as per<br>drawing and Technical specification | tonne | 303022.00 |
| 17.17 | Lifting of super-structure of bridge for reseting if required/refixing of new bearings including removal of old bearings and inluding all higher and running expenses of all plants, jacks, machines and equipments temporary supports, required for keeping the super-structure in lifted position for completing the operation, lowering of super-structure on bearings, without causing any detrimental effect to any part of the bridge structure complete but excluding cost of replaced bearings. | | |
| | a) Span upto 20m | Per Span | 62590.00 |
| | b) Span beyond 20m and upto 30m | Per Span | 75790.00 |
| | c) Span beyond 30m | Per Span | 88990.00 |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| 17.18 | Providing and fixing collapsible M.S. railing of channel of size 75x40x6mm length 2.50 m. at bottom and two nos. vertical post height 0.90 m. with top angle size 50x50x6mm and cross flat size 40x06mm diagonally fixed and braced with nut and bolts dully painted in two coats with synthetic enamel paint and grouted on existing kerbs i/c necessary construction of RCC M-20 block of size 0.50x0.45x0.20 m. complete as per approved drawings | RM | 3001.00 |
| 17.19 | Replacement of expansion Joint complete as per drawing. Cost of expansion joint to be paid separately. | RM | 292.00 |
| 17.20 | Removal and refixing of existing Pipe railing before and after rains including transportation of material from bridge to store place which is provided by the department as per under guidance of department. | RM | 200.00 |
| 17.21 | **Insitu Soil Reinforcement for Slope Restoration (Soil Nailing)** Designing, Providing drawings and installation of fully threaded (only Milled or Hot thread bars are allowed to avoid any loss of strength), Hot-dip galvanized solid geotechnical bars as soil nail having yield strength ≥ 670 Mpa, ultimate tensile strength ≥ 800 Mpa and minimum galvanization of 500 g/m2 required for construction of soil nailed stabilized slope including supply of galvanized nail plates, dome shape nuts, tapper washer, coupler (if required), centralizer and all accessories, including all lead and lifts as per detailed specification (Conforming to BS 8006 & AFNOR standards) | | |
| | a) 20 mm dia | metre | 2651.00 |
| | b) 25 mm dia | metre | 2778.00 |
| | c) 28 mm dia | metre | 2905.00 |
| | d) 32 mm dia | metre | 3095.00 |
| | e) 35 mm dia | metre | 3349.00 |
| 17.22 | Replacement of damaged concrete railing. | metre | 331.00 |
| 17.23 | Replacement of crash barrier. | metre | 603.00 |
| 17.24 | Replacement of damaged mild steel railing | metre | 277.00 |
| 17.25 | Repair of crash barrier (Repair of concrete crash barrier with cement concrete of M-30 grade by cutting and trimming the damaged portion to a regular shape, cleaning the area to be repaired thoroughly, applying cement concrete after erection of proper form work.) | metre | 16349.00 |
| 17.26 | Repair of RCC Railing (Carrying out repair of RCC M30 railing to bring it to the original shape.) | metre | 115.00 |
| 17.27 | Repair of steel Railing (Repair of steel railing to bring it to the original shape) | metre | 266.00 |
| 17.28 | Precasting and Installation of precast concrete arch units followed by stage wise filling with selected fill materials on both side of the arch units. This includes construction of reinforced earth head walls and wing walls to provide confinement to the selected fill material, complete as per approved drawings and standard specifications to be precasted, stacked, transported and installed as per the methodology, design and drawings of system suppliers or their authorised representative ensuring compliance to the system supplier's | metre | 719088.00 |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| | instruction. | | |
| 17.29 | Supply and installation of In-Situ Soil reinforcement with fully threaded hot-dip galvanized solid geotechnical bars as soil nails (galvanization minimum 500 grams per sqm) of minimum 25 mm diameter, having yield strength > 670N/mm2 and tensile strength > 800N/mm2 as per technical specifications and drawings etc. complete including drilling, flushing, grouting, and all Supply and installation of all components listed as per technical specifications and drawings etc. and considering all lead, lift and machinery. | metre | 4230.00 |
| 17.30 | Slope stabilization, widening of existing structures:-Designing, providing drawings, supply and fixing of fully threaded (only mill or hot thread bars are allowed to avoid any loss of strength), hot-dip galvanized solid geotechnical bars of Required diameter of Soil Nail as per design having yield strength ≥ 670 Mpa, ultimate tensile strength ≥ 800 Mpa and minimum galvanization of 500 grams per sqm required for construction of soil nailed stabilized slope including supply of galvanized nail plates, dome shape nuts, tapper washer, coupler (if required), centralizer and all accessories, including all lead and lifts as per detailed specification and including the cost of installation, drilling and grouting. Confirming to AFNOR NF P 94270, FHWA NHI, FHWA SA, IS 4759 (For Galvanizing) etc complete. As Directed by Engineer-in-Charge. | | |
| | (i) 20mm diameter of Soil Nail | RM | 3705.00 |
| | (ii) 25mm diameter of Soil Nail | RM | 3855.00 |
| | (iii) 28mm diameter of Soil Nail | RM | 3915.00 |
| | (iv) 32mm diameter of Soil Nail | RM | 4310.00 |
| | (v) 35mm diameter of Soil Nail | RM | 4815.00 |
| 17.31 | Repair of steel railing to bring it to original shape cleaning and repainting as per drawing etc. complete | RM | 435.00 |
| 17.32 | Removal of steel railing and refixing of railing to submersible bridges including cleaning and repainting etc complete | RM | 506.00 |
| 17.33 | White washing two coats on parapet walls and white washing and geru lime painting to tree trunks including preparation of surface by cleaning scraping etc. complete | sqm | 26.00 |
| 17.34 | Repair of RCC railing to bring it to the original shape, cleaning and repainting as per drawings etc. complete | RM | 835.00 |

# CHAPTER: 18.0
# Survey & Investigation, Preparation of D.P.R.

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| 18.1 | Survey and investigation for road work with chain and compass, auto level, theodolite or total station i/c fixing of permanent benchmark and also fixing of bench mark on all the permanent structures, along the alignment, like boundary wall, electrical poles etc. Also marking of locations of boundary wall, electric poles, telephone poles, trees etc. in the road plan, collection and submission of existing inventory of the road all along the alignment conducting survey@20 metre interval for L-section and for single lane X-section interval will be @ 0.75, 1.25, 1.875, 2.60, 3.75, 4.50, 5.50 and 6.50 metre on both side of centre line for two lane/ four lane interval for x-section shall be as specified in MORT&H Specifications. Data collected as specified above are required to be submitted in both hard and soft copies, L-section, X-section and plan is required to be submitted in the shape of drawing sheets drawn with the help of auto plotter. Soil samples @ every 500 metre or wherever soil appears to change its properties are required to be collected and soil investigation for LL, PL, swelling index and CBR are to be conducted and result too be submitted along with the project report. Job also includes collection of data for traffic census, fixing of RTL, getting it approved from Engineer-in-Charge and accordingly submission of pavement design in accordance with relevant clauses of IRC. Preparation of estimate complete and submission of same in eight copies duly spiral binded. | | |
| | i) For single lane road | km. | 8250.00 |
| | ii) For intermediate lane road | km | 8775.00 |
| | iii) For two lane road | km. | 9125.00 |
| | iv) For four lane road | km. | 10250.00 |
| 18.2 | Carrying out catchment area traverse survey of minor stream (not traceable in toposheet) below 100 acres | Sq Km | 3460.00 |
| 18.3 | Survey of C.D. works including taking L-Section of road, catchment area plan by Traverse method and taking trial pits of size 1.50M x 1.50M x 1.50M including preparation of detailed plans and estimates in 5 copies and all types of compliance if required from the department. | each | 5000.00 |
| 18.4 | Survey of Minor bridge including taking L section of existing road upto 500 m both side and two defined cross sections, one of 300 m U/s and one of 300 m D/s at the bridge site, one vergin cross section on alternate proposed cross sections and bed slope of stream upto sufficient long distance as required with chain and compass. Taking three trial pits of size 1.00x1.00x3.0 m and collecting samples of materials at site for ascertaining the silt factor, unit weight test, specific gravity from Govt. approved laboratories (testing charges to be borne by the agency) and preparation of detailed survey data, submission of the project duly typed, site plan and other drawings etc. complete as required by the Department. | each | 15000.00 |
| 18.5 | Survey of Major bridge including taking L-section of existing road upto 500 m each side and two defined cross sections, one of 300 m U/s and one of 300 m D/s and one virgin cross section on alternate proposed cross sections and bed slope of stream upto sufficient long distance as required with chain and compass. Taking 5 trial pits, three in bed, two on banks of size 2.00x2.00x3.00 m for collecting samples of materials at site for ascertaining the silt factor, unit weight test, specific gravity from Govt. approved laboratories | each | 25000.00 |

| Item No. | Description | Unit | Rate |
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| | (testing charges to be borne by the agency) and preparation of detailed survey data, submission of the project duly typed and coloring of ammonia prints, site plan and other drawings etc. complete. And all types of compliance if required from Department. (Excluding taking trial bores and contouring) | | |
| 18.6 | Detailed topo graphy Survey of ROB/Fly Over by using total station along with the existing or new road including L- Section of alignment, Cross section @ 20 m inteval, picking up all existing detailed surrounding data like electric telephone line, encroachment, existing structure etc. connecting all the primary total station control point by GIS Bench mark RL also correlate with MSL and preparation of detailed survey data, submission of the project duly typed and coloring of ammonia prints, site plan and other drawings etc. complete. And all types of compliance if required from Department. (Excluding taking trial bores and contouring) all autocad plan by using above data taking out two setof colour printout with required size. | each | 35000.00 |
| 18.7 | Demarcation of Road boundary (ROW)including fixing the boundary stones of standard size and shape including fixingin block of standard size of C.C. of 1:4:8 white washing at 30 M interval on both sides | km. | 41261.00 |
| 18.8 | Deploying Automated Traffic Count and classification system on Road (Two Lane) in both direction at designated traffic count post by using High Definition Cameras. Collecting the Non-Editable Data in DVR System with necessary electronic and electrical arrangements or inventor of required capacity Classification of data as per vehicle category as required with the help of software and submitting the the reports to the concerned department in hard copies as well as in soft copy, including using necessary technical manpower etc. complete. A) Up to 7 Days (Two Lane) | Per Day | 34487.00 |
| 18.9 | Subsurface Utility Engineering (SUE) Survey Conducting SUE Survey in a corridor of 4-6-meter width to detect buried utilities like pipes, cables etc. in such corridor. Marking of the detected utilities on the map of corridor with information of locations and depths to the top of various utilities detected. Onsite detection to be conducted using minimum tools of Electronic Pipe Locator (EPL) and Ground Penetrating Radar (GPR). GPR work to be conducted using 500MHz and 300MHz antenna for best possible resolution and penetration. The specifications are SUE survey should be in accordance with 'Code of Practice for Subsurface Utility Engineering Suiting Indian Conditions Guidelines & Specifications for Detection, Verification & Depiction of Existing Subsurface Utility Data'. | | |
| | Along the Road, 6m wide corridor | metre | 79.00 |
| | Road Crossings | | |
| | Roads Crossings without dividers and upto 30m width | metre | 58243.00 |
| | Roads Crossings with dividers upto 50m width | Crossings | 116486.00 |
| | Roads Crossings with dividers 60m width | crossings | 154927.00 |
| | For each additional increment over and above 60m width | crossings | 3788.00 |
| | **Generic Definition:** SUE is the process for Detection, Verification & Depiction of Existing Subsurface Utility Data, which combined conventional practices like background data search, appurtenances mapping etc. with modem geophysical techniques like EPL, GPR, Magnetics etc. These geophysical techniques allow the inspection | | |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| | of objects or geological features which lie beneath an optically opaque surface. Use of SUE process leads to detection and mapping of subsurface utilities with a known confidence level and utility quality attribute assigned to each utility. | | |
| 18.10 | Providing detailed geological reports of proposed site by maintaining geotechnical investigation of structure stratification collecting soil, rock and ground water samples for laboratory tests to arrive the foundation design parameters. Rock properties such as type of rock, jointing fractures, etc. complete with suggestion about the site foundation and remedies. | report | 38720.00 |
| 18.11 | Preparation of initial drawing of Bridge/ ROB/Fly Over with preparing General Arrangement drawing with dimensions etc. complete and preparation of rough estimate for all components of bridge. Including supplying four copies of coloured drawings and copy of drawing on C.D. (excluding geotechnical and structural design part.) | | |
| | (i) Slab Culvert | each | 1000.00 |
| | (ii) Minor Bridge | each | 2500.00 |
| | (iii) Major Bridge | each | 4000.00 |
| | (iv) ROB | each | 10000.00 |
| | (v) Flyover upto 2.00 km | each | 15000.00 |
| | (vi) Flyover 2.00 to 5 km | each | 20000.00 |
| | (vii) Flyover above 5.00 km | each | 25000.00 |
| 18.12 | Preparation of roadway, drawing (Plan and Profile) on computer by using Autocad of required scale showing all topo feature and proposed construction details such as carriage way, median service road, toe line, retaining walls, cross drainage works, major structures, etc. complete. Including indexing, level tables, legend, basic information of project and design details etc. | km. | 11201.00 |
| 18.13 | Geometrical design of road horizontal and vertical curves for design speed specified by IRC including carrying out trail for optimizing of civil cost minimizing land acquisition and maximum use of existing facility and preparation of widening arrangement drawing for insisting structures stake out date for horizontal alignment and working levels including super elevation effect etc. complete. | km. | 12738.00 |
| 18.14 | Preparation of Engineering drawing as strip plan, widening scheme, index / key map, typical cross section, retaining walls, road furniture, general arrangement drawings for bus bays / truck lay-bays / rest areas / toll plaza / booths / and type drawing of pipe/ slab drains etc. on computer using Auto CAD. | km. | 9982.00 |
| 18.15 | Design of junctions including detail layout of traffic signs, informatory boards traffic islands. Breaks in median verge including working levels at junctions as per IRC standards etc. complete. a) Major Junction. | each | 29951.00 |
| 18.16 | Preparation of Land Acquisition proposal including collection of 7/12 and 8/A documents marking alignment on village maps, area calculation, preparation of proposal, printing, binding, etc. complete and submission of 5 sets more ever liaisoning and coordination with revenue authorities with compliance if any until finalisation of award including Joint measurement of the alignment. For new alignment. | km. | 27486.00 |

# CHAPTER : 19.0

# VIDEO & PHOTOGRAPHY

| Item No. | | Description | Unit | Rate |
|---|---|---|---|---|
| 19.1 | | Providing and carrying out video shooting of roads, bridges, buildings, other programmes including hiring of vehicle equipment's for video shooting, titling, lighting, mixing, lettering, editing, including cost of two colour CDs of approved makes and quality etc. complete. | | |
| | i) | Within 50 Kms. | Per day | 2500.00 |
| | ii) | Beyond 50 kms. | Per day | 2700.00 |
| | iii) | Work upto 5 hours per day only. | Per day | 1500.00 |
| 19.2 | | Providing minimum 5 Nos. soft copies of Walk through presentation with detail project facilities as per DPR provision in the direction of Engineer-in-Charge | Per Min | 99000.00 |
| 19.3 | | Shooting of still photographs outdoor and indoor of construction andother works of building, bridges, roads etc. at any distances. | | |
| (A) | i) | Post Card size | 3 Nos. | 60.00 |
| | ii) | 4"x6" size | 3 Nos. | 85.00 |
| | iii) | 5"x7" size | 3 Nos. | 135.00 |
| | iv) | 8"x12" Size (A-4 Size) | 3 Nos. | 250.00 |
| | v) | 20"x30" size (Banner Size) | 1 No. | 600.00 |
| (B) | | Extra Copies : | | |
| | i) | Post Card size | 1 No. | 8.00 |
| | ii) | 4"x6" size | 1 No. | 13.00 |
| | iii) | 5"x7" size | 1 No. | 15.00 |
| | iv) | 8"x12" Size (A-4 Size) | 1 No. | 50.00 |
| | v) | 20'x30' size (Banner Size) | 1 No. | 450.00 |
| 19.4 | | Providing and performing video shooting and still photographs minimum 10 per sqm km by drone in 4k resolution (FHD 1920*1080,24/25/30/48/50/60 frames to UHD 4096*2160,24/25 frames) with 200m horizontal coverage on both side of the alignment of road, proposed site of bridge river, ground, including hiring of vehicle equipment's for video shooting, tilting, lightening, mixing, lettering, editing, including cost of digital data (4k video) cost of hard copies in two sets at least in A0/ Al sheet | | |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| | complete & soft copy. | | |
| | Within 0 Kms | Per sq. km. | 8700.00 |
| | (i) within 50 Kms of bridge division | | |
| | (ii) Beyond 50 Kms from bridge division add extra for travelling to reach the site. | Per sq. km. | 8950.00 |
| | (iii) add extra for every additional km horizontal measurement at same site. | Per sq. km. | 5500.00 |
| | Note: Above items are based on currend Building SOR w.e.f. 01-01-2024 | | |
| 19.5 | AI based Road Condition Assessment and Defect Identification using a Mobile phone and Cloud based software complete with a dashboard for viewing all results and customizable reports downloadable from the dashboard. Reports with defects mapped should be generatable up to 5 meter intervals and all data should be processed and available for viewing on the dashboard within 2 hours of survey completion. The Ai software used should be validated by a third party for up to 99% accuracy. The following defects and assets need to be mapped along with severity: Potholes, Edge Deterioration, Alligator Cracking, Rutting, Subsidence, Cracking, Bleeding, Fretting, Repaired Potholes, Line marking condition, Storm Drain, Manholes, Guard Rails & Traffic Signage Inventory for 1000 km to be surveyed twice in a year | Per km | 1250.0 |

# CHAPTER : 20.0

# ENVIRONMENTAL MANAGEMENT & BIO ENGINEERING

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| 20.1 | Construction of bamboo crib wall complete as per drawing and Technical specification approved by the Engineer-in-charge. | Rm. | 652.00 |
| 20.2 | Construction of Fascines with a bundle of sticks complete as per drawing and Technical specification approved by the Engineer-in-charge. | Rm. | 176.00 |
| 20.3 | Contruction and laying of brush Layers across the slope complete as per drawing and Technical specification approved by the Engineer-in-charge. | sqm. | 39.00 |
| 20.4 | Construction of vegetative Palisades in Rills using hard wood cutting complete as per drawing and Technical specification approved by the Engineer-in-charge. | Rm. | 143.00 |
| 20.5 | Laying of Palisades in Slopes complete as per drawing and Technical specification approved by the Engineer-in-charge. | sqm. | 39.00 |
| 20.6 | Providing Supplying and Sprinkling of water with mechanical sprayer (Dust free chemical compound shall be mixed @ 5% of the water) in the settlement and working area ONCE A WEEK as per instruction of Engineer-in-charge (July to Sept month of each year is excluded for any claim by Contractor Sprinkling of water. in case of additional requirment of sprinkling of water within a week time to be done by the contractor and shall not be paid separately. Any isolated water sprinkling required shall be carried by Contractor and shall be considered as incidental to the Contract) | sqm. | 18.00 |

# CHAPTER: 21.0

## ELECTRIFICATION ON ROADS AND BRIDGES

| Item No. | | Description | Unit | Rate |
|---|---|---|---|---|
| 21.1 | MCCB 3Ph, 3 Pole, 50Hz, 415V | Supply of ISI marked and approved make of moulded case circuit breaker (MCCB) suitable for 3 phase,3 pole, 50 Hz, 415 Volts, AC supply with respective interrupting capacity (KA) at 415 Volts cited against their range standard conforming to IS – 8828 | | |
| | (i) MCCB upto 63,10kA, 3P | MCCB current rating upto 63 Amp, 10kA, 3P 415V AC | each | 2721.00 |
| | (ii) MCCB 80 to 100A,10kA, 3P | MCCB current rating 80 to 100Amp, 10kA, 3P 415V AC | each | 3256.00 |
| | (iii) MCCB 25 to 100A, 25kA, 3P | MCCB 25kA current rating, 3P 25 to 100Amp & 80% -100% adjustable | each | 7369.00 |
| | (iv) MCCB 125A, 25kA, 3P | MCCB 25kA current rating, 3P 125Amp & 80% -100% adjustable | each | 9953.00 |
| | (v) MCCB 160A, 25kA, 3P | MCCB 25kA current rating, 3P 160Amp & 80% -100% adjustable | each | 12701.00 |
| | (vi) MCCB 200A, 25kA, 3P | MCCB 25kA current rating, 3P 200Amp & 80% -100% adjustable | each | 16061.00 |
| | (vii) MCCB 160A, 36kA, 3P | MCCB 36kA current rating, 3P 160Amp &adjustable: 80% -100% thermal | each | 14499.00 |
| | (viii) MCCB 200A, 36kA, 3P | MCCB 36kA current rating, 3P, 3P 200Amp &adjustable: 80% -100% thermal | each | 17555.00 |
| | (ix) MCCB 250A, 36kA, 3P | MCCB 36kA current rating, 3P 250Amp &adjustable: 80% -100% thermal | each | 21243.00 |
| | (x) MCCB 315A, 36kA, 3P | MCCB 36kA current rating, 3P 315Amp &adjustable: 80% -100% thermal | each | 23636.00 |
| | (xi) MCCB 400A, 36kA, 3P | MCCB 36kA current rating, 3P 400Amp &adjustable: 80% -100% thermal | each | 24413.00 |
| 21.2 | MCCB 3Ph, 3 Pole, 50Hz, 415V (with Thermal - Magnetic Release) | Supplying of ISI marked and approved make of moulded case circuit breaker (MCCB) suitable for 3 phase,3 pole, 50 Hz, 415 Volts, (Ics=100%Icu) AC supply with thermal - magnetic release having respective breaking capacity (KA) at 415 Volts cited against their range standard conforming to IS/IEC 60947-2 | | |
| | (i) MCCB 63 to 100A, 25kA, 3P | MCCB 25kA current rating 63 to100Amp & adjustable overload setting 80% - 100% of In, adjustable short circuit setting 6-10 In 3 Pole | each | 7538.00 |
| | (ii) MCCB 125A, 25kA, 3P | MCCB 25kA current rating 125Amp & adjustable overload setting 80% - 100% of In, adjustable short circuit setting 6-10 In 3 Pole | each | 10160.00 |
| | (iii) MCCB 160A, 25kA, 3P | MCCB 25kA current rating 160Amp & adjustable overload setting 80% - 100% of In, adjustable short circuit setting 6-10 In 3 Pole | each | 13327.00 |
| | (vi) MCCB 200A, 25kA, 3P | MCCB 25kA current rating 200Amp & adjustable overload setting 80% - 100% of In, adjustable short circuit setting 6-10 In 3 Pole | each | 17154.00 |
| | (v) MCCB 160A, 36kA, 3P | MCCB 36kA current rating 160Amp & adjustable overload setting 80% - 100% of In, adjustable short circuit setting 6-10 In 3 Pole | each | 15400.00 |
| | (vi) MCCB 200A, 36kA, 3P | MCCB 36kA current rating 200Amp & adjustable overload setting 80% - 100% of In, | each | 19291.00 |

| Item No. | | Description | Unit | Rate |
|---|---|---|---|---|
| | | adjustable short circuit setting 6-10 In 3 Pole | | |
| | (vii)MCCB 250A, 36kA, 3P | MCCB 36kA current rating 250Amp & adjustable overload setting 80% - 100% of In, adjustable short circuit setting 6-10 In 3 Pole | each | 22839.00 |
| | (viii) MCCB 315A, 36kA, 3P | MCCB 36kA current rating 315Amp & adjustable overload setting 80% - 100% of In, adjustable short circuit setting 6-10 In 3 Pole | each | 25946.00 |
| | (ix) MCCB 400A, 36kA, 3P | MCCB 36kA current rating 400Amp & adjustable overload setting 80% - 100% of In, adjustable short circuit setting 6-10 In 3 Pole | each | 26837.00 |
| | (x) MCCB 500A, 36kA, 3P | MCCB 36kA current rating 500Amp & adjustable overload setting 80% - 100% of In, adjustable short circuit setting 6-10 In 3 Pole | each | 44709.00 |
| 21.3 | MCCB 3Ph & N, 4Pole, 50Hz, 415V (thermal - magnetic release) | Supplying of ISI marked and approved make of moulded case circuit breaker (MCCB) suitable for 3 phase & N, 4 pole, 50 Hz, 415 Volts, (Ics=100%Icu) AC supply with thermal-magnetic release having respective breaking capacity (KA) at 415 Volts cited against their range standard conforming to IS/IEC 60947-2 | | |
| | (i) MCCB 63 to 100A, 25kA, 4P | MCCB 25kA current rating 63 to100Amp & adjustable overload setting 80% - 100% of In, adjustable short circuit setting 6 - 10 In 4 Pole | each | 9781.00 |
| | (ii) MCCB 125A, 25kA, 4P | MCCB 25kA current rating 125Amp & adjustable overload setting 80% - 100% of In, adjustable short circuit setting 6-10 In 4 Pole | each | 11593.00 |
| | (iii) MCCB 160A, 25kA, 4P | MCCB 25kA current rating 160Amp & adjustable overload setting 80% - 100% of In, adjustable short circuit setting 6-10 In 4 Pole | each | 15835.00 |
| | (iv) MCCB 200A, 25kA, 4P | MCCB 25kA current rating 200 Amp & adjustable overload setting 80% - 100% of In, adjustable short circuit setting 6-10 In 4 Pole | each | 19695.00 |
| | (v) MCCB 160A, 36kA, 4P | MCCB 36kA current rating 160 Amp & adjustable overload setting 80% - 100% of In, adjustable short circuit setting 6-10 In 4 Pole | each | 18923.00 |
| | (vi) MCCB 200A, 36kA, 4P | MCCB 36kA current rating 200Amp & adjustable overload setting 80% - 100% of In, adjustable short circuit setting 6-10 In 4 Pole | each | 21950.00 |
| | (vii) MCCB 250A, 36kA, 4P | MCCB 36kA current rating 250Amp & adjustable overload setting 80% - 100% of In, adjustable short circuit setting 6-10 In 4 Pole | each | 27693.00 |
| | (viii) MCCB 315A, 36kA, 4P | MCCB 36kA current rating 315Amp & adjustable overload setting 80% - 100% of In, adjustable short circuit setting 6-10 In 4 Pole | each | 30528.00 |
| | (ix) MCCB 400A, 36kA, 4P | MCCB 36kA current rating 400Amp & adjustable overload setting 80% - 100% of In, adjustable short circuit setting 6-10 In 4 Pole | each | 31907.00 |
| | (x)MCCB 500A, 36kA, 4P | MCCB 36kA current rating 500Amp & adjustable overload setting 80% - 100% of In, adjustable short circuit setting 6-10 In 4 Pole | each | 53891.00 |
| 21.4 | MCCB 3Ph, 3Pole, 50Hz, 415V | Supplying of ISI marked and approved make of moulded case circuit breaker (MCCB) | | |

| Item No. | | Description | Unit | Rate |
|---|---|---|---|---|
| | (Micro-processor release) | suitable for 3 phase,3 pole, 50 Hz, 415 Volts,(Ics=100%Icu) AC supply with Micro-Processor release having respective breaking capacity (KA) at 415 Volts cited against their range standard conforming to IS/IEC 60947-2 | | |
| | (i)MCCB 40 to 100A, 25kA, 3P | MCCB 25kA current rating 40 & 100Amp 3 Pole &micro-processor release, over load 40% - 100% x In. & short circuit 1.5 - 8 x Ir. Adjustable | each | 18389.00 |
| | (ii)MCCB 160A, 25kA, 3P | MCCB 25kA current rating 160Amp 3 Pole &micro-processor release, over load 40% - 100% x In. & short circuit 1.5 - 8 x Ir. Adjustable | each | 23257.00 |
| | (iii)MCCB 250A, 25kA, 3P | MCCB 25kA current rating 250Amp 3 Pole &micro-processor release, over load 40% - 100% x In. & short circuit 1.5 - 8 x Ir. Adjustable | each | 29146.00 |
| | (iv)MCCB 40 to 100A, 36kA, 3P | MCCB 36kA current rating 40 & 100Amp 3 Pole &micro-processor release, over load 40% - 100% x In. & short circuit 1.5 - 8 x Ir. Adjustable | each | 18716.00 |
| | (v)MCCB 160A, 36kA, 3P | MCCB 36kA current rating 160Amp 3 Pole &micro-processor release, over load 40% - 100% x In. & short circuit 1.5 - 8 x Ir. Adjustable | each | 21383.00 |
| | (vi)MCCB 250A, 36kA, 3P | MCCB 36kA current rating 250Amp 3 Pole &micro-processor release, over load 40% - 100% x In. & short circuit 1.5 - 8 x Ir. Adjustable | each | 26116.00 |
| | (vii)MCCB 400A, 36kA, 3P | MCCB 36kA current rating 400Amp 3 Pole &micro-processor release, over load 40% - 100% x In. & short circuit 1.5 - 8 x Ir. Adjustable | each | 37640.00 |
| | (viii)MCCB 630A, 36kA, 3P | MCCB 36kA current rating 630Amp 3 Pole &micro-processor release, over load 40% - 100% x In. & short circuit 1.5 - 8 x Ir. Adjustable | each | 47274.00 |
| | (ix)MCCB 800A, 50kA, 3P | MCCB 50kA current rating 800Amp 3 Pole &micro-processor release, over load 40% - 100% x In. & short circuit 1.5 - 8 x Ir. Adjustable | each | 72254.00 |
| | (x)MCCB 1250A, 50kA, 3P | MCCB 50kA current rating 1250Amp 3 Pole &micro-processor release, over load 40% - 100% x In. & short circuit 1.5 - 8 x Ir. Adjustable | each | 232253.00 |
| 21.5 | (MCCB) suitable for 3 phase & N, 4 pole, AC supply with micro-processor release | Supplying of ISI marked and approved make of moulded case circuit breaker (MCCB) suitable for 3 phase & N, 4 pole, 50 Hz, 415 Volts, (Ics=100%Icu) AC supply with microprocessor release with inbuilt instantaneous protection & variable trip class 7.2 Ir & 6 Ir having respective breaking | | |

| Item No. | | Description | Unit | Rate |
|---|---|---|---|---|
| | | capacity (KA) at 415 Volts cited against their range standard conforming to IS/IEC 60947-2 | | |
| | (i)MCCB, 25kA, 40 & 100 Amps, 4P | MCCB 25kA current rating 40 & 100 Amps 4 Pole & micro-processor release, over load 40% - 100% x In. & short circuit 1.5 - 8 x Ir. Adjustable | each | 21576.00 |
| | (ii)MCCB, 25kA, 160 Amps, 4P | MCCB 25kA current rating 160 Amps 4 Pole & micro-processor release, over load 40% - 100% x In. & short circuit 1.5 - 8 x Ir. Adjustable | each | 26273.00 |
| | (iii)MCCB, 25kA, 250 Amps, 4P | MCCB 25kA current rating 250 Amps 4 Pole & micro-processor release, over load 40% - 100% x In. & short circuit 1.5 - 8 x Ir. Adjustable | each | 38922.00 |
| | (iv)MCCB, 36kA, 40 & 100 Amps, 4P | MCCB 36kA current rating 40 & 100 Amps 4 Pole & Micro-Processor Release, over load 40% -100% x In. & short circuit 1.5 - 8 x Ir. Adjustable | each | 20196.00 |
| | (v)MCCB, 36kA, 160 Amps, 4P | MCCB 36kA current rating 160 Amps 4 Pole &micro-processor release, over load 40% - 100% x In. & short circuit 1.5 - 8 x Ir. Adjustable | each | 26747.00 |
| | (vi)MCCB, 36kA, 250 Amps, 4P | MCCB 36kA current rating 250 Amps 4 Pole &micro-processor release, over load 40% - 100% x In. & short circuit 1.5 - 8 x Ir. Adjustable | each | 39272.00 |
| | (vii)MCCB, 36kA, 400 Amps, 4P | MCCB 36kA current rating 400 Amps 4 Pole &micro-processor release, over load 40% - 100% x In. & short circuit 1.5 - 8 x Ir. Adjustable | each | 47659.00 |
| | (viii)MCCB, 36kA, 630 Amps, 4P | MCCB 36kA current rating 630 Amps 4 Pole & micro-processor release, over load 40% - 100% x In. & short circuit 1.5 - 8 x Ir. Adjustable. | each | 70459.00 |
| | (ix)MCCB, 50kA, 800 Amps, 4P | MCCB 50kA current rating 800 Amps 4 Pole & Micro-Processor Release, over load 40% - 100% x In. & short circuit 1.5 - 8 x Ir. Adjustable. | each | 90200.00 |
| | (x)MCCB, 50kA, 1250 Amps, 4P | MCCB 50kA current rating 1250 Amps 4 Pole & Micro-Processor Release, over load 40% - 100% x In. & short circuit 1.5 - 8 x Ir. Adjustable. | each | 315130.00 |
| 21.6 | MCCB 3Ph, 3/4Pole, 50Hz, 415V (without protection release) | Supplying of approved make moulded case circuit breaker (MCCB) without protection release suitable for 3 phase, 3/4 pole, 50 Hz, 415 Volts, AC supply standard conforming to IS – 8828. | | |
| | (i)MCCB 200A 3Pole | MCCB switch disconnector rating 200Amp 3 Pole without protection release. | each | 9411.00 |
| | (ii)MCCB 400A 3Pole | MCCB switch disconnector rating 400Amp 3 Pole without protection release. | each | 13560.00 |
| | (iii)MCCB 600A | MCCB switch disconnector rating 600Amp 3 | each | 16437.00 |

| Item No. | | Description | Unit | Rate |
|---|---|---|---|---|
| | 3Pole | Pole without protection release. | | |
| | (iv)MCCB 200A 4Pole | MCCB switch disconnector rating 200Amp 4 Pole without protection Release. | each | 11877.00 |
| | (v)MCCB 400A 4Pole | MCCB switch disconnector rating 400Amp 4 Pole without protection release. | each | 16942.00 |
| | (vi)MCCB 600A 4Pole | MCCB switch disconnector rating 600Amp 4 Pole without protection Release. | each | 22030.00 |
| 21.7 | Air circuit breaker (ACB) (Microprocessor Release) | Supply of ISI marked and approved make of air circuit breaker (ACB) with microprocessor release with overload, short circuit, neutral protection, instantaneous, ground fault protection (LSING) Ics=Icu=Icw for 1sec for total selectivity, ambient temperature 50deg.C, In with breaking capacity of 50 kA/60kA/75kA, 4 pole, 415 Volts conforming to IS/IEC : 60947-2 | | |
| | (i) 630A, Fixed 50kA | 630Amp, manual, fixed 50kA | each | 152207.00 |
| | (ii) 800A, Fixed 50kA | 800Amp, manual, fixed 50kA | each | 152207.00 |
| | (iii) 1000A, Fixed 50kA | 1000Amp, manual, fixed 50kA | each | 156559.00 |
| | (iv) 1250A, Fixed 50kA | 1250Amp, manual, fixed 50kA | each | 190353.00 |
| | (v) 1600A, Fixed 50kA | 1600Amp, manual, fixed 50kA | each | 215711.00 |
| | (vi) 630A, Drawout 50kA | 630Amp, manual, drawout 50kA | each | 233563.00 |
| | (vii) 800A, Drawout 50kA | 800Amp, manual, drawout 50kA | each | 237659.00 |
| | (viii) 1000A, Drawout 50kA | 1000Amp, manual, drawout 50kA | each | 274243.00 |
| | (ix) 1250A, Drawout 50kA | 1250Amp, manual, drawout 50kA | each | 295733.00 |
| | (x) 1600A, Drawout 50kA | 1600Amp, manual, drawout 50kA | each | 365740.00 |
| | (xi) 2000A, Drawout 50kA | 2000Amp, manual, drawout 50kA | each | 432582.00 |
| | (xii) 2500A, Drawout 60kA | 2500Amp, manual, drawout 60kA | each | 508365.00 |
| | (xiii) 3200A, Drawout 75kA | 3200Amp, manual, drawout 75kA | each | 548886.00 |
| | (xiv) 4000A, Drawout 75kA | 4000Amp, manual, drawout 75kA | each | 607592.00 |
| 21.8 | MCB 'C' Series, 240/415V, 50 Cycle, 10kA/15kA | Supply of ISI marked and accepted standard of miniature circuit breaker (MCB) of 'C' series with short circuit indication, suitable for 240/415 Volts,50 Cycle, 10 kA/15 kA Value AC supply conforming to IS : 8828 : 1996, IEC : 60898 :2002 & 60947-2 but without enclosures :- | | |
| | Single pole | **Single pole** | | |
| | (i) MCB SP 0.5A | MCB SP 0.5Amp to 5Amp 10kA rating | each | 439.00 |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| | to 5A, 10kA | | |
| | (ii) MCB SP 6A to 32A, 10kA | MCB SP 6Amp to 32Amp 10kA rating | each | 263000 |
| | (iii) MCB SP 40A, 10kA | MCB SP For 40Amp 10kA rating | each | 611.00 |
| | (iv) MCB SP 50A to 63A, 10kA | MCB SP 50Amp to 63Amp 10kA rating | each | 651.00 |
| | (v) MCB SP 80A, 15kA | MCB SP 80Amp 15kA rating | each | 2409.00 |
| | (vi) MCB SP 100A, 15kA | MCB SP 100Amp 15kA rating | each | 2701.00 |
| | (vii) MCB SP 125A, 15kA | MCB SP 125Amp 15kA rating | each | 2921.00 |
| | Double Pole | **Double pole** | | |
| | (i) MCB DP 0.5A to 5A, 10kA | MCB DP 0.5Amp to 5Amp 10kA rating | each | 1243.00 |
| | (ii) MCB DP 6A to 32A, 10kA | MCB DP 6Amp to 32Amp 10kA rating | each | 887.00 |
| | (iii) MCB DP 40A, 10kA | MCB DP For 40Amp 10kA rating | each | 1329.00 |
| | (iv) MCB DP 50A to 63A, 10kA | MCB DP 50Amp to 63Amp 10kA rating | each | 1478.00 |
| | (v) MCB DP 80A, 15kA | MCB DP 80Amp 15kA rating | each | 5344.00 |
| | (vi) MCB DP 100A, 15kA | MCB DP 100Amp 15kA rating | each | 5709.00 |
| | (vii) MCB DP 125A, 15kA | MCB DP 125Amp 15kA rating | each | 6274.00 |
| | Triple pole | **For Triple pole** | | |
| | (i) MCB TP 0.5A to 5A, 10kA | MCB TP 0.5Amp to 5Amp 10kA rating | each | 1756.00 |
| | (ii) MCB TP 6A to 32A, 10kA | MCB TP 6Amp to 32Amp 10kA rating | each | 1411.00 |
| | (iii) MCB TP 40A, 10kA | MCB TP For 40Amp 10kA rating | each | 2099.00 |
| | (iv) MCB TP 50A to 63A, 10kA | MCB TP 50Amp to 63Amp 10kA rating | each | 2157.00 |
| | (v) MCB TP 80A, 15kA | MCB TP 80Amp 15kA rating | each | 8436.00 |
| | (vi) MCB TP 100A, 15kA | MCB TP 100Amp 15kA rating | each | 8981.00 |
| | (vii) MCB TP 125A, 15kA | MCB TP 125Amp 15kA rating | each | 9976.00 |
| | Four pole | **Four pole** | | |
| | (i) MCB FP 0.5A to 5A, 10kA | MCB FP 0.5Amp to 5Amp 10kA rating | each | 2399.00 |
| | (ii) MCB FP 6A to 32A, 10kA | MCB FP 6Amp to 32Amp 10kA rating | each | 1965.00 |
| | (iii) MCB FP 40A, 10kA | MCB FP For 40Amp 10kA rating | each | 2714.00 |
| | (iv) MCB FP 50A to 63A, 10kA | MCB FP 50Amp to 63 Amp 10kA rating | each | 2795.00 |
| | (v) MCB FP 80A, 15kA | MCB FP 80Amp 15kA rating | each | 10806.00 |

| Item No. | | Description | Unit | Rate |
|---|---|---|---|---|
| | (vi) MCB FP 100A, 15kA | MCB FP 100Amp 15kA rating | each | 11264.00 |
| | (viii) MCB FP 125A, 15kA | MCB FP 125Amp 15kA rating | each | 12724.00 |
| 21.9 | MCB 'D' Series, 240/415V, 50 Cycle, 10kA/15kA | Supply of ISI marked and accepted standard of miniature circuit breaker (MCB) of 'D' series with short circuit indication, suitable for 240/415 Volts,50 Cycle, 10 kA/15 kA Value AC supply conforming to IS : 8828 : 1996, IEC : 60898 :2002 & 60947-2 but without enclosures :- | | |
| | Single pole | **Single pole** | | |
| (i) | MCB SP 0.5A to 5A, 10kA | MCB SP 0.5Amp to 5Amp 10kA rating | each | 481.00 |
| (ii) | MCB SP 6A to 32A, 10kA | MCB SP 6Amp to 32Amp 10kA rating | each | 383.00 |
| (iii) | MCB SP 40A, 10kA | MCB SP For 40Amp 10kA rating | each | 691.00 |
| (iv) | MCB SP 50A to 63A, 10kA | MCB SP 50Amp to 63Amp 10kA rating | each | 767.00 |
| | Double Pole | **Double pole** | | |
| (v) | MCB DP 0.5A to 5A, 10kA | MCB DP 0.5Amp to 5Amp 10kA rating | each | 1296.00 |
| (vi) | MCB DP 6A to 32A, 10kA | MCB DP 6Amp to 32Amp 10kA rating | each | 1038.00 |
| (vii) | MCB DP 40A, 10kA | MCB DP For 40Amp 10kA rating | each | 1453.00 |
| (viii) | MCB DP 50A to 63A, 10kA | MCB DP 50Amp to 63Amp 10kA rating | each | 1555.00 |
| | Triple pole | **For Triple pole** | | |
| (ix) | MCB TP 0.5A to 5A, 10kA | MCB TP 0.5Amp to 5Amp 10kA rating | each | 1913.00 |
| (x) | MCB TP 6A to 32A, 10kA | MCB TP 6Amp to 32Amp 10kA rating | each | 1519.00 |
| (xi) | MCB TP 40A, 10kA | MCB TP For 40Amp 10kA rating | each | 2306.00 |
| (xii) | MCB TP 50A to 63A, 10kA | MCB TP 50Amp to 63Amp 10kA rating | each | 2382.00 |
| | Four pole | **Four pole** | | |
| (xiii) | MCB FP 0.5A to 5A, 10kA | MCB FP 0.5Amp to 5Amp 10kA rating | each | 2525.00 |
| (xiv) | MCB FP 6A to 32A, 10kA | MCB FP 6Amp to 32Amp 10kA rating | each | 2275.00 |
| (xv) | MCB FP 40A, 10kA | MCB FP For 40Amp 10kA rating | each | 2855.00 |
| | MCB FP 50A to 63A, 10kA | MCB FP 50Amp to 63 Amp 10kA rating | each | 3073.00 |
| 21.10 | MCB Isolators | Supplying of MCB Isolators suitable for 240/415 Volts, 50 Hz AC supply with kA value rating 10 kA of approved make conforming to IS:13947-Part III :1993 & IEC:60947- 3:2001 (without enclosures ) :- | | |
| | Single pole | **Single pole** | | |
| (i) | Upto 40A | Current rating upto 40Amp | each | 410.00 |

| Item No. | | Description | Unit | Rate |
|---|---|---|---|---|
| (ii) | Upto 63A | Current rating upto 63Amp | each | 451.00 |
| | Double pole | **Double pole** | | |
| (iii) | Upto 40A | Current rating upto 40Amp | each | 575.00 |
| (iv) | Upto 63A | Current rating upto 63Amp | each | 733.00 |
| | Triple pole | **Triple pole** | each | |
| (v) | Upto 40A | Current rating upto 40Amp | each | 1103.00 |
| (vi) | Upto 63A | Current rating upto 63Amp | each | 1236.00 |
| (vii) | Upto 80A | Current rating upto 80Amp | each | 1386.00 |
| (viii) | Upto 100A | Current rating upto 100Amp | each | 1455.00 |
| | Four pole | **Four pole** | | |
| (ix) | Upto 40A | Current rating upto 40Amp | each | 1097.00 |
| (x) | Upto 63A | Current rating upto 63Amp | each | 1398.00 |
| (xi) | Upto 80A | Current rating upto 80Amp | each | 1665.00 |
| (xii) | Upto 100A | Current rating upto 100Amp | each | 1747.00 |
| 21.11 | SPN MCB DB | Supply of approved make powder coated sheet steel encloser SPN MCB DB inclusive of busbar, neutral bar, earth bar, connection copper wire not less than 16 sq.mm & two earth terminals etc. complete as per IS:13032(exclusive of MCB & isolator)- | | |
| | (i) 2 Way single door | 2 Way single door | each | 447.00 |
| | (ii)4 Way single door | 4 Way single door | each | 1097.00 |
| | (iii)4 Way double door | 4 Way double door IP 43 protection | each | 1699.00 |
| | (iv)6 Way double door | 6 Way double door IP 43 protection | each | 1846.00 |
| | (v)8 Way double door | 8 Way double door IP 43 protection | each | 2127.00 |
| | (vi)12 Way double door | 12 Way double door IP 43 protection | each | 2738.00 |
| | (vii)16 Way double door | 16 Way double door IP 43 protection | each | 3415.00 |
| 21.12 | TPN MCB DB double door | Supplying of approved make TPN MCB DB metal double door IP 43 protection with provision for FP MCB/ Isolator/ RCCB/ RCBO as incomer and SP MCBs as outgoing inclusive of busbar, neutral bar, earth bar, connection copper wire not less than 16 sq.mm & two earth terminals etc. complete as per IS:13032(exclusive of MCB & isolator): | | |
| (i) | 4 Way (4+12) | 4 Way (4+12) | each | 5126.00 |
| (ii) | 4 Way (8+12) | 4 Way (8+12) | each | 4444.00 |
| (iii) | 6 Way (8+18) | 6 Way (8+18) | each | 5526.00 |
| (iv) | 8 Way (8+24) | 8 Way (8+24) | each | 6716.00 |
| (v) | 12 Way (8+36) | 12 Way (8+36) | each | 9519.00 |
| 21.13 | Vertical TPN MCB DB double door | Supplying of approved make vertical TPN MCB DB metal double door IP 43 protection with provision for FP MCB/Isolator/RCCB/RCBO as incomer and | | |

| Item No. | | Description | Unit | Rate |
|---|---|---|---|---|
| | | SP/TP MCBs as outgoing inclusive of busbar, neutral bar, earth bar, & two earth terminals etc. complete as per IS:13032(exclusive of MCB & isolator) | | |
| (i) | 4 Way (8+12) | 4 Way (8+12) | each | 13577.00 |
| (ii) | 6 Way (8+18) | 6 Way (8+18) | each | 15436.00 |
| (iii) | 8 Way (8+24) | 8 Way (8+24) | each | 17263.00 |
| (iv) | 12 Way (8+36) | 12 Way (8+36) | each | 23179.00 |
| 21.14 | Vertical TPN MCB DB, with provision for MCCB upto 160A TP/FP 36kA, double door | Supplying of approved make, powder coated metal double door vertical TPN MCB DB IP 43 protection with provision for MCCB upto160A TP/FP 36kA as incomer and space for SP/TP MCBs as outgoing (without MCCB/MCBs) inclusive of busbar & connections etc | | |
| (i) | 4 Way without MCCB | 4 Way without MCCB | each | 13822.00 |
| (ii) | 6 Way without MCCB | 6 Way without MCCB | each | 15671.00 |
| (iii) | 8 Way without MCCB | 8 Way without MCCB | each | 15989.00 |
| (iv) | 12 Way without MCCB | 12 Way without MCCB | each | 21911.00 |
| 21.15 | Vertical TPN MCB DB, with provision for MCCB upto 250A TP/FP 50kA, double door | Supply of approved make, powder coated metal double door vertical TPN MCB DB IP 43 protection with provision for MCCB upto 250A TP/FP 50kA as incomer and space for SP/TP MCBs as outgoing inclusive of busbar, neutral bar, earth bar & two earth terminals etc. complete as per IEC:61439-1 & 3(without MCCB/MCBs) | | |
| (i) | 4 Way without MCCB | 4 Way without MCCB | each | 16825.00 |
| (ii) | 6 Way without MCCB | 6 Way without MCCB | each | 20725.00 |
| (iii) | 8 Way without MCCB | 8 Way without MCCB | each | 22840.00 |
| (iv) | 12 Way without MCCB | 12 Way without MCCB | each | 27727.00 |
| 21.16 | Vertical TPN MCCB sub-main DB, with provision for MCCB upto 630A, 36kA/50kA | Supply of approved make, powder coated metal double door vertical TPN MCCB sub-main DB IP 43 protection with provision for MCCB upto 630A, 36kA/50kA, TP as incomer and space for TP MCCBs as outgoing inclusive of tin-plated electrolytic copper busbar 250A/400A/630A as per incoming MCCB rating, neutral bar 100% of phase, earth bar 50% of phase & two earth terminals etc. complete as per IEC:61439-2(without MCCBs) | | |
| (i) | 4 Way, I/C 250A, O/G 100A | 4 Way without MCCB I/C 250Amp, O/G 100Amp | each | 38547.00 |
| (ii) | 8 Way, I/C 250A, O/G 100A | 8 Way without MCCB I/C 250Amp, O/G | each | 46657.00 |

| Item No. | | Description | Unit | Rate |
|---|---|---|---|---|
| | | 100Amp | | |
| (iii) | 12 Way, I/C 250A, O/G 100A | 12 Way without MCCB I/C 250Amp, O/G 100Amp | each | 59846.00 |
| (iv) | 16 Way, I/C 250A, O/G 100A | 16 Way without MCCB I/C 250Amp, O/G 100Amp | each | 69987.00 |
| (v) | 4 Way, I/C 400A, O/G 100A | 4 Way without MCCB I/C 400Amp, O/G 100Amp | each | 43614.00 |
| (vi) | 8 Way, I/C 400A, O/G 100A | 8 Way without MCCB I/C 400Amp, O/G 100Amp | each | 49703.00 |
| (vii) | 12 Way, I/C 400A, O/G 100A | 12 Way without MCCB I/C 400Amp, O/G 100Amp | each | 63899.00 |
| (viii) | 16 Way, I/C 400A, O/G 100A | 16 Way without MCCB I/C 400Amp, O/G 100Amp | each | 76076.00 |
| (ix) | 8 Way, I/C 630A, O/G 100A | 8 Way without MCCB I/C 630Amp, O/G 100Amp | each | 78098.00 |
| (x) | 12 Way, I/C 630A, O/G 100A | 12 Way without MCCB I/C 630Amp, O/G 100Amp | each | 105483.00 |
| (xi) | 16 Way, I/C 630A, O/G 100A | 16 Way without MCCB I/C 630Amp, O/G 100Amp | each | 115626.00 |
| (xii) | 4 Way, I/C 630A, O/G 250A | 4 Way without MCCB I/C 630Amp, O/G 250Amp | each | 76076.00 |
| (xiii) | 8 Way, I/C 630A, O/G 250A | 8 Way without MCCB I/C 630Amp, O/G 250Amp | each | 85198.00 |
| 21.17 | Vertical TPN MCCB sub-main DB accessories | Supply & fixing of double door vertical TPN MCCB sub-main DB accessories as per OEM of approved make powder coated metal box for metering, side cabling box and corner box on surface/concealed complete as required mentioned below :-. | | |
| (i) | Metering ext. box | Metering extension box 250Amp/400Amp | each | 6692.00 |
| (ii) | Metering ext. box | Metering extension box 630Amp | each | 7106.00 |
| (iii) | Bottom cabling box | Bottom cabling box 250Amp/400Amp | each | 6269.00 |
| (iv) | Bottom cabling box | Bottom cabling box 630Amp | each | 6692.00 |
| (v) | Corner box | Corner box 250Amp/400Amp/630Amp | each | 2097.00 |
| (vi) | Mounting base | Mounting base 250Amp/400Amp/630Amp | each | 3142.00 |
| (vii) | Side cabling box 250A-100A | Side cabling box 250Amp - 100Amp left/right side for - | | |
| (viii) | 4 Way | 4 Way | each | 6070.00 |
| (ix) | 6 Way | 6 Way | each | 7843.00 |
| (x) | 8 Way | 8 Way | each | 9405.00 |
| (xi) | 12 Way | 12 Way | each | 10971.00 |
| | Side cabling box 400A-100A | Side cabling box 400Amp - 100Amp left/right side for - | | |
| (xii) | 4 Way | 4 Way | each | 9405.00 |
| (xiii) | 6 Way | 6 Way | each | 10971.00 |
| (xiv) | 8 Way | 8 Way | each | 12019.00 |
| (xv) | 12 Way | 12 Way | each | 13584.00 |
| | Side cabling box 630A-2500A + 100A | Side cabling box 630Amp - 250Amp +100Amp left/right side for - | | |
| (xvi) | 4 Way/2 + 2 | 4 Way/2 + 2 | each | 10762.00 |

| Item No. | | Description | Unit | Rate |
|---|---|---|---|---|
| (xvii) | 6 Way/2 + 6 | 6 Way/2 + 6 | each | 12019.00 |
| (xviii) | 8 Way/2 + 10 | 8 Way/2 + 10 | each | 13998.00 |
| (xix) | 12 Way/2 + 14 | 12 Way/2 + 14 | each | 15466.00 |
| 21.18 | Industrial type metal plug & socket DBs, SPN sheet encloser | Supplying & fixing of approved make Industrial type metal plug & socket DBs (without MCB) SPN sheet encloser (dust protected) inclusive of 2 pole and earth metal plug and socket and space to incorporate SP MCB complete as per specification as required. | | |
| (i) | 10A | 10 Amp | each | 1037.00 |
| (ii) | 20A | 20 Amp | each | 1095.00 |
| 21.19 | Industrial type metal plug & socket DBs, TPN sheet encloser | Supplying & fixing of ISI marked Industrial type metal plug socket D.B.s (without MCBs) for TPN encloser three pole and earth metal plug and socket including space to incorporate TP MCB complete as per specification as required. | | |
| (i) | 20A | 20 Amp | each | 2249.00 |
| (ii) | 30/32A | 30/32 Amp | each | 2543.00 |
| 21.20 | Industrial type metal plug & socket, 3 Pin (2P+E) | Supply & fixing of approved make Industrial type metal plug & socket, 3 pin (2P + E) conforming to IEC : 60309 -1&3 complete as per specification | | |
| (i) | 10A Plug only | 10Amp Plug only | each | 153.00 |
| (ii) | 10A socket only | 10Amp socket only | each | 153.00 |
| (iii) | 20A Plug only | 20Amp Plug only | each | 179.00 |
| (iv) | 20A socket only | 20Amp socket only | each | 194.00 |
| 21.21 | RCBOs | Supplying and installing of RCBOs (Residual current circuit breaker with overload and short circuit protection) ISI marked complete as per I.S. standard conforming to IEC:61009-1 & IS:12640-2, 240/415V 50 Hz with 10 kA short circuit withstand capacity for earth leakage, overload & short circuit protection including connection in existing enclosure in approved manner as per specification. | | |
| (A) | DP (4 Module) | DP (4 module) | | |
| (i) | 2 Pole 6A- 25A, 30mA | 2 pole 6Amp to 25Amp, 30 mA sensitivity | each | 4491.00 |
| (ii) | 2 Pole 32A, 30mA | 2 pole 32Amp, 30mA sensitivity | each | 4586.00 |
| (iii) | 2 Pole 40A, 30mA | 2 pole 40Amp, 30mA sensitivity | each | 5010.00 |
| (iv) | 2 Pole 63A, 30mA | 2 pole 63Amp, 30mA sensitivity | each | 6372.00 |
| (v) | 2 Pole 6A- 25A, 100mA | 2 pole 6Amp to 25Amp, 100mA sensitivity | each | 4603.00 |
| (vi) | 2 Pole 32A, 100mA | 2 pole 32Amp, 100mA sensitivity | each | 4666.00 |
| (vii) | 2 Pole 40A, 100mA | 2 pole 40Amp, 100mA sensitivity | each | 5081.00 |
| (viii) | 2 Pole 63A, 100mA | 2 pole 63Amp, 100mA sensitivity | each | 6475.00 |
| (B) | FP (8 Module) | FP (8module) | | |

| Item No. | | Description | Unit | Rate |
|---|---|---|---|---|
| (i) | 4 Pole 16A, 30mA | 4 pole 16Amp, 30mA sensitivity | each | 6173.00 |
| (ii) | 4 Pole 25A-32A, 30mA | 4 pole 25-32Amp, 30mA sensitivity | each | 6366.00 |
| (iii) | 4 Pole 40A, 30mA | 4 pole 40Amp, 30mA sensitivity | each | 7077.00 |
| (iv) | 4 Pole 63A, 30mA | 4 pole 63Amp, 30mA sensitivity | each | 7908.00 |
| (v) | 4 Pole 16A, 100/300mA | 4 pole 16Amp, 100/300mA sensitivity | each | 6356.00 |
| (vi) | 4 Pole 25A-32A, 100A/300mA | 4 pole 25-32Amp, 100/300mA sensitivity | each | 6476.00 |
| (vii) | 4 Pole 40A, 100/300mA | 4 pole 40Amp, 100/300mA sensitivity | each | 7180.00 |
| (viii) | 4 Pole 63A, 100/300mA | 4 pole 63Amp, 100/300mA sensitivity | each | 7926.00 |
| 21.22 | SPDs, Type 2 maximum discharge current (Imax) upto70kA &type 1+2 (Imax) upto100kA | Supplying and installing of SPDs (surge protection device) ISI marked complete as per I.S. standard conforming to IEC:61543 - 11 for direct & indirect lightning surges protection type 2 maximum discharge current (Imax) up to 70kA &type 1+2 (Imax) up to 100KA including connection in existing enclosure in approved manner as per specification. | | |
| (i) | Type 2, 40kA, 1P+N | Type 2 : 40kA, 1P+N protection against indirect lightning & switching surges | each | 6642.00 |
| (ii) | Type 2, 70kA, 1P+N | Type 2 : 70kA, 1P+N protection against indirect lightning & switching surges | each | 10630.00 |
| (iii) | Type 2, 40kA, 3P+N | Type 2 : 40kA, 3P+N protection against indirect lightning & switching surges | each | 13282.00 |
| (iv) | Type 2, 70kA, 3P+N | Type 2 : 70kA, 3P+N protection against indirect lightning & switching surges | each | 17713.00 |
| (v) | Type 1+2, 50kA, 1P+N | Type 1+2 : 50kA, 1P+N protection against direct & indirect lightning surges | each | 9561.00 |
| (vi) | Type 1+2, 100kA, 1P+N | Type 1+2 : 100kA, 1P+N protection against direct & indirect lightning surges | each | 10630.00 |
| (vii) | Type 1+2, 50kA, 3P+N | Type 1+2 : 50kA, 3P+N protection against direct & indirect lightning surges | each | 18770.00 |
| (viii) | Type 1+2, 100kA, 3P+N | Type 1+2 : 100kA, 3P+N protection against direct & indirect lightning surges | each | 19477.00 |
| 21.23 | Steel enclosure, neutral bar suitable for MCB/RCBOs/MCCB | Supply & fixing of approved make accepted standard sheet steel enclosure (dust protected) duly powder coated paint neutral bar suitable for MCB/RCBOs/MCCB of capacity as required etc. complete as per specification (exclusive of M.C.B./ MCCB/RCBOs) | | |
| | For MCB | For MCB | | |
| (i) | SP enclosure | SP enclosure | each | 285.00 |
| (ii) | DP/TP/FP enclosure | DP/TP/FP enclosure | each | 341.00 |
| (iii) | FP RCBO enclosure | FP RCBO enclosure | each | 602.00 |
| (iv) | For MCCB | For MCCB | | |
| (v) | 63A/100A, 3P/4P enclosure | 63Amp/100Amp, 3P/4P enclosure | each | 1225.00 |
| (vi) | 160A, 3P/4P enclosure | 160Amp, 3P/4P enclosure | each | 2285.00 |

| Item No. | | Description | Unit | Rate |
|---|---|---|---|---|
| (vii) | 250A, 3P/4P enclosure | 250Amp, 3P/4P enclosure | each | 4058.00 |
| (viii) | 630A, 3P/4P enclosure | 630Amp, 3P/4P enclosure | each | 6149.00 |
| | Cable end box | Supplying and fixing cable end box (loose wire box) suitable for triple pole and neutral, sheet steel, MCB/MCCB distribution board, 415 V, on surface/recess, complete with testing and commissioning etc. as required. | | |
| (ix) | TPN DB MCB | TPN DB MCB SP O/G, (V) SP/TP O/G | each | 696.00 |
| (x) | TPN DB MCCB | TPN DB MCCB (V) SP/TP O/G | each | 2461.00 |
| **FIXING OF SFUs, DBs, BUSBAR, MCCBs, MCBs, ISOLATORs & RCBOs** | | | | |
| 21.25 | Fixing of metal clad iron clad enclosure on 25 X 5 mm flat | Fixing of metal clad iron clad enclosure on 25 X 5 mm flat iron clamp including supplying and fixing of clamp as required duly embedded in masonry including cable connection and other required material as per specification :- | | |
| (i) | 15/16A SP/DP switch/SPN | 15/16Amp SP/DP switch/SPN, DB up to 4 way | each | 476.00 |
| (ii) | 32A DP switch/SPN | 32Amp DP switch/SPN DB up to 8 way/16 Amp TPN DB up to 3 way per pole or 16 Amp TP or TPN switch/16 Amp 415Volt Isolator | each | 475.00 |
| (iii) | 32A TP or TPN Switch | 32 Amp TP or TPN switch/32 Amp 415Volt Isolator/16 Amp TPN DB up to 6 way per pole/32 Amp 415Volt bus bar 4 way | each | 477.00 |
| | Fixing of metal clad iron clad enclosure on 25 X 5 mm flat | Fixing of metal clad iron clad enclosure on 25 X 5 mm flat iron clamp including supplying and fixing of clamp as required duly embedded in masonry including cable connection and other required material as per specification :- | | |
| 21.26 | Fixing of metal clad, 25x25x5 mm angle | Fixing of metal clad enclosure on 25X 25 X 5 mm angle iron clamp including supplying and fixing of clamp as required duly embedded in masonry including cable connection and other required material as per specification :- | | |
| (i) | Fixing of 63A TP or TPN | Fixing of 63Amp TP or TPN main switches/DB 63Amp 240/415 Volt 4 to 8 way or 63 Amp change over switch/63Amp 415Volt Isolator/30 Amp 415Volt TPN DB up to 6 way per pole and bus bar 4 way | each | 484.00 |
| (ii) | Fixing of 100A TP or TPN | Fixing of 100Amp TP or TPN main switch or 100Amp change over switch/100 Amp 415Volt Isolators and bus bar 4 way | each | 510.00 |
| (iii) | Fixing of 200A TP or TPN | Fixing of 200 Amp TP or TPN main switch or 200 Amp change over switch/200 Amp 415 Volt Isolators and busbar 4 way | each | 521.00 |
| 21.27 | Fixing of metal clad, 35x35x5 mm angle | Fixing of metal clad enclosure on 35X 35 X 5 mm angle iron clamp including supplying and fixing of clamp as required duly embedded in masonry including cable connection and other required material as per specification :- | | |
| (i) | Fixing of 300/400A | Fixing of 300/400Amp TP or TPN main | each | 459.00 |

| Item No. | | Description | Unit | Rate |
|---|---|---|---|---|
| | TP or TPN main switch or change over switch | switch or 300/400Amp change over switch/300/400Amp 415 Volt Isolators and busbar 4 way | | |
| (ii) | Fixing of 500/600A TP Or TPN main switch or change over switch | Fixing of 500/600Amp TP or TPN main switch or 500/600Amp change over switch 500/600Amp 415 Volt Isolators and busbar 4 way | each | 468.00 |
| 21.28 | Fixing of MCB/MCCB/Isolator, (labour only) | Fixing of MCB/MCCB/Isolator in sheet steel enclosure as required as per accepted practice, including mounting on busbar and cable connection etc. complete (labour only) | | |
| (i) | MCB/Isolator SP/DP | MCB/Isolator SP/DP | each | 22.00 |
| (ii) | MCB/MCCB Isolator TP/TPN/FP | MCB/MCCB Isolator TP/TPN/FP | each | 28.00 |
| 21.29 | Labour charges | Labour charges for fixing sheet steel enclosure, MCB DB surface mounting type, as per accepted practice on 25x5mm flat iron clamp duly embedded in wall, cable connection etc. complete. | | |
| | For Item | For item no :- | | |
| (i) | Item SOR no. | 21.11.(i) to 21.11.(iv) | each | 517.00 |
| (ii) | Item SOR no. | 21.11.(v) & 21.12.(i) to 21.12.(ii) | each | 545.00 |
| (iii) | Item SOR no. | 21.12.(iii) to 21.12.(iv) & 21.13.(i) | each | 572.00 |
| 21.30 | Labour charges | Labour charges for fixing sheet steel enclosures, MCB DB surface mounting type, as per accepted practice on 25x25x 5 mm angle iron clamp, including supplying and fixing of clamp duly embedded in wall, cable connection etc. complete:- | | |
| | For Item Item SOR no. | For item no :- 21.11.(vi)& 21.11.(vii); 21.12.(v) ; 21.13.(ii) to 21.13.(vi) & 21.14.(i) to 21.14.(iv) ; 21.15.(i) to 21.15.(iv) | each | 414.00 |
| 21.31 | Labour charges | Labour charges for fixing sheet steel enclosures, MCCB DB surface mounting type, as per accepted practice on 35x35x 5 mm angle iron clamp, including supplying and fixing of clamp duly embedded in wall, cable connection etc. complete :- | | |
| (i) | Item SOR no. | 21.16.(i) & 21.16.(ii) ; 21.16.(v) & 21.16.(vi) | each | 3100.00 |
| (ii) | Item SOR no. | 21.16.(iii) & 21.16.(iv) ; 21.16.(vii) to 21.16.(ix), 21.16.(xii), 21.16.(xiii) | each | 3318.00 |
| (iii) | Item SOR no. | 21.16.(x) to 21.16.(xi) | each | 3991.00 |
| 21.32 | Labour charges | Labour charges for fixing sheet steel enclosures, MCB/MCCBDB flush mounting type, as per accepted practice, duly embedded and end plate completely flushed in wall, cable | | |

| Item No. | | Description | Unit | Rate |
|---|---|---|---|---|
| | | connection etc. complete :- | | |
| | For item | For item no :- | | |
| (i) | Item SOR no. | 21.11.1 to 21.11.7 ; 21.12.1 to 21.12.3 & 21.13.1 | each | 268.00 |
| (ii) | Item SOR no. | 21.12.4 & 21.12.5 ; 21.13.2 to 21.13.4 ; 21.14.1 to 21.14.4 & 21.15.1 to 21.15.4 | each | 297.00 |
| (iii) | Item SOR no. | 21. 16.1 & 21.16.2 ; 21.16.5 & 21.16.6 | each | 2788.00 |
| (iv) | Item SOR no. | 21.16.3 & 21.16.4 ; 21.16.7 to 21.16.9, 21.16.12, 21.16.13 | each | 3527.00 |
| (v) | Item SOR no. | 21.16.10 to 21.16.11 | each | 3532.00 |
| | | **ACCESSORIES** | | |
| 21.33 | Multi core flexible cables | Supply and fixing of multi core round FRLSH PVC insulated copper (flexible) conductor & PVC sheathed cables 1100Volts as per IS:694-1990 of approved make | | |
| (i) | Three core 0.5 sq.mm. | (16/0.2 mm)Three core 0.5 sq.mm. | metre | 20.00 |
| (ii) | Three core 0.75 sq.mm. | (24/0.2 mm) Three core 0.75 sq.mm. | metre | 25.00 |
| (iii) | Three core 1.5 sq.mm. | (30/0.25 mm) Three core 1.5 sq.mm. | metre | 50.00 |
| (iv) | Three core 2.5 sq.mm. | (50/0.25 mm) Three core 2.5 sq.mm. | metre | 86.00 |
| (v) | Three core 4.0 sq.mm. | (56/0.3 mm) Three core 4.0 sq.mm. | metre | 132.00 |
| 21.34 | APP heavy duty capacitor 440V, 3 Ph. | Supply and fixing of mixed die-electric ultra-heavy duty APP capacitor 440Volt, 3 Ph., peak inrush current upto 500 x In, over current upto 3 x In(normal current), operating losses total not more than 0.35 W/kVAR, operating life not less than 300000 hrs, switching 20000 operation/year, as per IS:13340-1993, IS:13341-1992, IS:13585-1994, of approved make as required as per specification | | |
| | 5 - 25 kVAR | 5 to 25 kVAR bank | Per kVAR | 849.00 |
| 21.35 | MPP H heavy duty capacitor 415V, 3 Ph. | Supply and fixing of MPP - H heavy duty capacitor 415 Volt, 3 Ph., peak inrush current upto 250 x In, over current upto 1.8 x In(normal current), operating losses total not more than 0.35 W/kVAR, switching 8000 operation/year, as per IS:13340-1993, IS:13341-1992 of approved make as required as per specification | | |
| | 1 - 4 kVAR | 1 to 4 kVAR bank | Per kVAR | 309.00 |
| | 5 - 25 kVAR | 5 to 25 kVAR bank | Per kVAR | 325.00 |
| 21.36 | LT panel | Supply, fixing, testing & commissioning wall/floor mounted LT panel primer coated with powder coated paint & provided with required gasket for dust/vermin proof with degree of protection IP42 suitable for 415Volt | kg | 397.00 |

| Item No. | | Description | Unit | Rate |
|---|---|---|---|---|
| | | 3 phase, 50 Hz, 4 wire system fabricated out of CRCA sheet upto 2 mm thick (1.6 mm for doors) duly compartmentalized for incomer, bus section, outgoings, cable alleys & CT, PT Ampere Metre, Volt Metre, selector switches, frequency Metre, phase indicating lamp, Energy Metre complete including cost of busbar supports, detachable cable gland plates, 2 earthing terminals, internal wiring & fixing of separately supplied MCBs, MCCBs, ACB, panel mounted changeover switch/SFUs, etc. as required but excluding cost of busbar strips, Ampere Metre, Volt Metre, selector switch as per approved design & specification | | |
| 21.38 | LT panel accessories | Supply and fixing of LT panel accessories of approved make in existing LT panel including connections etc. as required as per specification. | | |
| (i) | Digital Ampere Metre | Digital Ampere Metre with CTs with selector switch | Per Set | 2125.00 |
| (ii) | Digital Volt Metre | Digital Volt Metre with selector switch & HRC fuse | Per Set | 2398.00 |
| (iii) | Frequency Metre | Frequency Metre | each | 1435.00 |
| (iv) | Multifunction Meter | Multifunction Meter | each | 3053.00 |
| (v) | Copper bus bar strips | Copper bus bar strips with PVC sleeves | kg | 788.00 |
| (vi) | Aluminium bus bar strips | Aluminium bus bar strips with PVC sleeves | kg | 284.00 |
| (vii) | LED lamp indicator | LED lamp indicator | each | 255.00 |
| 21.39 | Capacitor Duty Contactor | Supplying and fixing of capacitor duty contactor of approved make in existing power factor/LT panel including connections etc.as required as per specification. | | |
| (i) | Upto 3 kVAR | Upto 3 kVAR 440V,3Ph,50Hz, Aux contact-1NO | each | 1431.00 |
| (ii) | For 5 kVAR | For 5 kVAR 440V,3Ph,50Hz, Aux contact-1NO | each | 1495.00 |
| (iii) | For 8.5 kVAR | For 8.5 kVAR 440V,3Ph,50Hz, Aux contact-1NO | each | 1596.00 |
| (iv) | For 10 kVAR | For 10 kVAR 440V,3Ph,50Hz, Aux contact-1NO | each | 1721.00 |
| (v) | For 12.5 kVAR | For 12.5 kVAR 440V,3Ph,50Hz, Aux contact-1NO | each | 1831.00 |
| (vi) | For 15 kVAR | For 15 kVAR 440V,3Ph,50Hz, Aux contact-1NO | each | 2512.00 |
| (vii) | For 20 kVAR | For 20 kVAR 440V,3Ph,50Hz, Aux contact-1NO | each | 2948.00 |
| (viii) | For 25 kVAR | For 25 KVAR 440V,3Ph,50Hz, Aux contact-1NO | each | 3261.00 |
| (ix) | For 30 kVAR | For 30 kVAR 440V,3Ph,50Hz, Aux contact-1NO | each | 4353.00 |

| Item No. | | Description | Unit | Rate |
|---|---|---|---|---|
| 21.40 | APFC relay | Supplying and fixing of APFC relay 1A/5 A site selectable, measurement of individual current and voltage harmonic (THD), capacitor failure indication, In-built temperature sensor of approved make in existing power factor Panel including connections etc. as required as per specification as mentioned below : | | |
| | 8 Step relay | 8 Step APFC relay (144 x144mm) | each | 13351.00 |
| | 12 Step relay | Automatic power factor relay 12 Steps 230V 144 x 144mm | each | 17505.00 |
| 21.41 | CT | Supplying and fixing of CT 1A/5A as per site selectable busbar dimension of approved make in existing main panel for power factor Panel including connections etc. as required as per specification. | each | 1664.00 |
| 21.42 | Analog time switch for automatic ON - OFF street light | Supplying and fixing of analog time switch for automatic ON - OFF street light of approved make in existing LT panel including connections etc. as required as per specification | each | 2689.00 |
| 21.43 | Contractor 2 Pole rating 240/415V | Supplying and fixing of contractor 2 Pole rating 240/415V :for ON - OFF street light of approved make in existing LT panel including connections etc. as required as per specification | | |
| (i) | 20A | 20A | each | 559.00 |
| (ii) | 25A | 25A | each | 666.00 |
| (iii) | 40A | 40A | each | 832.00 |
| | | **LUMINAIRES** | | |
| 21.44 | CFL Roadway lighting | Supplying, fixing & testing of approved make of non - integral type CFL roadway lighting consisting of powder coated sheet steel housing with epoxy white powder coated CRCA sheet steel gear tray, clear acrylic cover complete with all accessories (without lamp) including fixing on existing G.I. pipe, pole with connection and all required material suitable for – | | |
| | 1 x 85W Retrofit CFL/LED | For 1 x 85W Retrofit CFL/LED (Without control gear) | each | 1666.00 |
| 21.45 | Post top lantern | Supply, fixing & testing of approved make globe type polycarbonate post top lantern, consisting of cast aluminium spigot complete with all accessories, lamp holder, (without lamp) including fixing on gate pillars/exiting G.I. pipe, pole with connection & all required materials on existing required size of G.I. pipe, suitable for CFL/LED lamp for mentioned below sizes :- | | |
| | 200 mm (8") | 200 mm (8") | each | 569.00 |
| | 250 mm (10") | 250 mm (10") | each | 791.00 |
| | 300 mm (12") | 300 mm (12") | each | 971.00 |

| Item No. | | Description | Unit | Rate |
|---|---|---|---|---|
| 21.46 | LED tube rod fixture | Supply and fixing Led tube rod comprising of LED tube with non-integral/integral driver, upto 6500K color temp having 40000 burning hrs life with minimum @ L 70, system lumen output should be minimum with system efficacy> 100 lm/Watt. LED driver PF> 0.95 & THD < 20%. The colour rendering index of LED light should be more than 70. Submission LM 79-08/IS16106 (2012), IEC60598, IEC61347i/c connection wire, testing etc. to complete the job. | | |
| | Tube light LED 1 X 8/10W | Tube light LED 1 X 8/10Watt Integral i/c batten aluminium body, PC diffuser. | each | 502.00 |
| | Tube light LED 1 X 12W | Tube light LED 1 X 12Watt Integral i/c batten aluminium body, PC diffuser. | each | 605.00 |
| | Tube light LED 1 X 18W | Tube light LED 1 X 18Watt, Integral i/c batten aluminium body, PC diffuser. | each | 713.00 |
| | Tube light LED 1 X 20/22W | Tube light LED 1 X 20/22Watt, Integral i/c batten aluminium body, PC diffuser. | each | 782.00 |
| | LED retrofit glass tube | LED retrofit glass tube i/c driver only 20/22Watt,color temp 3000 - 6000K without box suitable for replacing existing 36/40Watt FTL | each | 763.00 |
| 21.47 | LED Panel (recessed mounting) | Supply and fixing of recessed mounting type Led light fixture, LED of 1 to 3 Watt each assembled on single MCPCB, having color temp upto 6500K & having 50000 burning hrs life with minimum @ L 70, system lumen output should be minimum with efficacy>100 lm/Watt. LED driver PF> 0.95, THD < 20% & surge protection 4KV. The colour rendering index of LED light should be more than 70. Housing made of CRCA powder coated frame with glare free diffused polycarbonate cover. Submission LM 79-08/IS16106 (2012), IEC60598, IEC61347i/c connection wire, testing etc. to complete the job. | | |
| | 2' X 2', 36W, 3000-6500k | LED luminiare 2' X 2', 36Watt, color temp 3000-6500k as required. | each | 3103.00 |
| | 2' X 2', 42/45W, 3000-6500k | LED luminiare 2' X 2', 42/45Watt, color temp 3000-6500k as required. | each | 4215.00 |
| 21.48 | LED light (surface mounting) | Supply and fixing of surface mounting type LED light fixture, LED of 1 to 3 Watt each assembled on single MCPCB, having color temp upto 6500K & having 50000 burning hrs. life with minimum @ L 70, system lumen output should be minimum with efficacy>100 lm/Watt. LED driver, PF> 0.95, THD < 20% & surge protection 4KV. The colour rendering index of LED light should be more than 70. Housing made of CRCA powder coated frame with glare free diffused polycarbonate cover. | | |

| Item No. | | Description | Unit | Rate |
|---|---|---|---|---|
| | | Submission LM 79-08/IS16106 (2012), IEC60598, IEC61347i/c connection wire, testing etc. to complete the job. | | |
| | 2' X 2', 36W, 3000-6500k | LED luminiare 2' X 2', 36Watt, color temp 3000-6500k as required. | each | 4384.00 |
| | 2' X 2', 42/45W, 3000-6500k | LED luminiare 2' X 2', 42/45Watt, color temp 3000-6500k as required. | each | 4472.00 |
| 21.49 | LED down lighter (recessed mounting) | Supply and fixing recessed mounting LED down lighter, LED of 1 to 3 Watt each assembled on single MCPCB, having color temp upto 6500K & having 50000 burning hrs. life with minimum @ L 70, system lumen output should be minimum with efficacy>100 lm/Watt. LED driver PF > 0.95, THD < 20% & surge protection 4KV. The colour rendering index of LED light should be more than 70. Housing made of pressure die cast aluminium/CRCA powder coated frame with glare free diffused polycarbonate cover. Submission LM 79-08/IS16106 (2012), IEC60598, IEC61347i/c connection wire, testing etc. to complete the job.. | | |
| (i) | 3W | 3 Watt, color temp 3000-6500k as required. | each | 426.00 |
| (ii) | 5/6W | 5/6 Watt, color temp 3000-6500k as required. | each | 719.00 |
| (iii) | 7/8/10W | 7/8/10 Watt, color temp 3000-6500k as required. | each | 761.00 |
| (iv) | 12W | 12 Watt, color temp 3000-6500k as required. | each | 846.00 |
| (v) | 14/15W | 14/15 Watt, color temp 3000-6500k as required. | each | 1150.00 |
| (vi) | 18/20W | 18/20Watt, color temp 3000-6500k as required. | each | 1651.00 |
| (vii) | 22/24W | 22/24Watt, color temp 3000-6500k as required. | each | 2377.00 |
| 21.50 | LED down lighter (surface mounting) | Supply and fixing surface mounting LED down lighter, LED of 1 to 3Watt each assembled on single MCPCB, having color temp upto 6500K & having 50000 burning hrs. life with minimum @ L 70, system lumen output should be minimum with efficacy>100 lm/Watt. LED driver PF > 0.95, THD < 20% & surge protection 4KV. The colour rendering index of LED light should be more than 70. Housing made of pressure die cast aluminium/CRCA powder coated frame with glare free diffused polycarbonate cover. Submission LM 79-08/IS16106 (2012), IEC60598, IEC61347i/c connection wire, testing etc. to complete the job... | | |
| (i) | 3W | 3 Watt, color temp 3000-6500k as required. | each | 687.00 |
| (ii) | 5/6W | 5/6 Watt, color temp 3000-6500k as required | each | 704.00 |
| (iii) | 7/8/10W | 7/8/10 Watt, color temp 3000-6500k as required. | each | 726.00 |

| Item No. | | Description | Unit | Rate |
|---|---|---|---|---|
| (iv) | 14/15W | 14/15 Watt, color temp 3000-6500k as required. | each | 1400.00 |
| (v) | 18/20W | 18/20 Watt, color temp 3000-6500k as required. | each | 1783.00 |
| (vi) | 21/22/24W | 21/22/24Watt, color temp 3000-6500k as required. | each | 2809.00 |
| 21.51 | COB LIGHT | Supply and fixing of recessed mounting LED swivel type COB downlighter, having color temp upto 6500K & having 20000 burning hrs life with minimum @ L 70, system lumen output should be minimum with efficacy>80lm/W. LED driver PF > 0.9, THD < 20% & surge protection 5KV. The colour rendering index of LED light should be more than 70. Housing made of pressure die cast aluminium / CRCA powder coated frame with glare free diffused polycorbonate cover. Submission LM 80-08 Form LED Source Manufacturer & LM79-08 / IS16106 from NABL approved lab. Manufacturer manadatory. i/c connection wire, testing etc. to complete the job. Guarantee as per tender agreement condition. | | |
| (i) | COB LIGHT | 3W | each | 1228.00 |
| (ii) | COB LIGHT | 6/7W | each | 1229.00 |
| (iii) | COB LIGHT | 9W | each | 1544.00 |
| (iv) | COB LIGHT | 12/15W | each | 2022.00 |
| (v) | COB LIGHT | 18/20W | each | 2925.00 |
| (vi) | COB LIGHT | 22/24W | each | 3758.00 |
| 21.52 | Linear profile suspended LED light | Supply and fixing of Linear profile suspended LED lumanaire, having color temp upto 6500K & having 20000 burning hrs life with minimum @ L 70, system lumen output should be minimum with efficacy>80lm/W. LED driver PF > 0.9, THD < 20% & surge protection 5KV. The colour rendering index of LED light should be more than 80. Housing made of pressure die cast aluminium / CRCA powder coated frame with high transmission diffuser. Submission LM 80-08 Form LED Source Manufacturer & LM79-08 / IS16106 from NABL approved lab. Manufacturer manadatory. i/c mounting arrangment, connection wire, testing etc. to complete the job. Guarantee as per tender agreement condition. | | |
| | | 32/40W | each | 8712.00 |
| | | 72/80W | each | 17316.00 |
| 21.53 | Linear profile recessed LED light | Supply and fixing of Linear profile suspended LED lumanaire, having color temp upto 6500K & having 20000 burning hrs life with minimum @ L 70, system lumen output | | |

| Item No. | | Description | Unit | Rate |
|---|---|---|---|---|
| | | should be minimum with efficacy>80lm/W. LED driver PF > 0.9, THD < 20% & surge protection 5KV. The colour rendering index of LED light should be more than 80. Housing made of pressure die cast aluminium / CRCA powder coated frame with high transmission diffuser. Submission LM 80-08 Form LED Source Manufacturer & LM79-08 / IS16106 from NABL approved lab. Manufacturer manadatory. i/c mounting arrangment, connection wire, testing etc. to complete the job. Guarantee as per tender agreement condition. | | |
| | | 32/40W | each | 7478.00 |
| | | 72/80W | each | 14146.00 |
| 21.54 | Post top lantern LED fitting | Supply and fixing integral post top lantern LED fitting comprises of copper dust finish cast aluminium spigot and spun aluminium canopy fixed with opal polycarbonate, pipe arrangement for vertical mounting, open construction driver and accessories are wired upto terminal block. LED of 1 to 3 Watt each assembled on single MCPCB, having color temp upto 6500K & having 50000 burning hrs. life with minimum @ L 70, system lumen output should be minimum with efficacy>100lm/Watt. LED driver PF > 0.95 & surge protection 10KV. The colour rendering index of LED light should be more than 70. Submission LM 79-08/IS16106 (2012), IEC60598, IEC61347i/c connection wire, testing etc. to complete the job. | | |
| | 40W | 40 Watt LED, color temp 3000-6500k as required. | each | 7768.00 |
| | 25W | 25 Watt LED, color temp 3000-6500k as required. | each | 4106.00 |
| 21.55 | Street light with high power LED system | Supply and fixing street light with high power LED of 3 to 6 W each on existing bracket assembled on single MCPCB, system lumens output with efficacy>120 lm/W. luminiare having color temp upto 6500K & 50000 burning hrs. life with minimum @ L 70, The colour rendering index of LED light should be more than 70. Luminiare comprises of driver, PF> 0.95 & surge protection 10KV. Housing made of pressure die cast aluminium with heat resistant flat glass/Lens type, IP65 protection. Submission LM 79-08/IS16106 (2012), IEC60598, IEC61347i/c connection wire, testing etc. to complete the job. | | |
| | 15-20W | 15Watt to 20Watt, color temp 3000-6500k as required. | each | 2316.00 |

| Item No. | | Description | Unit | Rate |
|---|---|---|---|---|
| | 24-25W | 24Watt to 25Watt, color temp 3000-6500k as required. | each | 2737.00 |
| 21.56 | Street light with high power LED system | Supply and fixing street light with high power LED of 3 to 6 Watt each on existing bracket assembled on single MCPCB and additional unique peanut lens on each LED, system lumens output with efficacy>150 lm/Watt. luminiare having color temp upto 6500K & 50000 hrs. burning life with minimum @ L 70, The colour rendering index of LED light should be more than 70. Luminiare comprises of driver, PF> 0.95 & surge protection 10KV. Housing made of pressure die cast aluminium with heat resistant flat glass, IP66 protection. Submission LM 79-08/IS16106 (2012), IEC60598, IEC61347 i/c connection wire, testing etc. to complete the job. | | |
| (i) | 40W | 40Watt to 48Watt, color temp 3000-6500k as required | each | 6105.00 |
| (ii) | 60W | 60Watt, color temp 3000-6500k as required. | each | 9791.00 |
| (iii) | 90W | 90Watt, color temp 3000-6500k as required. | each | 14173.00 |
| (iv) | 120W | 120Watt, color temp 3000-6500k as required. | each | 21183.00 |
| (v) | 150W | 150Watt, color temp 3000-6500k as required. | each | 22936.00 |
| (vi) | 180W/177Watt | 180W/177Watt, color temp 3000-6500k as required. | each | 33452.00 |
| (vii) | 250/240W | 250/240Watt, color temp 3000-6500k as required. | each | 34766.00 |
| 21.57 | LED flood light | Supplying and fixing flood light with high power LED of 3 to 6 Watt each assembled on single MCPCB and additional unique peanut lens on each LED, system lumens output with efficacy>170 lm/Watt. luminiare having color temp upto 6500K & 50000 hrs. burning life with minimum @ L 70, The colour rendering index of LED light should be more than 70. Luminiare comprises of driver, PF > 0.95 & surge protection 10KV. Housing made of pressure die cast aluminium with heat resistant flat glass, IP66 protection. Submission LM 79-08/IS16106 (2012), IEC60598, IEC61347i/c connection wire, testing etc. to complete the job. | | |
| (i) | 10W | 10Watt, color temp 3000-6500k as required. | each | 1285.00 |
| (ii) | 30W | 30Watt, color temp 3000-6500k as required. | each | 5178.00 |
| (iii) | 60W | 60Watt, color temp 3000-6500k as required. | each | 7872.00 |
| (iv) | 90/100W | 90/100Watt, color temp 3000-6500k as required. | each | 11553.00 |
| (v) | 120W | 120Watt, color temp 3000-6500k as required. | each | 16022.00 |
| (vi) | 150W | 150Watt, color temp 3000-6500k as required. | each | 21572.00 |
| (vii) | 180W | 180Watt, color temp 3000-6500k as required. | each | 27268.00 |
| (viii) | 250W | 250Watt, color temp 3000-6500k as required. | each | 34279.00 |
| (ix) | 300W | 300Watt, color temp 3000-6500k as required. | each | 34717.00 |

| Item No. | | Description | Unit | Rate |
|---|---|---|---|---|
| 21.58 | Aviation light LED | Supply, Installation testing and commissioning of red colour aviation light LED integral type comprising of aluminium powder coated housing with polycarbonate enclosure IP 65 on existing pipe/support including connection, testing etc. as required. Guarantee as per tender agreement condition. | | |
| | 20 watt(300 LED) | 20 watt (300 LED) | each | 3958.00 |
| 21.59 | Bollard | Supply, fixing & testing of approved make of LED integral type bollard cylindrical/square shape housing cast aluminium dome shape top cover fixed with cylindrical shape acrylic cover having base plate with holes for direct mounting complete with all accessories including preparation of foundation, fixing, connection etc. as required. LED of 1 to 3 Watt each assembled on single MCPCB, system lumens output with efficacy>90 lm/W. luminaire having color temp upto 6500K & 50000 burning hrs. life with minimum @ L 70, The colour rendering index of LED light should be more than 70. Luminiare, comprises of driver PF> 0.95 & surge protection 10KV. IP65 protection. Submission LM 79-08/IS16106 (2012), IEC60598, IEC61347i/c connection wire, testing etc. to complete the job. | | |
| | 8-10W cylindrical shape | Bollard LED 8-10 Watt cylindrical shape height > 700mm, color temp 3000-6000k | each | 4614.00 |
| | 8-10W mushroom shape | Bollard LED 8-10 Watt mushroom shape height > 600mm, color temp 3000-6000k | each | 4344.00 |
| 21.60 | LED strip | Supplying, fixing & testing of approved make flexible LED strip comprising of 60 LED per meter W/O driver for direct mounting complete with all accessories including connection lead, fixing, connection as required. Submission LM 79-08/IS16106 (2012), IEC60598, IEC61347i/c connection wire, testing etc. to complete the job. | | |
| | 24/25W | Flexible LED strip 24Watt/25Watt - 5 M indoor cove light W/O Driver IP 20 -33 | each | 1236.00 |
| | 72/75W | Flexible LED strip 72Watt/75Watt - 5 M indoor light W/O driver IP 20 - 33 | each | 1859.00 |
| | 24/25W | Flexible LED strip 24Watt/25Watt - 5 M outdoor cove light W/O driver IP65 | each | 1994.00 |
| | 72/75W | Flexible LED strip 72Watt/75Watt - 5 M outdoor light W/O driver IP65 | each | 2805.00 |
| 21.61 | Power supply for LED strip | Supplying, fixing & testing of approved make Power supply for LED strip | | |
| | 25/30W | Power supply 25Watt/30Watt for LED strip IP25/IP33 | each | 533.00 |
| | 48W | Power supply 48Watt for LED strip IP33 | each | 755.00 |

| Item No. | | Description | Unit | Rate |
|---|---|---|---|---|
| | 72/75W | Power supply 72Watt/75Watt for LED strip IP25/IP33 | each | 1020.00 |
| | 25/30W | Power supply 25Watt/30Watt for LED strip IP65 | each | 1331.00 |
| | 48W | Power supply 48Watt for LED strip IP65 | each | 1598.00 |
| | 72/75W | Power supply 72Watt/75Watt for LED strip IP65 | each | 3905.00 |
| 21.62 | LED Bulk Head | Supplying, fixing and testing of approved make LED bulkhead ABS body, PC diffuser including fixing on wall as required, with necessary material complete | | |
| | 6W | 6W IP 54 | each | 1484.00 |
| | 10W | 10W IP 55 | each | 1805.00 |
| 21.63 | Integral step/skirt LED light | Supply, fixing and testing of approved make integral step/skirt LED light suitable for 3 module box and plate including fixing on wall as required, with necessary material to complete the job. Guarantee as per tender agreement condition. | each | 676.00 |
| 21.64 | LED Fire signage, single side hanged | Supplying and fixing LED fire signage i/c 6V. 9AH, lithium battery sealed 2hrs backup single side, hanged/fixed on wall or ceiling etc. as required with necessary material to complete the job. Guarantee as per tender agreement condition. | each | 5953.00 |
| 21.65 | LED Fire signage, double side hanged | Supplying and fixing LED fire signage i/c 6V. 9AH, lithium battery sealed 2hrs backup double side hanged/fixed on wall or ceiling etc. as required with necessary material to complete the job. Guarantee as per tender agreement condition. | each | 7093.00 |
| 21.66 | Fixing of street light fitting | Fixing of street light bracket of required angular shape and size pipe bracket on wall/pole including M.S. clamp not less than 40X5mm flat, nut, bolt 12mm of required length with necessary material including connection 3 core flexible wire etc. complete as required excluding cost of pipe paid separately. | each | 498.00 |
| 21.67 | Panel LED driver | Supplying Fixing and testing of approved and suitable make LED driver for an existing LED luminary panel. | | |
| | 1'x1' 24W | LED Luminiare 1'x1', 24W Panel Driver | each | 558.00 |
| | 2'x2', 36W | LED Luminiare 2'x2', 36W Panel Driver | each | 670.00 |
| 21.68 | Street Light LED Driver | Supplying fixing and testing of approved and suitable make LED driver for an existing LED street light. | | |
| | 24-25W | 24Watt to 25Watt, color temp 3000-6500k | each | 537.00 |
| | 40W | 40Watt to 48Watt, color temp 3000-6500k | each | 903.00 |
| | 60W | 60Watt, color temp 3000-6500k | each | 1135.00 |
| | 90W | 90Watt, color temp 3000-6500k | each | 1683.00 |
| | 120W | 120Watt, color temp 3000-6500k | each | 2279.00 |

| Item No. | | Description | Unit | Rate |
|---|---|---|---|---|
| | 150W | 150Watt, color temp 3000-6500k | each | 2582.00 |
| 21.69 | Flood Light LED Driver | Supplying, fixing and testing of approved and suitable make LED driver for an existing LED Flood light. | | |
| | 60W | 60Watt, color temp 3000-6500k | each | 808.00 |
| | 90W | 90Watt, color temp 3000-6500k | each | 1420.00 |
| | 120W | 120Watt, color temp 3000-6500k | each | 1546.00 |
| | 150W | 150Watt, color temp 3000-6500k | each | 2046.00 |
| 21.70 | Decorative Street Light | Supply & Fixing of Decorative LED Street Light of 1 to 3 watt with each , system lumens output with Efficacy>=135lm/W(@6500k), It's body made of Die cast aluminium using pressure die-casting process. Hole for coupling on pipes Ø 60 mm, for lateral installation. Silicon gasket enusures the IP protection. IP Rating - IP66. Power LED Module with high efficacy LED's on single Printed Circuit Board with metal core plate. Colour Rendering Index: Ra > 70. Electronic Power Supply for LED Module, which offers Protection against Short Circuit, Over- Voltage & Over- Current, with in-built surge protection upto 4kv (an additional surge protection device of 10kV also provided in-built the fixture). PF >0.95 and THD<10%. Lens cover in PC (IK09). Estimated LED Lifetime is L70 @ 1,00,000 hours minimum. LM79-80 report from LED manufaturer should be submitted. Safety test report as per IEC 60598-2-3; IEC 60598-1 / IS 10322 IEC 61347 i/c connection wire , testing etc-Supporting Test Report from NABL approved lab, with 5 year on site warranty. | | |
| | 40 W | 40Watt, Color Temp 3000-6500k as required | each | 6578.00 |
| | 60 W | 60Watt, Color Temp 3000-6500k as required | each | 9927.00 |
| | 80/90 W | 80/90Watt, Color Temp 3000-6500k as required | each | 14113.00 |
| | 100 W | 100Watt, Color Temp 3000-6500k as required | each | 15548.00 |
| | 120W | 120Watt, Color Temp 3000-6500k as required | each | 21289.00 |
| | 150 W | 150Watt, Color Temp 3000-6500k as required | each | 23083.00 |
| | 180 W | 180Watt, Color Temp 3000-6500k as required | each | 29900.00 |
| | 200 W | 200Watt, Color Temp 3000-6500k as required | each | 30259.00 |
| | 250 W | 250Watt, Color Temp 3000-6500k as required | each | 34684.00 |
| | | **EARTHING AND LOOP EARTHING** | | |
| 21.71 | With G.I. earth pipe | Earthing with G.I. earth pipe 4.5 metre long and 40 mm dia with masonry encloser in cement mortar, cover plate having locking arrangement on the top etc. (but without charcoal or coke and salt) complete as required | each | 3312.00 |
| 21.72 | Add extra for using salt and charcoal/coke for pipe earth electrode | Add extra for using salt and charcoal/coke for pipe earth electrode as required including excavation & refilling | | |

| Item No. | | Description | Unit | Rate |
|---|---|---|---|---|
| | Excavation | Excavation 4.5 m by manual labour | each | 1942.00 |
| | Excavation | Excavation by making hole with auger | each | 1622.00 |
| 21.73 | Earthing with G.I. earth plate 600x600x6 mm | Earthing with G.I. earth plate 600mm X 600mm X 6mm thick including accessories and providing masonry encloser in cement mortar, cover plate having locking arrangement on the top and G.I. watering pipe 20mm dia 2.7 metre long etc. (but without charcoal or coke and salt) complete as required | each | 7487.00 |
| 21.74 | Earthing with copper earth plate 600x600x3 mm | Earthing with copper earth plate 600mm X 600mm X 3mm thick including accessories and providing masonry encloser in cement mortar, cover plate having locking arrangement on the top and G.I. watering pipe 20mm dia 2.7 metre long etc. (but without charcoal or coke and salt) complete as required | each | 16341.00 |
| 21.75 | Add extra for using salt and charcoal/ coke for G.I /copper plate (murrum) | Add extra for using salt and charcoal/coke for G.I. plate or copper plate earth electrode as required including excavation and refilling the pit in all type of soil, murrum | each | 1655.00 |
| 21.76 | Add extra for using salt and charcoal/coke for G.I/copper plate (black soil) | Add extra for using salt and charcoal/coke for G.I. plate or copper plate earth electrode as required including excavation in hard rock and refilling the pit with black soil. | each | 2783.00 |
| 21.77 | GI wire conductor earth electrode | Supply and laying 4.0mm (8 SWG) dia. G.I. wire at 0.5 metre below ground level as conductor earth electrode including jointing etc. as required. | metre | 21.00 |
| 21.78 | Cu wire conductor earth electrode | Supply and laying 4mm dia. copper wire at 0.5 metre below ground level as conductor earth electrode including soldering etc. as required. | metre | 114.00 |
| 21.79 | G.I. wire conductor earth electrode | Supply and laying 4.87mm 6 SWG G.I. wire at 0.5 metre below ground level as conductor earth electrode including jointing etc. as required. | metre | 25.00 |
| 21.80 | 25x5 mm G.I. strip | Supply and laying 25mm X 5mm G.I. strip at 0.5 metre below ground level as strip earth electrode including jointing etc. as required. | metre | 130.00 |
| 21.81 | 25x6 mm G.I. strip | Supply and laying 25mm X 6mm G.I. strip at 0.5 metre below ground level/surface as strip earth electrode including jointing etc. as required. | metre | 147.00 |
| 21.82 | 25x5 mm Cu strip | Providing and fixing 25mm X 5mm copper strip in 40 mm dia G.I. Pipe from earth electrode as required | metre | 1490.00 |
| 21.83 | 25x5 G.I. strip | Providing and fixing 25mm X 5mm G.I. strip in 40 mm dia G.I. pipe from earth electrode as required | metre | 928.00 |
| 21.84 | Earth connections from earth electrode with 4.00 | Providing and laying earth connections from earth electrode with 4.00 mm dia G.I. wire in 15mm dia G.I. Pipe from earth electrode as | metre | 222.00 |

| Item No. | | Description | Unit | Rate |
|---|---|---|---|---|
| | mm diaG.I. wire | required. | | |
| 21.85 | Earth connections from earth electrode with 4.00 mm dia cu. Wire | Providing and laying earth connections from earth electrode with 4.00 mm dia copper wire in 15mm dia G.I. Pipe from earth electrode as required. | metre | 308.00 |
| 21.86 | 25x5 mm Cu strip on surface | Providing and fixing 25mm X 5mm copper strip on surface or in recessed for connection etc. as required | metre | 1137.00 |
| 21.87 | 25x5 mm G.I. strip on surface | Providing and fixing 25mm X 5mm G.I. strip on surface or in recessed for connection etc. as required | metre | 130.00 |
| 21.88 | 6 SWG dia G.I. wire on surface/recessed | Providing and fixing 4.0mm (8 SWG) dia G.I. wire on surface or in recessed for loop earthing as required | metre | 25.00 |
| 21.89 | Loop earthing | Providing and fixing 4.87mm (6 SWG) dia. G.I. wire on surface or in recessed for loop earthing as required | metre | 30.00 |
| 21.90 | 4 mm cu. wire on surface/recessed | Providing and fixing 4.00mm copper wire on surface or in recessed for loop earthing as required | metre | 127.00 |
| 21.91 | 50x6 mm G.I. strip earth electrode | Supply and laying 50mm X 6mm G.I. strip at 0.5 metre below ground level/surface as strip earth electrode including jointing etc. as required. | metre | 319.00 |
| | | **LIGHTNING CONDUCTOR** | | |
| 21.92 | 18mm dia 500mm long bright nickel plated copper tube | Providing and fixing of lightning conductor finial made of 18mm dia 500mm long bright nickel plated copper tube tapered having single prong at top with bright nickel plated brass base M10 for connection and intersection of flat or round conductor including holes, clamp complete as required. | each | 3947.00 |
| 21.93 | 18mm dia 500mm long bright nickel plated GI tube | Providing and fixing of lightning conductor finial made of 18mm dia 500mm long bright nickel plated GI tube tapered having single prong at top with bright nickel plated GI base M10 for connection and intersection of flat or round conductor including holes, clamp complete as required. | each | 1285.00 |
| 21.94 | Fixing of lightning conductor, 25 mm dia 300 mm long cu./G.I. tube | Fixing of lightning conductor finial made of 25mm dia 300mm long copper tube/G.I. tube having single prong at top with base plate etc. complete as required. | each | 535.00 |
| 21.95 | Riveting/sweating and soldering | Riveting/sweating and soldering of copper/G.I. Tape (with another copper/G.I. Tape, base of finial or any other metallic object) as required | each | 107.00 |
| 21.96 | Cu. tape 20x3 mm (for horizontal runs) | Providing and fixing copper tape 20mm X 3mm thick on parapet and surface of wall of lightning conductor as required (for horizontal runs) | metre | 567.00 |
| 21.97 | Cu. tape 20x3 mm (for vertical runs) | Providing and fixing copper tape 20mm X 3mm thick on parapet and surface of wall of | metre | 591.00 |

| Item No. | | Description | Unit | Rate |
|---|---|---|---|---|
| | | lightning conductor as required (for vertical runs) | | |
| 21.98 | G.I. tape 20x3 mm (for horizontal runs) | Providing and fixing G.I. tape 20mm X 3mm thick on parapet and surface of wall of lightning conductor as required (for horizontal runs) | metre | 74.00 |
| 21.99 | G.I. tape 20x3 mm (for vertical runs) | Providing and fixing G.I. tape 20mm X 3mm thick on parapet and surface of wall of lightning conductor as required (for vertical runs) | metre | 85.00 |
| 21.100 | Fixing cu./G.I. tape 20x3 mm (for horizontal runs) | Fixing of copper/G.I. tape 20mm X 3mm thick on parapet and surface of wall for lightning conductor as required (for horizontal runs) | metre | 23.00 |
| 21.101 | Fixing cu./G.I. tape 20x3 mm (for vertical runs) | Fixing of copper/G.I. tape 20mm X 3mm thick on parapet and surface of wall for lightning conductor as required (for vertical runs) | metre | 27.00 |
| 21.102 | Testing joint 20x3 mm cu. strip | Providing and fixing "Testing Joint" made by 20mm X 3mm thick copper strip 125mm long with 4 Nos of tinned Brass bolts, Nut, check nuts and spring washers etc. complete as required. | each | 525.00 |
| 21.103 | Cu. tape 32x6 mm | Providing and laying copper tape 32mm X 6mm thick for earth electrode directly in ground as required | metre | 1686.00 |
| 21.104 | G.I. tape 38x6 mm | Providing and laying G.I. tape 38mm X 6mm thick for earth electrode directly in ground as required | metre | 222.00 |
| 21.105 | Laying cu./G.I. tape 32x6mm | Laying of copper/G.I. tape 32mm X 6mm thick for earth electrode directly in ground as required | metre | 23.00 |
| | **EARTHING FOR DOMESTIC ELECTRICAL INSTALLATION** | | | |
| 21.106 | Earthing 16 mm dia 2.5 metre long galvanized iron or steel rod electrode | Providing and fixing earthing arrangement with 16mm dia 2.5 metre long galvanized iron or steel rod electrode including packing of charcoal powder and salt as per specification watering pipe 19mm dia G.I. connection, cover plate having locking arrangement on the top etc. complete with refilling the pit as required, including excavation of earth pit in soil/ soft rock. | each | 3318.00 |
| 21.107 | Earthing 38.1 mm dia 2.5 metre long galvanized iron pipe electrode | Providing and fixing earthing arrangement with 38.1mm dia 2.5 metre long galvanized iron pipe electrode including packing of charcoal powder and salt as per specification watering pipe 19mm dia G.I. connection, cover plate having locking arrangement on the top etc. complete with refilling the pit as required, including excavation of earth pit in soil / soft rock | each | 3670.00 |
| 21.108 | G.I. earth lead | Supply and laying G.I. earth lead duly installed on wall or in G.I. Pipe including connection etc. complete as required. | | |

| Item No. | | Description | Unit | Rate |
|---|---|---|---|---|
| | 13 sq. mm. | 13 sq. mm. ( 4 mm dia ) | metre | 26.00 |
| | 25 sq. mm. | 25 sq. mm. ( 5.6 mm dia ) | metre | 38.00 |
| | 40 sq. mm. | 40 sq. mm. ( 7.1 mm dia ) | metre | 48.00 |
| 21.109 | G.I. earth conductor | G.I. or steel earth continuity conductor | | |
| | 6 sq. mm. | 6 sq. mm. | metre | 22.00 |
| | 10 sq. mm. | 10 sq. mm. | metre | 29.00 |
| 21.110 | Earth conductor of 4 sq. mm. al. | Providing and laying earth continuity conductor of 4 sq. mm. aluminium along the run of final sub circuit in PVC conduit in an approved manner. | metre | 28.00 |
| 21.111 | PVC insulated cu. cable earth continuity conductor | Supply and drawing PVC insulated copper cable (earth continuity conductor) of green colour FRLS with copper multi strand conductor ISI mark in existing conduit along with other wires as required as per specification : | | |
| | 4 sq. mm. | 4 sq. mm. | metre | 52.00 |
| | 6 sq. mm. | 6 sq. mm. | metre | 71.00 |
| | 10 sq. mm. | 10 sq. mm. | metre | 102.00 |
| | 16 sq. mm. | 16 sq. mm. | metre | 156.00 |
| | 25 sq. mm. | 25 sq. mm. | metre | 234.00 |
| 21.112 | Safe earthing electrode consisting pipe-in-pipe technology | Supply and erection of approved make safe earthing electrode consisting pipe-in-pipe technology as per IS 3043-1987 made of corrosion free G.I. pipes with constant ohmic value surrounded by highly conductive compound with high charge dissipation suitable for effective and maintenance free earthing in earth pit of minimum bore dia. 150mm size, as mentioned below | | |
| | 2 Metre G.I. pipe, outer dia 50 mm | With 2 metre G.I. pipes having outer pipe dia of 50mm having 80-200 micron galvanizing, Inner pipe dia. of 25mm having 200-250 micron galvanizing, connection terminal dia. of 12mm in nominal soil with 25 kg (one Bag) back filling bentonite compound UL listed and confirm to relevant International/Indian standards around electrode upto ground level. | each | 8972.00 |
| | 3 Metre G.I. pipe, outer dia 50 mm | With 3 metre G.I. pipes having outer pipe dia of 50mm having 80-200 micron galvanizing, Inner pipe dia. of 25mm having 200-250 Micron galvanizing, connection terminal dia. of 12mm in nominal soil with 50 kg (two bag) back filling bentonite compound UL listed and confirm to relevant International/Indian standards around electrode upto ground level. | each | 13140.00 |
| | 2 Metre pipe, outer dia 80 mm | With 2 metre pipe of 80mm outer dia., 40mm inner dia and 14mm terminal dia. In soft/hard rock/marshy soil with 25 kgs. (one bag) back filling bentonite compound UL listed and confirm to relevant International/Indian | each | 18262.00 |

| Item No. | | Description | Unit | Rate |
|---|---|---|---|---|
| | | standards around electrode upto ground level. | | |
| | 3 Metre pipe, outer dia 80 mm | With 3 metre pipe of 80mm outer dia., 40mm inner dia and 14mm terminal dia. In soft/hard rock/marshy soil with 50 kgs. (two bag) back filling bentonite compound UL listed and confirm to relevant International/Indian standards around electrode upto ground level. | each | 21419.00 |
| 21.113 | Back filling bentonite compound | Providing back filling bentonite compound suitable for safe earthing device. The product should be UL listed and confirm to relevant International/Indian standards. | kg | 134.00 |
| 21.114 | Polycarbonate heavy duty pit cover | Supply and installation of polycarbonate heavy duty pit cover tested for 5000Kg SWL for earth electrode at ground level size 300x300x155 mm with lockable lid. | each | 1501.00 |
| | | **HT AND LT CABLE** | | |
| 21.115 | XLPE insulated power cable, al. conductor | Supply of XLPE insulated power cable (conforming IS-7098 Part-I) 1100 Volt grade, 1 core/2 core/3½ core/4 core ISI marked with alu. stranded /solid conductor | | |
| (A) | Unarmoured1 core | Unarmoured 1 core | | |
| (i) | 16 sq. mm. | 16 sq. mm. | metre | 46.00 |
| (ii) | 25 sq. mm. | 25 sq. mm. | metre | 69.00 |
| (iii) | 35 sq. mm. | 35 sq. mm. | metre | 85.00 |
| (iv) | 50 sq. mm. | 50 sq. mm. | metre | 109.00 |
| (v) | 70 sq. mm. | 70 sq. mm. | metre | 143.00 |
| (vi) | 95 sq. mm. | 95 sq. mm. | metre | 187.00 |
| (vii) | 120 sq. mm. | 120 sq. mm. | metre | 223.00 |
| (viii) | 150 sq. mm. | 150 sq. mm. | metre | 266.00 |
| (ix) | 300 Sq. Mm. | 300 Sq. Mm. | metre | 483.00 |
| | Armoured 1 core | Armoured 1 core | | |
| (x) | 16 sq. mm. | 16 sq. mm. | metre | 82.00 |
| (xi) | 25 sq. mm. | 25 sq. mm. | metre | 105.00 |
| (xii) | 35 sq. mm. | 35 sq. mm. | metre | 124.00 |
| (xiii) | 50 sq. mm. | 50 sq. mm. | metre | 149.00 |
| (xiv) | 70 sq. mm. | 70 sq. mm. | metre | 194.00 |
| (xv) | 95 sq. mm. | 95 sq. mm. | metre | 212.00 |
| (xvi) | 120 sq. mm. | 120 sq. mm. | metre | 256.00 |
| (xvii) | 150 sq. mm. | 150 sq. mm. | metre | 302.00 |
| (xviii) | 300 sq. mm. | 300 sq. mm. | metre | 572.00 |
| (B) | Unarmoured 2 core | Unarmoured 2 core | metre | |
| (i) | 2.5 sq. mm. | 2.5 sq. mm. | metre | 42.00 |
| (ii) | 4 sq. mm. | 4 sq. mm. | metre | 49.00 |
| (iii) | 6 sq. mm. | 6 sq. mm. | metre | 63.00 |
| (iv) | 10 sq. mm. | 10 sq. mm. | metre | 87.00 |
| (v) | 16 sq. mm. | 16 sq. mm. | metre | 104.00 |
| (C) | Armoured 2 core | Armoured 2 core | | |
| (i) | 2.5 sq. mm. | 2.5 sq. mm. | metre | 73.00 |
| (ii) | 4 sq. mm. | 4 sq. mm. | metre | 80.00 |
| (iii) | 6 sq. mm. | 6 sq. mm. | metre | 100.00 |
| (iv) | 10 sq. mm. | 10 sq. mm. | metre | 121.00 |
| (v) | 16 sq. mm. | 16 sq. mm. | metre | 131.00 |

| Item No. | | Description | Unit | Rate |
|---|---|---|---|---|
| (D) | Unarmoured 3 core | Unarmoured 3 core | | |
| (i) | 6 sq. mm. | 6 sq. mm. | metre | 78.00 |
| (ii) | 10 sq. mm. | 10 sq. mm. | metre | 104.00 |
| (iii) | 16 sq. mm. | 16 sq. mm. | metre | 116.00 |
| (iv) | 25 sq. mm | 25 sq. mm | metre | 167.00 |
| (v) | 35 sq. mm | 35 sq. mm | metre | 228.00 |
| (vi) | 50 sq. mm | 50 sq. mm | metre | 285.00 |
| (vii) | 70 sq. mm | 70 sq. mm | metre | 401.00 |
| (viii) | 95 sq. mm. | 95 sq. mm. | metre | 497.00 |
| (E) | Armoured 3 core | Armoured 3 core | | |
| (i) | 6 sq. mm. | 6 sq. mm. | metre | 112.00 |
| (ii) | 10 sq. mm. | 10 sq. mm. | metre | 142.00 |
| (iii) | 16 sq. mm. | 16 sq. mm. | metre | 146.00 |
| (iv) | 25 sq. mm | 25 sq. mm | metre | 192.00 |
| (v) | 35 sq. mm | 35 sq. mm | metre | 252.00 |
| (vi) | 50 sq. mm | 50 sq. mm | metre | 326.00 |
| (vii) | 70 sq. mm | 70 sq. mm | metre | 433.00 |
| (viii) | 95 sq. mm. | 95 sq. mm. | metre | 540.00 |
| (F) | 3½ core unarmoured | 3½ core unarmoured | | |
| (i) | 25 sq. mm | 25 sq. mm | metre | 210.00 |
| (ii) | 35 sq. mm | 35 sq. mm | metre | 250.00 |
| (iii) | 50 sq. mm | 50 sq. mm | metre | 338.00 |
| (iv) | 70 sq. mm | 70 sq. mm | metre | 458.00 |
| (v) | 95 sq. mm. | 95 sq. mm. | metre | 577.00 |
| (vi) | 120 sq. mm. | 120 sq. mm. | metre | 739.00 |
| (vii) | 150 sq. mm. | 150 sq. mm. | metre | 866.00 |
| (viii) | 185 sq. mm. | 185 sq. mm. | metre | 1100.00 |
| (ix) | 240 sq. mm. | 240 sq. mm. | metre | 1327.00 |
| (x) | 300 sq. mm. | 300 sq. mm. | metre | 1697.00 |
| (xi) | 400 sq. mm. | 400 sq. mm. | metre | 2142.00 |
| (G) | 3½ core armoured | 3½ core armoured | | |
| (i) | 25 sq. mm | 25 sq. mm | metre | 231.00 |
| (ii) | 35 sq. mm | 35 sq. mm | metre | 280.00 |
| (iii) | 50 sq. mm | 50 sq. mm | metre | 373.00 |
| (iv) | 70 sq. mm | 70 sq. mm | metre | 500.00 |
| (v) | 95 sq. mm. | 95 sq. mm. | metre | 625.00 |
| (vi) | 120 sq. mm. | 120 sq. mm. | metre | 780.00 |
| (vii) | 150 sq. mm. | 150 sq. mm. | metre | 904.00 |
| (viii) | 185 sq. mm. | 185 sq. mm. | metre | 1150.00 |
| (ix) | 240 sq. mm. | 240 sq. mm. | metre | 1460.00 |
| (x) | 300 sq. mm. | 300 sq. mm. | metre | 1792.00 |
| (xi) | 400 sq. mm. | 400 sq. mm. | metre | 2288.00 |
| (H) | 4 core unarmoured | 4 core unarmoured | | |
| (i) | 6 sq. mm. | 6 sq. mm. | metre | 92.00 |
| (ii) | 10 sq. mm. | 10 sq. mm. | metre | 119.00 |
| (iii) | 16 sq. mm. | 16 sq. mm. | metre | 148.00 |
| (iv) | 25 sq. mm | 25 sq. mm | metre | 221.00 |
| (v) | 35 sq. mm | 35 sq. mm | metre | 278.00 |
| (vi) | 50 sq. mm | 50 sq. mm | metre | 372.00 |
| (vii) | 70 sq. mm | 70 sq. mm | metre | 527.00 |
| (viii) | 95 sq. mm. | 95 sq. mm. | metre | 627.00 |

| Item No. | | Description | Unit | Rate |
|---|---|---|---|---|
| (ix) | 120 sq. mm. | 120 sq. mm. | metre | 755.00 |
| (x) | 150 sq. mm. | 150 sq. mm. | metre | 955.00 |
| (xi) | 185 sq. mm. | 185 sq. mm. | metre | 1172.00 |
| (xii) | 240 sq. mm. | 240 sq. mm. | metre | 1536.00 |
| (xiii) | 300 sq. mm. | 300 sq. mm. | metre | 1883.00 |
| (xiv) | 400 sq. mm. | 400 sq. mm. | metre | 2474.00 |
| (I) | 4 core armoured | 4 core armoured | | |
| (i) | 6 sq. mm. | 6 sq. mm. | metre | 151.00 |
| (ii) | 10 sq. mm. | 10 sq. mm. | metre | 181.00 |
| (iii) | 16 sq. mm. | 16 sq. mm. | metre | 218.00 |
| (iv) | 25 sq. mm | 25 sq. mm | metre | 279.00 |
| (v) | 35 sq. mm | 35 sq. mm | metre | 358.00 |
| (vi) | 50 sq. mm | 50 sq. mm | metre | 474.00 |
| (vii) | 70 sq. mm | 70 sq. mm | metre | 646.00 |
| (viii) | 95 sq. mm. | 95 sq. mm. | metre | 791.00 |
| (ix) | 120 sq. mm. | 120 sq. mm. | metre | 985.00 |
| (x) | 150 sq. mm. | 150 sq. mm. | metre | 1175.00 |
| (xi) | 185 sq. mm. | 185 sq. mm. | metre | 1483.00 |
| (xii) | 240 sq. mm. | 240 sq. mm. | metre | 1895.00 |
| (xiii) | 300 sq. mm. | 300 sq. mm. | metre | 2435.00 |
| (xiv) | 400 sq. mm. | 400 sq. mm. | metre | 3017.00 |
| 21.116 | High-tension XLPE cable, 1 core armoured al. conductor | Supply of approved high-tension XLPE cable (conforming IS-7098/II/85) as per ISI standard single core armoured with alu. solid/stranded conductor ISI marked as required | | |
| (A) | Single core XLPE cable 11 Kv grade | Single core XLPE cable 11 Kv grade | | |
| (i) | 50 sq. mm | 50 sq. mm | metre | 302.00 |
| (ii) | 70 sq. mm | 70 sq. mm | metre | 354.00 |
| (iii) | 95 sq. mm. | 95 sq. mm. | metre | 477.00 |
| (iv) | 120 sq. mm. | 120 sq. mm. | metre | 544.00 |
| (v) | 150 sq. mm. | 150 sq. mm. | metre | 592.00 |
| (vi) | 185 sq. mm. | 185 sq. mm. | metre | 696.00 |
| (vii) | 240 sq. mm. | 240 sq. mm. | metre | 774.00 |
| (viii) | 300 sq. mm. | 300 sq. mm. | metre | 885.00 |
| (B) | Single core XLPE cable 33 Kv grade | Single core XLPE cable 33 Kv grade | | |
| (i) | 95 sq. mm. | 95 sq. mm. | metre | 675.00 |
| (ii) | 120 sq. mm. | 120 sq. mm. | metre | 800.00 |
| (iii) | 150 sq. mm. | 150 sq. mm. | metre | 920.00 |
| (iv) | 185 sq. mm. | 185 sq. mm. | metre | 1013.00 |
| (v) | 240 sq. mm. | 240 sq. mm. | metre | 1158.00 |
| (vi) | 300 sq. mm. | 300 sq. mm. | metre | 1321.00 |
| 21.117 | High-tension XLPE cable, 3 core armoured al. conductor | Supply of approved high-tension XLPE cable (conforming IS-7098/II/85) as per ISI standard 3 core armoured with alu. solid/stranded conductor ISI marked as required | | |
| (A) | 3 core XLPE cable 11 Kv grade | 3 core XLPE cable 11 Kv grade | | |
| (i) | 50 sq. mm | 50 sq. mm | metre | 786.00 |
| (ii) | 70 sq. mm | 70 sq. mm | metre | 1016.00 |
| (iii) | 95 sq. mm. | 95 sq. mm. | metre | 1197.00 |

| Item No. | | Description | Unit | Rate |
|---|---|---|---|---|
| (iv) | 120 sq. mm. | 120 sq. mm. | metre | 1392.00 |
| (v) | 150 sq. mm. | 150 sq. mm. | metre | 1560.00 |
| (vi) | 185 sq. mm. | 185 sq. mm. | metre | 1782.00 |
| (vii) | 240 sq. mm. | 240 sq. mm. | metre | 2118.00 |
| (viii) | 300 sq. mm. | 300 sq. mm. | metre | 2470.00 |
| (B) | 3 core XLPE cable 33 Kv grade | 3 core XLPE cable 33 Kv grade | | |
| (i) | 95 sq. mm. | 95 sq. mm. | metre | 1672.00 |
| (ii) | 120 sq. mm. | 120 sq. mm. | metre | 1927.00 |
| (iii) | 150 sq. mm. | 150 sq. mm. | metre | 2418.00 |
| (iv) | 185 sq. mm. | 185 sq. mm. | metre | 2675.00 |
| (v) | 240 sq. mm. | 240 sq. mm. | metre | 3050.00 |
| (vi) | 300 sq. mm. | 300 sq. mm. | metre | 3400.00 |
| 21.118 | XLPE insulated FRLS heavy duty power cable | Supply of XLPE insulated FRLS heavy duty power cable conforming IS-7098 (Part-1) 1988, 1100 Volt grade, 2/3/4/8/10/12 core ISI marked with copper stranded/solid conductor. | | |
| (A) | Armoured 2 core | Armoured 2 core | | |
| (i) | 2.5 sq. mm. | 2.5 sq. mm. | metre | 136.00 |
| (ii) | 4 sq. mm. | 4 sq. mm. | metre | 183.00 |
| (iii) | 6 sq. mm. | 6 sq. mm. | metre | 241.00 |
| (iv) | 10 sq. mm. | 10 sq. mm. | metre | 369.00 |
| (B) | Armoured 3 core | Armoured 3 core | | |
| (i) | 2.5 sq. mm. | 2.5 sq. mm. | metre | 168.00 |
| (ii) | 4 sq. mm. | 4 sq. mm. | metre | 229.00 |
| (iii) | 6 sq. mm. | 6 sq. mm. | metre | 314.00 |
| (iv) | 10 sq. mm. | 10 sq. mm. | metre | 504.00 |
| (v) | 16 sq. mm. | 16 sq. mm. | metre | 723.00 |
| (vi) | 25 sq. mm. | 25 sq. mm. | metre | 1144.00 |
| (C) | Armoured 3½ core | Armoured 3½ core | | |
| (i) | 25 Sq. mm | 25 Sq. mm (10 Sq. mm 1/2 core). | metre | 1338.00 |
| (ii) | 35 Sq. mm | 35 Sq. mm (16 Sq. mm 1/2 core) | metre | 1759.00 |
| (iii) | 50 Sq. mm | 50 Sq. mm (25 Sq. mm 1/2 core) | metre | 2469.00 |
| (iv) | 70 Sq. mm | 70 Sq. mm (35 Sq. mm 1/2 core) | metre | 3317.00 |
| (v) | 95 Sq. mm | 95 Sq. mm (50 Sq. mm 1/2 core) | metre | 4816.00 |
| (vi) | 120 Sq. mm | 120 Sq. mm (70 Sq. mm 1/2 core) | metre | 6190.00 |
| (D) | Armoured 4 core | Armoured 4 core | | |
| (i) | 1.5 sq. mm. | 1.5 sq. mm. | metre | 148.00 |
| (ii) | 2.5 sq. mm. | 2.5 sq. mm. | metre | 217.00 |
| (iii) | 4 sq. mm. | 4 sq. mm. | metre | 302.00 |
| (iv) | 6 sq. mm. | 6 sq. mm. | metre | 430.00 |
| (v) | 10 sq. mm. | 10 sq. mm. | metre | 625.00 |
| (vi) | 16 sq. mm. | 16 sq. mm. | metre | 909.00 |
| (vii) | 25 sq. mm. | 25 sq. mm. | metre | 1455.00 |
| (E) | Armoured 8 core | Armoured 8 core | | |
| | 2.5 Sq. Mm. | 2.5 Sq. Mm. | metre | 394.00 |
| (F) | Armoured 10 core | Armoured 10 core | | |
| | 2.5 Sq. Mm. | 2.5 Sq. Mm. | metre | 469.00 |

| Item No. | | Description | Unit | Rate |
|---|---|---|---|---|
| (G) | Armoured 12 core | Armoured 12 core | | |
| | 2.5 Sq. Mm. | 2.5 Sq. Mm. | metre | 540.00 |
| (H) | Unarmoured 2 core | Unarmoured 2 core | | |
| | 2.5 sq. Mm. | 2.5 sq. Mm. | metre | 108.00 |
| (I) | Unarmoured 4 core | Unarmoured 4 core | | |
| | 2.5 sq. Mm. | 2.5 sq. Mm. | metre | 187.00 |
| (J) | Unarmoured 8 core | Unarmoured 8 core | | |
| | 2.5 sq. Mm. | 2.5 sq. Mm. | metre | 357.00 |
| (K) | Unarmoured 12 core | Unarmoured 12 core | | |
| | 2.5 sq. Mm. | 2.5 sq. Mm. | metre | 490.00 |
| 21.119 | Gland | Brass compression gland | | |
| | Gland for P.V.C. insulated armoured cable | Supplying and fixing heavy duty cable gland for P.V.C. insulated armoured cable with brass washer, rubber ring complete erected with cable and lead connection etc. as per specification complete. | | |
| (i) | 22 mm | Gland size 22mm suitable for cable 2,3,3½ & 4 x upto 6 sq. mm | each | 79.00 |
| (ii) | 22 mm | Gland size 22mm suitable for cable 2/3, 3½, 2/4 x 10 sq. mm or 2x 16 sq. mm | each | 97.00 |
| (iii) | 28 mm | Gland size 28mm for 3/4 x 16 sq. mm | each | 138.00 |
| (iv) | 32 mm | Gland size 32 mm for 2/3, 3½, 4 x 25 sq. mm OR 2/3, 3½ x 35 sq. mm or 2/3/3½ x 50 sq. mm. | each | 158.00 |
| (v) | 38 mm | Gland size 38 mm 3½ x 70 sq. mm, 3x95 sq. mm | each | 219.00 |
| (vi) | 45 mm | Gland size 45 mm 3/3½ x 120 sq. mm, 3½ x 95 sq. mm, 3 x 150 sq. mm. | each | 260.00 |
| (vii) | 50 mm | Gland size 50 mm ,3½ x 150 sq. mm,3 x 185 sq. mm | each | 290.00 |
| (viii) | 57 mm | Gland size 57 mm, 3 x 225 sq. mm, 3 ½ x 185 sq. mm | each | 440.00 |
| (ix) | 70 mm | Gland size 70 mm, 3 x 240 sq. mm, 3½ x 300 sq. mm | each | 605.00 |
| (x) | 82 mm | Gland size 82 mm 3½ x 400 sq. mm | each | 864.00 |
| 21.120 | Ferrules | Supplying and fixing ferrules as per IS - specification suitable for following size of cable for circuit identification including connection as required complete | | |
| | For conductor size- | For conductor size- | | |
| (i) | 2.5-6.00 sq. mm | 2.5 to 6.00 sq. mm | each | 3.00 |
| (ii) | 10.00 sq. mm | 10.00 sq. mm | each | 4.00 |
| (iii) | 16.00 sq. mm | 16.00 sq. mm | each | 4.00 |
| (iv) | 25.00 sq. mm | 25.00 sq. mm | each | 6.00 |
| (v) | 35.00 sq. mm | 35.00 sq. mm | each | 7.00 |
| (vi) | 50.00 sq. mm | 50.00 sq. mm | each | 11.00 |
| (vii) | 70.00 sq. mm | 70.00 sq. mm | each | 14.00 |
| (viii) | 95.00 sq. mm | 95.00 sq. mm | each | 16.00 |

| Item No. | | Description | Unit | Rate |
|---|---|---|---|---|
| (ix) | 120.00 sq. mm | 120.00 sq. mm | each | 18.00 |
| (x) | 150.00 sq. mm | 150.00 sq. mm | each | 25.00 |
| (xi) | 185.00 sq. mm | 185.00 sq. mm | each | 32.00 |
| (xii) | 240.00 sq. mm | 240.00 sq. mm | each | 46.00 |
| (xiii) | 300.00 sq. mm | 300.00 sq. mm | each | 57.00 |
| (xiv) | 400.00 sq. mm | 400.00 sq. mm | each | 70.00 |
| 21.121 | Crimping type alum. lugs | Supplying and fixing crimping type alum. lugs as per I.S.S. specification suitable for following size of cable with alu./copper solid/stranded conductor evenly crimped with high/pressure tool and connected to switch gear/bus/M.C.C.B./M.C.B. etc. as required complete. | | |
| | For conductor size- | For conductor size- | | |
| (i) | 6 -16 sq. mm | 6mm to 16 sq. mm | each | 8.00 |
| (ii) | 25 sq. mm | 25 sq. mm | each | 14.00 |
| (iii) | 35 sq. mm | 35 sq. mm | each | 22.00 |
| (iv) | 50 sq. mm | 50 sq. mm | each | 22.00 |
| (v) | 70 sq. mm | 70 sq. mm | each | 35.00 |
| (vi) | 95 sq. mm. | 95 sq. mm. | each | 48.00 |
| (vii) | 120 sq. mm. | 120 sq. mm. | each | 58.00 |
| (viii) | 150 sq. mm. | 150 sq. mm. | each | 66.00 |
| (ix) | 180 sq. mm | 180 sq. mm | each | 76.00 |
| (x) | 240 sq. mm. | 240 sq. mm. | each | 116.00 |
| (xi) | 300 sq. mm. | 300 sq. mm. | each | 138.00 |
| (xii) | 400 sq. mm. | 400 sq. mm. | each | 213.00 |
| 21.122 | Pin terminal lugs | Supplying & fixing pin terminal lugs as per ISS specification suitable for cable evenly cramped with high pressure tool & connection to switch gear terminal as required complete. | | |
| (A) | Aluminum lugs pin type | Aluminum lugs pin type | | |
| (i) | Upto 16 sq. mm conductor | Upto 16 sq. mm conductor size | each | 19.00 |
| (ii) | Upto 25 sq. mm conductor | Upto 25 sq. mm conductor size | each | 22.00 |
| (iii) | 35 sq. mm conductor | 35 sq. mm conductor size | each | 26.00 |
| (B) | Copper lugs pin type | Copper lugs pin type | | |
| (i) | Upto 16 sq. mm conductor | Upto 16 sq. mm conductor size | each | 30.00 |
| (ii) | Upto 25 sq. mm conductor | Upto 25 sq. mm conductor size | each | 42.00 |
| (iii) | 35 sq. mm conductor | 35 sq. mm conductor size | each | 50.00 |
| (C) | Aluminum lugs tube type | Aluminum lugs tube type | | |
| (i) | Upto 16 sq. mm conductor | Upto 16 sq. mm conductor size | each | 6.00 |
| (ii) | 25 sq. mm | 25 sq. mm conductor size | each | 10.00 |

| Item No. | | Description | Unit | Rate |
|---|---|---|---|---|
| | conductor | | | |
| (iii) | 35 sq. mm conductor | 35 sq. mm conductor size | each | 13.00 |
| (iv) | 50 sq. mm. conductor | 50 sq. mm. conductor size | each | 17.00 |
| (D) | Copper lugs tube type | Copper lugs tube type | | |
| (i) | Upto 16 sq. mm conductor | Upto 16 sq. mm conductor size | each | 13.00 |
| (ii) | 25 sq. mm conductor | 25 sq. mm conductor size | each | 22.00 |
| (iii) | 35 sq. mm conductor | 35 sq. mm conductor size | each | 32.00 |
| (iv) | 50 sq. mm. conductor | 50 sq. mm. conductor size | each | 43.00 |
| 21.123 | Cable end termination with heat shrinkable jointing kit | Providing &making cable end termination with heat shrinkable jointing kit complete with all accessories including lugs suitable for 33 KV/11 KV single core XLPE alum. conductor cable as required as per specification and as per accepted standard including connection testing complete. | | |
| (A) | 11 KV XLPE single core | Heat shrinkable joint, kit 11 KV XLPE single core | | |
| | 11 KV XLPE cable O.D. termination | Heat shrinkable Jointing kit 11 KV XLPE cable O.D. termination | | |
| (i) | 1 x 16-50 sq. mm. | 1 x 16-50 sq. mm. | each | 3214.00 |
| (ii) | 1 x 70-120 sq. mm. | 1 x 70-120 sq. mm. | each | 3511.00 |
| (iii) | 1 x 150-300 sq. mm. | 1 x 150-300 sq. mm. | each | 3960.00 |
| (iv) | 1 x 400-630 sq. mm. | 1 x 400-630 sq. mm. | each | 5714.00 |
| (B) | 11 KV XLPE cable I.D. termination | Heat shrinkable jointing kit 11 KV XLPE cable I.D. termination | | |
| (i) | 1 x 16-50 sq. mm. | 1 x 16-50 sq. mm. | each | 2940.00 |
| (ii) | 1 x 70-120 sq. mm. | 1 x 70-120 sq. mm. | each | 3177.00 |
| (iii) | 1 x 150-300 sq. mm. | 1 x 150-300 sq. mm. | each | 3523.00 |
| (iv) | 1 x 400-630 sq. mm. | 1 x 400-630 sq. mm. | each | 4893.00 |
| (C) | 11 KV XLPE cable straight through termination | Heat shrinkable jointing kit 11 KV XLPE cable straight through termination | | |
| (i) | 1 x 16-35 sq. mm. | 1 x 16-35 sq. mm. | each | 9790.00 |
| (ii) | 1 x 50-95 sq. mm. | 1 x 50-95 sq. mm. | each | 10701.00 |
| (iii) | 1 x 120-185 sq. mm. | 1 x 120-185 sq. mm. | each | 11221.00 |
| (iv) | 1 x 300-400 sq. mm. | 1 x 300-400 sq. mm. | each | 12674.00 |
| (v) | 1 x 400-500 sq. mm. | 1 x 400-500 sq. mm. | each | 13921.00 |
| (D) | 33 KV XLPE single core | Heat shrinkable joint, kit 33 KV XLPE single core | | |
| | 33 KV XLPE cable O.D. termination | Heat shrinkable Jointing kit 33 KV XLPE cable O.D. termination | | |

| Item No. | | Description | Unit | Rate |
|---|---|---|---|---|
| (i) | 1 x 25-50 sq. mm. | 1 x 25-50 sq. mm. | each | 5372.00 |
| (ii) | 1 x 70-185 sq. mm. | 1 x 70-185 sq. mm. | each | 5809.00 |
| (iii) | 1 x 240-400 sq. mm. | 1 x 240-400 sq. mm. | each | 7519.00 |
| (E) | 33 KV XLPE cable I.D. termination | Heat shrinkable jointing kit 33 KV XLPE cable I.D. termination | | |
| (i) | 1 x 25-50 sq. mm. | 1 x 25-50 sq. mm. | each | 4853.00 |
| (ii) | 1 x 70-185 sq. mm. | 1 x 70-185 sq. mm. | each | 5320.00 |
| (iii) | 1 x 240-400 sq. mm. | 1 x 240-400 sq. mm. | each | 7018.00 |
| (F) | 33 KV XLPE cable straight through termination | Heat shrinkable jointing kit 33 KV XLPE cable straight through termination | | |
| (i) | 1 x 35-70 sq. mm. | 1 x 35-70 sq. mm. | each | 16877.00 |
| (ii) | 1 x 95-150 sq. mm. | 1 x 95-150 sq. mm. | each | 18791.00 |
| (iii) | 1 x 185-300 sq. mm. | 1 x 185-300 sq. mm. | each | 20726.00 |
| (iv) | 1 x 400 sq. mm. | 1 x 400 sq. mm. | each | 24469.00 |
| 21.124 | Cable end termination with heat shrinkable jointing kit | Providing & Making cable end termination with heat shrinkable jointing kit complete with all accessories including lugs suitable for 33 KV/11 KV 3 core XLPE alum. conductor cable as required as per specification and as per accepted standard including connection testing complete. | | |
| | 11 KV XLPE, 3 core | Heat shrinkable joint, kit 11 KV XLPE, 3 core | | |
| (A) | 11 KV XLPE cable O.D. termination | Heat shrinkable Jointing kit 11 KV XLPE cable O.D. termination | | |
| (i) | 3 x 16-50 sq. mm. | 3 x 16-50 sq. mm. | each | 9194.00 |
| (ii) | 3 x 70-120 sq. mm. | 3 x 70-120 sq. mm. | each | 9488.00 |
| (iii) | 3 x 150-225 sq. mm. | 3 x 150-225 sq. mm. | each | 10963.00 |
| (iv) | 3 x 240-400 sq. mm. | 3 x 240-400 sq. mm. | each | 11389.00 |
| (B) | 11 KV XLPE cable I.D. termination | Heat shrinkable jointing kit 11 KV XLPE cable I.D. termination | | |
| (i) | 3 x 16-50 sq. mm. | 3 x 16-50 sq. mm. | each | 8342.00 |
| (ii) | 3 x 70-120 sq. mm. | 3 x 70-120 sq. mm. | each | 8431.00 |
| (iii) | 3 x 150-225 sq. mm. | 3 x 150-225 sq. mm. | each | 9734.00 |
| (iv) | 3 x 240-400 sq. mm. | 3 x 240-400 sq. mm. | each | 9470.00 |
| (C) | 11 KV XLPE cable straight joint | Heat shrinkable jointing kit 11 KV XLPE cable straight joint | | |
| (i) | 3 x 16-50 sq. mm. | 3 x 16-50 sq. mm. | each | 18520.00 |
| (ii) | 3 x 70-95 sq. mm. | 3 x 70-95 sq. mm. | each | 19986.00 |
| (iii) | 3 x 120-185 sq. mm. | 3 x 120-185 sq. mm. | each | 21585.00 |
| (iv) | 3 x 240-300 sq. mm. | 3 x 240-300 sq. mm. | each | 23710.00 |

| Item No. | | Description | Unit | Rate |
|---|---|---|---|---|
| (v) | 3 x 400-500 sq. mm. | 3 x 400-500 sq. mm. | each | 29159.00 |
| (D) | 33 KV XLPE, 3 core | Heat shrinkable joint, kit 33 KV XLPE, 3 core | | |
| | 33 KV XLPE cable O.D. termination | Heat shrinkable jointing kit 33 KV XLPE cable O.D. termination | | |
| (i) | 3 x 35-50 sq. mm. | 3 x 35-50 sq. mm. | each | 15549.00 |
| (ii) | 3 x 70-95 sq. mm. | 3 x 70-95 sq. mm. | each | 16311.00 |
| (iii) | 3 x 120-185 sq. mm. | 3 x 120-185 sq. mm. | each | 16600.00 |
| (iv) | 3 x 240-400 sq. mm. | 3 x 240-400 sq. mm. | each | 20808.00 |
| (E) | 33 KV XLPE cable I.D. termination | Heat shrinkable Jointing kit 33 KV XLPE cable I.D. termination | | |
| (i) | 3 x 35-50 sq. mm. | 3 x 35-50 sq. mm. | each | 13614.00 |
| (ii) | 3 x 70-95 sq. mm. | 3 x 70-95 sq. mm. | each | 14302.00 |
| (iii) | 3 x 120-185 sq. mm. | 3 x 120-185 sq. mm. | each | 15295.00 |
| (iv) | 3 x 240-400 sq. mm. | 3 x 240-400 sq. mm. | each | 18182.00 |
| (F) | 33 KV XLPE cable straight joint | Heat shrinkable Jointing kit 33 KV XLPE cable straight joint | | |
| (i) | 3 x 35-70 sq. mm. | 3 x 35-70 sq. mm. | each | 36816.00 |
| (ii) | 3 x 95-150 sq. mm. | 3 x 95-150 sq. mm. | each | 39937.00 |
| (iii) | 3 x 185 sq. mm. | 3 x 185 sq. mm. | each | 44644.00 |
| (iv) | 3 x 240 sq. mm. | 3 x 240 sq. mm. | each | 46543.00 |
| (v) | 3 x 300 sq. mm. | 3 x 300 sq. mm. | each | 47749.00 |
| (vi) | 3 x 400 sq. mm. | 3 x 400 sq. mm. | each | 53732.00 |
| 21.125 | Cable straight joint/end termination with heat shrinkable jointing kit | Supply and making cable straight joint/end termination with heat shrinkable jointing kit complete with all accessories including lugs suitable for 1.1 KV XLPE/HD alum. conductor cable as required as per specification and as per accepted standard including connection testing complete for (I.D../O.D.) termination. | | |
| (A) | 1.1 KV XLPE/HD cable end termination jointing kit | Heat shrinkable jointing kit 1.1 KV XLPE/HD cable end termination jointing kit (I.D./O.D.) | | |
| (i) | 16-35 sq. mm. 1 core | 16-35 sq. mm. 1 core | each | 446.00 |
| (ii) | 50-95 sq. mm. 1 core | 50-95 sq. mm. 1 core | each | 497.00 |
| (iii) | 120-185 sq. mm. 1 core | 120-185 sq. mm. 1 core | each | 595.00 |
| (iv) | 225-300 sq. mm. 1 core | 225-300 sq. mm. 1 core | each | 991.00 |
| (v) | 400-500 sq. mm. 1 core | 400-500 sq. mm. 1 core | each | 1477.00 |
| (B) | 1.1 KV XLPE/HD cable straight through jointing kit | Heat shrinkable jointing kit 1.1 KV XLPE/HD cable straight through jointing kit (I.D./O.D.) | | |
| (i) | 16-35 sq. mm. 1 core | 16-35 sq. mm. 1 core | each | 462.00 |

| Item No. | | Description | Unit | Rate |
|---|---|---|---|---|
| (ii) | 50-95 sq. mm. 1 core | 50-95 sq. mm. 1 core | each | 559.00 |
| (iii) | 120-185 sq. mm. 1 core | 120-185 sq. mm. 1 core | each | 768.00 |
| (iv) | 225-300 sq. mm. 1 core | 225-300 sq. mm. 1 core | each | 1073.00 |
| (v) | 400-500 sq. mm. 1 core | 400-500 sq. mm. 1 core | each | 1663.00 |
| (C) | 1.1 KV XLPE/HD cable I.D./O.D. end termination | Heat shrinkable Jointing kit 1.1 KV XLPE/HD cable I.D./O.D. end termination | | |
| (i) | 4-16 sq. mm. 3½/4 core | 4-16 sq. mm. 3½/4 core | each | 3009.00 |
| (ii) | 25-50 sq. mm. 3½/4 core | 25-50 sq. mm. 3½/4 core | each | 3677.00 |
| (iii) | 70-150 sq. mm. 3½/4 core | 70-150 sq. mm. 3½/4 core | | 3840.00 |
| (iv) | 185-300 sq. mm. 3½/4 core | 185-300 sq. mm. 3½/4 core | each | 6329.00 |
| (v) | 400-500 sq. mm. 3½/4 core | 400-500 sq. mm. 3½/4 core | each | 8447.00 |
| (D) | 1.1 KV XLPE/HD cable straight through jointing kit | Heat shrinkable jointing kit 1.1 KV XLPE/HD cable straight through jointing kit (I.D../O.D.) | | |
| (i) | 10-16 sq. mm. 3½/4core | 10-16 sq. mm. 3½/4core | each | 1991.00 |
| (ii) | 25-50 sq. mm. 3½/4core | 25-50 sq. mm. 3½/4core | each | 2505.00 |
| (iii) | 70 -150 sq. mm. 3½/4core | 70 -150 sq. mm. 3½/4core | each | 3495.00 |
| (iv) | 185-240 sq. mm. 3½/4core | 185-240 sq. mm. 3½/4core | each | 5822.00 |
| (v) | 300-400 sq. mm. 3½/4core | 300-400 sq. mm. 3½/4core | each | 6496.00 |
| 21.126 | G.I. pipe for protection of underground cable | Supplying & installing G.I. pipe for protection of underground cable fixed on wall/support/in trench/fixed between two rigid existing support of wall/beam for erection of ceiling Fan/down rod for stiff pendent for light luminaries /fan/protective for earthing, lightening conductor down strip/overhead service line/for submersible cable or centrifugal pump for water supply with necessary iron clamp coupler, bend, tee, elbow, nuts and bolts etc. complete in an approved manner as required to complete excluding cost of excavation/dismantling & other finished masonry Item complete. | | |
| (A) | For 'B' class pipe | For 'B' class pipe ISI marked (IS-1161-68) | | |
| (i) | 15.00mm | 15.00mm | metre | 167.00 |
| (ii) | 20.00mm | 20.00mm | metre | 201.00 |
| (iii) | 25.00mm | 25.00mm | metre | 291.00 |

| Item No. | | Description | Unit | Rate |
|---|---|---|---|---|
| (iv) | 32.00mm | 32.00mm | metre | 368.00 |
| (v) | 40.00mm | 40.00mm | metre | 423.00 |
| (vi) | 50.00mm | 50.00mm | metre | 550.00 |
| (vii) | 65.00mm | 65.00mm | metre | 693.00 |
| (viii) | 80.00mm | 80.00mm | metre | 913.00 |
| (ix) | 100.00mm | 100.00mm | metre | 1256.00 |
| (x) | 150.00mm | 150.00mm | metre | 2062.00 |
| (B) | For 'A' class G.I. pipe | For 'A' class G.I. pipe ISI marked(IS-1161-69) | | |
| (i) | 15.00mm | 15.00mm | metre | 142.00 |
| (ii) | 19/20mm | 19/20mm | metre | 194.00 |
| (iii) | 25.00mm | 25.00mm | metre | 256.00 |
| (iv) | 32.00mm | 32.00mm | metre | 331.00 |
| (v) | 40.00mm | 40.00mm | metre | 390.00 |
| (vi) | 50.00mm | 50.00mm | metre | 503.00 |
| (vii) | 65.00mm | 65.00mm | metre | 661.00 |
| (viii) | 80.00mm | 80.00mm | metre | 790.00 |
| (ix) | 100.00mm | 100.00mm | metre | 1151.00 |
| 21.127 | Rigid PVC single socket and, non-metallic pipe | Supply and laying rigid PVC single socket and, non-metallic pipe ISI marked (IS-4985-2000) 10 Kg/sq.cm class 5 for protection of underground cable/or for water pump delivery side, for water supply complete with necessary clamp coupler, band, tee, elbow etc. in an approved manner excluding cost of excavation etc. | | |
| (i) | 50.00mm O.D | 50.00mm O.D | metre | 163.00 |
| (ii) | 75.00mm O.D | 75.00mm O.D | metre | 357.00 |
| (iii) | 110.00mm O.D | 110.00mm O.D | metre | 753.00 |
| (iv) | 160.00mm O.D | 160.00mm O.D | metre | 1617.00 |
| (v) | 200.00mm O.D | 200.00mm O.D | metre | 2585.00 |
| 21.128 | Double wall corrugated pipes (DWC) of HDPE | Supplying and installing double wall corrugated pipes (DWC) of HDPE (IS 14930 Part II -marked) for cable laid underground with necessary material and at required depth upto 90cm. below road/ground surface, excluding excavation, back filling with excavated material, ramming and making the surface good. | | |
| (i) | 50.00mm outside dia. | 50.00mm outside dia. | metre | 133.00 |
| (ii) | 63.00mm outside dia. | 63.00mm outside dia. | metre | 175.00 |
| (iii) | 90.00mm outside dia. | 90.00mm outside dia. | metre | 214.00 |
| (iv) | 110.00mm outside dia. | 110.00mm outside dia. | metre | 252.00 |
| (v) | 120.00mm outside dia. | 120.00mm outside dia. | metre | 279.00 |
| (vi) | 180.00mm outside dia. | 180.00mm outside dia. | metre | 390.00 |
| (vii) | 210.00mm outside | 210.00mm outside dia. | metre | 532.00 |

| Item No. | | Description | Unit | Rate |
|---|---|---|---|---|
| | dia. | | | |
| (B) | Cable laying | Cable laying | | |
| (i) | Laying, 1.1KV grade of size not exceeding 25 sq. mm | Laying of one number armoured / unarmoured power cable 1.1KV grade of size not exceeding 25 sq. mm direct in ground including excavation in all soil/ murrum /soft rock, sand cushioning, protective covering and refilling the pit etc. as required. | metre | 230.00 |
| (ii) | Laying, 1.1.KV grade of size exceeding 25sq. mm but not exceeding 120 sq. mm | Laying of one number armoured / unarmoured power cable 1.1.KV grade of size exceeding 25sq. mm but not exceeding 120 sq. mm direct in ground including excavation in all soil/ murrum /soft rock, sand cushioning, protective covering and refilling the pit etc. as required. | metre | 246.00 |
| (iii) | Laying, 1.1.KV grade of size exceeding 120 sq. mm. but not exceeding 400 sq. mm | Laying of one number armoured / unarmoured cable 1.1.KV grade of size exceeding 120 sq. mm. but not exceeding 400 sq. mm direct in ground including excavation in all soil/ murrum /soft rock, and sand cushioning protective covering and refilling the pit etc. as required. | metre | 266.00 |
| (iv) | Laying, 1.1KV grade of size not exceeding 25sq. mm, additional cable | Laying of one number additional armoured / unarmoured power cable 1.1KV grade of size not exceeding 25sq. mm direct in ground in the same trench in one tier horizontal formation including excavation in all soil/murrum/soft rock, sand cushioning protective covering and refilling the pit etc. as required. | metre | 136.00 |
| (v) | Laying, 1.1KV grade of size exceeding 25 sq. mm but not exceeding 120 sq. mm, additional cable | Laying of one number additional armoured / unarmoured power cable 1.1KV grade of size exceeding 25 sq. mm but not exceeding 120 sq. mm direct in ground in the same trench in one tier horizontal formation including excavation, sand cushioning, protective covering and refilling the pit etc. as required. | metre | 146.00 |
| (vii) | Laying, 1.1KV grade of size exceeding 120 sq. mm. but not exceeding 400 sq. mm., additional cable | Laying of one number additional armoured / unarmoured power cable 1.1KV grade of size exceeding 120 sq. mm. but not exceeding 400 sq. mm. direct in ground in the same trench in one tier horizontal formation including excavation, sand cushioning, protective covering and refilling the pit etc. as required. | metre | 159.00 |
| (viii) | Laying, 1:1 KV grade of size not exceeding 25 sq. mm in the existing item | Laying of one number armoured / unarmoured power cable 1:1 KV grade of size not exceeding 25 sq. mm in the existing RCC hume/stone ware/G.I. pipe/ DWC Pipe/ surface in existing trench as required. | metre | 28.00 |
| (ix) | Laying, 1.1 KV grade of size exceeding 25 sq. mm but not exceeding 120 sq. | Laying of one number armoured / unarmoured power cable 1.1 KV grade of size exceeding 25 sq. mm but not exceeding 120 sq. mm. in the existing RCC Hume /Stone ware/G.I. Pipe/ | metre | 35.00 |

| Item No. | | Description | Unit | Rate |
|---|---|---|---|---|
| | mm. in the existing item | DWC Pipe/ surface in existing trench as required. | | |
| (x) | Laying, 1.1 KV grade of size exceeding 120 sq. mm but not exceeding 400 sq. mm. in the existing item | Laying of one number armoured / unarmoured power cable 1.1 KV grade of size exceeding 120 sq. mm but not exceeding 400 sq. mm. in the existing RCC Hume /Stone ware/G.I. Pipe/ DWC Pipe/ surface in existing trench as required. | metre | 54.00 |
| (xi) | Laying, 1.1 KV grade of size not exceeding 25 sq. mm in the existing masonry open duct | Laying of one number armoured / unarmoured power cable 1.1 KV grade of size not exceeding 25 sq. mm in the existing masonry open duct as required. | metre | 23.00 |
| (xii) | Laying, 1.1KV grade of size exceeding 25sq. mm but not exceeding 120 sq. mm | Laying of one number armoured/unarmoured power cable 1.1KV grade of size exceeding 25sq. mm but not exceeding 120 sq. mm in the existing masonry open duct as required. | metre | 36.00 |
| (xiii) | Laying, 1.1KV grade of size exceeding 120 sq. mm but not exceeding 400 sq. mm | Laying of one number armoured/unarmoured power cable 1.1KV grade of size exceeding 120sq. mm but not exceeding 400 sq. mm in the existing masonry open duct as required. | metre | 50.00 |
| (xiv) | Laying, 1.1KV but not exceeding 11KV of size not exceeding 300 sq. mm | Laying of one number single core H.T armoured/unarmoured power cable of grade exceeding 1.1KV but not exceeding 11KV of size not exceeding 300 sq. mm direct in ground including excavation in all soil/murrum/soft rock, sand cushioning, excluding protective covering and refilling the trench etc. as required. | metre | 126.00 |
| (xv) | Laying, 33KV of size not exceeding 300 sq. mm | Laying of one number single core H.T armoured/unarmoured power cable of grade 33KV of size not exceeding 300 sq. mm direct in ground including excavation sand cushioning, excluding protective covering and refilling the trench etc. as required | metre | 125.00 |
| (xvi) | Laying, 1.1KV but not exceeding 11KV of size not exceeding 300 sq. mm | Laying of one number single core H.T armoured/unarmoured power cable of grade exceeding 1.1KV but not exceeding 11KV of size not exceeding 300 sq. mm in the existing RCC/Hume/Stoneware/G.I. Pipe/ DWC Pipe/surface in existing trench as required. | metre | 28.00 |
| (xvii) | Laying, 1.1KV but not exceeding 11KV of size not exceeding 300 sq. mm. | Laying of one number single core H.T armoured/unarmoured power cable of grade exceeding 1.1KV but not exceeding 11KV of size not exceeding 300 sq. mm. in the existing masonry open duct as required. | metre | 33.00 |
| (xviii) | Laying, 33KV grade of size not exceeding 300 sq. mm | Laying of one number single core H.T armoured/unarmoured power cable 33KV grade of size not exceeding 300 sq. mm in the | metre | 34.00 |

| Item No. | | Description | Unit | Rate |
|---|---|---|---|---|
| | | existing RCC/Hume/Stoneware/G.I. Pipe/ DWC Pipe/surface in existing trench as required. | | |
| (xix) | Laying, 33KV grade of size not exceeding 300 sq. mm. | Laying of one number single core H.T armoured/unarmoured power cable 33KV grade of size not exceeding 300 sq. mm. in the existing masonry open duct as required. | metre | 38.00 |
| (xx) | Laying, 11KV grade of size not exceeding 300 sq. mm | Laying of one number single core H.T armoured/unarmoured power cable 11KV grade of size not exceeding 300 sq. mm on wall/pole etc. as required. | metre | 92.00 |
| (xxi) | Laying, 33KV grade of size not exceeding 300 sq. mm | Laying of one number single core H.T armoured/unarmoured power cable 33KV grade of size not exceeding 300 sq. mm on wall/pole etc. as required. | metre | 91.00 |
| (xxii) | Laying, 1.1KV but not exceeding 11KV of size not exceeding 300 sq. | Laying of one number 3 core H.T armoured/unarmoured power cable of grade exceeding 1.1KV but not exceeding 11KV of size not exceeding 300 sq. mm direct in ground including excavation in all soil/murrum/soft rock, sand cushioning, excluding protective covering and refilling the trench etc. as required. | metre | 221.00 |
| (xxiii) | Laying, 33KV of size not exceeding 300 sq. mm | Laying of one number 3 core H.T armoured/unarmoured power cable of grade 33KV of size not exceeding 300 sq. mm direct in ground including excavation in all soil/murrum/soft rock, sand cushioning, excluding protective covering and refilling the trench etc. as required | metre | 230.00 |
| (xxiv) | Laying, 1.1KV but not exceeding 11KV of size not exceeding 300 sq. mm | Laying of one number additional 3 core H.T armoured/unarmoured power cable of grade exceeding 1.1KV but not exceeding 11KV of size not exceeding 300 sq. mm direct in ground in the same trench in one tier horizontal formation including sand cushioning, excluding protective covering and refilling the trench as required. | metre | 144.00 |
| (xxv) | Laying, 33KV grade of size not exceeding 300 sq. mm | Laying of one number additional 3 core H.T armoured/unarmoured power cable 33KV grade of size not exceeding 300 sq. mm direct in ground in the same trench in one tier horizontal formation including sand cushioning, excluding protective covering and refilling the trench etc. as required. | metre | 143.00 |
| (xxvi) | Laying, 1.1KV but not exceeding 11KV of size not exceeding 300 sq. mm | Laying of one number 3 core H.T armoured/unarmoured power cable of grade exceeding 1.1KV but not exceeding 11KV of size not exceeding 300 sq. mm in the existing RCC/hume/stoneware/G.I. pipe/ DWC Pipe/surface in existing trench as required. | metre | 50.00 |
| (xxvii) | Laying, 1.1KV but | Laying of one number 3 core H.T | metre | 49.00 |

| Item No. | | Description | Unit | Rate |
|---|---|---|---|---|
| | not exceeding 11KV of size not exceeding 300 sq. mm. | armoured/unarmoured power cable of grade exceeding 1.1KV but not exceeding 11KV of size not exceeding 300 sq. mm. in the existing masonry open duct as required. | | |
| (xxviii) | Laying, 11KV grade of size not exceeding 300 sq. mm | Laying of one number 3 core H.T armoured/unarmoured power cable 11KV grade of size not exceeding 300 sq. mm on wall/pole etc. as required. | metre | 150.00 |
| (xxix) | Laying, 33KV grade of size not exceeding 300 sq. mm | Laying of one number 3 core H.T armoured/unarmoured power cable 33KV grade of size not exceeding 300 sq. mm in the existing RCC/hume/stoneware/G.I. pipe/ DWC Pipe/surface in existing trench as required. | metre | 50.00 |
| (xxx) | Laying, 33KV grade of size not exceeding 300 sq. mm. | Laying of one number 3 core H.T armoured/unarmoured power cable 33KV grade of size not exceeding 300 sq. mm. in the existing masonry open duct as required. | metre | 49.00 |
| (xxxi) | Laying, 33KV grade of size not exceeding 300 sq. mm | Laying of one number 3 core H.T armoured/unarmoured power cable 33KV grade of size not exceeding 300 sq. mm on wall/pole etc. as required. | metre | 152.00 |
| (xxxii) | Laying, 1.1KV grade of size not exceeding 25 sq. mm | Laying of one number armoured/unarmoured cable 1.1KV grade of size not exceeding 25 sq. mm on surface in existing trench etc. as required. | metre | 25.00 |
| (xxxiii) | Laying,1.1KV grade of size exceeding 25 sq. mm but not exceeding 120 sq. mm | Laying of one number armoured/unarmoured cable 1.1KV grade of size exceeding 25 sq. mm but not exceeding 120 sq. mm on surface in existing trench etc. as required. | metre | 33.00 |
| (xxiv) | Laying, 1.1KV grade of size exceeding 120 sq. mm but not exceeding 400 sq. mm | Laying of one number armoured/unarmoured cable 1.1KV grade of size exceeding 120 sq. mm but not exceeding 400 sq. mm on surface in existing trench etc. as required. | metre | 41.00 |
| 21.130 | Excavation of the trenches in hard rock | Excavation of the trenches in hard rock not exceeding 1.5 metre in width and lift upto 1.5 metre including getting out the excavated metal and disposal of surplus excavated soil/metal as directed within the lead of 50 metre (without blasting) | Cu. Mtr. | 691.00 |
| 21.131 | Laying, 1.1KV grade of size not exceeding 25 sq. mm | Laying of one number armoured/unarmoured cable 1.1KV grade of size not exceeding 25 sq. mm on wall/truss with approved type of iron clamp etc. as required. | metre | 25.00 |
| 21.132 | Laying, 1.1KV grade of size exceeding 25 sq. mm but not exceeding 120 sq. mm | Laying of one number armoured/unarmoured cable 1.1KV grade of size exceeding 25 sq. mm but not exceeding 120 sq. mm on wall/truss with approved type of iron clamp etc. as required. | metre | 71.00 |

| Item No. | | Description | Unit | Rate |
|---|---|---|---|---|
| 21.133 | Laying, 1.1KV grade of size exceeding 120 sq. mm but not exceeding 400 sq. mm | Laying of one number armoured/unarmoured cable 1.1KV grade of size exceeding 120 sq. mm but not exceeding 400 sq. mm on wall/truss with approved type of iron clamp etc. as required. | metre | 112.00 |
| 21.134 | Laying of underground cable | Laying of underground cable armoured./unarmoured as per specification in air suspended on existing approved type of iron clamp, G.I wire 7/1.6mm & shackle insulator/insulated strip complete. | | |
| (i) | 16 sq. mm. | 1/2/3/4 Core cable upto 16 sq. mm | metre | 25.00 |
| (ii) | 25-120 sq. mm. | 1/2/3/3½/4 Core cable 25 sq. mm to 120 sq. mm | metre | 40.00 |
| (iii) | 150 sq. mm. & above | 1/2/3/3½/4 Core cable 150 sq. mm and above | metre | 66.00 |
| 21.135 | Laying, cable cover/flag stone over | Laying of cement concrete or approved type of cable cover/flag stone over L.T./H.T. cable & refill trench as per specification. | metre | 16.00 |
| 21.136 | cable cover flag stone 200mm width & 32-50mm thick | Supply of approved type cable cover flag stone 200mm width & 32mm to 50mm thick of required length inclusive of transport as required. | metre | 71.00 |
| 21.137 | Cable route marker round/square | Supply &fixing cable route marker round/square size 100mm, 5mm. thick, G.I. Plate duly bolted/welded on 35mm x 35mm x 5mm angle iron 60cm long including excavation, grouting in cement concrete 1:2:4 (1cement and 2 coarse sand: 4 graded stone aggregate 20mm nominal size) foundation of size 20cm x 20cm x 30cm duly embedded in ground 30cm depth or as required. | each | 299.00 |
| 21.138 | FRP/ GRP Cable Tray (Ladder Type) | Supply and Erecting of Ladder type Fiber Glass Reinforced Polymer (FRP) / Glass Fiber Reinforced Polymer (GRP) cable tray of specific dimensions. The cable tray shall be hung from ceiling/ fixed to wall with necessary angle/ flat iron ceiling suspensors, clamps, nuts, bolts, washers complete with required tees, bends etc. as per specification to complete the job. The tray shall be as follows :- | | |
| (i) | 150mm | 150mm width with straight coupler plate size: 200x70x4mm & 8 sets of SS 304 Hardware size: M6x20mm | metre | 1228.00 |
| (ii) | 300mm | 300mm width with straight coupler plate size: 200x70x4mm & 8 sets of SS 304 Hardware size: M6x20mm | metre | 1371.00 |
| (iii) | 450mm | 450mm width with straight coupler plate size: | metre | 1535.00 |

| Item No. | | Description | Unit | Rate |
|---|---|---|---|---|
| | | 200x70x4mm & 8 sets of SS 304 Hardware size: M6x20mm | | |
| (iv) | 600mm | 600mm width with straight coupler plate size: 200x70x4mm & 8 sets of SS 304 Hardware size: M6x20mm | metre | 1695.00 |
| (v) | 750mm | 750 mm width with straight coupler plate size: 200x70x4mm & 8 sets of SS 304 Hardware size: M6x20mm | metre | 1878.00 |
| 21.138 | FRP/ GRP Cable Tray (Perforated Type) | Supply and Erecting of Perforated type Fiber Glass Reinforced Polymer (FRP) / Glass Fiber Reinforced Polymer (GRP) cable tray of specific dimensions. The cable tray shall be hung from ceiling/ fixed to wall with necessary angle/ flat iron ceiling suspensors, clamps, nuts, bolts, washers complete with required tees, bends etc. as per specification to complete the job. The tray shall be as follows :- | | |
| (i) | 50mm | 50 mm width with straight coupler plate size: 200x25x4mm & 8 sets of SS 304 Hardware size: M5x20mm | metre | 819.00 |
| (ii) | 100mm | 100 mm width with straight coupler plate size: 200x25x4mm & 8 sets of SS 304 Hardware size: M5x20mm | metre | 875.00 |
| (iii) | 150mm | 150 mm width with straight coupler plate size: 200x25x4mm & 8 sets of SS 304 Hardware size: M5x20mm | metre | 1018.00 |
| (iv) | 200mm | 200 mm width with straight coupler plate size: 200x25x4mm & 8 sets of SS 304 Hardware size: M5x20mm | metre | 1357.00 |
| (v) | 300mm | 300 mm width with straight coupler plate size: 200x25x4mm & 8 sets of SS 304 Hardware size: M5x20mm | metre | 1757.00 |
| 21.139 | FRP/GRP Flat Cover | Supply and Fixing of Fiber Glass Reinforced Polymer (FRP) / Glass Fiber Reinforced Polymer (GRP) flat cover of minimum thickness 3mm as per required size of cable tray | | |
| (i) | 50mm | Flat Cover of 50 mm width | metre | 129.00 |
| (ii) | 100mm | Flat Cover of 100 mm width | metre | 211.00 |
| (iii) | 150mm | Flat Cover of 150 mm width | metre | 289.00 |
| (iv) | 200mm | Flat Cover of 200 mm width | metre | 379.00 |
| (v) | 300mm | Flat Cover of 300 mm width | metre | 527.00 |
| (vi) | 450mm | Flat Cover of 450 mm width | metre | 813.00 |
| (vii) | 600mm | Flat Cover of 600 mm width | metre | 1040.00 |
| | 750mm | Flat Cover of 750 mm width | metre | 1817.00 |

| Item No. | | Description | Unit | Rate |
|---|---|---|---|---|
| 21.140 | Fire Survival Cable | Supply of XLPE insulated 1.5 sq. mm. Armoured 2 core FRLSZH fire alarm cable (RED) Fire Survival Cable conforming IS-7098 (Part-1) 1988, 1100 Volt grade, 2/3/4/8/10/12 core ISI marked with copper stranded/solid conductor. | metre | 154.00 |
| | | **EXTERNAL ELECTRIFICATION AND OVERHEAD** | | |
| 21.141 | Rail pole | Supply of support for overhead line rail pole of I.S. standard including base plate, welding, drilling of required hole etc. complete as required. | | |
| (i) | 60 kg/metre | Rail pole standard weight 60 kg/metre | metre | 4712.00 |
| (ii) | 52 kg/metre | Rail pole standard weight 52 kg/metre | metre | 4063.00 |
| 21.142 | RS Joist/H-Beam | Supply of support for overhead line RS joist of I.S. standard including drilling of required hole etc. complete as required. | | |
| (i) | 100x200-25.4 kg/metre | R.S. Joist 100x200 - 25.4 kg/metre | metre | 1874.00 |
| (ii) | 175x85 - 19.3 kg/metre | R.S. Joist 175x85 - 19.3 kg/metre | metre | 1467.00 |
| (iii) | 150x100 - 17 kg/metre | R.S. Joist 150x100 - 17 kg/metre | metre | 1290.00 |
| (iv) | 150x80 - 14.9 kg/metre | R.S. Joist 150x80 - 14.9 kg/metre | metre | 1256.00 |
| (v) | 125x75 - 13 kg/metre | R.S. Joist 125x75 - 13 kg/metre | metre | 1007.00 |
| (vi) | H - Beam 152x152mm | H - Beam 152x152mm, Std weight 37.1 Kg/Mtr | metre | 2843.00 |
| 21.143 | Steel tubular pole | Supply of steel tubular pole swaged type as per IS:2713-1980 complete with base plate and top Canopy | | |
| (i) | SP-2 - 7.00 metre | 410 SP-2 - 7.00 metre | each | 7799.00 |
| (ii) | SP-5 - 7.50 metre | 410 SP-5 - 7.50 metre | each | 8302.00 |
| (iii) | SP-8 - 7.50 metre | 410 SP-8 - 7.50 metre | each | 11025.00 |
| (iv) | SP-11 - 8.00 metre | 410 SP-11 - 8.00 metre | each | 8701.00 |
| (v) | SP-14 - 8.00 metre | 410 SP-14 - 8.00 metre | each | 11911.00 |
| (vii) | SP-17 - 8.50 metre | 410 SP-17 - 8.50 metre | each | 9356.00 |

| Item No. | | | Description | Unit | Rate |
|---|---|---|---|---|---|
| (viii) | SP-20 - 8.50 metre | 410 | SP-20 - 8.50 metre | each | 12140.00 |
| (ix) | SP-23 - 8.50 metre | 410 | SP-23 - 8. 50 metre | each | 15852.00 |
| (x) | SP-26 - 9.00 metre | 410 | SP-26 - 9.00 metre | each | 9933.00 |
| (xi) | SP-29 - 9.00 metre | 410 | SP-29 - 9.00 metre | each | 13005.00 |
| (xii) | SP-32 - 9.00 metre | 410 | SP-32 - 9.00 metre | each | 16340.00 |
| (xiii) | SP-35 - 9.50 metre | 410 | SP-35 - 9.50 metre | each | 13872.00 |
| (xiv) | SP-38 - 9.50 metre | 410 | SP-38 - 9.50 metre | each | 17102.00 |
| (xv) | SP-41 - 10.00 metre | 410 | SP-41 - 10.00 metre | each | 14168.00 |
| (xvi) | SP-44 - 10.00 metre | 410 | SP-44 - 10.00 metre | each | 17623.00 |
| (xvii) | SP-47 - 10.00 metre | 410 | SP-47 - 10.00 metre | each | 23251.00 |
| (xviii) | SP-50 - 11.00 metre | 410 | SP-50 - 11.00 metre | each | 15284.00 |
| (xix) | SP-53 - 11.00 metre | 410 | SP-53 - 11.00 metre | each | 19204.00 |
| (xx) | SP-56 - 11.00 metre | 410 | SP-56 - 11.00 metre | each | 25335.00 |
| (xxi) | SP-59 - 12.00 metre | 410 | SP-59 - 12.00 metre | each | 20686.00 |
| (xxii) | SP-62 - 12.00 metre | 410 | SP-62 - 12.00 metre | each | 27363.00 |
| (xxiii) | SP-65 - 12.00 metre | 410 | SP-65 - 12.00 metre | each | 32227.00 |
| (xxiv) | SP-68 - 13.00 metre | 410 | SP-68 - 13.00 metre | each | 28934.00 |
| (xxv) | SP-71 - 13.00 metre | 410 | SP-71 - 13.00 metre | each | 33847.00 |
| (xxvi) | SP-73 - 14.50 metre | 410 | SP-73 - 14.50 metre | each | 34642.00 |
| (xxvii) | SP-76 - 14.50 metre | 410 | SP-76 - 14.50 metre | each | 40968.00 |
| (xxviii) | SP-77 - 16.00 metre | 410 | SP-77 - 16.00 metre | each | 27112.00 |
| (xxix) | SP-80 - | 410 | SP-80 - 16.00 metre | each | 28220.00 |

| Item No. | | Description | Unit | Rate |
|---|---|---|---|---|
| | 16.00 metre | | | |
| (xxx) | SP-10 (Bend type) 8.00 metre | 410 SP-10 (Bend type) 8.00 metre | each | 7514.00 |
| (xxxi) | SP-16 (Bend type) 8.50 metre | 410 SP-16 (Bend type) 8.50 metre | each | 8067.00 |
| (xxxii) | SP-25 (Bend type) 9.00 metre | 410 SP-25 (Bend type) 9.00 metre | each | 8338.00 |
| (xxxiii) | SP-34 (Bend type) 9.50 metre | 410 SP-34 (Bend type) 9.50 metre | each | 13236.00 |
| (xxiv) | SP-40 (Bend type) 10.00 metre | 410 SP-40 (Bend type) 10.00 metre | each | 13884.00 |
| (xxv) | SP-49 (Bend type) 11.00 metre | 410 SP-49 (Bend type) 11.00 metre | each | 14617.00 |
| 21.144 | PCC pole | Supplying of pre stressed cement concrete (PCC) pole conforming to IS: 785 at work site including required transportation. | | |
| (i) | 8 meter | PCC pole 8 meter Long 140 kg | each | 3731.00 |
| (ii) | 9.1 meter | PCC pole 9.1 meter Long 280 kg | each | 5099.00 |
| 21.145 | AAAC conductors | Supplying and drawing all aluminium alloy conductor (AAAC) of approved make conforming to IS 398-1979 Pt. IV, including binding at existing insulator, jointing, jumpering, tearing off, connecting etc. as required including clearing of obstacles (if any) | | |
| (i) | Weasel | AAAC 0.03 sq inch (34 sq.mm Al. EQ.)- (Weasel) | km. | 39115.00 |
| (ii) | Rabbit | AAAC 0.05 sq inch (55 sq.mm Al. EQ.)- (Rabbit) | km. | 61509.00 |
| (iii) | Raccoon | AAAC 0.075 sq inch (80 sq.mm Al. EQ.)- (Raccoon) | km. | 79735.00 |
| (iv) | Dog | AAAC 0.1 sq inch (100 sq.mm Al. EQ.)- (Dog) | km. | 104339.00 |
| (v) | Panther | AAAC 0.2 sq inch (232 sq.mm Al. EQ.)- (Panther) | km. | 250382.00 |
| 21.146 | ACSR conductor | Supplying and drawing of stranded aluminium conductor steel reinforced (ACSR) conforming to IS:398-1976 of approved make, stringing, making off complete with binding at existing insulator, jointing, jumpering, tearing off, connecting etc. as required and clearing of obstacles (if any) etc. | | |
| (i) | Squirrel | ACSR 0.02 sq.inch (20 sq.mm Al. EQ.)- (Squirrel) | km. | 27345.00 |
| (ii) | Weasel | ACSR 0.03 sq inch (34 sq.mm Al. EQ.)- (Weasel) | km. | 38982.00 |
| (iii) | Rabbit | ACSR 0.05 sq inch (55 sq.mm Al. EQ.)- (Rabbit) | km. | 64148.00 |

| Item No. | | Description | Unit | Rate |
|---|---|---|---|---|
| (iv) | Raccoon | ACSR 0.075 sq inch (80 sq.mm Al. EQ.)- (Raccoon) | km. | 88530.00 |
| (v) | Dog | ACSR 0.1 sq inch (100 sq.mm Al. EQ.)- (Dog) | km. | 119836.00 |
| (vi) | Panther | ACSR 0.2 sq inch (261 sq.mm Al. EQ.)- (Panther) | km. | 341832.00 |
| 21.147 | Covered aluminium conductor | Supplying, drawing and testing of XLPE insulated covered aluminium conductor of approved make, stringing, making off complete with binding at existing insulator, jointing, jumpering, tearing off, connecting etc. as required and clearing of obstacles (if any) etc. | | |
| (i) | 70 sqmm | 70 Sqmm XLPE 11KV grade | km. | 541000.00 |
| (ii) | 99 Sqmm | 99 Sqmm XLPE 11KV grade | km | 729410.00 |
| (iii) | 99 Sqmm | 99 Sqmm XLPE 33KV grade | km. | 927480.00 |
| (iv) | 104 Sqmm | 104.98 Sqmm XLPE 33 KV grage | km. | 945000.00 |
| 21.148 | Guard insulator or split insulator | Supplying and fixing guard insulator or split insulator complete: | | |
| (i) | 25x25 mm | 25x25 mm | Pair | 42.00 |
| (ii) | 37x37 mm | 37x37 mm | Pair | 48.00 |
| (iii) | 50x50 mm | 50x50 mm | Pair | 60.00 |
| 21.149 | Guard wire/earth wire/bearer wire 13 sq mm | Supplying and drawing guard wire/earth wire/bearer wire 13 sq mm (4mm. dia) G.I. including stringing, binding at existing insulators or brackets, jointing, jumpering, connecting & cradle etc. as required and clearing of obstacles (if any) | km. | 15675.00 |
| 21.150 | Guard wire/earth wire/bearer wire 18.62 sq mm | Supplying and drawing guard wire/earth wire/bearer wire 18.62 sq mm (4.87mm. dia/6swg) G.I. including stringing, binding at existing insulators or brackets, jointing, jumpering, connecting & cradle etc. as required and clearing of obstacles (if any) | km. | 18761.00 |
| 21.151 | G.I. Bearer wire 7/1.6mm | Supplying and drawing G.I. bearer wire 7/1.6mm including stringing, binding at existing brackets/pole, etc. for cable support in air as required and clearing of obstacles (if any) | PM | 23.00 |
| 21.152 | Guards for existing overhead lines | Supplying and erection of guards for existing overhead lines as required including split insulator required as per specification: | | |
| (i) | Cradle type | Cradle type | each | 38.00 |
| (ii) | Hexagonal type | Hexagonal type | each | 30.00 |
| (iii) | Ring type | Ring type | each | 27.00 |
| 21.153 | Water tight terminal box (M.S. Sheet) | Supplying and erection of water tight terminal box made of 16 SWG M.S. Sheet (1.6mm thick) i/c rubber gasket of required size suitable for kit kat/MCB/switch including 2 nos. pole clamp of M.S. flat iron size 32x3mm, nut and bolts, The box shall be powder coated excluding the cost of the kit kat/MCB/Switch 4 Pole epoxy connector nut & bolt type, 30Amp, 500Volt complete as per specification as mentioned below. | | |

| Item No. | | Description | Unit | Rate |
|---|---|---|---|---|
| (i) | 20x15x10 Cms. | 20x15x10 Cms. | each | 910.00 |
| (ii) | 30x20x10 Cms. | 30x20x10 Cms. | each | 1158.00 |
| (iii) | 4 Pole epoxy connector | Supplying and fixing of 4 Pole epoxy connector nut & bolt type, 30Amp. 500Volt. in existing box as required as per specification: | each | 195.00 |
| 21.154 | Water tight terminal box (Polycarbonate) | Supplying and erection of water tight terminal box made of PC-GFS (polycarbonate),IP 66 /IP 67 (cable entry systems LES) in accordance with IEC 60670-22, resistant to weather influences (UV due to solar radiation, protected against rainwater, temperature resistant, impact resistant, etc. insulation voltage 1000 V a.c./d.c., wall thickness of the bottom part 2 / 2.7 mm of required size suitable for kit kat/MCB/switch including 2 nos. pole clamp of M.S. flat iron size 32x3mm, nut and bolts, excluding the cost of the kit kat/MCB/Switch, 4 Pole epoxy connector nut & bolt type, 30Amp, 500Volt complete as per specification as mentioned below. | | |
| (i) | 104 x 104 x 70 mm | 104 x 104 x 70 mm | each | 452.00 |
| (ii) | 130 x 180 x 77 mm | 130 x 180 x 77 mm | each | 889.00 |
| (iii) | 155 x 210 x 92 mm | 155 x 210 x 92 mm | each | 1384.00 |
| (iv) | 205 x 255 x 112 mm | 205 x 255 x 112 mm | each | 2163.00 |
| 21.155 | Cable Gland | Supplying and erection of cable glands of Polyamide material for knockouts as required as mentioned below: | | |
| (i) | M20 | M20 | each | 47.00 |
| (ii) | M25 | M25 | each | 71.00 |
| (iii) | M32 | M32 | each | 100.00 |
| 21.156 | MCB/Connector mounting | Supplying and fixing of MCB/Connector mounting Corrosion proof mild steel DIN Rail of thickness 2.5mm to 4mm in existing box as required as per specification: | Per Metre | 107.00 |
| 21.157 | Single pole connector DIN Rail mounted | Supply and fixing of single pole connector DIN Rail mounted Insulating body made from Polyamide 6.6 conforms to UL 94 standard, IEC 60947-7-1, 1000Volt in existing box as required as per specification: | | |
| (i) | Upto 4sq mm cable | Suitable for upto 4sq mm cable | each | 52.00 |
| (ii) | For 6sqmm-16sq mm cable | Suitable for 6sqmm-16sq mm cable | each | 68.00 |
| 21.158 | Terminal set polyamide | Supplying and fixing of terminal set polyamide with copper connector 5 pole including fixing arrangement for below mentioned size: | | |
| (i) | 2.5 to 4 sq mm | Terminal set 2.5 to 4 sq mm | each | 90.00 |
| (ii) | 6 sq mm | Terminal set 6 sq mm | each | 208.00 |

| Item No. | | Description | Unit | Rate |
|---|---|---|---|---|
| 21.159 | Erection of poles, length 7 metres but not exceeding 10 metres | Erection of RS Joist pole/'H' beam/steel tubular/steel rail pole of length 7 metres but not exceeding 10 metres in cement concrete 1:3:6 (1 cement : 3 coarse sand: 6 graded stone aggregate 40mm nominal size) base, foundation, muffing (concreting 0.65 Cu Mtr. per pole) including excavation and back refilling etc. as required. | each | 4074.00 |
| 21.160 | Erection of poles, length exceeding 10 metres but not exceeding 15 metres | Erection of RS Joist pole/'H' beam/steel tubular/steel rail pole of length exceeding 10 metres but not exceeding 15 metres in cement concrete M-10 (1 cement:3 core sand: 6 graded stone aggregate 40mm nominal size) base, foundation, muffing (concreting 0.65 Cu Mtr. per pole) including excavation and back refilling etc. as required. | each | 5289.00 |
| 21.161 | Erection of R.C.C./P.C .C. pole | Erection of R.C.C./P.C.C. pole of length 8 meters but not exceeding 11 meters in brick ballast and ramming the foundation (concreting 0.65 Cu Mtr. per pole) including excavation and refilling etc.as required. | each | 2282.00 |
| 21.162 | Erection of metallic street light pole length upto 2 metres | Erection of metallic street light pole length upto 2 metres in cement concrete M-10 (1cement:3coarse sand:6 graded stone aggregate 40mm nominal size) foundation with cement concrete collar of size 0.4 metres dia x 0.3 metres height to accommodate the looping type cable end box including excavation and refilling etc. as required. | each | 1980.00 |
| 21.163 | Erection of metallic street light pole length exceeding 3 metres but not exceeding 6 metres | Erection of metallic street light pole length 3 metres but not exceeding 6 metres in cement concrete M-10 (1cement:3coarse sand:6 graded stone aggregate 40mm nominal size) foundation with cement concrete collar of size 0.4 metres dia x 0.5 metres height to accommodate the looping type cable end box including excavation and refilling etc. as required. | each | 2421.00 |
| 21.164 | Erection of pole strut in cement concrete 1:3:6 | Erection of RS Joist pole/'H' beam/steel tubular/steel rail pole strut in cement concrete 1:3:6 (1cement:3coarse sand: 6 aggregate 40mm nominal size) foundation (concreting 0.65 Cu Mtr. per pole) including excavation and refilling and secured with holding clamp, nuts and bolts, etc. as required. | each | 3854.00 |
| 21.165 | Erection of RCC/PCC pole strut in brick ballast | Erection of RCC/PCC pole strut in brick ballast and ramming the foundation including excavation and refilling and secured with holding clamp nuts and bolts etc. as required. | each | 2374.00 |
| 21.166 | Stay set complete (galvanized) 19mm. dia x 1.8 metre long stay rod | Supplying and erection of stay set complete (galvanized) 19mm. dia x 1.8 metre long stay rod anchor plate of size 300 mm x 300 mm x7.5mm thimble stay clamp, turn buckle (19mm dia x 600mm), 7/4.00mm dia G.I. stay wire etc. in cement concrete M-10 (1 Cement: 3 Coarse sand: 6 graded stone aggregate 40mm nominal size) (concreting 0.30 Cu Mtr. per stay) foundation including excavation and refilling etc. as required. | each | 4600.00 |

| Item No. | | Description | Unit | Rate |
|---|---|---|---|---|
| 21.167 | Stay set complete (galvanized) with 16mm dia x1.8 metre long stay | Supplying and erection of stay set complete (galvanized) with 16mm dia x1.8 metre long stay anchor plate of size 200mm x 200mm x 6.0mm thimble stay clamp turn buckle 19mmx600mm, 7/3.15mm dia G.I. stay wire etc. in cement concrete M-10 (1 cement: 3 coarse and: 6 grinded stone aggregate 40mm nominal size) (concreting 0.30 Cu Mtr. per stay) foundation including excavation and refilling etc. as required. | each | 3780.00 |
| 21.168 | Stay set complete with 19/20mm.dia 1.8 metre long stay rod | Supplying and erection of stay set complete with 19/20mm.dia 1.8 metre long stay rod, turn buckle, anchor plate of size 300mm x 300mm x 7.5mm thimble stay clamp, bow tightener, 7/4.00 dia G.I. stay wire etc. in cement concrete M-10 (1 cement: 3 coarse and: 6 granded stone aggregate 40mm nominal size) (concreting 0.30 Cu Mtr. per stay) foundation including excavation and back refilling etc. as required. | each | 4856.00 |
| 21.169 | Stay set complete with 16mm.dia 1.8 metre long stay rod | Supplying and erection of stay set complete with 16mm.dia 1.8 metre long stay rod, turn buckle, anchor plate of size 200mm x 200mm x 6.0mm, stay clamp, bow tightener, 7/3.15 dia G.I. stay wire etc. in cement concrete M-10 (1 cement: 3 coarse and: 6 granded stone aggregate 40mm nominal size) (concreting 0.30 Cu Mtr. per stay) foundation including excavation and back refilling etc. as required. | each | 3600.00 |
| 21.170 | Erection of stay set complete in cement concrete M-10 | Erection of stay set complete in cement concrete M-10 (1 cement: 3 coarse sand: 6 graded stone aggregate 40mm nominal size) (concreting 0.30 Cu Mtr. per stay) foundation including excavation and back refilling etc. as required. | each | 2165.00 |
| 21.171 | Bow stay set complete (galvanized) with 19mm dia x 1.8m long stay rod | Supplying and erection of bow stay set complete (galvanized) with 19mm dia x 1.8m long stay rod, anchor plate of size 300mmx300mm x 7.5mm, stay clamp, turn buckle 19mmx600mm, 7/4.00mm dia G.I. stay wire and bow tightener, etc. in cement concrete M-10 ( 1 cement : 3 coarse sand : 6 graded stone aggregate 40mm nominal size) (concreting 0.30 Cu Mtr. per stay) foundation including excavation and back refilling and also with 0.6m long brace of size 50mmx 50mmx6mm angle iron with 50mm dia pulley fixed at one end of the brace as required. | each | 5033.00 |
| 21.172 | Bow stay set complete (galvanized) with 16mm dia. X 1.8m long stay rod | Supplying and erection of bow stay set complete (galvanized) with 16mm dia. X 1.8m long stay rod, anchor plate of size 200mmx200mmx6.0mm, stay clamp, turn buckle 19mmx600mm, 7/3.15 mm dia G.I. Stay wire and bow tightener, etc. in cement concrete M-10 (1cement : 3 coarse sand : 6 graded stone aggregate 40mm nominal size) (concreting 0.30 Cu Mtr. per stay) foundation including excavation and back refilling and also with 0.6m long brace of size 50mmx50mmx6mm angle iron with 50mm dia pulley fixed at one end of the brace as required. | each | 4251.00 |
| 21.173 | Bow stay set | Supplying and erection of bow stay set complete with 19mm dia x 1.8m long stay rod, anchor plate of size | each | 4947.00 |

| Item No. | | Description | Unit | Rate |
|---|---|---|---|---|
| | complete with 19mm dia x 1.8m long stay rod | 300mmx300mm x 7.5mm, stay clamp, turn buckle 19mmx600mm, 7/4.00mm dia G.I. stay wire and bow tightener, etc. in cement concrete M-10 ( 1 cement : 3 coarse sand : 6 graded stone aggregate 40mm nominal size) (concreting 0.30 Cu Mtr. per stay) foundation including excavation and back refilling and also with 0.6m long brace of size 50mmx 50mmx6mm angle iron with 50mm dia pulley fixed at one end of the brace as required. | | |
| 21.174 | Bow stay set complete with 16mm dia. X 1.8m long stay rod | Supplying and erection of bow stay set complete with 16mm dia. X 1.8m long stay rod, anchor plate of size 200mmx200mmx6.0mm thimble stay clamp, turn buckle 19mmx600mm, 7/3.15 mm dia G.I. Stay wire and bow tightener, etc. in cement concrete M-10 (1cement: 3 coarse sand: 6 graded stone aggregate 40mm nominal size) (concreting 0.30 Cu Mtr. per stay) foundation including excavation and back refilling and also with 0.6m long brace of size 50mmx50mmx6mm angle iron with 50mm dia pulley fixed at one end of the brace as required. | each | 4030.00 |
| 21.175 | 50x65mm shackle insulator | Supplying and erection of 50mm x 65mm shackle insulator with G.I. bolts, nuts and straps etc. as required. | each | 92.00 |
| 21.176 | 75x90mm shackle insulator | Supplying and erection of 75mm x 90mm shackle insulator with G.I. bolts, nuts and straps etc. as required. | each | 102.00 |
| 21.177 | 100x110mm shackle insulator | Supplying and erection of 100mmx110mm shackle insulator with G.I. bolts, nuts and straps etc. as required. | each | 109.00 |
| 21.178 | Aluminium knob/bobbin | Supplying and erection of aluminium knob/bobbin with nut, bolts & straps etc. as required. | each | 119.00 |
| 21.179 | 19/20mm dia. M.S. pipe 1 meter in length for lights | Supplying and fixing of 19mm/20mm dia. M.S. pipe 1 meter in length for CFL/LED street light/bulkhead/water tight bracket including bending to shape and wiring with 1.5sqmm W.P./PVC wire etc. as required. | each | 279.00 |
| 21.180 | 19mm/20 mm dia. G.I. pipe 2 meter long for lights | Supplying and fixing of 19mm/20mm dia. G.I. pipe 2 meter long for CFL/LED street light/bulkhead/water tight bracket including bending to shape and wiring with 1.5sqmm W.P./PVC wire etc. as required. | each | 542.00 |
| 21.181 | 32mm dia. X 2.00 meter long G.I. pipe (medium class) bracket | Supplying and fixing of 32mm dia. X 2.00 meter long G.I. pipe (medium class) bracket for mounting of fluorescent/S.V/M.H/L.E.D lamp street light fittings on poles including bending the pipe to the required shape and wiring with 1.5sq.mm W.P./PVC wire etc. as required. | each | 791.00 |
| 21.182 | Angle/channel/flat iron fitting for overhead | Supply, fabrication and erection of angle/channel/flat iron fitting for overhead line & sub-station etc such as 'D' bracket, cross arms, top clamp, 'V' cross arms, back/support clamp or other similar work etc. including nut bolts of required size, making holes, fabrication, welding, cutting, | kg. | 105.00 |

| Item No. | | Description | Unit | Rate |
|---|---|---|---|---|
| | line & sub-station etc. | etc. and painting with one coat of red oxide paint & two coat of aluminium paint as required as per specification. | | |
| 21.183 | Labour charges for erection of angle/channel/flat iron fitting for overhead line & sub-station etc | Labour charges for erection of angle/channel/flat iron fitting for overhead line & sub-station etc such as 'D' bracket, cross arms, top clamp, 'V' cross arms, back/support clamp or other similar work etc. including nut bolts of required size, as required as per specification. | kg. | 23.00 |
| 21.184 | 11KV porcelain pin insulator | Supplying and erection of 11KV porcelain pin insulator complete with long steel head G.I. pin, nut, washer etc. as required. | each | 375.00 |
| 21.185 | 11KV porcelain disc insulator | Supplying and erection of 11KV porcelain disc insulator for 11KV overhead lines with galvanized insulator fittings, ball and socket type, and complete with galvanized strain clamp, bolts, nuts washer etc. as required. | each | 750.00 |
| 21.186 | 33KV porcelain pin insulator | Supplying and erection of 33KV porcelain pin insulator complete with long steel head G.I. pin, nut, washer etc. as required | each | 788.00 |
| 21.187 | Set of three 11KV porcelain disc insulators for 33KV overhead lines | Supplying and erection of a set of three 11KV porcelain disc insulators for 33KV overhead lines with galvanized insulator fittings, ball and socket type, and complete with galvanized strain clamp, bolts, nuts, washers etc. as required. | each | 1928.00 |
| 21.188 | Set of four 11KV porcelain disc insulators for 33KV overhead lines | Supplying and erection of a set of four 11KV porcelain disc insulators for 33KV overhead lines with galvanized insulator fittings, ball and socket type, and complete with galvanized strain clamp, bolts, nuts, washers etc. as required. | each | 2639.00 |
| 21.189 | Labour, 11/33KV porcelain pin insulator | Labour for erection of 11/33KV porcelain pin insulator complete with hardware as required. | each | 169.00 |
| 21.190 | Labour, 11KV porcelain disc insulator for 11KV overhead lines | Labour for erection of 11KV porcelain disc insulator for 11KV overhead lines complete with hardware as required | each | 169.00 |

| Item No. | | Description | Unit | Rate |
|---|---|---|---|---|
| 21.191 | Labour, 11KV porcelain disc insulators for 33KV overhead lines | Labour for erection of a set of three/ four 11KV porcelain disc insulators for 33KV overhead lines complete with hardware as required. | each | 404.00 |
| 21.192 | 11KV (5 KN) polymer pin insulator | Supplying and erection of 11KV (5 KN) polymer pin insulator complete with long steel head G.I. pin, nut, washer etc. as required. | each | 402.00 |
| 21.193 | 11KV (45 KN) Polymer disc insulator for 11KV overhead lines | Supplying and erection of 11KV (45 KN) Polymer disc insulator for 11KV overhead lines with galvanized insulator fittings, ball and socket type, and complete with galvanized strain clamp, bolts, nuts washer etc. as required. | each | 502.00 |
| 21.194 | 33KV (10 KN) polymer pin insulator | Supplying and erection of 33KV (10 KN) polymer pin insulator complete with long steel head G.I. pin, nut, washer etc. as required | each | 595.00 |
| 21.195 | 33KV (45 KN) polymer disc insulators | Supplying and erection of 33KV (45 KN) polymer disc insulators for 33KV overhead lines with galvanized insulator fittings, ball and socket type, and complete with galvanized strain clamp, bolts, nuts, washers etc. as required. | each | 1018.00 |
| 21.196 | Labour, 11/33KV polymer pin insulator | Labour for erection of 11/33KV polymer pin insulator complete with hardware as required. | each | 160.00 |
| 21.197 | Labour, 11/33KV polymer disc insulators | Labour for erection of 11/33KV polymer disc insulators for 11/33KV overhead lines complete with hardware as required. | each | 259.00 |
| 21.198 | Single piece nonlinear resistor type lighting arrestor (set of 3 nos) for 3 wire, 11KV | Supplying and erection of single piece nonlinear resistor type lighting arrestor (set of 3 nos) for 3 wire, 11KV overhead lines/sub-station with rated voltage of 9KV (rms) with a nominal discharge current rating of 5 KA and complete with galvanized clamping arrangement, G.I. bolts, nuts, washer etc. as required. | Set | 3832.00 |
| 21.199 | Single piece non-linear | Supplying and erection of single piece non-linear resistor type lightning arrestor (set of 3 nos) for 3 wire, 33KV overhead lines/sub-station with rated voltage of 30KV (rms) | Set | 32662.00 |

| Item No. | | Description | Unit | Rate |
|---|---|---|---|---|
| | resistor type lightning arrestor (set of 3 nos) for 3 wire, 33KV | with a nominal discharge current rating of 5 KA and complete with galvanized clamping arrangement, G.I. bolts, nuts, washer etc. as required. | | |
| 21.200 | Single piece nonlinear resistor polymer type lighting arrestor (set of 3 nos) for 3 wire, 11KV | Supplying and erection of single piece nonlinear resistor polymer type lighting arrestor (set of 3 nos) for 3 wire, 11KV overhead lines/sub-station with rated voltage of 9KV (rms) with a nominal discharge current rating of 10 KA and complete with galvanized clamping arrangement, G.I. bolts, nuts, washer etc. as required. | Set | 3617.00 |
| 21.201 | Single piece non-linear resistor polymer type lightning arrestor (set of 3 nos) for 3 wire, 33KV | Supplying and erection of single piece non-linear resistor polymer type lightning arrestor (set of 3 nos) for 3 wire, 33KV overhead lines/ sub-station with rated voltage of 27KV (rms) with a nominal discharge current rating of 10 KA and complete with galvanized clamping arrangement, G.I. bolts, nuts, washer etc. as required. | Set | 35697.00 |
| 21.202 | Outdoor H.T., AB switch assembly set | Supplying installing, testing & commissioning of outdoor H.T., AB switch assembly set gang operated with brass contact parts, including required GI pipe operating rod, handle & locking arrangement on On-Off position conforming to IS complete with required material and installing on existing DP structure to complete the job as required as per specification.(Set of 3 nos.) | | |
| (i) | 11 KV, 400Amp | 11 KV, 400Amp | set | 8989.00 |
| (ii) | 33KV, 600Amp | 33KV, 600Amp | set | 31525.00 |
| 21.203 | D.O. fuse | Supplying installing, testing & commissioning D.O. fuse assembly with brass part contact for 33 or 11/0.4 KV DP Structure set of 3 Nos. with fuse barrel i/c required fuse element & other materials as per specification on existing D.P. structure as required. | | |
| (i) | 11KV | 11KV | set | 4721.00 |
| (ii) | 33KV | 33KV | set | 16000.00 |
| 21.204 | Replacement of D.O. fuse barrel with brass parts | Supplying & replacement of D.O. fuse barrel with brass parts for 33 or 11 KV i/c required fuse element according to load as per specification on existing D.P. structure as required. | | |
| (i) | 11KV | 11KV | each | 910.00 |

| Item No. | | Description | Unit | Rate |
|---|---|---|---|---|
| (ii) | 33KV | 33KV | each | 1895.00 |
| 21.205 | Replacement of D.O. fuse barrel for 33 or 11 KV | Supplying & replacement of D.O. fuse barrel for 33 or 11 KV as per specification on existing D.P. structure as required. | | |
| (i) | 11KV | 11KV | each | 274.00 |
| (ii) | 33KV | 33KV | each | 314.00 |
| 21.206 | Replacement of D.O. fuse element | Supplying & replacement of D.O. fuse element according to load for 33 or 11 KV as per specification on existing D.P. structure as required. | | |
| (i) | 11KV | 11KV | each | 102.00 |
| (ii) | 33KV | 33KV | each | 119.00 |
| 21.207 | Anti-climbing device | Supply & fixing anti climbing device with 2 ply G.I. barbed wire 1 Kg. per pole complete as per MPMKVVCO. Specification. | each | 231.00 |
| 21.208 | G.I. earth coil of 4mm dia | Supply, laying and fixing of G.I. earth coil of 4mm dia G.I. wire having 120 turns of nearly 50mm dia. and 3 mt. long tail in existing pit duly earthed with pole etc complete as required as per specification. | each | 293.00 |
| 21.209 | Danger board | Supplying and fixing 33/11/0. 4 KV enamel coated danger board size 200x250mm with clamp on existing HT/LT structure/poles. | each | 205.00 |
| 21.210 | Transformers, 11/0.4 KV 3 Phase | Supplying, installing, testing and commissioning of 11/0.4 KV 3 Phase 50 cycle oil immersed, naturally cooled, outdoor type transformer connected delta on H.T. side and star on L.T. side, dehydrating silica gel breather on eye level, rating and diagram plate, two earthing terminal, lifting lugs, oil level gauge, drain valve with plug, temperature not exceeding 50ºC on load, oil filling hole with plug, four unidirectional roller, arcing horns, explosion vent, terminal arrangement bushing on H.V. side and cable box on LV side, first filling of oil upto desired level and transformer installing on existing structure with all required materials arrangement as required ISI marked & as per IS specification, hand operated off load tap changer switch above 100 KVA, oil conservator with drain valve, plug 63 KVA and above. | | |
| (A) | Level-2 Al. wound | Energy efficiency level -2 (New level -1) Aluminium wound | | |
| (i) | 25 KVA | 25 KVA | each | 120963.00 |
| (ii) | 63 KVA | 63 KVA | each | 228764.00 |
| (iii) | 100 KVA | 100 KVA | each | 316509.00 |
| (iv) | 200 KVA | 200 KVA | each | 607948.00 |
| (B) | Cu. wound | Copper wound | | |
| (i) | 315 KVA | 315 KVA | each | 1096813.00 |
| (ii) | 500 KVA | 500 KVA | each | 1472863.00 |
| (C) | Level-3 Al. wound | Energy efficiency level -3 (New level-2) Aluminium wound | | |
| (i) | 25 KVA | 25 KVA | each | 169223.00 |

| Item No. | | Description | Unit | Rate |
|---|---|---|---|---|
| (ii) | 63 KVA | 63 KVA | each | 266369.00 |
| (iii) | 100 KVA | 100 KVA | each | 348160.00 |
| (iv) | 200 KVA | 200 KVA | each | 714495.00 |
| (D) | Cu. wound | Copper wound | | |
| (i) | 315 KVA | 315 KVA | each | 1219029.00 |
| (ii) | 500 KVA | 500 KVA | each | 1764301.00 |
| 21.211 | Transformers, 33/0.4 KV 3 Phase | Supplying, installing, testing and commissioning of 33/0.4 KV 3 Phase 50 cycle oil immersed, naturally cooled, outdoor type transformer connected delta on H.T. side and star on L.T. side, hand operated off load Tap changer switch above 100 KVA, rating and diagram plate, two earthing terminal, lifting lugs, oil level gauge, drain valve with plug, temperature not exceeding 50ºC on load, oil conservator with drain plug, oil filling hole with plug, dehydrating silica gel breather on eye level, four unidirectional roller, arcing horns, explosion vent, terminal arrangement bushing on H.V. side and cable box on LV side, first filling of oil upto desired level and transformer installing on existing structure with all required materials arrangement as required ISI marked & as per IS specification. | | |
| (A) | Level-2 Cu. wound | Energy efficiency level -2 (New level -1) Copper wound | | |
| (i) | 100 KVA | 100 KVA | each | 651820.00 |
| (ii) | 200 KVA | 200 KVA | each | 842979.00 |
| (iii) | 315 KVA | 315 KVA | each | 1153220.00 |
| (iv) | 500 KVA | 500 KVA | each | 1595079.00 |
| (B) | Level-3 Cu. wound | Energy efficiency level -3 (New level-2) Copper wound | | |
| (i) | 100 KVA | 100 KVA | each | 745833.00 |
| (ii) | 200 KVA | 200 KVA | each | 968329.00 |
| (iii) | 315 KVA | 315 KVA | each | 1407054.00 |
| (iv) | 500 KVA | 500 KVA | each | 1783104.00 |
| 21.212 | Transformer, 33/0.4 KV 3 phase | Supplying, installing, testing and commissioning of 33/0.4 KV 3 Phase 50 cycle oil immersed, naturally cooled, outdoor type transformer connected delta on H.T. side and star on L.T. side, hand operated off load Tap changer switch, rating and diagram plate, two earthing terminal, lifting lugs, oil level gauge, drain valve with plug, temperature not exceeding 50ºC on load, temperature dial gauge, Bucholz relay, oil conservator with drain plug, oil filling hole with plug, dehydrating silica gel breather on eye level, four unidirectional roller, arcing horns, explosion vent, terminal arrangement bushing on H.V. side and cable box on LV side, first filling of oil upto desired level and transformer installing on existing structure with all required materials arrangement as required ISI marked & as per IS specification. | | |
| (A) | Level-2 Cu. wound | Energy efficiency level -2 (New level -1) Copper wound | | |
| (i) | 630 KVA | 630 KVA | each | 1723563.00 |
| (ii) | 1000 KVA | 1000 KVA | each | 2130950.00 |
| (iii) | 1250 KVA | 1250 KVA | each | 2312708.00 |

| Item No. | | Description | Unit | Rate |
|---|---|---|---|---|
| (iv) | 1600 KVA | 1600 KVA | each | 3002133.00 |
| (v) | 2000 KVA | 2000 KVA | each | 4255633.00 |
| (vi) | 2500 KVA | 2500 KVA | each | 5007733.00 |
| (B) | Level-3 Cu. wound | Energy efficiency level -3 (New level-2) Copper wound | | |
| (i) | 630 KVA | 630 KVA | each | 1848913.00 |
| (ii) | 1000 KVA | 1000 KVA | each | 2344045.00 |
| (iii) | 1250 KVA | 1250 KVA | each | 2550873.00 |
| (iv) | 1600 KVA | 1600 KVA | each | 3315508.00 |
| (v) | 2000 KVA | 2000 KVA | each | 5139350.00 |
| (vi) | 2500 KVA | 2500 KVA | each | 5640750.00 |
| 21.213 | Shifting, erection, & commissioning of 100/200/315/500 KVA (11/0.433 KV) distribution transformer | Shifting, erection, & commissioning of 25/63/100/200/315/500 KVA (11/0.433 KV) distribution transformer on existing structure including Bi-metallic clamp, ACSR/AAAC Conductor i/c colour code sleeve, for connection from H.T. Line to transformer as required as per specification. | each | 10846.00 |
| 21.214 | Limit switch with accessories | Supplying and fixing limit switch with accessories & arrangement on existing HT/LT structure/poles AB switch handle for inter locking between distribution panel main incoming LT switch gear/ system as required to complete the job. | each | 4851.00 |
| 21.215 | Fencing of chain link mesh 100x100 mm, 12 SWG GI wire | Supply & fixing fencing of chain link mesh 100 x 100 mm, 12 SWG GI wire for for D.P. structure electric sub-station switch yard including single post of angle iron size 50x50x5mm, 2.5 mt long should be provided upto 2.00 mt. apart or as required. The panel frame with required angle iron/ flat iron members/cross members including mesh welded in frame panel shall be erected between the posts. The angle iron post shall be erected in cement concrete M-15 (1:2:4) (20 mm graded metal) foundation, including excavation of pit and refilling the same. The fencing shall be provided with one entrance gate including locking arrangement as required as per specification. The height of fencing shall be 2 mt. above ground level. Fencing shall be painted with required shade. | sq.Mt | 1821.00 |
| 21.216 | Fencing of chain link mesh 100x100 mm, 12 SWG GI wire | Supply & fixing fencing of chain link mesh 100 x 100 mm, 12 SWG GI wire for electric sub-station switch yard including single post of angle iron size 50x50x5mm, 2.5 mt long should be provided upto 2.00 mt. apart or as required with angle iron/ flat iron members /cross members including mesh fix on frame and erected between the post. The angle iron post shall be erected in cement concrete M-15 (1:2:4) (20 mm graded metal) foundation, cement concrete, excavation of pit and refilling the same. The fencing shall be provided with one entrance gate including | sq.mt | 1491.00 |

| Item No. | | Description | Unit | Rate |
|---|---|---|---|---|
| | | locking arrangement as required as per specification. The height of fencing shall be 2 mt. above ground level. Fencing shall be painted with required shade. | | |
| 21.217 | Rubber mat | Supply of rubber mat 2 x 1.0Mtr, 1100V tested | each | 1913.00 |
| 21.218 | Hand gloves 11KV | Supply of hand gloves 11KV tested | each | 1077.00 |
| 21.219 | Hand gloves 33KV | Supply of hand gloves 33KV tested | each | 1291.00 |
| 21.220 | Fiber discharge rod | Supply of fiber discharge rod I/C connection copper lead tested | each | 4042.00 |
| 21.221 | D.O. operating rod | Supply of fiber D.O operating rod for 11/33KV tested | each | 3854.00 |
| 21.222 | First aid box | Supply of first aid box complete with medicine & bandage as per specification. | set | 1646.00 |
| 21.223 | Shock instruction chart | Supply of shock instruction chart duly framed with glass, as reqd. | each | 539.00 |
| 21.224 | Transformer breather with silica gel | Supply and fixing transformer breather with silica gel suitable for 100 -500 KVA as reqd. | each | 601.00 |
| 21.225 | New transformer oil | Supply and filling of new transformer oil of desired break down value for topping as reqd. | Ltr. | 222.00 |
| 21.226 | Filtration of transformer oil | Filtration of transformer oil for 33/0.4 KV, 11/0.4 KV 3 phase 50 cycle oil immersed, naturally cooled, distribution transformer upto desired break down value. | Ltr. | 30.00 |
| 21.227 | Construction of pedestal (plinth) for transformer | Designing and construction of pedestal (plinth) for transformer of desired size including excavation in hard rock/soft rock, base concrete M-7.5 (1:4:8), 20mm metal and pedestal of RCC M-15 (1:2:4) including required tar steel reinforcement for required height as per IE rules from base concrete including shuttering and finishing with 10mm thick plaster in cement mortar 1:3 curing & white washing etc. complete as per specifications. | Cu. Mt. | 7379.00 |
| 21.228 | Spreading of metal dust, murum | Collection and spreading of metal dust, murum and 40mm graded hard metal in sub-station yard or as required as mentioned below - | | |
| (i) | Metal dust | Metal dust | Cu. Mt. | 1274.00 |
| (ii) | Murum | Murum | Cu. Mt. | 847.00 |
| (iii) | Metal 40 mm | Metal 40 mm | Cu. Mt. | 1478.00 |
| 21.229 | PVC insulated | Supplying & drawing/laying of PVC insulated XLPE aerial bunch power cable conductor H2/H4 grade aluminium | | |

| Item No. | | Description | Unit | Rate |
|---|---|---|---|---|
| | XLPE aerial bunch power cable | solid/stranded conforming to IS-7098 Part-ll, Alu. alloy messenger of approved make in air with necessary material as per specification of required size as mentioned below – | | |
| (A) | 11KV grade | 11KV grade arial bunch cable :- | | |
| (i) | 3x50 + 1x70 sq.mm. | 3 x50 + 1 x 70 sq.mm. | P.Mt. | 764.00 |
| (ii) | 3x70 + 1x70 sq.mm. | 3 x70 + 1 x 70 sq.mm. | P.Mt. | 887.00 |
| (iii) | 3x95 + 1x80 sq.mm. | 3 x95 + 1 x 80 sq.mm. | P.Mt. | 980.00 |
| (iv) | 3x150 + 1x120 sq.mm. | 3 x150 + 1 x 120 sq.mm. | P.Mt. | 1306.00 |
| (v) | 3x150 + 1x120 sq.mm. | 4 x150 + 1 x 120 sq.mm. | P.Mt. | 1307.00 |
| (vi) | 50 sq.mm. | 50 sq.mm. | P.Mt. | 1549.00 |
| (vii) | 95 sq.mm. | 95 sq.mm. | P.Mt. | 2081.00 |
| (viii) | 150 sq.mm. | 150 sq.mm. | P.Mt. | 2394.00 |
| 21.230 | PVC insulated XLPE aerial bunch power cable | Supplying & drawing/laying of PVC insulated XLPE aerial bunch power cable conductor H2/H4 grade aluminium solid/stranded conforming to IS-14255- 1995, alu. alloy insulated messenger 1100 volts grade cable of approved make in air with necessary material as per specification of required size as mentioned below – | | |
| (A) | Single core cable | Single core cable(insulated neutral messenger) | | |
| (i) | 2x25 sq.mm, 1x16 | 2x25 sq.mm, 1x16(insulated neutral messenger) | P.Mt. | 202.00 |
| (ii) | 2x35 sq.mm, 1x16 | 2x35 sq.mm, 1x16(insulated neutral messenger) | P.Mt. | 248.00 |
| (iii) | Single core cable | Single core cable(insulated neutral messenger) | | |
| (iv) | 3x25 sq.mm | 3x25 sq.mm, 1x16(st. light), 1x35(insulated neutral messenger) | P.Mt. | 246.00 |
| (v) | 3x35 sq.mm | 3x35 sq.mm, 1x16(st. light), 1x35(insulated neutral messenger) | P.Mt. | 311.00 |
| (vi) | 3x50 sq.mm | 3x50 sq.mm, 1x16(st. light), 1x35(insulated neutral messenger) | P.Mt. | 321.00 |
| (vii) | 3x70 sq.mm | 3x70 sq.mm, 1x16(st. light), 1x35(insulated neutral messenger) | P.Mt. | 525.00 |
| (viii) | 3x95 sq.mm | 3x95 sq.mm, 1x16(st. light), 1x50(insulated neutral messenger) | P.Mt. | 647.00 |
| (ix) | 3x120 sq.mm | 3x120 sq.mm, 1x16(st. light), 1x70(insulated neutral messenger) | P.Mt. | 808.00 |
| (x) | 3x150 | 3x150 sq.mm, 1x35(st. light), 1x95(insulated neutral | P.Mt. | 922.00 |

| Item No. | | Description | Unit | Rate |
|---|---|---|---|---|
| | sq.mm | messenger) | | |
| 21.231 | Aerial bunch cable accessories suspension clamp | Supplying & fixing of aerial bunch cable accessories suspension clamp i/c required attachment for single core cable of approved make on pole with clamp, necessary material as per specification for required size. | each | 427.00 |
| 21.232 | Aerial bunch cable accessories tension clamp (dead end) | Supplying &fixing of aerial bunch cable accessories tension clamp (dead end) i/c required attachment for single core cable of approved make in air with clamp, necessary material as per specification for required size. | each | 446.00 |
| 21.233 | Piercing connector | Supplying & fixing of waterproof insulation piercing connector for aerial bunch cable i/c required attachment for single core cable of approved make in air with necessary material as per specification of required size as mentioned below – | | |
| (i) | Main:16-95sq.mm | Main:16-95sq.mm, tap:1.5-50sq.mm | each | 172.00 |
| (ii) | Main:50-150sq.mm | Main:50-150sq.mm, tap:6-50sq.mm | each | 220.00 |
| 21.234 | Spring loaded busbar | Supplying &fixing of spring loaded busbar screwless connector with enclosure of approved make including required clamp, nut, bolt, connection, testing as per specification to complete the job. | | |
| (A) | Single phase | Single phase | | |
| (i) | 1Ph, 1In+2Out | 1Ph, 1In+2Out | each | 1972.00 |
| (ii) | 1Ph, 1In+3Out | 1Ph, 1In+3Out | each | 2023.00 |
| (iii) | 1Ph, 1In+4Out | 1Ph, 1In+4Out | each | 2989.00 |
| (iv) | 1Ph, 1In+5Out | 1Ph, 1In+5Out | each | 3080.00 |
| (v) | 1Ph, 1In+7Out | 1Ph, 1In+7Out | each | 3226.00 |
| (vi) | 1Ph, 1In+8Out | 1Ph, 1In+8Out | each | 3449.00 |
| (B) | Three Phase | Three Phase | | |
| (i) | 3Ph, 1In+2Out | 3Ph, 1In+2Out | each | 4746.00 |
| (ii) | 3Ph, 1In+4Out | 3Ph, 1In+4Out | each | 5839.00 |
| (iii) | 3Ph, 1In+6Out | 3Ph, 1In+6Out | each | 7276.00 |
| (iv) | 3Ph, 1In+7Out | 3Ph, 1In+7Out | each | 7404.00 |
| (v) | 3Ph, 1In+8Out | 3Ph, 1In+8Out | each | 7538.00 |
| 21.235 | 12.5 meter high mast lighting | Providing and erection on existing foundation12.5 meter high mast lighting system suitable to install 6 No. 2 x 400 w M.H/S.V/LED flood light fitting and control gear with | each | 205322.00 |

| Item No. | | Description | Unit | Rate |
|---|---|---|---|---|
| | | integral Power tool, weight 340 kg. comprising of 2 section of hot dipped galvanized materials as per BSEN ISO 1461 thickness 3 mm, dia 100mm & 360mm for top and bottom respectively, stress fitting arrangement on site with 350mm overlap dynamic loading to withstand max wind pressure as per -IS 875 part III, parameters for structural & foundation design must be taken from wind tunnel test. Lightning protection of GI single spike 800mm at top and at base inside compartment with double internal lock with adequate size of MCB erected on PVC board complete with base plate of 25mm thick 520mm dia. and foundation bolts having 4 nos. bolts of 24mm dia, 750mm long (EN8 grade), Anchor plate 445 PCD, including accessories viz.<br><br>(1) Lantern carriage of 50 NB ERW class-B, MS pipe covered with PVC sleeve suitable to carry 250kg. load and upto 6 fittings symmetrically.<br>(2) Trailing copper cable 5x2.5 sq.mm, EPR insulated PCP sheathed.<br>(3) Double drum/350kg winch having gear 53:1, oil bath (SAE90/140) arrangement.<br>(4) 2 nos. stainless steel wire ropes 5mm dia (7/19) breaking load capacity 1450 kg.x 2.<br>(5) Integral power tool 3-phase, 0.75 HP 2m/min single speed.<br>(6) Feeder pillar fabricated out of 14 SWG CRCA sheet and comprise of incoming MCB 32 A TPN switch, HRC fuses, single dial timer, suitable size of contactors for lighting and power tool, 2 nos. outgoing, reversing switch for motor.<br>(7) Foundation drawing, test certificate and guarantee certificate shall be provided by manufacturer. (without fixture and lamp) | | |
| 21.236 | 16 meter high mast lighting | Providing and erection on existing foundation 16 meter high mast lighting system suitable to install upto 8 No. 2 x 250/ 400 w M.H/S.V/LED flood light and control gear with integral power tool, weight 450 kg. comprising of 2 sections of hot dipped galvanized materials as per BSEN ISO 1461 thickness 3 mm & 4 mm, dia 150mm & 360mm for top and bottom respectively, stress fitting arrangement on site with 500mm overlap dynamic loading to withstand max wind pressure as per -IS 875 part III, parameters for structural & foundation design must be taken from wind tunnel test. Lightning protection of GI single spike 1200mm at top and at base inside compartment having double internal lock with adequate size of MCB erected on PVC board complete with base plate of 25mm thick 520mm dia. and foundation bolts having 6 nos. bolts of 24mm dia 750 mm long (EN.8 grade) anchor plate 445 PCD, including accessories viz.<br>(1) Lantern carriage of 50 NB ERW class-B, MS pipe covered with PVC sleeve suitable to carry 500kg. load and upto 8 fittings symmetrically.<br>(2) Trailing copper cable 5x2.5 sq.mm, PVC insulated PVC Sheathed<br>(3) Double drum 350Kg Winch having gear 53:1, oil bath (SAE90/140) arrangement | each | 266303.00 |

| Item No. | | Description | Unit | Rate |
|---|---|---|---|---|
| | | (4) 2 nos. Stainless steel wire ropes 5 mm dia (7/19) breaking load capacity 1450 kg x 2,<br>(5) Integral power tool 3-phase, 0.75HP 2m/min single speed .torque limiter upto 500 kg adjustable<br>(6) Single LED 300 Nos. aviation light<br>(7) Feeder pillar fabricated out of 14 SWG CRCA sheet and comprise of incoming MCB 32 A TPN switch, HRC fuses, single dial timer, suitable size of contactors for lighting and power tool, 2 nos. outgoing, reversing switch for motor.<br>(8) Foundation drawing, test certificate and guarantee certificate shall be provided by manufacturer. (without fixture and lamp) | | |
| 21.237 | 20 meter high mast lighting | Providing and erection on existing foundation 20 meter high mast lighting system suitable to install upto 10 No. 2 x 250/ 400 w M.H/S.V/LED flood light and control gear with integral Power tool, weight 450 kg. comprising of 2 sections of hot dipped galvanized materials as per BSEN ISO 1461 thickness 3 mm & 4 mm, dia 150mm & 410mm for top and bottom respectively, stress fitting arrangement on site with 500mm overlap dynamic loading to withstand max wind pressure as per -IS 875 part III, parameters for structural & foundation design must be taken from wind tunnel test. Lightning protection of GI single spike 1200mm at top and at base inside compartment having double thickness 3 mm & 4 mm, dia 150mm & 410mm for top and bottom respectively, stress fitting arrangement on site with 500mm overlap dynamic loading to withstand max wind pressure as per -IS 875 part III, parameters for structural & foundation design must be taken from wind tunnel test.<br>Lightning protection of GI single spike 1200mm at top and at base inside compartment having double internal lock with adequate size of MCB erected on PVC board complete with base plate of 25mm thick 570mm dia. and foundation bolts having 8 nos. bolts of 30mm dia 850 mm long (EN.8 grade) Anchor plate 490 PCD, including accessories viz.<br><br>(1) Lantern carriage of 50 NB ERW class-B, MS pipe covered with PVC sleeve suitable to carry 500kg. load and upto 10 fittings symmetrically.<br>(2) Trailing copper cable 5x4.0 sq.mm, PVC insulated PVC sheathed.<br>(3) Double drum 750Kg Winch having gear 53:1, oil bath (SAE90/140) arrangement.<br>(4) 2 nos. stainless steel wire ropes 6 mm dia (7/19) breaking load capacity 2350 kg. x 2,<br>(5) Integral power tool 3-phase, 1HP 2m/min single speed .torque limiter upto 500 kg adjustable<br>(6) 2 Nos. LED 300 nos. aviation light<br><br>(7) Feeder pillar fabricated out of 14 SWG CRCA sheet and comprise of incoming MCB 63 A TPN switch, HRC fuses, single dial timer, suitable size of contactors for lighting and power tool, 2 nos. outgoing, reversing switch for motor.<br>(8) Foundation drawing, test certificate and guarantee certificate shall be provided by manufacturer. (without | each | 372547.00 |

| Item No. | | Description | Unit | Rate |
|---|---|---|---|---|
| | | fixture and lamp) | | |
| 21.238 | Octagonal poles | Providing and erection of hot dipped galvanized octagonal poles in single section made from 3mm thick sheet having lockable weather proof flush door junction box, suitable size & type of foundation bolts 4 nos. 'J' type (EN8 grade) complete, erected in an approved manner on provided foundation. | | |
| | 3 meter | 3 meter height 130 X 70mm A/F, 200x200x12mm base plate having weight 28 kg ± 3% , 4 nos X 16 x 450mm bolt having weight 3.2 kg ± 3%. | each | 4691.00 |
| | 4 meter | 4 meter height 130 X 70mm A/F, 200x200x12mm base plate having weight 36 kg ± 3%, 4 nos X 16 x 450mm bolt having weight 3.2 kg ± 3%. | each | 5899.00 |
| | 5 meter | 5 meter height 130 X 70mm A/F, 200x200x12mm base plate having weight 44 kg ± 3%, 4 nos X 16 x 450mm bolt having weight 3.2 kg ± 3%. | each | 6943.00 |
| | 6 meter | 6 meter height 130 X 70mm A/F, 220x220x12mm base plate having weight 52 kg ± 3%, 4 nos. X 20 x 600mm bolt having weight 6.7 kg ± 3%. | each | 7956.00 |
| | 7 meter | 7 meter height, 130 X 70mm A/F, 220x220x12mm base plate having weight 60 kg ± 3%, 4 nos. X 20 x 700mm bolt having weight 7.7 kg ± 3%. | each | 9863.00 |
| | 8 meter | 8 meter height, 135 X 70mm A/F, 225x225x16mm base plate having weight 70 kg ± 3%, 4 nos. X 20 x 750mm bolt having weight 8.2 kg ± 3%. | each | 13415.00 |
| | 9 meter | 9 meter height, 155 X 70mm A/F, 250x250x16mm base plate having weight 87 kg ± 3%, 4 nos. X 24 x 750mm bolt having weight 12 kg ± 3%. | each | 15447.00 |
| | 10 meter | 10 meter height, 175 X 70mm A/F, 275x275x16mm base plate having weight 105 kg ± 3%, 4 nos. X 24 x 750mm bolt having weight 12 kg ± 3%. | each | 17114.00 |
| | 11 meter | 11 meter height, 175 X 70mm A/F, 275x275x16mm base plate having weight 115 kg ± 3%, 4 nos. X 24 x 750mm bolt having weight 12 kg ± 3%. | each | 18587.00 |
| | 11 meter | 11 meter height, 210 X 90mm A/F, 300x300x16mm base plate having weight 141 kg ± 3%, 4 nos. X 24 x 750mm bolt having weight 12 kg ± 3%. | each | 21737.00 |
| | 12 meter | 12 meter height, 210 X 90mm A/F, 300x300x16mm base plate having weight 153 kg ± 3%, 4 nos. X 24 x 750mm bolt having weight 12 kg ± 3%. | each | 24579.00 |
| 21.239 | Bakelite sheet MCBSP & stud type 4 way epoxy connector for decorative /octagonal pole box. | Providing and fixing of bakelite sheet 10mm thick, one MCB 6Amp-32Amp SP & 16sq.mm stud type 4 way epoxy connector for decorative designer poles /octagonal pole box. | each | 323.00 |
| 21.240 | Sword canopy type single arm GI bracket | Providing and fixing of sword canopy type single arm GI bracket upto 2 meter Long for 70mm A/F top octagonal pole | each | 1658.00 |
| 21.241 | Sword | Providing and fixing of sword canopy type single arm GI | each | 2071.00 |

| Item No. | | Description | Unit | Rate |
|---|---|---|---|---|
| | canopy type single arm GI bracket | bracket upto 2 meter Long for 90mm A/F top octagonal pole | | |
| 21.242 | Sword canopy type double arm GI bracket | Providing and fixing of sword canopy type double arm GI bracket upto 2 meter Long for 70mm A/F top octagonal pole | each | 2200.00 |
| 21.243 | Sword canopy type double arm GI bracket | Providing and fixing of sword canopy type double arm GI bracket upto 2 meter Long for 90mm A/F top octagonal pole | each | 3496.00 |
| 21.244 | Single arm standard bracket | Single arm standard bracket | | |
| | | Providing and fixing of 1000mm long Single Arm GI Hot Dip Galvanized Bracket with 42.4mm OD and 3.2mm thick Pipe. Canopy pipe should be 500mm with 4 Nos 10mm dia Pinch bolts. | each | 2059.00 |
| | | Providing and fixing of 1500mm long Single Arm GI Hot Dip Galvanized Bracket with 42.4mm OD and 3.2mm thick Pipe. Canopy pipe should be 500mm with 4 Nos 10mm dia Pinch bolts. | each | 3084.00 |
| | | Providing and fixing of 2000mm long Single Arm GI Hot Dip Galvanized Bracket with 42.4mm OD and 3.2mm thick Pipe. Canopy pipe should be 500mm with 4 Nos 10mm dia Pinch bolts. | each | 4178.00 |
| 21.245 | Double arm standard bracket | Double arm standard bracket | | |
| | | Providing and fixing of 1000mm long Double Arm GI Hot Dip Galvanized Bracket with 42.4mm OD and 3.2mm thick Pipe. Canopy pipe should be 500mm with 4Nos 10mm dia Pinch bolts. | each | 3695.00 |
| | | Providing and fixing of 1500mm long Double Arm GI Hot Dip Galvanized Bracket with 42.4mm OD and 3.2mm thick Pipe. Canopy pipe should be 500mm with 4 Nos 10mm dia Pinch bolts.. | each | 4583.00 |
| | | Providing and fixing of 2000mm long Double Arm GI Hot Dip Galvanized Bracket with 42.4mm OD and 3.2mm thick Pipe. Canopy pipe should be 500mm with 4 Nos 10mm dia Pinch bolts. | each | 5341.00 |
| 21.246 | Ring carriage dia | Providing and fixing of ring carriage dia. not less than 750mm made of 50mm dia. GI pipe Light 'A' class for 70mm A/F top octagonal pole with required clamps, nut, bolt etc as required. | each | 2468.00 |
| 21.247 | Decorative designer poles | Providing and erection of black painted galvanized decorative designer poles in different section, top 60mm dia., bottom 140mm dia. with gold colour painted ornamental cast aluminium ring having lockable weather proof flush door junction box in bottom section, base plate dia. 290mm having 4 holes of 22mm dia. complete erected | | |

| Item No. | | Description | Unit | Rate |
|---|---|---|---|---|
| | | in an approved manner on existing foundation. Suitable size & type of foundation bolts 4 nos. X 16mm X 450mm 'J' type (EN8 grade) | | |
| (i) | 3 meter | 3 meter Height | each | 17737.00 |
| (ii) | 4 meter | 4 meter Height | each | 19831.00 |
| (iii) | 5 meter | 5 meter height | each | 23654.00 |
| 21.248 | Casting of open raft shallow footing foundation for 12.5meter high mast | Designing & casting of open raft shallow footing foundation with M-20 cement concrete suitable for 12.5meter High Mast considering the safe soil bearing capacity at site as 10 T/sqm at 2meter depth including excavation, foundation nut bolts in an approved manner as per manufacturer drawing or Engineer-in-charge. | each | 33771.00 |
| 21.249 | Casting of open raft shallow footing foundation for 16meter high mast | Designing & casting of open raft shallow footing foundation with M-20 cement concrete suitable for 16meter high mast considering the safe soil bearing capacity at site as 10 T/sqm at 2meter depth including excavation, foundation nut bolts in an approved manner as per manufacturer drawing or engineer-in-charge. | each | 41931.00 |
| 21.250 | Casting of open raft shallow footing foundation for 20meter high mast | Designing & casting of open raft shallow footing foundation with M-20 cement concrete suitable for 20meter high mast considering the safe soil bearing capacity at site as 10 T/sqm at 2meter depth including excavation, foundation nut bolts in an approved manner as per manufacturer drawing or engineer-in-charge. | each | 42707.00 |
| 21.251 | Foundation for octagonal pole | Designing and construction of footing foundation for octagonal pole of desired size including excavation in soil/soft rock, base concrete M-7.5 (1:4:8), 15mm thick and pedestal of RCC M-20 (1:1.5:3), required tar steel reinforcement for required size from base concrete, shuttering, curing etc. with existing foundation bolts complete as per specifications in an approved manner as per manufacturer drawing or engineer-in-charge. | Cu. Mt. | 6770.00 |
| 21.252 | Dismantling of existing pole accessories | Dismantling of existing H- Beam/RS Joist/PCC/Rail pole i/c accessories such as cross arms, top clamp, V-cross arms, strut, stay & conductor/cable/guarding span complete. The material should be transport & handed over to MPMKVVCL/PWD store. | each pole | 1457. 00 |
| 21.253 | KWh meter with LCD display (for energy metering) 5 - 30A , 1 phase 2 wire, 240 Volts | Supply, installation, testing & commissioning of KWh meter with LCD display (for energy metering) 5 - 30A , 1 phase 2 wire, 240 Volts, accuracy class 1 duly tested from electricity board with uPVC BOX . | each | 3732.00 |

| Item No. | | Description | Unit | Rate |
|---|---|---|---|---|
| 21.254 | KWh meter with LCD display (for energy metering) 5 - 20A, 3 phase 4 wire | Supply, installation, testing & commissioning of KWh meter with LCD display (for energy metering) 5 - 20A, 3 Phase 4 Wire, 440 Volts, accuracy class 1 duly tested from electricity board with uPVC BOX. | each | 5675.00 |
| 21.255 | KWh meter with LCD display (for energy metering) 5 - 10A, CT operated, 3 phase 4 wire | Supply, installation, testing & commissioning of KWh Meter with LCD display (For Energy Metering) 5 - 10A, CT Operated, 3 Phase 4 Wire, 440 Volts, and Accuracy Class 1 duly tested from electricity board with uPVC BOX. | each | 8301.00 |
| 21.256 | LT trivector energy meter | Supply, installation, testing & commissioning of LT trivector energy meter, DLMS compliant category-A (for substation/feeder metering) , 3 phase 4 wire, 440 Volts, 5 Amp, accuracy class 0.5S with GSM(GPRS compatible) modem suitable for required C.T. 5A duly tested from electricity board with uPVC box . | each | 14882.00 |
| 21.257 | HT trivector energy meter | Supply, installation, testing & commissioning of HT trivector energy meter, 3 phase 4 wire, 110 Volts, 5 Amp, accuracy class 0.5S with GSM (GPRS compatible) intelligent modem for consumer (store N forward type) suitable for required C.T. including uPVC box/sheet metal box with TTB duly tested from electricity board. | each | 21686.00 |
| 21.258 | HT trivector energy meter | Supply, installation, testing & commissioning of HT trivector energy meter, DLMS compliant category-A (for substation/feeder metering energy audit), 3 phase 4 wire, 110 Volts, 5 Amp, accuracy class 0.5S with GSM (GPRS compatible) modem suitable for required C.T. including uPVC box/sheet metal box with TTB duly tested from electricity board. | each | 22454.00 |
| 21.259 | CT/PT ratio 300-150/5 Amp | Supply, installation, testing & commissioning of HT 11 KV/110 V 3P4W CT/PT ratio 300-150/5 Amp outdoor oil immersed type duly tested from electricity board. | each | 55469.00 |
| 21.260 | CT/PT ratio 400-200/5 Amp | Supply, installation, testing & commissioning of HT 33 KV/110 V 3P4W CT/PT ratio 400-200/5 Amp outdoor oil immersed type duly tested from electricity board. | each | 72425.00 |
| 21.261 | CT ratio 500-200/5 Amp | Supply, installation, testing & commissioning of LT CT ratio 500-200/5 Amp epoxy coated Indoor Type duly tested from electricity board. | each | 992. 00 |
| 21.262 | Vacuum circuit breaker | Supplying, installation, testing and commissioning of triple pole outdoor type 11KV/33KV vacuum circuit breaker with outdoor control & relay panel with numerical static relay for | | |

| Item No. | | Description | Unit | Rate |
|---|---|---|---|---|
| | | feeder protection (1F). The installation of VCB shall be carried out in accordance with IS: 3072. The VCB shall be mounted on suitable size duly painted MS channel/angle iron structure with nut bolts and washer. The structure shall be fixed on existing cement concrete foundation. The complete work should be done as per IS specification. | | |
| (i) | 11 KV 630A VCB | 11 KV 630A VCB | each | 190130.00 |
| (ii) | 33 KV, 1250A VCB | 33 KV, 1250A VCB | each | 319644.00 |
| 21.263 | Single phase C.T./ P.T. | Supplying, installation, testing and commissioning of single phase C.T./P.T. outdoor type 11KV/33KV with terminal box. The C.T./P.T. shall be mounted on existing structure fabricated with suitable size MS channel/angle iron structure with nut bolts and washer duly painted. The structure shall be fixed on existing cement concrete foundation as per IS specification as mentioned below. | | |
| (i) | 11KV CT, 200-100/5 Amp | 11 KV CT outdoor type 200-100/5 Amp | each | 13044.00 |
| (ii) | 33KV CT, 200-100/5 Amp | 33 KV CT outdoor type 200-100/5 Amp | each | 18294.00 |
| 21.264 | Single phase C.T./ P.T. | Supplying, installation, testing and commissioning of single phase C.T./P.T. outdoor type 11KV/33KV with terminal box. The C.T./P.T. shall be mounted on existing structure fabricated with suitable size MS channel/angle iron structure with nut bolts and washer duly painted. The structure shall be fixed on existing cement concrete foundation as per IS specification as mentioned below. | | |
| (i) | 11 KV/110V, PTs | 11 KV/110V single phase PTs oil immersed | each | 15989.00 |
| (ii) | 33 KV/110V, PTs | 33 KV/110V single phase PTs oil immersed | each | 20488.00 |
| 21.265 | Indoor control relay panel | Supplying, installation, testing and commissioning of indoor control relay panel shall consisting of ammeter, voltmeter, selector switch, auxiliary relays for anti-pumping device, trip and alarm of buchholz relay, oil and winding temperature circuit, electronic hooter, 230 V AC to 110 V DC power pack with charging equipment for closing, tripping and indication, triple pole IDMTL two O/C and one E/F delay with current setting range 50-200% and 10-50%, push button high speed trip relay etc. complete as per IS specification. | each | 129039.00 |
| 21.266 | Sub-station Compact outdoor sub-station comprising of HV compartment for | Compact outdoor sub-station :<br>Design, supply, installation, testing and commissioning of compact outdoor sub-station comprising of HV compartment for 33/.433KV, DYn II, dry type resin cast transformer with standard accessories 1 no. of 33 KV, 630A, 25KA, 3 phase 50 HZ vacuum circuit breaker and 1 no. of 33 KV, 100A, 50KA, 3 phase 50 HZ Isolator (non - extensible RMU) connection on H.T side with single core | | |

| Item No. | | Description | Unit | Rate |
|---|---|---|---|---|
| | 33/.433KV | H.T. cable and standard accessories, connection on LV side insulated aluminium busbar of required rating, compartment comprising of MCCB's 4 pole, and standard voltmeter 0 - 500 V with VSS, ammeter with ASS of required rating, LED indicator (RYB) 22.5mm dia. accessories as mentioned below :-<br>The incomer VCB and Isolator shall be interlocked mechanically and VCB with necessary interconnection wiring, ferrule properly, earthing terminal, etc. The layout dimensional drawings, electrical wiring diagram and operation maintenance manuals in duplicate shall be supplied with VCB. The VCB/Isolator shall be supplied in conformity with relevant ISS & IEC with up to date amendments along with manufacturer's test certificate.<br>The housing of compact outdoor sub - station base frame and detachable cover plates, walls, roof, doors and other components are made of 2 mm thick galvanized sheet steel with at least 20 & 40 µm zinc layer. Hinges, rivets are made of stainless steel. Wide door opening angle of 180 degrees, lock into top and bottom frame. Transformer roof has detachable facility to enable easy installation and maintenance without affecting other equipment's, MV & LV compartments roofs are separated from transformer roof. Roof slope for water runoff. air intake in bottom of doors for natural chimney effect and smooth air flow and sufficient self-cooling of transformer. One common earthing terminal. Type tested as per latest IEC standard 62271-202, IEC 62271-202 high-voltage/low-voltage prefabricated substation. MV & LV compartments have IP class - IP 54 & IP23 degree of protection for transformer compartment. The enclosure exterior shall be painted with epoxy based powder paint (colour RAL 7032) excluding platform, incoming/outgoing cable, earthing. | | |
| | Cu. wound | Copper wound | | |
| (i) | 100KVA | 100KVA. 33/0.4KV O/G MCCB 4P - 2 Nos 160A 25KA | each | 3418688.00 |
| (ii) | 200KVA | 200KVA, 33/0.4KV O/G MCCB 4P - 1 Nos 200A 25KA, 1 No 300A 36KA | each | 3948566.00 |
| (iii) | 315KVA | 315KVA, 33/0.4KV O/G MCCB 4P - 1 Nos 630A 36KA, 1 No 300A 36KA | each | 4230091.00 |
| 21.267 | Monopole | Design, Manuafcturing, Supply & Erection Commissioning of Hot Dip Galvanized Monopole made of High Tensile Steel sheet, suitable upto 100Mtr Span. Monopole Manufacturing plant should have the In house NABL Accreditted Laboratory with Standard ISO/IEC 17025:2017 to Testing and calibration of Monopoles Structure and their Components. Galvanizing bath length should not be less than 14.3Mtr to maintain Quality of Galvanizing.<br>Following ISO Certificates required alongwith material:- ISO 9001:2015, ISO 14001:2015, ISO 45001:2018. Design & Drawing<br>shoulb be vetted from SVNIT/IIT. | | |
| 21.267.1 | | 14M 33KV Double Circuit PA Type (0-2D) Monopole made in Single Section alongwith Cross Arms, Stringing, Bird Gaurd and hardware. It's also includes Foundation Bolts, Anchor, Template Plate, Number Plate, Danger Plate & Circuit Plate, 3M Long<br>ISI Marked B Class 40MM Dia GI Pipe Earthing along with | each | 579890 |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| | GI 50X6 MM Strip etc | | |
| 21.267.2 | 14M 33KV Double Circuit PB Type (2-30D) Monopole made in Single Section alongwith Cross Arms, Stringing, Bird Gaurd and hardware. It's also includes Foundation Bolts, Anchor, Template Plate, Number Plate, Danger Plate & Circuit Plate, 3M Long Earthing ISI Marked B Class 40MM Dia GI Pipe along with GI 50X6 MM Strip etc | each | 631216 |
| 21.267.3 | 14M 33KV Double Circuit PC Type (30-60D) Monopole made in Single Section alongwith Cross Arms, Stringing, Bird Gaurd and hardware. It's also includes Foundation Bolts, Anchor, Template Plate, Number Plate, Danger Plate & Circuit Plate, 3M Long Earthing ISI Marked B Class 40MM Dia GI Pipe along with GI 50X6 MM Strip etc | each | 752890 |
| 21.267.4 | 14M 33KV Double Circuit PD Type (60-90D) Monopole made in Single Section alongwith Cross Arms, Stringing, Bird Gaurd and hardware. It's also includes Foundation Bolts, Anchor, Template Plate, Number Plate, Danger Plate & Circuit Plate, 3M Long Earthing ISI Marked B Class 40MM Dia GI Pipe along with GI 50X6 MM Strip etc | each | 887539 |
| 21.267.5 | 17M 33KV Double Circuit PA Type (0-2D) Monopole made in Two Sections alongwith Cross Arms, Stringing, Bird Gaurd and hardware. It's also includes Foundation Bolts, Anchor, Template Plate, Number Plate, Danger Plate & Circuit Plate, 3M Long ISI Marked B Class 40MM Dia GI Pipe Earthing along with GI 50X6 MM Strip etc. | each | 1077689 |
| 21.267.6 | 17M 33KV Double Circuit PB Type (2-30D) Monopole made in Two Sections alongwith Cross Arms Stringing, Bird Gaurd and hardware. It's also includes Foundation Bolts, Anchor, Template Plate, Number Plate, Danger Plate & Circuit Plate, 3M Long ISI Marked B Class 40MM Dia GI Pipe Earthing along with GI 50X6 MM Strip etc. | each | 1133549 |
| 21.267.7 | 17M 33KV Double Circuit PC Type (30-60D) Monopole made in Two Sections alongwith Cross Arms Stringing, Bird Gaurd and hardware. It's also includes Foundation Bolts, Anchor, Template Plate, Number Plate, Danger Plate & Circuit Plate, 3M Long ISI Marked B Class 40MM Dia GI Pipe Earthing along with GI 50X6 MM Strip etc. | each | 1269346 |
| 21.267.8 | 17M 33KV Double Circuit PD Type (60-90D) Monopole made in Two Sections alongwith Cross Arms Stringing, Bird Gaurd and hardware. It's also includes Foundation Bolts, Anchor, Template Plate, Number Plate, Danger Plate & Circuit Plate, 3M Long ISI Marked B Class 40MM Dia GI Pipe Earthing along with GI 50X6 MM Strip etc. | Set | 1387909 |
| 21.267.9 | 17M 33KV Multi Circuit PA Type (0-2D) Monopole made in Two Sections alongwith Cross Arms, Stringing, Bird Gaurd and hardware. It's also includes Foundation Bolts, Anchor, Template Plate, Number Plate, Danger Plate & Circuit Plate, 3M Long ISI Marked B Class 40MM Dia GI Pipe Earthing along with GI 50X6 MM Strip etc. | Set | 1252689 |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| 21.267.9 | 17M 33KV Multi Circuit PB Type (2-30D) Monopole made in Two Sections alongwith Cross Arms Stringing, Bird Gaurd and hardware. It's also includes Foundation Bolts, Anchor, Template Plate, Number Plate, Danger Plate & Circuit Plate, 3M Long<br>ISI Marked B Class 40MM Dia GI Pipe Earthing along with GI 50X6 MM Strip etc. | Set | 1333549 |
| 21.267.10 | 17M 33KV Multi Circuit PC Type (30-60D) Monopole made in Two Sections alongwith Cross Arms Stringing, Bird Gaurd and hardware. It's also includes Foundation Bolts, Anchor, Template Plate, Number Plate, Danger Plate & Circuit Plate, 3M Long<br>ISI Marked B Class 40MM Dia GI Pipe Earthing along with GI 50X6 MM Strip etc. | Set | 1494346 |
| 21.267.11 | 17M 33KV Multi Circuit PD Type (60-90D) Monopole made in Two Sections alongwith Cross Arms Stringing, Bird Gaurd and hardware. It's also includes Foundation Bolts, Anchor, Template Plate, Number Plate, Danger Plate & Circuit Plate, 3M Long<br>ISI Marked B Class 40MM Dia GI Pipe Earthing along with GI 50X6 MM Strip etc. | Set | 1617909 |
| 21.267.12 | 20M 33KV Double Circuit PA Type (0-2D) Monopole made in Two Sections alongwith Cross Arms, Stringing, Bird Gaurd and hardware. It's also includes Foundation Bolts, Anchor, Template Plate, Number Plate, Danger Plate & Circuit Plate, 3M Long<br>ISI Marked B Class 40MM Dia GI Pipe Earthing along with GI 50X6 MM Strip etc. | Set | 1234412 |
| 21.267.13 | 20M 33KV Double Circuit PB Type (2-30D) Monopole made in Two Sections alongwith Cross Arms Stringing, Bird Gaurd and hardware. It's also includes Foundation Bolts, Anchor, Template Plate, Number Plate, Danger Plate & Circuit Plate, 3M Long<br>ISI Marked B Class 40MM Dia GI Pipe Earthing along with GI 50X6 MM Strip etc. | Set | 1349367 |
| 21.267.14 | 20M 33KV Double Circuit PC Type (30-60D) Monopole made in Two Sections alongwith Cross Arms Stringing, Bird Gaurd and hardware. It's also includes Foundation Bolts, Anchor, Template Plate, Number Plate, Danger Plate & Circuit Plate, 3M Long<br>ISI Marked B Class 40MM Dia GI Pipe Earthing along with GI 50X6 MM Strip etc. | Set | 1477129 |
| 21.267.15 | 20M 33KV Double Circuit PD Type (60-90D) Monopole made in Two Sections alongwith Cross Arms Stringing, Bird Gaurd and hardware. It's also includes Foundation Bolts, Anchor, Template Plate, Number Plate, Danger Plate & Circuit Plate, 3M Long<br>ISI Marked B Class 40MM Dia GI Pipe Earthing along with GI 50X6 MM Strip etc. | Set | 1591891 |
| 21.267.16 | 20M 33KV Multi Circuit PA Type (0-2D) Monopole made in Two Sections alongwith Cross Arms, Stringing, Bird Gaurd and hardware. It's also includes Foundation Bolts, Anchor, Template Plate, Number Plate, Danger Plate & Circuit Plate, 3M Long<br>ISI Marked B Class 40MM Dia GI Pipe Earthing along with GI 50X6 MM Strip etc. | Set | 1384412 |
| 21.267.17 | 20M 33KV Multi Circuit PB Type (2-30D) Monopole made | Set | 1524367 |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| | in Two Sections alongwith Cross Arms Stringing, Bird Gaurd and hardware. It's also includes Foundation Bolts, Anchor, Template Plate, Number Plate, Danger Plate & Circuit Plate, 3M Long<br>ISI Marked B Class 40MM Dia GI Pipe Earthing along with GI 50X6 MM Strip etc. | | |
| 21.267.18 | 20M 33KV Multi Circuit PC Type (30-60D) Monopole made in Two Sections alongwith Cross Arms Stringing, Bird Gaurd and hardware. It's also includes Foundation Bolts, Anchor, Template Plate, Number Plate, Danger Plate & Circuit Plate, 3M Long<br>ISI Marked B Class 40MM Dia GI Pipe Earthing along with GI 50X6 MM Strip etc. | Set | 1667129 |
| 21.267.19 | 20M 33KV Multi Circuit PD Type (60-90D) Monopole made in Two Sections alongwith Cross Arms Stringing, Bird Gaurd and hardware. It's also includes Foundation Bolts, Anchor, Template Plate, Number Plate, Danger Plate & Circuit Plate, 3M Long<br>ISI Marked B Class 40MM Dia GI Pipe Earthing along with GI 50X6 MM Strip etc. | Set | 1816891 |
| 21.267.20 | 24M 33KV Double Circuit PA Type (0-2D) Monopole made in Three Sections alongwith Cross Arms Stringing, Bird Gaurd and hardware. It's also includes Foundation Bolts, Anchor, Template Plate, Number Plate, Danger Plate & Circuit Plate, 3M Long<br>ISI Marked B Class 40MM Dia GI Pipe Earthing along with GI 50X6 MM Strip etc. | Set | 1534412 |
| 21.267.21 | 24M 33KV Double Circuit PD Type (2-30D) Monopole made in Two Sections alongwith Cross Arms Stringing, Bird Gaurd and hardware. It's also includes Foundation Bolts, Anchor, Template Plate, Number Plate, Danger Plate & Circuit Plate, 3M Long<br>ISI Marked B Class 40MM Dia GI Pipe Earthing along with GI 50X6 MM Strip etc. | Set | 1727129 |
| 21.267.22 | 24M 33KV Double Circuit PC Type (30-60D) Monopole made in Two Sections alongwith Cross Arms Stringing, Bird Gaurd and hardware. It's also includes Foundation Bolts, Anchor, Template Plate, Number Plate, Danger Plate & Circuit Plate, 3M Long<br>ISI Marked B Class 40MM Dia GI Pipe Earthing along with GI 50X6 MM Strip etc. | Set | 1927129 |
| 21.267.23 | 24M 33KV Double Circuit PD Type (60-90D) Monopole made in Two Sections alongwith Cross Arms Stringing, Bird Gaurd and hardware. It's also includes Foundation Bolts, Anchor, Template Plate, Number Plate, Danger Plate & Circuit Plate, 3M Long<br>ISI Marked B Class 40MM Dia GI Pipe Earthing along with GI 50X6 MM Strip etc. | Set | 2133176 |
| 21.267.24 | 17M 33KV Double Circuit PD Type (60-90D) Monopole with Cable Termination made in Two Sections alongwith Cross Arms, Cable Tray Stringing, Bird Gaurd and hardware. It's also includes Foundation Bolts, Anchor, Template Plate, Number Plate, Danger Plate & Circuit Plate, 3M Long ISI Marked B Class 40MM Dia GI Pipe Earthing along with GI 50X6 MM Strip etc. | Set | 1766307 |
| 21.267.25 | 17M 33KV Multi Circuit PD Type (60-90D) Monopole with Cable Termination made in Two Sections alongwith | Set | 2063210 |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| | Cross Arms, Cable Tray Stringing, Bird Gaurd and hardware. It's also includes Foundation Bolts, Anchor, Template Plate, Number Plate, Danger Plate & Circuit Plate, 3M Long ISI Marked B Class 40MM Dia GI Pipe Earthing along with GI 50X6 MM Strip etc. | | |
| 21.267.26 | 20M 33KV Double Circuit PD Type (60-90D) Monopole with Cable Termination made in Two Sections alongwith Cross Arms, Cable Tray Stringing, Bird Gaurd and hardware. It's also includes Foundation Bolts, Anchor, Template Plate, Number Plate, Danger Plate & Circuit Plate, 3M Long ISI Marked B Class 40MM Dia GI Pipe Earthing along with GI 50X6 MM Strip etc. | Set | 2390304 |
| 21.267.27 | 20M 33KV Multi Circuit PD Type (60-90D) Monopole with Cable Termination made in Two Sections alongwith Cross Arms, Cable Tray Stringing, Bird Gaurd and hardware. It's also includes Foundation Bolts, Anchor, Template Plate, Number Plate, Danger Plate & Circuit Plate, 3M Long ISI Marked B Class 40MM Dia GI Pipe Earthing along with GI 50X6 MM Strip etc. | Set | 2519265 |
| 21.268 | Solar street light | | |
| 21.268.1 | Supply of all in one integrated LED solar streetlight consists of solar photovoltaic module, battery, battery management system, control electronics, LED driver, PIR sensor, inter-connecting cables and pole mounting piece integrated into the luminaire as a single unit. Solar street light with IP65 ingress protection having a system wattage of 17W with 3000 lumen output, 35Wp high efficiency monocrystalline solar panel, highly efficient MPPT based charge controller and 12.8 V, 20 Ah IP68 Lithium Ferro Phosphate(LiFePO4) battery with proper battery management system. LED lumen efficacy shall be at least 175lm/W and LED driver should have >90% efficiency. Typical correlated colour temperature should be 5700K with CRI>70. It should have a PIR sensor to ensure battery optimisation with helpa of a suitable dimming profile. The luminaire should be housed in IK 08 impact resistance certified aluminum pressure diecast housing with anti-corrosive coating and must have a UV stabilized polycarbonate cover. The luminaire shall have a specially designed pole mounting piece that enables lateral as well as post top mounting and installation at varying tilts from 0 to 15 degrees for optimal light distribution. | each | 92750 |
| 21.268.2 | Supply of all in one integrated LED solar streetlight consists of solar photovoltaic module, battery, battery management system, control electronics, LED driver, PIR sensor, inter- | each | 102000 |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| | connecting cables and pole mounting piece integrated into the luminaire as a single unit. Solar street light with IP65 ingress protection having a system wattage of 27w with 4500 lumen output, 60Wp high efficiency monocrystalline solar panel, highly efficient MPPT based charge controller and 12.8 V, 30 Ah IP68 Lithium Ferro Phosphate (LiFePO4) battery with proper battery management system. LED lumen efficacy shall be at least 175lm/W and LED driver should have >90% efficiency. Typical correlated colour temperature should be 5700K with CRI>70. It should have a PIR sensor to ensure battery optimisation with help of a suitable dimming profile.The luminaire should be housed in IK 08 impact resistance certified aluminium pressure diecast housing with anti-corrosive coating and must have a UV stabilized polycarbonate cover. The luminaire shall have a specially designed pole mounting piece that enables lateral as well as post top mounting and installation at varying tilts from 0 to 15 degrees for optimal light distribution. | | |
| 21.268.3 | Supply of all in one integrated LED solar streetlight consists of solar photovoltaic module, battery, battery management system, control electronics, LED driver, inter-connecting cables and pole mounting piece integrated into the luminaire as a single unit. Solar street light with IP65 ingress protection having a system wattage of 35W with 6000 lumen output, 60Wp high efficiency monocrystalline solar panel, highly efficient MPPT based charge controller and 12.8 V, 30 Ah IP68 Lithium Ferro Phosphate(LiFePO4) battery with proper battery management system. LED lumen efficacy shall be at least 175lm/W and LED driver should have >90% efficiency. Typical correlated colour temperature should be 5700K with CRI>70. It should ensure battery optimisation with help of a suitable dimming profile. The luminaire shall have inbuilt hybrid power supply option that can charge the battery during non sunny days automatically from the grid. The luminaire should be housed in IK 08 impact resistance certified aluminum pressure diecast housing with anti-corrosive coating and | each | 125000 |

| Item No. | | Description | Unit | Rate |
|---|---|---|---|---|
| | | must have a UV stabilized polycarbonate cover. The luminaire shall have a specially designed pole mounting piece that enables lateral as well as post top mounting and installation at varying tilts from 0 to 15 degrees for optimal light distribution. | | |
| 21.269 | CCMS panel | Supply of cloud-based CCMS Group control cum gateway communicating on RF Free band frequency with SLC and over cellular/ethernet with cloud server. Group controller cum gateway shall be placed in IP55 enclosure. Bidirectional communication over RF Free Band frequency of 865 MHz - 868 MHz. It shall have communication of GPRS/WIFI with internal antenna and IMEI inside. The enclosure should have class-1 energy meter, latching relay, bypass timer, MCB complete as required as per detail technical specification. | each | 198000 |

# CHAPTER : 22.0

# LABORATORY MACHINES , EQUIPMENT & TEST RATE

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| 22.1 | Providing and installation/fixing of 1000KN, microprocessor based digital Universal Testing Machine (Vertical Type) to test mechanical properties of deformed steel flat and bars up to 32 mm thick, including commissioning, providing training to the laboratory staff at site and calibration with following details:<br>Components: Complete equipment along with suitable Hydraulic unit, operation and control unit with programmed controller, electronic extensometer for gauge length 50 mm (min), component mounting unit, jig and fixtures etc. | each | 1995600.00 |
| 22.2 | Providing portable digital equipment for the measurement of the luminance coefficient under diffuse illumination Qd and coefficient of retro reflected luminance RL of a field of a road marking includes an illumination system, a photometer and means to define a horizontal reference plane with a reference centre i.e. Retroreflectrometer for road marking. This retroreflectrometer must be in accordance of the requirements of IRC 35:2015 including installation, commissioning, providing training to the laboratory staff at site and calibration by NABL accredited calibration lab with traceability. | each | 1293600.00 |
| 22.3 | Providing portable digital equipment for the measurement of the coefficient of retro-reflection RA and R' of retro reflective sheeting used on the sign boards i. e. Retroreflectrometer for sign boards. This retroreflectrometer must be in accordance of the requirements of IRC 67:2012 including installation, commissioning, providing training to the laboratory staff at site and calibration by NABL accredited calibration lab with traceability. | each | 590400.00 |
| 22.4 | Providing portable Digital Thickness Gauge, range of 0-25 mm with 0.01 mm accuracy, for the measurement of thickness of road markings and similar coatings. This Digital Thickness Gauge must be in accordance of the requirements of IRC 35:2015 including installation, commissioning, providing training to the laboratory staff at site and calibration by NABL accredited calibration lab with traceability. | each | 138000.00 |
| 22.5 | Providing and fixing electric motor driven mechanical Universal Automatic Soil Compactor with 2.6 and 4.9 kg rammers, for compaction of soil mould size 100 & 150 mm diameter to measure OMC MDD and CBR of soil as per IS 2720 (Part VII & VIII) including installation, commissioning, providing training to the laboratory staff at site and calibration by NABL accredited calibration lab with traceability. | each | 162000.00 |
| 22.6 | Providing compact platform type Class II Digital Weighing Balance of capacity 100 kg or more, least count 10 g or less with green LED display, inbuilt battery backup and adopter of 220 V AC, including installation, commissioning, providing training to the laboratory staff at site and calibration by NABL accredited calibration lab with traceability. Model of Balance must be approved by Legal Metrology Department, Government of India. | each | 26400.00 |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| 22.7 | Providing compact table top Class II Digital Weighing Balance of capacity 30 kg or more, least count 1 g or less with green LED display, inbuilt battery backup and adopter of 220 V AC, including installation, commissioning, providing training to the laboratory staff at site and calibration by NABL accredited calibration lab with traceability. Model of Balance must be approved by Legal Metrology Department, Government of India. | each | 8280.00 |
| 22.8 | Providing compact table top Class II Digital Weighing Balance of capacity 3 kg or more, least count 0.1 g or less with green LED or LCD display supported by back light, inbuilt battery backup and adopter of 220 V AC, including installation, commissioning, providing training to the laboratory staff at site and calibration by NABL accredited calibration lab with traceability. Model of Balance must be approved by Legal Metrology Department, Government of India. | each | 7200 |
| 22.9 | Providing compact table top Class II Digital Weighing Balance of capacity 500 g or more, least count 0.01 g or less with LED or LCD display supported by back light, inbuilt battery backup and adopter of 220 V AC, including installation, commissioning, providing training to the laboratory staff at site and calibration by NABL accredited calibration lab with traceability. Model of Balance must be approved by Legal Metrology Department, Government of India. | each | 7680 |
| 22.10 | Providing compact pocket type battery operated Class II Digital Weighing Balance of capacity 50 g or more, least count 0.001 g or less with standard weight of 50 g, LCD display supported by back light, including installation, commissioning, providing training to the laboratory staff at site and calibration by NABL accredited calibration lab with traceability. Model of Balance must be approved by Legal Metrology Department, Government of India. | each | 8640.00 |
| 22.11 | Providing thermostatically controlled, digital display, hot air Electric Oven, range up to 250o C, internal body of stainless-steel size 450 x 450 x 600 mm, with forced air circulation by stainless steel fan and digital temperature control cum Indicator, including installation, commissioning, providing training to the laboratory staff at site and calibration by NABL accredited calibration lab with traceability. | each | 88800.00 |
| 22.12 | Providing single sheet pressed, jointless, round edge Enamel Trays for general laboratory purpose.<br>(a) Size 600 x 450 x 50 mm – 1No<br>**(b)** Size 450 x 300 x 40 mm – 2 Nos<br>**(c)** Size 300 x 250 x 40 mm – 3 Nos<br>**(d)** Size 200 x 200 x 30 mm – 9 Nos | each | 13200.00 |
| 22.13 | Providing and fixing of Los Angeles Abrasion Testing Machine as per IS: 2386 (Part – IV), with minimum 12 number of Abrasive charges, including installation, commissioning, providing training to the laboratory staff at site and calibration by NABL accredited | each | 96000.00 |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| | calibration lab with traceability. | | |
| 22.14 | Providing and fixing of Accelerated Polishing Machine as per BS 812-114, including installation, commissioning, providing training to the laboratory staff at site and calibration by NABL accredited calibration lab with traceability. | each | 196800.00 |
| 22.15 | Providing and fixing of Asphalt Theoretical Density Meter as per ASTM D 2041 to measure Theoretical Density Gmm of bituminous mixes, with digital vacuum indicator, two vacuum containers, vibratory table, control table, control panel and vacuum pump, including installation, commissioning, providing training to the laboratory staff at site and calibration by NABL accredited calibration lab with traceability. | each | 189600.00 |
| 22.16 | Providing and fixing of Buoyancy Balance equipment to measure density of bituminous core, aggregate etc, with density basket. The equipment consists a rigid support frame, appropriate size water tank mounted on a platform, a mechanical lifting device to raise the water tank through the frame height with Digital balance of Capacity 3 kg or more, least count 0.1 g or less, including installation, commissioning, providing training to the laboratory staff at site and calibration by NABL accredited calibration lab with traceability. | each | 73200.00 |
| 22.17 | Providing and fixing of Sand Equivalent Value Test Apparatus as per IS 2720 (Part 37) to measure the relative proportion of fine dust or clay like material in soils or fine aggregates, including installation, commissioning, providing training to the laboratory staff at site and calibration by NABL accredited calibration lab with traceability. | each | 18000.00 |
| 22.18 | Providing and fixing of Humidity Chamber for cement testing with following details –<br>(a) Relative humidity range 35% RH to 95% RH with an accuracy ± 3% RH.<br><br>(b) Temperature range 10°c to 60°c with an accuracy ± 0.5°c.<br><br>(c) Digital temperature and humidity controller with digital read out.<br><br>(d) Internal Size – 450 x 450 x 450 mm with modular removable shelves made of stainless steel.<br><br>(e) Automatic refrigeration system and low-pressure boilers that cause steam for chamber humidity.<br><br>(f) Double walled construction with inner chamber stainless steel and outer body powder coated mild steel. | each | 123600.00 |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| | (g) Vertical air circulation by stainless steel fan.<br><br>(h) A key lockable insulated door having gasket seals perfectly. including installation, commissioning, providing training to the laboratory staff at site and calibration by NABL accredited calibration lab with traceability. | | |
| 22.19 | Providing and fixing of portable digital plug and play mist type lab humidifier with humidifying capacity 1 litre/hr, inbuilt digital display, mist control option and a water tank having capacity 12 litter or more. The operating range of humidifier must be 5°C to 35°C for temperature and 10%-95% RH for humidity, including installation, commissioning, providing training to the laboratory staff at site and calibration by NABL accredited calibration lab with traceability. | each | 50400.00 |
| 22.20 | Providing and fixing of single-phase, 220 V AC operated Rock/Concrete cutting machine, capacity 2 kW or more, to cut top and bottom portion of Concrete Core up to 150 mm diameter, Stones, building materials and metallic specimens with two concrete cutting wheels and two iron cutting wheels, including installation, commissioning, providing training to the laboratory staff at site and calibration by NABL accredited calibration lab with traceability. | each | 58800.00 |
| 22.21 | Providing of Digital Infra-Red Moisture Balance to measure moisture content of soil up to 50 g as per IS:2720 (Part 2), including installation, commissioning, providing training to the laboratory staff at site and calibration by NABL accredited calibration lab with traceability. | each | 138000.00 |
| 2.22 | Providing Type 2 Pattern 'C' Desiccator of nominal size 250 mm as per IS:6128 to store and protect soil samples during their cooling period. Providing Type 2 Pattern 'C' Desiccator of nominal size 250 mm as per IS:6128 to store and protect soil samples during their cooling period. | each | 4680.00 |
| 22.23 | Providing Automatic Universal Penetrometer Apparatus with needle, transfer dish and container of 55 mm and 70 mm diameter to determine the penetration of semi-solid and solid bituminous materials and liquid limit of soil, as per IS:1448 (Part 60), IS:1203 and IS:11196 including installation, providing training to the laboratory staff at site and calibration by NABL accredited calibration lab with traceability.<br>The apparatus must be compact with built-in electronic timer to control duration of present penetration from 00 to 05 second, lead screw, gear arrangement, levelling screws, spirit level, in build focus light and magnifier. | each | 28800.00 |
| 22.24 | Providing and fixing of stainless-steel Digital Thickness Gauge, range up to 300 mm and more, least count 0.01 mm or less and accuracy of ±0.03 mm or more, including installation, | each | 28800.00 |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| | commissioning, providing training to the laboratory staff at site and calibration by NABL accredited calibration lab with traceability. | | |
| 22.25 | Providing brass mould set of three, as per IRC: SP:53-2023 for Elastic Recovery Test to measure elastic recovery of modified bitumen, including providing training to the laboratory staff at site and calibration by NABL accredited calibration lab with traceability. | each | 11640.00 |
| 22.26 | Providing and installation of a skid mounted, portable, self-contained, self-start 3kW Generator Set, equipped with controls, instruments, and accessories necessary for operation. The generator set consists of a petrol engine, permanent magnet 220 V AC generator, control box assembly with over load protection, output/load panel, enclosure cooling and ventilation system, engine exhaust assembly and a 12 V DC battery. The Supply of equipment is included training to the laboratory staff at site. | each | 151200.00 |
| 22.27 | Providing a portable 220 V AC operated core drilling machine to wet drilling by diamond core bit in reinforced concrete, pavement concrete, asphalt, natural stone etc with following details-<br>(a) Motor capacity 2500 W or more,<br><br>(b) Rig Size 850 x 220 x 185 mm Approx. and motor & rig as a complete unit.<br><br>(c) Over load protection for motor.<br><br>(d) Core bit up to 200 mm diameter and 400 mm long spindle connection size 1 ¼" UNC, can be used.<br><br>(e) With additional mild steel stand of Size 600 x 300 x 50 mm to achieve stability and safety without fastening the machine, during core drilling on pavement.<br><br>(f) With 6 m long power supply input cable and 15-amp plug.<br><br>(g) With 6 m long 5-layer robust water inlet hose of appropriate internal diameter with brass tap and clamps.<br><br>The supply of machine is included a diamond core bit of 150 x 400 mm, training to the laboratory staff at site and calibration by NABL accredited calibration lab with traceability. | each | 69600.00 |
| 22.28 | Providing a compression type, class 1 integral proving ring of 2000 kN to inhouse calibration purpose with micro meter (dial gauge), calibration charts and polished & ground pair of loading pads and a pair of shackles. Characteristics of proving ring must be as per IS:4169. The supply of Proving Ring is included training to the laboratory staff at site and calibration by NABL accredited calibration lab with traceability. | each | 115200.00 |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| 22.29 | Providing a compression type, class 1 integral proving ring of 100 kN to inhouse calibration purpose with micro meter (dial gauge), calibration charts and polished & ground pair of loading pads. Characteristics of proving ring must be as per IS:4169. The supply of Proving Ring is included training to the laboratory staff at site and calibration by NABL accredited calibration lab with traceability. | each | 94800.00 |
| 22.30 | Providing and fixing of a Sample Splitter with three collection pans, capacity to split up to 50 kg of aggregate at a time. Main body of Sample Splitter and chute bar must be made of Cadmium plated mild steel and stainless steel respectively. The width of Chute bar must be 12 mm so that openings of 12-24-36-48-60-72-84-96-108 mm can be possible. The supply of equipment is included training to the laboratory staff at site. | each | 146400.00 |
| 22.31 | Providing digital universal camber plate with 3 m long straight made up of aluminium rectangular tube of size 50 x 25 x 3 mm, to determine camber up to 15% least count 0.1%, gradient up to 45° least count 1°, and lateral surface irregularity up to ± 50 mm or more, of a road surface. The equipment must be equally supported at both ends to produce a set height between the road surface and the beam.<br>The supply of equipment is included incremented wedge, capacity 50 mm and least count 1 mm, training to the laboratory staff at site and calibration by NABL accredited calibration lab with traceability. | each | 58800.00 |
| 22.32 | Providing equipment and material in a compact suitable box for testing<br>of construction water for following details-<br>Property Range Min. number of tests<br>pH 2 to 10 100 tests<br>Total Alkalinity 25-750 mg/L 100 tests<br>Acidity 25-750 mg/L 100 tests<br>TDS (Tester) 0-9990 ppm 100 tests<br>Sulphate 0-250 mg/L 100 tests<br>Chloride 20-1000mg/L 100 tests<br>The supply of Testing Kit is included training to the laboratory staff at site and calibration by NABL accredited calibration lab with traceability, if required. | each | 13200.00 |
| 22.33 | Providing car axle mounted bump integrator (CAMBI) as per IRC: SP: 16-2019 and fixing it in lab vehicle to determine the road roughness.<br>The supply of equipment is included calibration at every 5000 Km run of equipment by NABL accredited calibration lab with traceability or by CRRI and training to the laboratory staff at site. | each | 2467200.00 |
| 22.34 | Providing free soil index testing apparatus, includes 425-micron IS sieve, two 100 ml capacity glass graduated cylinders as per Table 1 of IS 878:2008, digital weighing balance, capacity 50 g or more | each | 13200.00 |

| Item No. | Description | Unit | Rate |
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| | least count 0.01 g, 500 ml kerosene oil and 500 ml distilled water in LDPE wash bottles with flexible delivery tube to determine the free-soil index of soil as per IS-2720-P-40, comprehensively.<br><br>The supply of apparatus is included two extra 100 ml capacity glass graduated cylinders as mentioned above, training to the laboratory staff at site and calibration by NABL accredited calibration lab with traceability. | | |
| 22.35 | Providing silt content test apparatus, includes two 250 ml capacity glass graduated cylinders as per Table 1 of IS 878:2008, 500 ml distilled water in LDPE wash bottle with flexible delivery tube, a stainless-steel tea spoon, volume approx. 5 ml, 100 g salt in air tight container and a digital alarm watch, to determine the silt content of two samples of fine aggregate at a time as per Appendix 'C' (Clause 3.1.3.2) of CPWD specifications 2019, comprehensively.<br><br>The supply of apparatus is included two extra 250 ml capacity glass graduated cylinder as mentioned above, training to the laboratory staff at site and calibration by NABL accredited calibration lab with traceability. | each | 15600.00 |
| 22.36 | Providing Plastic Limit test apparatus, includes 425-micron IS sieve, a porcelain evaporating dish 12 cm or more in diameter, flexible spatula with blad of 8 cm long and 2 cm wide, a ground glass plate 20x15 cm or more with stainless-steel rods of 3 mm diameter pasted on each edge, a A4 size 3X Fresnel lens sheet of focal length 200 mm fixed on 20 cm above the glass plate, two air tight aluminium container capacity 100 ml, digital weighing balance, capacity 50 g or more least count 0.01 g and one stainless-steel rod of 3 mm diameter to determine the plastic limit of soil as per IS 2720 Part V. This apparatus must be well lit by white LED lights.<br><br>The supply of apparatus is included one extra a ground glass plate as mentioned above, training to the laboratory staff at site and calibration by NABL accredited calibration lab with traceability. | each | 21600.00 |
| 22.37 | Providing Pile Integrity Tester based on the reflected wave method and fit for testing drilled piles and hammer piles, for low strain non-destructive integrity testing of concrete pile (diameter up to 1.5 m) and identify the extant and location of the pile defect as per IS 14893:2021. It must also be useful for testing exposed top surface of construction such as pillars.<br><br>The supply of apparatus is included wireless sensors, Laptop/Notepad with appropriate software up to lifetime use, Hammer stocked with interchangeable tips and all other necessary accessories, training to the laboratory staff at site and calibration by NABL accredited calibration lab with traceability. | each | 1248000.00 |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| 22.38 | Providing portable and battery-operated Wall Scanner, based on Radar Technology, to detect objects like magnetic metals (e.g. iron), non-magnetic metals (e.g. copper), live cables, low-voltage & data cables (Three-phase power cables, cavities), non-live cables, wooden substructures, water-filled plastic pipes, Empty plastic pipes etc up to 200 mm deep with Object view, Spot view, Signal View, Signal View 2D, in wall or dry concrete. The equipment must be able to transfer data by USB-C, SD-Card or wirelessly for analysis and documentation.<br>The supply of equipment is included<br><br>(a) Necessary software, if needed, with life time validity.<br><br>(b) Training to the laboratory staff at site, and<br><br>(c) Calibration by NABL accredited calibration lab with traceability, if needed. | each | 364800.00 |
| 22.39 | Providing portable and battery-operated Dowel, Transvers, and reinforcement bar (diameter up to 50 mm) locator, based on Radar Technology, to measure the position and diameter of Dowel, transverse, and reinforcement bar, up to 240 mm depth in concrete. The equipment must be able to transfer data by USB-C, SD-Card or wirelessly for analysis and documentation.<br>The supply of equipment is included<br>(a) Necessary software, if needed, with life time validity.<br><br>(b) Training to the laboratory staff at site, and<br><br>(c) Calibration by NABL accredited calibration lab with traceability, if needed. | each | 415200.00 |
| 22.40 | Providing Digital Concrete Rebound Hammer of following specifications, to measure the compressive strength of concrete by measuring the rebound index after striking the concrete surface. The equipment must be able to transfer data by USB-C, SD-Card or wirelessly for analysis and documentation.<br>The supply of equipment is included<br><br>(a) Necessary software, if needed, with life time validity.<br><br>(b) Training to the laboratory staff at site, and<br><br>(c) Calibration by NABL accredited calibration lab with traceability, if needed. | each | 96000.00 |
| 22.41 | Providing and fixing of Aggregate Impact Value Test Apparatus as per IS: 2386 (Part – IV) including installation, commissioning, providing training to the laboratory staff at site and calibration by NABL accredited calibration lab with traceability. | each | 15600.00 |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| 22.42 | Providing Cylindrical Core Cutter to cut the core from fine-grained natural or compacted soils, made of steel, complete as per IS: 2720 (part 29) including providing training to the laboratory staff at site and calibration by NABL accredited calibration lab with traceability. | each | 4200.00 |
| 22.43 | Providing 150 mm diameter size Crushing Value Test Apparatus to determine crushing strength of aggregate, made of steel, complete with cylindrical cell, plunger, base plate, tamping rod, metal measure etc, as per IS: 2386 (part 4) including providing training to the laboratory staff at site and calibration by NABL accredited calibration lab with traceability. | each | 16800.00 |
| 22.44 | Providing Cylindrical Metal Measures of 3, 15 and 30 litres capacity with temping rods to determine bulk density and voids of aggregate, made of steel, complete as per IS: 2386 (part 3) including providing training to the laboratory staff at site and calibration by NABL accredited calibration lab with traceability. | each | 21600.00 |
| 22.45 | Providing and fixing electrically operated Centrifuge Extractor as per IS: 19008:2023 to determine the binder percentage of bituminous materials, including installation, providing training to the laboratory staff at site and calibration by NABL accredited calibration lab with traceability.<br>The unit must be compact with built-in dimmer stat for speed variation from 0 to 3600 rpm. | each | 40800.00 |
| 22.46 | Providing Ring and Ball Apparatus as per IS:1205 to determine the Softening Point of bitumen, including installation, providing training to the laboratory staff at site and calibration by NABL accredited calibration lab with traceability.<br><br>The apparatus must be compact with built-in magnetic stirrer with heating facility and digital display of temperature, the heating can be adjusted through knob and can be controlled through the PID control. | each | 39600.00 |
| 22.47 | Providing and fixing electrically operated Ductility Testing Machine with 3 moulds and base plates as per IS:1208 to determine the Ductility of bitumen, including installation, providing training to the laboratory staff at site and calibration by NABL accredited calibration lab with traceability.<br>The apparatus must consist of a carriage moving over a lead screw, an electric motor driven reduction gear unit, an electric pump for water circulation, a heater and a thermostatically temperature control unit. Moving speed of carriage must be 5 cm/min and 1 cm/min both. | each | 136800.00 |
| 22.48 | Providing Vicat Apparatus as per IS:5513 to determine the consistency of standard cement paste and initial and final setting times of cement, including installation, providing training to the laboratory staff at site and calibration by NABL accredited | each | 4560.00 |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| | calibration lab with traceability. | | |
| 22.49 | Providing and fixing electrically operated Laboratory Concrete Mixture of capacity 1 cube feet. The mixer consists of a steel vessel of 30 liters, mounted on a frame. The vessel is rotated at 20-24 RPM with the help of a motor and a pulling. The vessel can be tilted to any angle by a handle. This facilitates mixing and discharge. Blades are provided inside the vessel to mix the material thoroughly. The large handle strip facilitates manual rotation of the drum during power failure. The drum, handle and motor etc. are mounted on a steel frame. This model is provided with two large wheels for carting away. Supplied complete, with motor of 1 HP along with lead wire.<br>The supply of equipment is included installation, providing training to the laboratory staff at site and calibration by NABL accredited calibration lab with traceability. | each | 86400.00 |
| 22.50 | Providing Slump Test Apparatus as per IS:7320 to determine the slump of fresh concrete as a measure of its consistency.<br>The supply of equipment is included installation, providing training to the laboratory staff at site and calibration by NABL accredited calibration lab with traceability. | each | 4800.00 |
| 22.51 | Providing and fixing of Digital Compression Testing Machine, loading range up to 2000 kN, with electrical load indicators and a visual display, to fulfil the requirements of IS:14858-2000 and IS:516 to test the concrete and mortar.<br><br>The supply of machine is included installation, providing training to the laboratory staff at site and calibration by NABL accredited calibration lab with traceability. | each | 778800.00 |
| 22.52 | Providing of stainless-steel double wall Water Bath, Size 600 x 300 x 200 mm, with heating and cooling arrangements and thermostatically temperature control system.<br>The supply of equipment is included installation, providing training to the laboratory staff at site and calibration by NABL accredited calibration lab with traceability. | each | 93600.00 |
| 22.53 | Providing Sand Pouring Cylinder Apparatus to determine the dry density of natural or compacted soil as per IS:2720 (Part 28). The apparatus consists of: Sand Pouring Cylinder fitted with conical funnel & shutter - cap. 16.5 lt., Cylindrical Calibration container - 200 mm ID x 250 mm ht., Metal Tray - size 45x45x5 cm with central hole.<br>The supply of apparatus is included installation, providing training to the laboratory staff at site and calibration by NABL accredited calibration lab with traceability. | each | 21600.00 |
| 22.54 | Providing speedy moisture meter as per IS 2720: Section -5 to determine the moisture content of soil, sand and other fine-grained material. | each | 13200.00 |

| Item No. | Description | Unit | Rate |
|---|---|---|---|
| | The supply of instrument is included installation, providing training to the laboratory staff at site and calibration by NABL accredited calibration lab with traceability. | | |
| 22.55 | Providing Le-Chatelier Mould as per IS 5514 – 1996,2021. The supply of mould is included installation, providing training to the laboratory staff at site and calibration by NABL accredited calibration lab with traceability. | each | 840.00 |
| 22.56 | Providing mould Size 150 x 150 x 700 mm for flexural Prism as per IS:10086. The supply of mould is included providing training to the laboratory staff at site and calibration by NABL accredited calibration lab with traceability. | each | 7680.00 |
| 22.57 | Providing mould Size 150 x 150 x 150 mm for flexural Prism as per IS:10086. The supply of mould is included providing training to the laboratory staff at site and calibration by NABL accredited calibration lab with traceability. | each | 2280.00 |
| 22.58 | Providing mould, Size 70.6 x 70.6 x 70.6 mm for cement testing as per IS: 10080. The supply of mould is included providing training to the laboratory staff at site and calibration by NABL accredited calibration lab with traceability. | each | 4320.00 |
| 22.59 | Providing Controlled Temperature Curing Tank for curing of concrete cubes. Size 900 x 450 x 500 mm with a hinged lid, heating & cooling arrangements, thermostatically temperature control & setting unit and recirculation pump. There must be provision of two removeable racks allowing free circulation of water around each cube. The pump, drain valves and electrical equipment must be housed in a compartment located at one end of the tank. The supply of equipment is included installation, providing training to the laboratory staff at site and calibration by NABL accredited calibration lab with traceability. | each | 148800.00 |
| 22.60 | Providing Flash and Fire Apparatus as per IS:1209 method A, B and C to determine Flash Point and Fire Point of Tar and Bituminous Materials. The supply of apparatus is included installation, providing training to the laboratory staff at site and calibration by NABL accredited calibration lab with traceability. | each | 51600.00 |
| 22.61 | Providing Solubility Determination Apparatus as per IS 1206:2023 method A to determine solubility of asphaltic bitumen in trichloroethylene. The supply of apparatus is included installation, providing training to the laboratory staff at site and calibration by NABL accredited calibration lab with traceability. | each | 42000.00 |

Note:- (i)Prior Permission of Chief Engineer is mandatory for item no 21.01 to 21.61.

Note:- (ii) For this items from 21.01to 21.61 the agency will get calibrated and keep maintained the machines for 3 years. In case of failure of agency, the machines shall be repaired and calibrated from the P.G. of contractor.

# APPENDIX

# MPPWD SCHEDULE OF TESTING CHARGES

| S.No. | Material of Test | Specified Test Performed | Reference Test Method | Quantity of Sample | Testing Charges Per Test (Rs.) |
|---|---|---|---|---|---|
| | | | | | |
| 1 | 2 | 3 | 4 | 5 | 6 |
| 1 | Cement | Standared Consistency | IS:4031 (Part-4) - 1988 | Min. 1 ealed Bag of 50 Kgs. | 385.00 |
| | | Initial and Final Setting Time | IS:4031 (Part-5) - 1988 | | 385.00 |
| | | Compressive Strength of Cement (For 2 Ages) | IS:4031 (Part-6) - 1988 | | 855.00 |
| | | Soundness Test by Le Chatelier Method | IS:4031 (Part-3) - 1988 | | 620.00 |
| | | Fineness test by Blains Air Premiability method | IS:4031 (Part-2) - 1989 | | 650.00 |
| | | Adultration by Bromoform Method | - | | -665.00 |
| 2 | Aggregate | Sieve Analysis/Fineness Modulus (Coarse Aggregate) | IS:2386 (Part-1) - 1963 | Min.25 Kgs. | 690.00 |
| | | Sieve Analysis/Fineness Modulus (Fine Aggregate) | IS:2386 (Part-1) - 1963 | Min.5 Kgs. | 520.00 |
| | | Flakiness and Elongation Index | IS:2386 (Part-1) - 1963 | Min.50 Kgs. | 420.00 |
| | | Water Absorption (Course Aggregate) | IS:2386 (Part-3) - 1963 | Min.5 Kgs. | 265.00 |
| | | Water Absorption (Fine Aggregate) | IS:2386 (Part-3) - 1963 | Min.2 Kgs. | 265.00 |
| | | Specific Gravity (Coarse Aggregate) | IS:2386 (Part-3) - 1963 | Min.5 Kgs. | 265.00 |
| | | Specific Gravity (Fine Aggregate) | IS:2386 (Part-3) - 1963 | Min.2 Kgs. | 265.00 |
| | | Aggregate Impact Value | IS:2386 (Part-4) - 1963 | Min.10 Kgs. | 455.00 |
| | | Crushing Value | IS:2386 (Part-4) - 1963 | Min.20 Kgs. | 455.00 |
| | | Bulk Density (Coarse Aggregate) | IS:2386 (Part-3) - 1963 | Min.30 Kgs. | 200.00 |
| | | Bulk Density (Fine Aggregate) | IS:2386 (Part-3) - 1963 | Min.7 Kgs. | 265.00 |
| | | Soundness (Coarse Aggregate) (5 Cycle) | IS:2386 (Part-5) - 1963 | Min.10 Kgs. | 2125.00 |
| | | Soundness (Fine Aggregate) (5 Cycle) | IS:2386 (Part-5) - 1963 | Min.10 Kgs. | 1875.00 |
| | | Delitrious Material (Coarse Aggregate) | IS:2386 (Part-2) - 1963 | Min.10 Kgs. | 705.00 |
| | | Delitrious Material (Fine Aggregate) | IS:2386 (Part-2) - 1963 | Min.5 Kgs. | 590.00 |
| | | Material Finer Than 75 u | IS: 2386 (Part-1)-1963 | Min.25 Kgs. | 280.00 |
| | | 10% Fine Value | IS: 2386 (Part-4)-1963 | Min.20 Kgs. | 615.00 |
| | | Bulking of Fine Aggregate | IS: 2386 (Part-3)-1963 | Min.10 Kgs. | 460.00 |
| | | Los Angeles Abrasion | IS: 2386 (Part-4)-1963 | Min.20 Kgs. | 560.00 |
| | | Silt Content by Volume | CPWD Speci. Vol.1 | Min.2 Kgs. | 390.00 |
| | | Stripping Value of Agregate | IS:6241-1971 | Min. Agg. 5Kgs + Bit1 kg | 325.00 |
| 3 | Bricks | Compressive Strength | IS: 3495 (Part-1) - 1992 | Min.5 Kgs. | 550.00 |
| | | Water Absorption | IS: 3495 (Part-2) - 1992 | Min.5 Kgs. | 290.00 |
| | | Eflorescence | IS: 3495 (Part-3) - 1992 | Min.5 Kgs. | 235.00 |
| | | Dimensions | IS: 1077 - 1992 | Min.10 Kgs. | 180.00 |

| S.No. | Material of Test | Specified Test Performed | Reference Test Method | Quantity of Sample | Testing Charges Per Test (Rs.) |
|---|---|---|---|---|---|
| | | | | | |
| 4 | Soil | Sieve Analysis | IS: 2720 (Part - 4) 1985 | Min.7 Kgs. | 655.00 |
| | | Optimum Moisture Content and Maximum Dry Density | IS: 2720 (Part - 7) 1980<br>IS: 2720 (Part - 8) 1983 | Min.10 Kgs. | 1270.00 |
| | | Liquid Limit | IS: 2720 (Part - 5) 1985 | Min.5 Kgs. | 550.00 |
| | | Plastic Limit | IS: 2720 (Part - 5) 1985 | Min.5 Kgs. | 290.00 |
| | | Shrinkage Limit | IS: 2720 (Part - 6) 1972 | Min.5 Kgs. | 540.00 |
| | | C. B. R. (Souked) | IS: 2720 (Part - 16) 1987 | Min.35 Kgs. | 2560.00 |
| | | C. B. R. (Un Souked) | IS: 2720 (Part - 16) 1987 | Min.35 Kgs. | 1645.00 |
| | | Classification of Soil by Serve Analysis and Determination of Coefficient of Uniformity and Coefficient of Curvature | IS: 1498 - 1970 | Min.10 Kgs. | 1040.00 |
| | | Free Swelling Index | IS: 2720 (Part - 40) 1977 | Min.1 Kgs. | 310.00 |
| | | Specific Gravity | IS: 2720 (Part - 3) 1980 | Min.1 Kgs. | 310.00 |
| | | Natural Moisture Content | IS: 2720 (Part - 2) 1973 | Min.1 Kgs. | 310.00 |
| 5 (A) | Bitumen | Penetration | IS: 1203 - 1978 | Min.1 Kgs. | 480.00 |
| | | Absolute Viscosity | IS: 1206 (Part-2) | Min.1 Kgs. | 540.00 |
| | | Ductility | IS: 1208 - 1978 | Min.1 Kgs. | 800.00 |
| | | Specific Gravity | IS: 1202 - 1978 | Min.1 Kgs. | 295.00 |
| | | Softening Point | IS: 1205 - 1978 | Min.1 Kgs. | 480.00 |
| | | Flash Point & Fire Point | IS: 1209 - 1978 | Min.1 Kgs. | 480.00 |
| | | Rate of Spray of Binder | IS: 11 - 1976 | - | 365.00 |
| 5 (B) | Bitumen mix | Bitumen Binder Content | SP: 11-1988. ASTM D 2172. | Min.5 Kgs. | 1115.00 |
| | | Marshal Stability | ASTM D 1559 | Min.20 Kgs. | 1540.00 |
| | | Bituminus Mix Design Lob Mix Formula for Bituminus Mix) by Marshal Stability Method | IRC 29-1988 | Min Bit. 20 Kgs.+ Agg. 50 Kgs. | 10835.00 |
| 6 | Steel | Determination of Yeild Strength, Tensile Strength and Elongation of Steel Bar of All Diameter with the Help of UTM | IS:1608 2005 | 3 Nos, of Min 1000 mm Long | 2170.00 |
| | | Weight Per Meter | IS:1786-2008 | 3 Nos, of Min 1000 mm Long | 235.00 |
| | | Bend & Rebend Test | IS: 1599-1985, IS:1786-2008 | 3 Nos, of Min 1500 mm Long | 875.00 |

| S.No. | Material of Test | Specified Test Performed | Reference Test Method | Quantity of Sample | Testing Charges Per Test (Rs.) |
|---|---|---|---|---|---|
| | | | | | |
| 7 | Concrete | Mix Design Upto M25 Grade | IS: 10262 - 1982, IS 456, SP23 | Min. Cement 50 Kgs. +Sand 80 Kgs + Agg (10mm): 80 Kgs. + Agg (20mm): 120 Kgs. | 6890.00 |
| | | Mix Design Above M25 Grade | IS 10262 - 1982 | Min. Cement 50 Kgs. +Sand 80 Kgs + Agg (10mm): 80 Kgs. + Agg (20mm): 120 Kgs. | 7860.00 |
| | | Compressive Strength of Cube (15x15x15 CMS) | IS 516 - 1959 | 3 Nos. | 380.00 |
| | | Compressive Strength of Cylinderical, Concrete Specimen | IS 516 - 1960 | 3 Nos. | 380.00 |
| | | Determination of ratio of Cement, Sand and Aggregate in Hardened Concrete | IS: 1199 - 1959 | Min. Cement 10 Kgs. + Cement Used: 1Kg. + Sand Used 1Kg. | 3880.00 |
| 8 | Cement Mortar | Determination of ratio of Cement, Sand and Aggregate in Hardened Concrete | IS: 1199 - 1959 | Min. Cement 5 Kgs. + Cement Used: 1Kg. + Sand Used 1Kg. | 3500.00 |
| 9 | Concre Paver Block | Water Absorption | IS: 15658: 2006 | 3 Nos. | 290.00 |
| | | Abrasion Resistance Value | IS: 15658: 2006 | 8 Nos. | 1035.00 |
| | | Compressive Strength | IS: 15658: 2006 | 8 Nos. | 495.00 |
| 10 | Concrete/ Terrazo/ Chequered Flooring Tiles | Water Absorption | IS: 1237 - 1980/IS 13801 | 6 Nos. | 290.00 |
| | | Resistance to wear | IS: 1237 - 1980/IS 13801 | 6 Nos. | 800.00 |
| | | Wet. Transverse Strength | IS: 1237 - 1980/IS 13801 | 6 Nos. | 480.00 |
| 11 | Glazed Ceramic Vetrified Tiles | Water Absorption | IS: 13630 (Part - 2) 2006 | 5 Nos. | 290.00 |
| | | Modulus of Rupture | IS: 13630 (Part - 6) 2006 | 5 Nos. | 480.00 |
| | | Scratch Hardnees According To MOHS | IS: 13630 (Part - 13) 2006 | 3 Nos. | 760.00 |
| | | Dimensions/Surfase Quality | IS: 13630 (Part - 1) 2006 | 10 Nos. | 265.00 |
| | | Abrasion | IS: 13630 (Part - 11) 2006 | 15 Nos. | 800.00 |
| | | Chemical Resistance For Stains | IS: 13630 (Part - 2) 2006 | 5 Nos. | 690.00 |
| 12 | Marble/Granite/ Kota Stone | Scratch Hardnees According To MOHS | IS: 13630 (Part - 13) 2006 | 3 Nos. | 760.00 |
| 13 | Timber | Moisture Content | IS: 1708 (Part-1) 1986 | 1 No. of 5x5x2.5 Cms | 415.00 |
| | | Flush Door END Emersion test | IS: 4020 (Part-13) 1986 | 1 Door Shutter | 520.00 |
| | | Flush Door Knife Test | IS: 4020 (Part-14) 1986 | 1 Door Shutter | 520.00 |
| | | Fluse Door Adhesion Test | IS: 4020 (Part-15) 1986 | 1 Door Shutter | 520.00 |
| 14 | Hollow Concerete Block | Compressive Strength | IS: 2185 (Part-1) 1979 | 8 Nos. | 495.00 |
| | | Water Absorption | IS: 2185 (Part-1) 1979 | 3 Nos. | 290.00 |
| | | Block Density | IS: 2185 (Part-1) 1979 | 3 Nos. | 345.00 |

| S.No. | Material of Test | Specified Test Performed | Reference Test Method | Quantity of Sample | Testing Charges Per Test (Rs.) |
|---|---|---|---|---|---|
| | | | | | |
| 15 | Field Test | Plate Load Test (SBC) (Without Lab Conveyance) | IS: 1888 - 1982 | - | 15225.00 |
| | | Standard Penetration Test (Without Lab Conveyance) | IS: 2131 - 1981 | - | 8330.00 |
| | | Field density by Sand Replacement Method (Without Lab Conveyance) | IS: 2720 (Part-28) 1974 | - | 790.00 |
| | | Field density by Core Cutter Method (Without Lab Conveyance) | IS: 2720 (Part-29) 1975 | - | 420.00 |
| | | Slump Test (3 Trials) | IS: 1199 - 1959 | | 380.00 |
| | | 3 Nos. Core of 15 Cms Dia and Max 30 Cms in Length Obtained from Cutting of Bituminus Pavement (Without Lab Conveyance) | - | - | 3000.00 |
| | | 3 Nos. Core of 15 Cms Dia and Max 30 Cms in Length Obtained from Cutting of Concrete Pavement (Without Lab Conveyance) | - | - | 3000.00 |
| 16 | Collection of Sample | Upto 8 Kms (Without Lab Conveyance) | - | - | 800.00 |
| | | Beyond 8 Kms (Without Lab Conveyance) | - | - | 1600.00 |

# BASIC QUALITY CONTROL

# TESTS & THEIR FREQUENCIES

## BASIC QUALITY CONTROL TESTS& THEIR FREQUENCIES

## FLEXIBLE PAVEMENT.

### A) Sub-grades

| S. No. | Type of Construction | Test | Frequency (Min.) | Reference |
|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 |
| 1) | **Subgrade /Earth work. & Earthen Shoulders** | i) CBR.<br>ii) Plastic Index.<br>iii) Moisture Content Prior to compaction.<br>iv) Density of compacted layer. | One test (average of 3 specimens) per 3000 cu.m.<br>2 test per 3000 cum.<br>2 test per 3000 cum.<br>One set of 10 measurements of Density for each 3000 sqm. on each layer. | MOR T & H Specifications 2013 Section 900 ( 903.2) |

### B) Sub-base & base work.

| S. No. | Type of Construction | Test | Frequency (min.) | Reference |
|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 |
| 1) | **Lime/Cement /Fly AshStabilised Soil Sub Base.** | A)Quality of cement<br>a) Consistency.<br>b) Initial & Final setting time.<br>c) Fineness.<br>d) Specific Gravity.<br>e) Soundness.<br>f) Compressive strength. | One test for each consignment subject to a minimum of one test per 5 tonnes. | MORT & H Specifications 2013 Section 900 (Table 900-3) |

| | | | | |
|---|---|---|---|---|
| | | **B)**Quality of FlyAsh:-<br>i) Specific Gravity.<br><br>ii)Fineness.<br><br>iii) CBR<br><br>iv) Moisture Content prior to compaction.<br><br>v) Density of compacted layer. | One tests for each consignment.<br><br>One tests for each consignment.<br><br>Two test per every 3000 cum.<br><br>One test per every 250 cum.<br><br>At least one measurement of density for each 1000 sqm. | IS:2720 (Part -8) 1983<br><br>IS: 2720 (Part -8) 1983<br><br>IS: 2720 (Part -2) 1973<br><br>IS 1720 (Part : 28) |
| 2) | **Granular Sub-base** | i) Gradation<br>ii) Atterberg limits<br>iii) Moisture content prior to compaction<br><br>iv)Densityof compacted layer<br><br>v) Wet aggregate Impact value if water absorption of the aggregate is more than 2%<br><br>vi) CBR | One test per 400 cu.m<br>One test per 400 cu.m<br>One test per 400 cu.m<br><br>One test per 1000 sq.m.<br><br>One test per source.<br><br>As required. | MORTH Specifications 2013 section 900 (Table -900-3) |
| 3) | **Water Bound Macadam** | i) Aggregate Impact Value.<br><br>ii)Grading of aggregate<br><br>iii)Combined flakiness and ElongationIndices.<br><br>iv) Atterberg limits of binding material.<br><br>v) Atterberg limits of screenings. | One test per 1000 cu.m. of aggregate.<br><br>One tests per 250 cu.m<br><br>One test per 500 cu.m. of aggregate.<br><br>One test per 50 cu.m of binding material.<br><br>One test per 100 cu.m. of | MORTH Specifications 2013 section 900 (Table -900-3) |

| | | | | |
|---|---|---|---|---|
| 4) | Wet Mix Macadam | i) Aggregate Impact Value.<br><br>ii)Grading of aggregate.<br><br>iii)Combined flakiness andElongation Indices.<br><br>iv) Atterberg limits of portion of passing 425 micron sieve.<br><br>v) Density of compacted layer | One test per 1000 cu.m. of aggregate.<br><br>One test per 200 cu.m.of aggregate.<br><br>One test per 500 cu.m. of aggregate.<br><br>One test per 200 cu.m.of aggregate.<br><br>One set of three tests per 1000sqm. | MORTH Specifications 2013 section 900 (Table -900.4) |
| 5 | BBM / MPM . | i) Quality of binder<br><br>a) Specific Gravity<br><br>b) Flash and Fire point,<br><br>c) Softening point<br><br>d) Penetration Test<br><br>e) Ductility Test<br><br>f) Viscosity. (Absolute ,Kinematic)<br><br>g) Binder temperature for application<br><br>ii) Aggregate. Impact Value.<br><br>iii)Grading of aggregate.<br><br>iv)Combined flakiness andElongation Indices.<br><br>v)Stripping Value<br><br>v) Water absorption of | Two Sample per lot to be subjected to all or some test as directed by engineer.<br><br>At regular close interval<br><br>One test per 2000cu.m. of aggregate.<br><br>One tests per 100cu.mof aggregate.<br><br>One tests per 200 cu.mof aggregate .<br><br>Initially One set of three representative specimens for each source of supply subsequently when warranted by changes in the quality of aggregate . | IRC:SP:20-2002 |

<table>
<tr><td></td><td></td><td>aggregate .<br><br><br><br><br>vi) Temperature of binder at application aggregate .<br><br>vii) Rate of spread of binder .</td><td>Initially One set of three representative specimens for each source of supply subsequently when warranted by changes in the quality of aggregate .<br><br>At regular close intervals.<br><br><br><br>One tests per 500sq.m of area.</td><td></td></tr>
<tr><td>6</td><td>BUSG</td><td><u>i) Quality of binder</u><br><br>a) Specific Gravity<br><br>b) Flash and Fire point,<br><br>c) Softening point<br><br>d) Penetration Test<br><br>e) Ductility Test<br><br>f) Viscosity. (Absolute ,Kinematic)<br><br>g) Binder temperature for application<br><br>ii) Aggregate. Impact Value. or Abrasion Value<br><br>iii)Grading of aggregate.<br><br>iv)Combined flakiness and Elongation Indices.<br><br>v)Stripping Value</td><td><table><tr><th>Lot size</th><th>No. of container</th></tr><tr><td>Up to 50 lots</td><td>3</td></tr><tr><td>51 to 150</td><td>5</td></tr><tr><td>151 to 500</td><td>7</td></tr><tr><td>501&above</td><td>10</td></tr></table><br><br><br><br><br><br><br><br><br>One test per 200 cu.m. of aggregate.<br><br>One tests per 100 cu.m of aggregate.<br><br>One tests per 200 cu.m of aggregate .<br><br>Initially One set of three representative specimens for each source of supply subsequentlywhen warranted</td><td>IRC:47-1972<br>IS -73<br>As frequency for BUSG is not mention in IRC:47-1972 hence frequency for BBM & MPM as perIRC:SP-20-2002 is adopted.<br><br><br><br><br><br><br><br><br><br><br>IS -2386 (Part-4)<br><br>IRC:47-1972<br><br>IS -2386 (Part-1)<br><br>IS -6241</td></tr>
</table>

| | | | | |
|---|---|---|---|---|
| | | vi) Water absorption of aggregate . | aggregate .<br><br>Initially One set of three representative specimens for each source of supply subsequently when warranted by changes in the quality of aggregate . | IS -2386 (Part-3) |
| 7 | Hard Murum | i)Atterbergs Limit | One test per 400 cu.m. | MORTH Specifications 2013 clause 900 (Table 900-3) |

## C)Bituminous works.

<table>
<tr><th>Sr. No.</th><th>Type of Constructio n</th><th>Test</th><th>Frequency (min.)</th><th>Reference</th></tr>
<tr><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr>
<tr><td></td><td>Prime coat/ Tack Coat</td><td><u>i) Quality of binder</u><br><br>a) Specific Gravity<br><br>b) Flash and Fire point,<br><br>c) Softening point<br><br>d) Penetration Test<br><br>e) Ductility Test<br><br>f) Viscosity. (Absolute ,Kinematic)</td><td><table><tr><th>Lot size</th><th>No. of container</th></tr><tr><td>Up to 50 lots</td><td>3</td></tr><tr><td>51 to 150</td><td>5</td></tr><tr><td>151 to 500</td><td>7</td></tr><tr><td>501&above</td><td>10</td></tr></table></td><td>MORT H Specifications 2013 section 900 (Table 900-4) IS:73 IS:217 & IS:8887.</td></tr>
<tr><td></td><td></td><td>ii) Binder temperature for application<br><br>iii) Rate of spread of Binder</td><td>At regular close intervals<br><br>Three tests per day</td><td></td></tr>
</table>

<table>
<tr>
<td>2)</td>
<td>Seal coat/ Surface Dressing</td>
<td><u>i) Quality of binder</u><br><br>a) Specific Gravity<br><br>b) Flash and Fire point,<br><br>c) Softening point<br><br>d) Penetration Test<br><br>e) Ductility Test<br><br>f) Viscosity. (Absolute ,Kinematic)</td>
<td>
<table>
<tr><th>Lot size</th><th>No. of container</th></tr>
<tr><td>Up to 50 lots</td><td>3</td></tr>
<tr><td>51 to 150</td><td>5</td></tr>
<tr><td>151 to 500</td><td>7</td></tr>
<tr><td>501&above</td><td>10</td></tr>
</table>
</td>
<td>MORT H Specifications 2013 section 900 (Table 900-4) IS:73 IS:217 & IS:8887.</td>
</tr>
<tr>
<td></td>
<td></td>
<td>ii) Aggregate Impact Value<br><br>iii)Combined flakiness and elongation Indices<br><br>iv) Stripping value of aggregate (Immersion Tray Test)<br><br>v) Water absorption of aggregate<br><br>vi)Grading of aggregate<br><br>vii) Temperature of binder in boiler, in dryer and mix at the time of laying andcompaction.<br><br>ix) Rate of spread of material</td>
<td>One test per 200 cu.m.of each source and whenever there is change in the quality of aggregate<br><br>One test per 100 cu.m. of aggregate for each source and whenever there is change in the quality of aggregate<br><br>One test of each source and whenever there is change in the quality of aggregate<br><br>One test of each source and whenever there is change in the quality of aggregate<br><br>Two tests per day.<br><br>At regular intervals<br><br>Three test per day.</td>
<td></td>
</tr>
</table>

<table>
<tr>
<td>3)</td>
<td>Open-graded Premix Surfacing/ Close-graded Premix Surfacing</td>
<td><u>i) Quality of binder</u><br><br>a) Specific Gravity<br><br>b) Flash and Fire point,<br><br>c) Softening point<br><br>d) Penetration Test<br><br>e) Ductility Test<br><br>f) Viscosity. (Absolute ,Kinematic)</td>
<td>
<table>
<tr><th>Lot size</th><th>No. of container</th></tr>
<tr><td>Up to 50 lots</td><td>3</td></tr>
<tr><td>51 to 150</td><td>5</td></tr>
<tr><td>151 to 500</td><td>7</td></tr>
<tr><td>501&above</td><td>10</td></tr>
</table>
</td>
<td>MORT H Specifications 2013 section 900 (Table 900-4) IS:73 IS:217 & IS:8887.</td>
</tr>
<tr>
<td></td>
<td></td>
<td>ii) Aggregate. Impact Value.<br><br>iii) Combined Flakiness and Elongation Indices<br><br>iv) Stripping value<br><br>v) Water absorption of aggregate.<br><br>vi) Grading of aggregate.<br><br>vii) Temperature of binder at application</td>
<td>One test per 200 cu.m.of each source and whenever there is change in the quality of aggregate.<br><br>One test per 100 cu.m. of for each source and whenever there is change in the quality of aggregate.<br><br>One test of each source and whenever there is change in the quality of aggregate.<br><br>One test of each source and whenever there is change in the quality of aggregate.<br><br>Two tests per day.<br><br>At regular intervals</td>
<td></td>
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| | | | | |
|---|---|---|---|---|
| 4) | **Bituminous Macadam** | i) Quality of binder<br><br>a) Specific Gravity<br><br>b) Flash and Fire point,<br><br>c) Softening point<br><br>d) Penetration Test<br><br>e) Ductility Test<br><br>f) Viscosity. (Absolute ,Kinematic) | Lot size / No. of container<br>Up to 50 lots — 3<br>51 to 150 — 5<br>151 to 500 — 7<br>501&above — 10 | MORT H Specifications 2013, section 900 (Table 900-4), IS:73, IS:217 & IS:8887. |
| | | ii) Aggregate Impact Value. | One test per 200 cu.m.of each source and whenever there is change in the quality of aggregate. | |
| | | iii) Combined Flakiness and Elongation Indices | One test per 350 cu.m. of aggregate for each source. | |
| | | iv) Stripping value | One test of each source and whenever there is change in the quality of aggregate. | |
| | | v) Water absorption of aggregate. | One test of each source and whenever there is change in the quality of aggregate. | |
| | | vi) Grading of aggregate. | Two tests per day. | |
| | | vii) Binder content. | Two tests per day per plant. | |
| | | viii)Control of temperature of binder &aggregate for mix aggregate of the mix at the time of laying & rolling. | At regular intervals. | |
| | | ix) Density of compacted Layer. | One test per 700 Sq.mt.area. | |
| | | x) Rate of Spread of | At regular intervals. | |

<table>
<tr>
<td>5)</td>
<td>Dense Bituminous Macadam/Bituminous Concrete</td>
<td><u>i) Quality of binder</u><br><br>a) Specific Gravity<br><br>b) Flash and Fire point,<br><br>c) Softening point<br><br>d) Penetration Test<br><br>e) Ductility Test<br><br>f) Viscosity. (Absolute ,Kinematic)</td>
<td>
<table>
<tr><th>Lot size</th><th>No. of container</th></tr>
<tr><td>Up to 50 lots</td><td>3</td></tr>
<tr><td>51 to 150</td><td>5</td></tr>
<tr><td>151 to 500</td><td>7</td></tr>
<tr><td>501&above</td><td>10</td></tr>
</table>
</td>
<td>MORT H Specifications 2013 section 900 (Table 900-4)<br>IS :73<br>IRC SP 53<br>IS:15462</td>
</tr>
<tr>
<td></td>
<td></td>
<td>ii) Aggregate Impact Value.</td>
<td>One test per 350 cu.m ofaggregate for each source and whenever there is change in the quality of aggregate.</td>
<td></td>
</tr>
<tr>
<td></td>
<td></td>
<td>iii) Flakiness and Elongation Indices</td>
<td>One test per 350 cu.m of aggregate for each source and whenever there is change in the quality of aggregate.</td>
<td></td>
</tr>
<tr>
<td></td>
<td></td>
<td>iv) Water absorption of aggregate.</td>
<td>One test for each source and whenever there is change in the quality of aggregate.</td>
<td></td>
</tr>
<tr>
<td></td>
<td></td>
<td>v) Mix grading</td>
<td>One set for individual constituent and mixed aggregate from dryer for each 400 tones of mix subject to minimum of two tests per day per plant</td>
<td></td>
</tr>
<tr>
<td></td>
<td></td>
<td>vi) Stability and voids analysis of mix including theoretical maximum specific of loose mix</td>
<td>Three tests for stability, flow value, density and void contents for each 400 tonnes of mix subject to minimum of two tests per day per plant.</td>
<td></td>
</tr>
<tr>
<td></td>
<td></td>
<td>vii) Temperature of binder in boiler,</td>
<td>At regular intervals.</td>
<td></td>
</tr>
</table>

<table>
<tr><td></td><td></td><td>indryer and<br>mix at the time of<br>laying and<br>compaction.<br><br>viii) Binder content<br><br><br><br>ix) Rate of spread of<br>mix<br><br>x) Density of<br>Compacted layer</td><td><br><br><br><br>One set for each 400<br>tonnesof mix subject to<br>minimum of two tests per<br>day per plant<br><br>After every 5th truck load<br><br><br>One test per 700 sq.m area</td><td></td></tr>
<tr><td>6)</td><td>Slurry seal<br>and Micro<br>surfacing.</td><td><u>i) Quality of binder</u><br><br>a) Specific Gravity<br><br>b) Flash and Fire<br>point,<br><br>c) Softening point<br><br>d) Penetration Test<br><br>e) Ductility Test<br><br>f) Viscosity.<br>(Absolute<br>,Kinematic)<br><br>ii) Aggregate<br>Moisture.<br><br>iii) Aggregate<br>Gradation.<br><br>iv) Binder Content.<br><br>v) Quantity of<br>Slurry.<br>(By weight of )</td><td><table><tr><th>Lot size</th><th>No. of container</th></tr><tr><td>Up to 50 lots</td><td>3</td></tr><tr><td>51 to 150</td><td>5</td></tr><tr><td>151 to 500</td><td>7</td></tr><tr><td>501&above</td><td>10</td></tr></table><br><br><br><br><br><br><br><br><br><br><br>Two test per day..<br><br>Two test per day at site.<br><br>Two test per day.<br><br>Daily (Travel time of<br>Machine)</td><td>MORTH<br>Specifications<br>2013 section<br>900<br>(Table 900-4)<br>IS:73<br>IS:217 &<br>IS:8887.<br><br>Remarks :<br>As per<br>MORTH,<br>SpecificationE<br>mulsion is to<br>usedin<br>slurryseal &<br>micro<br>surfacing.</td></tr>
</table>

| | | | | |
|---|---|---|---|---|
| 7) | **Mastic Asphalt** | <u>i) Quality of binder</u><br><br>a) Specific Gravity<br><br>b) Flash and Fire point,<br><br>c) Softening point<br><br>d) Penetration Test<br><br>e) Ductility Test<br><br>f) Viscosity. (Absolute ,Kinematic)<br><br>ii) Aggregate Impact Value.<br><br>iii) Combined Flakiness and Elongation Indices<br><br>iv) Striping value<br><br>v) Grading of aggregates.<br><br>vi) Water absorption of aggregates.<br><br>vii) Binder content and aggregates grading of mix material.<br><br>viii) Control of temperature of binder and aggregates for mixing and of the mix at the time of laying and rolling. | <table><tr><th>Lot size</th><th>No. of container</th></tr><tr><td>Up to 50 lots</td><td>3</td></tr><tr><td>51 to 150</td><td>5</td></tr><tr><td>151 to 500</td><td>7</td></tr><tr><td>501&above</td><td>10</td></tr></table><br>One tests per 350 cum.of aggregates for each source & whenever there is change in the quality of aggregates.<br><br>One tests per 350 cum.ofaggregates for each source & whenever there is change in the quality of aggregates.<br><br>One test of each source & whenever there is change in the quality of aggregates.<br><br>Two tests per day per plant on the individual constituent & mixed aggregate from the dryer.<br><br>One test for each source and whenever there is change in the quality of aggregates.<br><br>Two Test per day per plant.<br><br>At regular close intervals | MORT H Specifications 2013 section 900 (Table 900-4)<br>IS:73<br>IRC SP 53<br>IS: 15462. |

<table>
<tr><td></td><td></td><td>ix) Rate of Spread of MixedMaterial.</td><td>Regular control through check of layer thickness.</td><td></td></tr>
<tr><td>8)</td><td>Cold Mixes.</td><td><u>i) Quality of binder</u><br><br>a) Specific Gravity<br><br>b) Flash and Fire point,<br><br>c) Softening point<br><br>d) Penetration Test<br><br>e) Ductility Test<br><br>f) Viscosity. (Absolute ,Kinematic)<br><br>ii) Aggregate Impact Value.<br><br>iii) Combined Flakiness and Elongation Indices<br><br>iv) Water absorption of aggregate.<br><br>v) Grading of aggregate.<br><br>vi) Temperature of binder at application<br><br>vii) Binder content.</td><td><table><tr><th>Lot size</th><th>No. of container</th></tr><tr><td>Up to 50 lots</td><td>3</td></tr><tr><td>51 to 150</td><td>5</td></tr><tr><td>151 to 500</td><td>7</td></tr><tr><td>501&above</td><td>10</td></tr></table><br><br>One test per 350 cu.m.of each source and whenever there is change in the quality of aggregate.<br><br>One test per 350cu.m. of aggregates for each source and whenever there is change in the quality of aggregate.<br><br>One test of each source and whenever there is change in the quality of aggregate.<br><br>Two tests per day.<br><br>At regular intervals<br><br>Two tests per day.</td><td>MORT H Specifications 2013 section 900 (Table 900-4) IS:73 IRC SP 53 IS : 15462</td></tr>
</table>

## BASICQUALITY CONTROL TESTS & THEIR FREQUENCIES

## RIGID PAVEMENT

| Sr. No. | Material | Description of Test | Frequency Min. | Reference |
|---|---|---|---|---|
| 1) | Cement (Physical Tests) | i) Compressive Strength.<br>ii) Initial Setting Time.<br>iii) Final setting Time.<br>iv) SpecificGravity.<br>v) Soundness.<br>vi) Fineness. | Once for each source of supply & occasionally when called for in case of long/improper storage. Besides the contractor also will submit daily test data on cement released by the manufacturer. | M.O.R.T.H. Specification Section.903 (Table 900-6)<br>IS : 269<br>IS: 455<br>IS: 1489<br>IS: 8112<br>IS : 12269 |
| 2) | Aggregate | i) Aggregate Impact Value<br>iii) Gradation.<br>iii) Water absorption. | Once for each source of supply & subsequently on monthly basis.<br>one test for every days work of each fraction of course aggregate initially.<br>One test per day. | M.O.R.T.H. Specification 2013, Section 903. (Table 900-6)<br>IS 2386 (Part 4-IV)<br>IS 2386 (Part- III) |
| 3) | Sand | i) FinenessModulus.<br>ii)Water absorption. | At Beginning & if there is change in source.<br>Two tests per day. | PWD HAND BOOK Chapter-3,for RCC,&M.O.R.T.H. Specification 2013, Section 903. (Table 900-6)<br>IS : 2386 (Part- III) |